

TOWARDS OPTIMAL
MATRIX-VECTOR COMPLEXITY
IN NUMERICAL LINEAR ALGEBRA

DISSERTATION

Submitted in Partial Fulfillment of

the Requirements of

the Degree of

DOCTOR OF PHILOSOPHY (Computer Science)

at the

NEW YORK UNIVERSITY
TANDON SCHOOL OF ENGINEERING

by

Raphael Arkady Meyer

May 2024

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His work at NYU has been supported by an NYU SoE Fellowship, NSF grant 2045590, DOE Award DE-SC0022266, and a GAANN Fellowship from the US Department of Education.

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*This thesis is dedicated to all who supported me throughout my Ph.D.
Including, but not limited to, my family, friends, colleagues, and
partner.*

ABSTRACT

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by

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The world is full of important problems with linear structure. Many machine learning, data science, and scientific computing applications require subroutines that solve numerical linear algebra problems. By solving numerical linear algebra problems more quickly, we can scale up the models and datasets used in these applications, enabling bigger and better scientific computations.

In practice, most fast linear algebra algorithms interact with their input matrix only by computing matrix-vector products, so it is crucial to understand how many of these products are necessary and sufficient to solve numerical linear algebra problems. Motivated by this, we will explore the *Matrix-Vector Complexity* of such algorithms.

This thesis will cover recent progress in four fundamental problems in numerical linear algebra: trace estimation, Kronecker trace estimation, low-rank approximation, and low-rank approximation of matrix functions. We will examine fast and pragmatic randomized algorithms that solve these problems with low matrix-vector complexity. We then complement most of these algorithms with information-theoretic lower bounds on the number of matrix-vector products needed.

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Chapter 1

Intro

The world is full of important problems with linear-algebraic structure. There are tons of applications in scientific computing, data science, machine learning, and big-data that are well-described using vectors and matrices. Being able to solve numerical problems involving these vectors and matrices is therefore also really important. These numerical problems, roughly speaking, are referred to as problems in *Numerical Linear Algebra (NLA)*.

As datasets grow larger and machine learning models grow more complex, it becomes necessary to efficiently solve increasingly larger NLA problems. A more efficient NLA algorithm enables a more sophisticated machine learning model or scientific computing simulation. However, exactly solving most NLA problems on $n \times n$ matrices requires $O(n^3)$ time today¹. This is prohibitively expensive for large n , and we would instead like the dependence on n in the runtime of our algorithms to be at most quadratic since it takes $O(n^2)$ time to read the input matrix. Essentially all existing fast algorithms for NLA problems that have a runtime that scale as $O(n^2)$ can only achieve this because they are both *approximate* and *randomized* algorithms.

This thesis surveys recent progress in the theoretical understanding of randomized approximation algorithms for numerical linear algebra. We will focus on the results of four papers, each of which makes progress towards one of four specific NLA problems: *trace estimation*[83], *Kronecker trace estimation*[81], *low-rank approximation*[82], and $f(\mathbf{A})$ *low-rank approximation*[96].

There are common themes among all four papers. In the rest of this introduction, we will briefly discuss some of these problems and discuss the most central theme.

¹There exist a small but positive number of settings where the Strassen or Coppersmith-Winograd algorithms are used. We will focus on the typical case where the naive $O(n^3)$ time algorithm is most efficient.

1.1 The Matrix-Vector Oracle Model

One common computational problem that arises in practice is that of *spectral sum estimation*. That is, given a large symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and a function $f : \mathbb{R} \rightarrow \mathbb{R}$, we want to compute an (ε, δ) -approximation to $\text{tr}(f(\mathbf{A}))$. That is, we want an algorithm that returns a value t such that

$$|t - \text{tr}(f(\mathbf{A}))| \leq \varepsilon |\text{tr}(f(\mathbf{A}))|.$$

We typically assume that either $\mathbf{A}$ or $f(\mathbf{A})$ is PSD to make the problem more tractable. The problem of spectral sum estimation arises in many tasks. For instance:

- The log-determinant $\log \det(\mathbf{K}) = \text{tr}(\log(\mathbf{K}))$ of a kernel matrix $\mathbf{K}$ needs to be computed to evaluate the log-likelihood of a Gaussian Process when hyperparameter tuning its kernel function [123, 129].
- The Estrada Index $\text{tr}(e^{\mathbf{B}})$ of adjacency matrix $\mathbf{B}$ is used to measure the connectivity of a graph in public transit planning [126].
- The number of triangles in a graph, a fundamental quantity in theoretical computer science [11], can be computed as $\text{tr}(\frac{1}{6}\mathbf{B}^3)$, where $\mathbf{B}$ is the adjacency matrix of the graph [113].

We can abstract out all three cases by saying that we are interested in estimating $\text{tr}(\mathbf{A})$, where $\mathbf{A}$ is a function of a matrix stored in memory (either $\mathbf{K}$ or $\mathbf{B}$). Further, when $\mathbf{A}$ is a function of a matrix $\mathbf{B}$ stored in memory, we can often quickly compute matrix-vector products with $\mathbf{A}$ by running algorithms like *Lanczos-FA Iteration* with $\mathbf{B}$ [88]. As a simple example, this is certainly true for the third case above, where $\mathbf{A} = \frac{1}{6}\mathbf{B}^3$. In this case, we can easily compute $\mathbf{A}\mathbf{x}$ as $\frac{1}{6}\mathbf{B}(\mathbf{B}(\mathbf{B}\mathbf{x}))$ in just $O(n^2)$ time via repeated matrix-vector products. This is in contrast to directly computing the matrix $\mathbf{A}$ itself, using $O(n^3)$ or $O(n^\omega)$ time.

This setup, where we want to estimate the trace of $\mathbf{A}$ given only the ability to compute matrix-vector products with $\mathbf{A}$, sets up the first concrete theoretical problem studied in this thesis:

Definition 1 (Matrix-Vector Oracle). *Let $\mathbf{A} \in \mathbb{R}^{n \times d}$ be a matrix. Then, the Matrix-Vector Oracle for $\mathbf{A}$ is the oracle that, when given the vector $\mathbf{x}$ responds with the vector $\mathbf{A}\mathbf{x}$:*

$$\mathbf{x} \xrightarrow{\text{input}} \text{ORACLE} \xrightarrow{\text{output}} \mathbf{A}\mathbf{x}.$$

We say that an algorithm belongs to the Matrix-Vector Oracle Model if it only interacts with its input matrix via a matrix-vector oracle.

Problem 2 (Trace Estimation [7]). Fix $\varepsilon > 0$ and $\delta \in (0, 1)$. Suppose we have a Matrix-Vector Oracle to PSD matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$. Then, using as few queries to the oracle as possible, find an estimate $\tilde{t}$ such that

$$|\text{tr}(\mathbf{A}) - \tilde{t}| \leq \varepsilon \text{tr}(\mathbf{A})$$

holds with probability at least $1 - \delta$.

This abstracted problem of trace estimation from matrix-vector products has been well studied [43, 60, 6, 100, 83, 94, 38]. In [Chapter 2](#), we will show that it is possible to solve [Theorem 2](#) using $O(\frac{1}{\varepsilon})$ matrix-vector products, which results in a runtime of $O(n^2 \cdot \frac{1}{\varepsilon})$ in the common setting that computing a matrix-vector product with $\mathbf{A}$ takes $O(n^2)$ time.

Instead of proving this result, for now, we will instead examine the role of [Theorem 1](#) more closely. In the context of spectral sum estimation, the matrix-vector oracle arises naturally since algorithms like Lanczos-FA give us matrix-vector access to a matrix not stored in memory. Indeed, this is not the only setting where we care about designing efficient algorithms in the matrix-vector oracle model. For instance:

- In gradient-based optimization, we can often efficiently compute matrix-vector products with a Hessian matrix. By using this matrix-vector access, we can estimate the attributes of the Hessian which give information about the curvature of the loss function [2].
- When we know that $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a sparse matrix, we can compute $\mathbf{A}\mathbf{x}$ in $O(\text{nnz}(\mathbf{A}))$ time [46]. So, algorithms that can compute something about $\mathbf{A}$ just using a few matrix-vector products can run in time nearly linear in the description size of $\mathbf{A}$. That is, matrix-vector oracle algorithms are especially efficient on sparse matrices.
- Even when solving linear algebra problems on dense matrices stored in memory, we may prefer a matrix-vector algorithm to a non-matrix-vector algorithm. This is because of software packages like BLAS and parallelized hardware implementations of linear algebra primitives. Computing a matrix-vector product is very cheap compared to many other operations on a computer that would also hypothetically take $O(n^2)$ time.

Settings like these motivate why we want to design algorithms in the matrix-vector oracle model. This model has emerged as a convenient level of abstraction that produces algorithms that are efficient simultaneously in all of these downstream applications [7, 105, 112].

Further, independently of these constraints in downstream settings, most of the fast NLA algorithms designed in the past 20 years have been algorithms in the matrix-vector oracle model. This includes random projection methods like JL projections, iterative methods like Power iteration or Krylov Methods, as well as many sketching and streaming methods.

So, in summary, we find two complementary big-picture motivations that encourage us to study the matrix-vector complexity of NLA:

1. In many downstream settings, computing a matrix-vector product is the only (efficient) way to interact with the input matrix $\mathbf{A}$.
2. Even without constraints from downstream settings, most fast NLA algorithms designed in recent memory only interact with the input $\mathbf{A}$ via matrix-vector products.

The first point motivates research on upper bounds for matrix-vector complexity because it is clear that algorithms with low complexity are helpful in the real world. The second point motivates research on lower bounds for matrix-vector complexity because it is clear that *if no matrix-vector algorithm can estimate an important property of $\mathbf{A}$ efficiently, then we need a paradigm shift in how we design fast NLA algorithms.*

This thesis covers four papers that study the matrix-vector complexity of various numerical linear algebra problems. Each paper contains an algorithmic recommendation on how to solve an NLA problem with low matrix-vector complexity. These recommendations are supported by rigorous theoretical guarantees. Some of these results are further complemented by lower bounds on the minimum number of matrix-vector products needed. With the eventual goal of getting these upper and lower bounds to match, we can summarize the core goal of the research here:

The goal of this thesis is to design algorithms and lower-bound arguments that push us toward an understanding of the optimal matrix-vector complexity of solving numerical linear algebra problems.

Additionally, the algorithmic recommendations in this thesis are all very pragmatic: The Hutch++ algorithm from [Chapter 2](#) has been implemented in the SciPy

library, the Kronecker Girard-Hutchinson Estimator in [Chapter 3](#) is currently implemented by physicists studying the interactions of quantum particles, and the Single Vector Krylov method analyzed in [Chapter 4](#) is closely related to the implementation of the `EIGS()` command in MATLAB.

Lastly, we emphasize that the techniques used within the papers here are never too sophisticated. Despite how much research on NLA already exists, the research in this thesis betrays the fact that:

Many fundamental NLA tasks are not “fully solved”, and progress on these tasks does not necessitate building new complicated theoretical machinery.

where here we mean “fully solved” in the sense of having matching upper and lower bounds on the matrix-vector complexity of solving said NLA task.

1.2 Organization

The rest of the thesis will survey results from four papers on four distinct NLA tasks. In [Chapter 2](#), we will solve trace estimation ([Theorem 2](#)) with asymptotically optimal matrix-vector complexity. In [Chapter 3](#), we will study trace estimation under a heavily constrained matrix-vector oracle, and see that the best-known existing algorithm for trace estimation has a very slow exponential runtime. In [Chapter 4](#), we study low-rank approximation and provide the first theoretical results showing that a ubiquitous low-rank approximation method (single vector Krylov iteration) has unreasonably strong convergence guarantees and therefore only needs to use a few matrix-vector products. Lastly, in [Chapter 5](#), we show that finding a low-rank approximation to $f(\mathbf{A})$ is no harder than finding a low-rank approximation to $\mathbf{A}$ when f is a function that is sufficiently smooth and grows sufficiently slowly.

Chapter 2

Optimal Trace Estimation in the Matrix-Vector Oracle Model

2.1 Introduction

Data Scientists are designing new complex models of the world using larger and larger datasets. Unfortunately, these models are only usable if they are computationally efficient. Fortunately, they almost all rely on the same computational toolkit: *Numerical Linear Algebra*. Hence, faster linear algebra algorithms enable faster and more complex data science applications.

However, despite many decades of work in designing computationally efficient numerical linear algebra algorithms, we have remarkably little understanding of the *optimality* of the worst-case runtime of these algorithms. In this paper, we explore novel hardness results for a couple fundamental problems under a canonical class of linear algebra algorithms. We further complement these results with algorithms that have matching or otherwise state-of-the-art runtime.

The existing prior work on hardness results for numerical linear algebra mostly takes on one of three shapes: (1) Lower bounds tailored to specific algorithms [47]. (2) Reductions to discrete problems (like triangle detection or communication complexity). These usually only apply to heavily constrained classes of algorithms [72, 112]. (3) Statistical lower bounds in the *Matrix-Vector Oracle Model*.

We will focus on this third class of lower bound, which is motivated by an observation about the algorithms we use in practice:

Observation 3. *Most state-of-the-art linear algebra algorithms exclusively interact with the input matrix $\mathbf{A}$ by computing Matrix-Vector Products with $\mathbf{A}$. This includes*

Krylov Iteration, Lanczos Iteration, Random Projections, and much more.

This motivates us to count Matrix-Vector products as the measure of complexity of a linear algebra algorithm. Fast algorithms generally compute few products, while slow algorithms generally compute many. This perspective is inline with many Data Science domains, like sparse matrix methods and GPU-driven applicants, which view Matrix-Vector products as an atomic unit of computation [121].

Formally, we consider algorithms that access a matrix exclusively through its Matrix-Vector Oracle:

Definition 4. *The Matrix-Vector Oracle for a matrix $\mathbf{A}$ is the oracle that, when given any vector $\mathbf{x}$, returns the vector $\mathbf{Ax}$.*

$$\mathbf{x} \xrightarrow{\text{input}} \text{ORACLE} \xrightarrow{\text{output}} \mathbf{Ax}$$

Our goal is to design algorithmic upper bounds and statistical lower bounds for the number of queries an accurate algorithm makes to the oracle. As mentioned, many algorithms already give upper bounds for this setting. This has also been explicitly studied under the names of *Implicit Matrix Methods* or *Matrix-Free Methods*. In recent years a few lower bounds have also been developed, namely for approximate SVD, maximum eigenvalue estimation, and linear system solving [109, 17].

In this paper, we study various *Spectral Sum Estimation Problems*. At the most abstract level, we can state the problem as follows:

Problem 5. *Let $\mathbf{A}$ be a matrix with eigenvalues $\lambda_1, \dots, \lambda_d$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any function. Then, our goal is to estimate $\text{tr}(f(\mathbf{A})) := \sum_{i=1}^d f(\lambda_i)$.*

This includes many natural functions of interest in data science, like $\text{tr}(\mathbf{A})$, $\|\mathbf{A}\|_F^2$, $\log \det(\mathbf{A})$, and $\text{tr}(e^{\mathbf{A}})$. Spectral sums are also used in the computation of other properties of interest, like the spectral density or Schatten norm of a matrix [18].

Often, we can approximate this function $f(t)$ by a low-degree polynomial $p(t)$. When this is possible, we can then estimate $\text{tr}(f(\mathbf{A}))$ with $\text{tr}(p(\mathbf{A}))$. By linearity of trace, we can compute $\text{tr}(p(\mathbf{A}))$ as a linear combination of $\{\text{tr}(\mathbf{A}), \text{tr}(\mathbf{A}^2), \dots, \text{tr}(\mathbf{A}^q)\}$, where q is the degree of the polynomial p . So, in this paper we focus on two estimation problems: $\text{tr}(\mathbf{A})$ and $\text{tr}(\mathbf{A}^q)$ for $q \geq 2$. Being able to efficiently estimate these two terms enables estimating $\text{tr}(f(\mathbf{A}))$ for a wide family of functions f .

Note that we often interchangeably use the terms *query complexity*, *oracle complexity*, *number of matrix-vector products*, and *sample complexity* to refer to the number of queries submitted to a Matrix-Vector Oracle.

2.1.1 Adaptivity

Before stating our results, we also highlight one divide among Matrix-Vector Oracle Algorithms: some algorithms are *Adaptive*. That is, some algorithms use previous responses from the oracle to determine which queries to send next.

A natural example of this is Power Iteration: the user submits $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then $\mathbf{Ax}$, then $\mathbf{A}^2\mathbf{x}$, then $\mathbf{A}^3\mathbf{x}$, and so on. The algorithm must read the response from the oracle to determine which vector to submit next.

Algorithms that can instead submit all queries to the oracle simultaneously are called *Non-Adaptive Algorithms*. For example, random projections are non-adaptive, since they just involve multiplying many i.i.d. vectors with $\mathbf{A}$. Non-adaptive algorithms have many benefits. If computing individual matrix-vector products with $\mathbf{A}$ is very expensive, then GPUs or distributed hardware can often speed up the process by computing many products in parallel.

We will also see that non-adaptive algorithms can even be helpful in the design of adaptive algorithms, as we can instantiate many non-adaptive algorithms simultaneously and union-bound over their randomness sources to reduce the query complexity of bigger problems (see [Section 2.4](#) for details).

There are very many problems whose optimal sample complexities are unknown in the Matrix-Vector Oracle Model. So, it should not be surprising that we do not understand *when adaptivity is helpful*. For maximum eigenvalue estimation, adaptivity seems necessary. For trace estimation, adaptivity does not help.

To work towards this understanding, there also needs to be more work on unconditional lower bounds for non-adaptive algorithms in the Matrix-Vector Oracle Model. We view this as an important future direction in research. While we will not present this result in any detail, in [83], we provide an example of a non-adaptive lower bound proven via statistical hypothesis testing, giving a good proof of concept.

2.2 Preliminaries

We denote matrices with capital letters like $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{\Delta}$. We denote vectors as lower case bold letters like $\mathbf{x}$, $\mathbf{y}$, and $\boldsymbol{\alpha}$. All other letters denote scalars or functions. The norms $\|\mathbf{A}\|_2$, $\|\mathbf{A}\|_F$, and $\|\mathbf{A}\|_*$ denote the spectral, Frobenius, and nuclear norms, respectively. For square matrices, $\mathbf{A} \in \mathbb{R}^{d \times d}$, we have $\text{tr}(\mathbf{A}) = \sum_{i=1}^d A_{ii} = \sum_{i=1}^d \lambda_i(\mathbf{A})$, where $\lambda_1 \geq \dots \geq \lambda_d$ are the eigenvalues of $\mathbf{A}$. We focus on PSD matrices, which have all non-negative eigenvalues, so that $\text{tr}(\mathbf{A}) = \|\mathbf{A}\|_*$. Let $\mathbf{A}_k$ denote the best rank- k

approximation to $\mathbf{A}$. Let $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$.

2.3 Trace Estimation

(This work was published at SOSA 2021 [83].)

We start by studying the most fundamental spectrum sum estimation problem: Trace Estimation.

Problem 6. *Given access to the Matrix-Vector Oracle for PSD $\mathbf{A} \in \mathbb{R}^{d \times d}$, using as few matrix-vector products as possible, compute $\tilde{t}$ such that $|\tilde{t} - \text{tr}(\mathbf{A})| \leq \varepsilon \text{tr}(\mathbf{A})$.*

For indefinite matrices, $\text{tr}(\mathbf{A})$ can be arbitrarily close to zero, so the error guarantee becomes impossible to guarantee with $\Omega(d)$ queries. Instead, since $\text{tr}(\mathbf{A}) = \|\mathbf{A}\|_*$ for PSD $\mathbf{A}$, the generalized requirement $|\tilde{t} - \text{tr}(\mathbf{A})| \leq \varepsilon \|\mathbf{A}\|_*$ may be used in the indefinite case.

Prior work had shown two most notable results for the trace estimation problem:

1. Upper Bound: A simple algorithm known as *Hutchinson's Estimator* solves 6 with $O(\frac{\log(1/\delta)}{\varepsilon^2})$ matrix-vector products with probability $1 - \delta$ [7].
2. Lower Bound: A restricted class of algorithms including Hutchinson's Estimator must use $\Omega(\frac{\log(1/\delta)}{\varepsilon^2})$ matrix-vector products with probability $1 - \delta$ [131].

We improve on both of these points:

1. Upper Bounds: We provide a simple adaptation of Hutchinson's Estimator, called Hutch++, that uses $O(\frac{\sqrt{\log(1/\delta)}}{\varepsilon} + \log(\frac{1}{\delta}))$ matrix-vector products to succeed with probability $1 - \delta$. For constant failure probability, this improves the complexity from $O(\frac{1}{\varepsilon^2})$ to $O(\frac{1}{\varepsilon})$, and our code empirically outperforms the prior work.
2. Lower Bounds: We show that any algorithm must use $\Omega(\frac{1}{\varepsilon})$ matrix-vector products to succeed with probability at least $\frac{2}{3}$.

The original paper on Hutch++ provides two different lower bounds: a result for non-adaptive algorithms in the Real-RAM model via statistical hypothesis testing, and a result for adaptive algorithms in the finite-precision model via communication complexity. For the sake of brevity, we only present a new shorter proof in the adaptive case, which will also be useful in later sections.

Notably, all our lower bounds have the rate $\Omega(\frac{1}{\varepsilon})$, without an optimal dependence on the failure probability δ . We are aware of a follow-up work in submission that proves the rate $\Omega(\frac{\sqrt{\log(1/\delta)}}{\varepsilon})$ for adaptive algorithms, so the Hutch++ algorithm is very nearly asymptotically optimal. In fact, for the sake of legibility and brevity throughout the entire paper, we often forgo the analysis of δ and focus on rates with respect to ε .

2.3.1 Prior Work: Hutchinson's Estimator

The classical result on trace estimation is due to Michael Hutchinson in 1989 [60]. Hutchinson's Estimator starts with a simple observation:

Imported Lemma 7. (*Lemma 9 from [7]*) *Fix symmetric $\mathbf{A}$ and let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then, $\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{tr}(\mathbf{A})$ and $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = 2 \|\mathbf{A}\|_F^2$.*

Proof. To verify the expectation, just expand the sum:

$$\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \sum_{i=1}^d \sum_{j=1}^d \mathbf{A}_{i,j} \mathbb{E}[\mathbf{x}_i \mathbf{x}_j] = \sum_{i=1}^d \sum_{j=1}^d \mathbf{A}_{i,j} \mathbb{1}_{[i=j]} = \sum_{i=1}^d \mathbf{A}_{i,i} = \text{tr}(\mathbf{A})$$

Since linearity of variance only applies to independent terms, we cannot control the variance with the analysis above. Instead, we expand the spectral decomposition $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ and define $\mathbf{y} := \mathbf{U}^\top \mathbf{x}$. Since $\mathbf{U}$ is a rotation matrix and Gaussians are rotationally invariant, we know that $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then,

$$\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{Var}[(\mathbf{U}^\top \mathbf{x})^\top \mathbf{\Lambda} (\mathbf{U}^\top \mathbf{x})] = \text{Var}[\mathbf{y}^\top \mathbf{\Lambda} \mathbf{y}] = \sum_{i=1}^d \text{Var}[\lambda_i y_i^2] = \sum_{i=1}^d 2\lambda_i^2 = 2 \|\mathbf{A}\|_F^2$$

□

Often, Rademacher vectors are used instead of Gaussian vectors because Rademacher estimates have strictly lower variance. However, this improvement is non-asymptotic, so we will only use Gaussians in this analysis. From here can define Hutchinson's Estimator as the empirical mean of ℓ such samples:

$$\text{Hutchinson's Estimator:} \quad \mathbf{H}_\ell(\mathbf{A}) := \frac{1}{\ell} \sum_{i=1}^{\ell} \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_i = \frac{1}{\ell} \text{tr}(\mathbf{G}^\top \mathbf{A} \mathbf{G}) \quad (2.1)$$

where $\mathbf{G} = [\mathbf{x}_1, \dots, \mathbf{x}_\ell] \in \mathbb{R}^{d \times \ell}$ is just an i.i.d. $\mathcal{N}(0, 1)$ matrix. With this empirical mean, we can bound the sample complexity need to achieve relative error.

Lemma 8. *Let $\mathbf{A}$ be PSD. Then, $|H_\ell(\mathbf{A}) - \text{tr}(\mathbf{A})| \leq \varepsilon \text{tr}(\mathbf{A})$ with probability $1 - \delta$ if $\ell = O(\frac{\log(1/\delta)}{\varepsilon^2})$.*

Proof. We prove this for constant $\delta = \frac{1}{3}$. The proof for general δ uses a more nuanced analysis via the Hanson-Wright Inequality. For details, see the proof of Lemma 2 in Appendix A of [83].

By Chebyshev's Inequality, $|H_\ell(\mathbf{A}) - \text{tr}(\mathbf{A})| \leq \sqrt{3} \cdot \sqrt{\frac{2}{\ell}} \|\mathbf{A}\|_F$ with probability $\frac{2}{3}$. Further, since $\mathbf{A} \succeq \mathbf{0}$, we have $\|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})$. Then,

$$\begin{aligned} |H_\ell(\mathbf{A}) - \text{tr}(\mathbf{A})| &\leq \sqrt{\frac{6}{\ell}} \|\mathbf{A}\|_F && \text{(Chebyshev Ineq.)} \\ &\leq \sqrt{\frac{6}{\ell}} \text{tr}(\mathbf{A}) && (\|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})) \\ &\leq \varepsilon \text{tr}(\mathbf{A}) && (\ell \geq \Omega(\frac{1}{\varepsilon^2})) \end{aligned}$$

□

This provides a great starting point, as this simple non-adaptive estimator can be implemented in a couple lines of Matlab or Julia, and gives a strong theoretical guarantee. Further, prior work has shown that the *analysis* of Hutchinson's Estimator in [Theorem 8](#) is tight[131]. However, a more detailed look at the proof of [Theorem 8](#) reveals that the estimator itself can be improved.

2.3.2 Hutchinson versus The Top Few Eigenvalues

The proof of [Theorem 8](#) shows that in the worst-case, Hutchinson's Estimator needs $O(\frac{1}{\varepsilon^2})$ queries. However, the analysis also suggests that Hutchinson's Estimator only needs to use so many queries for certain kinds of matrices. We restate the algebra in the proof of [Theorem 8](#):

$$\begin{aligned} |H_\ell(\mathbf{A}) - \text{tr}(\mathbf{A})| &\leq \sqrt{\frac{6}{\ell}} \|\mathbf{A}\|_F && \text{(Line 1: Chebyshev Ineq.)} \\ &\leq \sqrt{\frac{6}{\ell}} \text{tr}(\mathbf{A}) && \text{(Line 2: } \|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})) \\ &\leq \varepsilon \text{tr}(\mathbf{A}) && \text{(Line 3: } \ell \geq \Omega(\frac{1}{\varepsilon^2})) \end{aligned}$$

Line 1 is tight for all matrices, as this is just the standard deviation of Hutchinson's Estimator, whose variance is *exactly* $\frac{2}{\ell} \|\mathbf{A}\|_F^2$ for all $\mathbf{A}$. Line 3 is also always tight, as $\frac{\sqrt{6}}{\sqrt{\ell}} = \varepsilon$ only if $\ell = O(\frac{1}{\varepsilon^2})$. Line 2, however, is only tight for matrices that have

$\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$. In other words, Hutchinson's Estimator only has the worst-case sample complexity of $O(\frac{1}{\varepsilon^2})$ for matrices with $\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$.

Notably, matrices with $\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$ have very strong structure: they have nearly-sparse eigenvalues. To see this, note that $\|\mathbf{A}\|_F$ is the ℓ_2 norm of the eigenvalues, while $\text{tr}(\mathbf{A})$ is the ℓ_1 norm of the eigenvalues (for PSD matrices). Then, it is just a property of vectors on $\mathbb{R}^d$ that a vector $\mathbf{v}$ has $\|\mathbf{v}\|_2 \approx \|\mathbf{v}\|_1$ only if $\mathbf{v}$ is nearly sparse. Gilbert et al. formally prove this relationship in the context of compressed sensing:

Imported Lemma 9. (*Lemma 7 from [42]*) Let $\mathbf{v} = [\lambda_1 \ \lambda_2 \ \dots \ \lambda_d]^\top$ with $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_d|$. Let $\mathbf{v}_k := [\lambda_1 \ \lambda_2 \ \dots \ \lambda_k \ 0 \ \dots \ 0]^\top$ be the best k -sparse approximation to $\mathbf{v}$. Then, $\|\mathbf{v} - \mathbf{v}_k\|_2 \leq \frac{1}{2\sqrt{k}} \|\mathbf{v}\|_1$.

Phrased in terms of PSD matrices, this becomes:

Corollary 10. Let $\mathbf{A}$ be a PSD matrix, and let $\mathbf{A}_k$ be the best rank- k approximation to $\mathbf{A}$. Then, $\|\mathbf{A} - \mathbf{A}_k\|_F \leq \frac{1}{2\sqrt{k}} \text{tr}(\mathbf{A})$.

To see why this reinforces our intuition, note that if $\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$, then **Theorem 10** ensures us that $\text{tr}(\mathbf{A}_k) \gtrsim (1 - \frac{1}{2k}) \text{tr}(\mathbf{A})$ ¹. That is, if $\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$ and $k = O(\frac{1}{\varepsilon})$, then the sum of the top k eigenvalues of $\mathbf{A}$ is a $(1 \pm \varepsilon)$ approximation of $\text{tr}(\mathbf{A})$. All of this serves the following guiding principle:

*If $\mathbf{A}$ converges slowly under Hutchinson's Estimator,
then the sum of the top k eigenvalues represents most of $\text{tr}(\mathbf{A})$.*

With this in mind, we propose an algorithm that works in three steps: (1) Estimate the sum of the top few eigenvalues with low-rank approximation tools. (2) Estimate the sum of the rest of the eigenvalues with Hutchinson's Estimator. (3) Return the sum of these two estimates.

If the target matrix has $\text{tr}(\mathbf{A}) \approx \|\mathbf{A}\|_F$, then the first step will capture most of the trace. Otherwise we know that $\text{tr}(\mathbf{A}) \gg \|\mathbf{A}\|_F$, and the Hutchinson's Estimator will approximate $\text{tr}(\mathbf{A})$ well and with few queries.

¹To see this, rearrange the expression $(\text{tr}(\mathbf{A}))^2 \approx \|\mathbf{A}\|_F^2 = \|\mathbf{A}_k\|_F^2 + \|\mathbf{A} - \mathbf{A}_k\|_F^2 \leq \|\mathbf{A}_k\|_F^2 + \frac{1}{2k}(\text{tr}(\mathbf{A}))^2$ to get $\|\mathbf{A}_k\|_F \gtrsim \sqrt{1 - \frac{1}{2k}} \text{tr}(\mathbf{A}) \geq (1 - \frac{1}{2k}) \text{tr}(\mathbf{A})$. Then, since $\text{tr}(\mathbf{A}_k) \geq \|\mathbf{A}_k\|_F$, we get $\text{tr}(\mathbf{A}_k) \gtrsim (1 - \frac{1}{2k}) \text{tr}(\mathbf{A})$.

2.3.3 The Hutch++ Estimator

Note that the intuition from the previous section does not clarify *how to compute* $\mathbf{A}_k$. This matrix is expensive to compute exactly, so we rely on approximations from the sketching and streaming literature. For instance, with $O(k)$ queries, we can find an orthonormal matrix $\mathbf{Q}$ whose orthogonal projection $\mathbf{P} := (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)$ enjoys $\|\mathbf{P}\mathbf{A}\mathbf{P}\|_F \leq 2\|\mathbf{A} - \mathbf{A}_k\|_F$ [87], which can then be bounded by [Theorem 10](#). This guarantee motivates the following theorem:

Theorem 11. *Let $\mathbf{A}$ be PSD, $\delta \in (0, 1/2)$, $\ell \in \mathbb{N}$, $k \in \mathbb{N}$. Let $\tilde{\mathbf{A}}$ and Δ be matrices such that both*

$$\text{tr}(\mathbf{A}) = \text{tr}(\tilde{\mathbf{A}}) + \text{tr}(\Delta) \quad \text{and} \quad \|\Delta\|_F \leq 2\|\mathbf{A} - \mathbf{A}_k\|_F$$

Then $Z := \text{tr}(\tilde{\mathbf{A}}) + H_\ell(\Delta)$ has $|Z - \text{tr}(\mathbf{A})| \leq O(\sqrt{\frac{\log(1/\delta)}{k\ell}}) \text{tr}(\mathbf{A})$ with probability $1 - \delta$.

Proof.

$$\begin{aligned} |Z - \text{tr}(\mathbf{A})| &\leq |H_\ell(\Delta) - \text{tr}(\Delta)| \quad (Z = \text{tr}(\mathbf{A}) + H_\ell(\Delta) \text{ and } \text{tr}(\mathbf{A}) = \text{tr}(\tilde{\mathbf{A}}) + \text{tr}(\Delta)) \\ &\leq C \sqrt{\frac{\log(1/\delta)}{\ell}} \|\Delta\|_F \quad (\text{Theorem 8}) \\ &\leq 2C \sqrt{\frac{\log(1/\delta)}{\ell}} \|\mathbf{A} - \mathbf{A}_k\|_F \quad (\|\Delta\|_F \leq \|\mathbf{A} - \mathbf{A}_k\|_F) \\ &\leq C \sqrt{\frac{\log(1/\delta)}{\ell}} \text{tr}(\mathbf{A}) \quad (\text{Theorem 10}) \end{aligned}$$

□

In the context of our prior discussion, take $\tilde{\mathbf{A}} = \mathbf{Q}\mathbf{Q}^\top\mathbf{A}$ and $\Delta = \mathbf{P}\mathbf{A}\mathbf{P}^\top$. [Theorem 11](#) formally shows how using both Hutchinson's Estimator and low-rank approximation can speed up the convergence rate:

Corollary 12. *In the setting of [Theorem 11](#), if $k = \ell = O(\frac{\sqrt{\log(1/\delta)}}{\varepsilon})$, then $|Z - \text{tr}(\mathbf{A})| \leq \varepsilon \text{tr}(\mathbf{A})$.*

Continuing with the example where $\tilde{\mathbf{A}} = \mathbf{Q}\mathbf{Q}^\top\mathbf{A}$ and $\Delta = \mathbf{P}\mathbf{A}\mathbf{P}$, by the cyclic property of trace, we can compute $\text{tr}(\tilde{\mathbf{A}})$ as $\text{tr}(\mathbf{Q}^\top\mathbf{A}\mathbf{Q})$ with k queries. So, taking $k = \ell$ as suggested by [Theorem 12](#), we realize the following algorithm:

Algorithm 1 Hutch++**input:** Matrix-Vector Oracle for $\mathbf{A} \in \mathbb{R}^{d \times d}$. Number of Queries m .**output:** Approximation to $\text{tr}(\mathbf{A})$.

- 1: Sample $\mathbf{S} \in \mathbb{R}^{d \times \frac{m}{3}}$ and $\mathbf{G} \in \mathbb{R}^{d \times \frac{m}{3}}$ with i.i.d. $\mathcal{N}(0, 1)$ entries.
- 2: Compute an orthonormal basis $\mathbf{Q} \in \mathbb{R}^{d \times \frac{m}{3}}$ for the span of $\mathbf{AS}$ (e.g., via QR decomposition).
- 3: **return** Hutch++($\mathbf{A}$) = $\text{tr}(\mathbf{Q}^T \mathbf{A} \mathbf{Q}) + \frac{3}{m} \text{tr}(\mathbf{G}^T (\mathbf{I} - \mathbf{Q} \mathbf{Q}^T) \mathbf{A} (\mathbf{I} - \mathbf{Q} \mathbf{Q}^T) \mathbf{G})$.

This algorithm uses a total of m matrix vector products: $\frac{m}{3}$ to compute $\mathbf{AS}$, $\frac{m}{3}$ to compute $\mathbf{AQ}$, and $\frac{m}{3}$ to compute $\mathbf{A} \cdot ((\mathbf{I} - \mathbf{Q} \mathbf{Q}^T) \mathbf{G})$. Further, the algorithm can be written with just a five lines of Matlab:

```

1 % Generate random Gaussian matrices
2 S = randn(d,ceil(m/3));
3 G = randn(d,floor(m/3));
4 % Compute the basis Q and project G away from it
5 [Q,~] = qr(A*S,0);
6 G = G - Q*(Q'*G);
7 % Return the Hutch++ estimate
8 trace_estimate = trace(Q'*A*Q) + 1/size(G,2)*trace(G'*A*G);

```

Since Algorithm 1 satisfies Theorem 12, we know that it enjoys a $O(\frac{1}{\epsilon})$ convergence rate.

2.3.4 Non-Adaptive Hutch++

Notice that Algorithm 1 is adaptive: it computes $\mathbf{AS}$ in order to compute $\mathbf{Q}$, which is then used to compute $\text{tr}(\mathbf{Q}^T \mathbf{A} \mathbf{Q})$ and $\mathbf{A}(\mathbf{I} - \mathbf{Q} \mathbf{Q}^T) \mathbf{G}$. Here, we show that trace estimation does not benefit from adaptivity. The generalized structure of Theorem 11 allows us to leverage a streaming result to satisfy the same conditions on $\tilde{\mathbf{A}}$ and $\mathbf{\Delta}$ without being adaptive.

In particular, a result of [24] shows that if $\mathbf{S} \in \mathbb{R}^{d \times \frac{m}{6}}$ and $\mathbf{R} \in \mathbb{R}^{d \times \frac{m}{3}}$ for $m = O(k \log(1/\delta))$, then $\tilde{\mathbf{A}} := \mathbf{AR}(\mathbf{S}^T \mathbf{AR})^+ (\mathbf{AS})^T$ and $\mathbf{\Delta} := (\mathbf{A} - \tilde{\mathbf{A}})$ enjoys $\|\mathbf{\Delta}\|_F \leq 2 \|\mathbf{A} - \mathbf{A}_k\|_F$ with probability $1 - \delta$. With the cyclic property of trace, we can efficiently compute $\text{tr}(\tilde{\mathbf{A}})$ as $\text{tr}((\mathbf{S}^T \mathbf{AR})^+ (\mathbf{AS})^T (\mathbf{AR}))$. So, using this constructure, we arrive at the Non-Adaptive variant of Hutch++:

Algorithm 2 NA-Hutch++ (Non-Adaptive Variant of Hutch++)**input:** Matrix-Vector Oracle for $\mathbf{A} \in \mathbb{R}^{d \times d}$. Number of Queries m .**output:** Approximation to $\text{tr}(\mathbf{A})$.

- 1: Sample $\mathbf{S} \in \mathbb{R}^{d \times \frac{m}{6}}$, $\mathbf{R} \in \mathbb{R}^{d \times \frac{m}{3}}$, and $\mathbf{G} \in \mathbb{R}^{d \times \frac{m}{2}}$ with i.i.d. $\mathcal{N}(0, 1)$ entries.
- 2: Compute $\mathbf{Z} = \mathbf{A}\mathbf{R}$ and $\mathbf{W} = \mathbf{A}\mathbf{S}$.
- 3: **return** $\text{NA-Hutch++}(\mathbf{A}) = \text{tr}((\mathbf{S}^T \mathbf{Z}) + (\mathbf{W}^T \mathbf{Z})) + \frac{2}{m} [\text{tr}(\mathbf{G}^T \mathbf{A} \mathbf{G}) - \text{tr}(\mathbf{G}^T \mathbf{Z} (\mathbf{S}^T \mathbf{Z}) + \mathbf{W}^T \mathbf{G})]$

Again, this matrix uses a total of m matrix-vector products: $\frac{m}{6}$ for $\mathbf{W}$, $\frac{m}{3}$ for $\mathbf{Z}$, and $\frac{m}{2}$ for $\text{tr}(\mathbf{G}^T \mathbf{A} \mathbf{G})$. Experimentally, NA-Hutch++ has consistently slightly slower convergence than Hutch++, but benefits from non-adaptivity.

Corollary 13. *If $m = O(\frac{\log(1/\delta)}{\varepsilon})$, then [Algorithm 2](#) returns a $(1 \pm \varepsilon)$ estimate of $\text{tr}(\mathbf{A})$ with probability $1 - \delta$.*

Proof. Combine [Theorem 12](#) with the observation that [24] requires a larger sample complexity of $m = O(\frac{\log(1/\delta)}{\varepsilon})$. $\square$

2.3.5 Lower Bound

We now show that, even in the adaptive model, every algorithm that achieves relative error ε with probability $\frac{2}{3}$ must use $\Omega(\frac{1}{\varepsilon})$ queries. This is achieved using a powerful result from [17]:

Definition 14. *Let $\mathbf{G} \in \mathbb{R}^{d \times d}$ have i.i.d. $\mathcal{N}(0, 1)$ entries. Then we say the matrix $\mathbf{W} := \mathbf{G}^T \mathbf{G}$ is distributed as a Wishart Matrix.*

Imported Lemma 15. *(Lemma 3.4 from [17]) Let $\mathbf{W} \in \mathbb{R}^{d \times d}$ be a Wishart Matrix. Suppose an algorithm accesses $\mathbf{W}$ with k (possibly adaptive) matrix-vector products. Then, there exists an orthogonal matrix $\mathbf{V}$ such that*

$$\mathbf{V}^T \mathbf{W} \mathbf{V} = \mathbf{\Delta} + \begin{bmatrix} 0 & 0 \\ 0 & \tilde{\mathbf{W}} \end{bmatrix}$$

where, *conditioned on all the algorithm's observations*, $\mathbf{\Delta} \in \mathbb{R}^{d \times d}$ is deterministically computable and $\tilde{\mathbf{W}} \in \mathbb{R}^{(d-k) \times (d-k)}$ is distributed as a Wishart Matrix.

Note that $\mathbf{\Delta}$ here is completely different from $\mathbf{\Delta}$ as defined in [Theorem 11](#). Further, since $\text{tr}(\mathbf{W}) = \|\mathbf{G}\|_F^2$ is the sum of d^2 many squared Gaussians, we have

$\text{tr}(\mathbf{W}) \sim \chi_{d^2}^2$. Because of this distribution, we first characterize the anti-concentration of Chi-Squared variables:

Lemma 16. *Let $X \sim \chi_k^2$, and let $t \in \mathbb{R}$ be independent of X . Then, with probability δ , $|X - t| \geq C_\chi \sqrt{k}$ for some constant C_χ depending only on δ .*

Proof. Corollary 3 from [134] says that $\Pr[X \geq k + \gamma\sqrt{k}] \geq C_0 e^{-\min\{C_1\gamma\sqrt{k}, \gamma^2\}} = C_0 e^{-\gamma^2}$ for small enough constant γ . If $t \leq k$, then with probability $C_0 e^{-\gamma^2}$ we have $|t - X| \geq \gamma\sqrt{k}$. The same corollary also says that $\Pr[X \leq k - \gamma\sqrt{k}] \geq C_0 e^{-C_2\gamma^2}$, so long as $\gamma\sqrt{k} \leq \frac{k}{2}$, which holds for small enough γ . If $t \geq k$, then with probability $C_0 e^{-C_2\gamma^2}$ we have $|t - X| \geq \gamma\sqrt{k}$. So, for some small enough γ , both directions hold with probability at least δ . Let C_χ be this value of γ . $\square$

Theorem 17. *Any algorithm in the Matrix-Vector Oracle Model that runs k queries and finds a value $\tilde{t}_0$ such that $|\tilde{t}_0 - \text{tr}(\mathbf{A})| \leq \varepsilon \text{tr}(\mathbf{A})$ must have $k = \Omega(\frac{1}{\varepsilon})$.*

Proof. Suppose that a Wishart Matrix $\mathbf{W} \in \mathbb{R}^{d \times d}$ is sampled and is placed inside the Matrix-Vector Oracle, and that an algorithm sends k queries to the oracle. Let $\mathbf{V}$, $\mathbf{\Delta}$, and $\tilde{\mathbf{W}}$ be the matrices as defined in Theorem 15. Then,

$$\text{tr}(\mathbf{W}) = \text{tr}(\mathbf{W}\mathbf{V}\mathbf{V}^\top) = \text{tr}(\mathbf{V}^\top\mathbf{W}\mathbf{V}) = \text{tr}(\mathbf{\Delta}) + \text{tr}(\tilde{\mathbf{W}})$$

Now let $\tilde{t}_0 \in (1 \pm \varepsilon) \text{tr}(\mathbf{W})$ be any trace estimate returned by the algorithm. Define $\tilde{t}_1 := \tilde{t}_0 - \text{tr}(\mathbf{\Delta})$. Then,

$$|\tilde{t}_0 - \text{tr}(\mathbf{W})| = |\tilde{t}_1 - \text{tr}(\tilde{\mathbf{W}})| \leq \varepsilon \text{tr}(\mathbf{W})$$

Notice that $\text{tr}(\mathbf{W}) = \text{tr}(\mathbf{G}^\top\mathbf{G}) = \|\mathbf{G}\|_F^2 \sim \chi_{d^2}^2$, and similarly that $\text{tr}(\tilde{\mathbf{W}}) \sim \chi_{(d-k)^2}^2$. So, by Markov's Inequality, $\text{tr}(\mathbf{W}) \leq C_1 d^2$ with probability $\frac{5}{6}$. Further, notice that $\tilde{t}_0$ is independent of $\text{tr}(\mathbf{W})$, so by Theorem 16 we have $|\tilde{t}_0 - \text{tr}(\mathbf{W})| \geq C_\chi(d - k)$ with probability $\frac{5}{6}$. We condition on the above two events occurring, which happens with probability $\frac{2}{3}$. The above inequality then becomes

$$C_\chi(d - k) \leq |\tilde{t}_1 - \text{tr}(\tilde{\mathbf{W}})| \leq \varepsilon \text{tr}(\mathbf{W}) \leq \varepsilon C_1 d^2$$

Taking the outermost terms, $C_\chi \cdot (d - k) \leq C_1 \cdot \varepsilon d^2$. Rearranging this in terms of k , we get $k \geq d - \frac{C_1}{C_\chi} \varepsilon d^2$. Taking the maximizer $d = \frac{C_\chi}{2C_1\varepsilon}$, we find $k \geq \frac{C_\chi}{4C_1} \cdot \frac{1}{\varepsilon} = \Omega(\frac{1}{\varepsilon})$. $\square$

Optimality of Hutchinson's Estimator

This already shows the optimality of Hutch++ for stochastic trace estimation. Recall that $|H_k(\mathbf{A}) - \text{tr}(\mathbf{A})| \leq \varepsilon \|\mathbf{A}\|_F$ for $k = O(\frac{1}{\varepsilon^2})$. Since this is a stronger error guarantee than a relative-error guarantee, we could also wonder if Hutchinson's Estimator is tight in this sense.

Corollary 18. *Any algorithm in the Matrix-Vector Oracle Model that runs k queries and finds a value $\tilde{t}_0$ such that $|\tilde{t}_0 - \text{tr}(\mathbf{A})| \leq \varepsilon \|\mathbf{A}\|_F$ must have $k = \Omega(\frac{1}{\varepsilon^2})$.*

Proof. This follows the proof of [Theorem 17](#), except we use the bound

$$C_\chi(d - k) \leq |\tilde{t}_1 - \text{tr}(\tilde{\mathbf{W}})| \leq \varepsilon \|\mathbf{W}\|_F \leq \varepsilon C_1 d^{3/2}$$

Where the right-hand side applies Markov's Inequality to $\mathbb{E}[\|\mathbf{W}\|_F^2] \leq 3d^2$ to get $\|\mathbf{W}\|_F \leq C_1 d^{3/2}$ for some C_1 . Keeping the outermost terms, $C_\chi(d - k) \leq \varepsilon C_1 d^{3/2}$. Taking the maximizer $d = \frac{2C_\chi^2}{3C_1\varepsilon^2}$, we find $k \geq \frac{4C_\chi}{27C_1} \cdot \frac{1}{\varepsilon^2} = \Omega(\frac{1}{\varepsilon^2})$. $\square$

So, for the stronger error guarantee of $\varepsilon \|\mathbf{A}\|_F$, Hutchinson's Estimator is already optimal. Notably, these two lower bounds both show that adaptivity is not helpful for estimating the trace of a matrix.

2.3.6 Experiments

We round off this section with some empirical verification that Hutch++ outperforms Hutchinson's Estimator, particularly for matrices where $\|\mathbf{A}\|_F \approx \text{tr}(\mathbf{A})$. We will simply discuss a simple synthetic experiment to illustrate this improvement, as a more full suite of experiments are available in the publication [83].

Here, we generated matrices in two steps: first we made a diagonal matrix $\mathbf{A}$ where $\mathbf{A}_{ii} = i^{-c}$ for some constant c , then we randomly generated an orthogonal matrix $\mathbf{Q}$, and built the matrix $\mathbf{A} = \mathbf{Q}\mathbf{A}\mathbf{Q}^\top$. This way, $\mathbf{A}$ has eigenvalues decaying exactly as i^{-c} . We built two 5000×5000 matrices this way, either with $c = 1.5$ or $c = 0.5$:

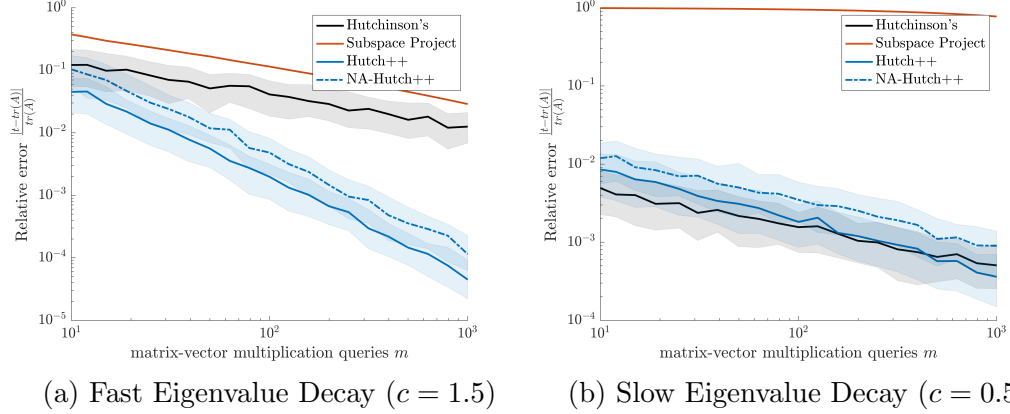


Figure 2.1: Confidence intervals show 25th and 75th quartiles after 200 trials. *Subspace Projection* is only estimating $\text{tr}(\mathbf{Q}^\top \mathbf{A} \mathbf{Q})$ as in [Algorithm 1](#). When $\mathbf{A}$ has fast decay, $\|\mathbf{A}\|_F \approx 0.42 \text{tr}(\mathbf{A})$, and Hutch++ vastly outperforms Hutchinson’s Estimator. When $\mathbf{A}$ has slow decay, $\|\mathbf{A}\|_F \approx 0.02 \text{tr}(\mathbf{A})$, and Hutchinson’s catches up to Hutch++’s speed.

2.4 Trace Monomials

(All results in this section represent ongoing unpublished research.)

Having bounded the complexity of estimating the trace of a matrix, we now turn our attention to the next building block needed for spectral-sum estimation: estimating $\text{tr}(\mathbf{A}^q)$ for some q . We can naively achieve a rate $O(\frac{q}{\varepsilon})$ by running Hutch++ with Matrix-Vector products of $\mathbf{A}^q$, calculated as $\mathbf{A}^q \mathbf{x} = \mathbf{A}(\mathbf{A}(\dots(\mathbf{A} \mathbf{x})))$. For $q = 1$, we already know that $O(\frac{1}{\varepsilon})$ is optimal, so we have confidence that the dependence on ε is tight.

However, the dependence on q is loose, and we often think of the degree q as a large value. For instance, when estimating $\text{tr}(e^{\mathbf{A}})$, we use a degree $q = \sqrt{\lambda_1(\mathbf{A})}$ polynomial. So, we would like to sharpen the dependence on q .

In this section, we provide two results: a $O(\frac{\sqrt{q}}{\varepsilon})$ algorithm and a $\Omega(\frac{1}{\sqrt{\varepsilon}})$ lower bound for all $q \geq 2$. Notably, we believe the lower bound is loose, and that the upper bound rate is optimal for all q .

2.4.1 Polynomial Acceleration

We first note a result from [104] on polynomial approximations to the monomial:

Imported Theorem 19 (Theorem 3.3 from [104]). *Fix $q \in \mathbb{N}$, $\Delta > 0$. Then for all*

$s \geq \sqrt{2q \ln(2/\Delta)}$, there exists a degree- s polynomial $p(t) = \sum_{j=0}^s u_j t^j$ such that

$$\sup_{t \in [-1, 1]} |p(t) - t^q| \leq \Delta \quad (2.2)$$

Or, equivalently, for any $s \geq 1$ there is a degree- s polynomial $p(t)$ such that

$$\sup_{t \in [-1, 1]} |p(t) - t^q| \leq 2e^{-s^2/2q} \quad (2.3)$$

Theorem 19 tells us that a $\tilde{O}(\sqrt{q})$ polynomial can approximate $f(t) = t^q$ very well. Since we can compute $\mathbf{A}^{\sqrt{q}}\mathbf{x}$ using only $\sqrt{q}$ queries to $\mathbf{A}$, we hope this provides a benefit over the naive method of computing products with $\mathbf{A}^q$ directly.

Since this approximation only holds on $[-1, 1]$, the core idea of the algorithm is to:
 1) Rescale the input matrix $\mathbf{A}$ such that it has eigenvalues in this range. 2) Compute the trace-polynomial of the scaled matrix with Hutch++. 3) Rescale the output of Hutch++ back to the original scale of $\mathbf{A}$.

At first, we might write down the following algorithm:

Algorithm 3 Trace-Monomial Estimation (Incorrect)

input: Matrix-Vector Oracle for $\mathbf{A}$. Degree q . Parameters $k, \ell \in \mathbb{N}$. Polynomial $p(t) = \sum_{j=0}^s u_j t^j$.

output: Approximation to $\text{tr}(\mathbf{A}^q)$.

- 1: Compute $\tilde{\lambda}_1 \approx \lambda_1(\mathbf{A})$ with k steps of Krylov Iteration.
 - 2: Let $\mathbf{B} := \frac{1}{\gamma} \mathbf{A}$ where $\gamma = q^{1/q} \tilde{\lambda}_1$
 - 3: Compute $t_j := \text{Hutch++}(\mathbf{B}^j)$ using ℓ matrix-vector products, for all $j = 1, \dots, s$.
 - 4: **return** $\gamma^q \cdot (\sum_{j=0}^s u_j \tilde{t}_j)$.
-

The parameter γ is chosen such that $\mathbf{B}$ has eigenvalues between 0 and 1, so the polynomial approximation holds for $s = \tilde{O}(\sqrt{q})$. However, there is an issue with the query complexity of this algorithm. Putting aside the Krylov iteration, just consider line 2: Computing $\text{Hutch++}(\mathbf{B}^s)$ uses $O(\frac{s}{\epsilon})$ queries, so computing $s = \tilde{O}(\sqrt{q})$ such trace estimates uses $O(\frac{s^2}{\epsilon}) = \tilde{O}(\frac{q}{\epsilon})$ products, which is already too expensive. We resolve this by making the various trace-estimates share their matrix-vector products, by using a non-adaptive estimator:

Algorithm 4 Trace-Monomial Estimation (Correct)

input: Matrix-Vector Oracle for $\mathbf{A}$. Degree q . Parameters $k, \ell \in \mathbb{N}$. Polynomial $p(t) = \sum_{j=0}^s u_j t^j$.

output: Approximation to $\text{tr}(\mathbf{A}^q)$.

- 1: Compute $\tilde{\lambda}_1 \approx \lambda_1(\mathbf{A})$ with k steps of Krylov Iteration.
- 2: Let $\mathbf{B} := \frac{1}{\gamma} \mathbf{A}$ where $\gamma = q^{1/q} \tilde{\lambda}_1$.
- 3: Sample vectors $\mathbf{x}_1, \dots, \mathbf{x}_\ell \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 4: Compute $\mathbf{B}\mathbf{x}_i, \mathbf{B}^2\mathbf{x}_i, \dots, \mathbf{B}^s\mathbf{x}_i$, for all $i = 1, \dots, \ell$
- 5: Compute $t_j := \text{NA-Hutch}++(\mathbf{B}^j)$ using $\mathbf{B}^j\mathbf{x}_1, \dots, \mathbf{B}^j\mathbf{x}_\ell$, for all $j = 1, \dots, s$
- 6: **return** $\gamma^q \cdot (\sum_{j=0}^s u_j \tilde{t}_j)$.

Every matrix-vector product now either appears in Line 1 with Krylov Iteration, or in Line 4 with $O(s\ell)$ matrix-vector products. With $s = \tilde{O}(\sqrt{q})$ and $\ell = O(\frac{1}{\varepsilon})$, the query-complexity issue has been resolved:

Theorem 20. *In [Algorithm 4](#), if $k = \tilde{O}(\sqrt{q})$, $\ell = \tilde{O}(\frac{1}{\varepsilon})$, and $p(t)$ is chosen to make [Equation \(2.3\)](#) hold for $s = \tilde{O}(\sqrt{q})$, then [Algorithm 4](#) returns a $(1 \pm \varepsilon)$ estimate to $\text{tr}(\mathbf{A}^q)$ for any $\mathbf{A} \succeq \mathbf{0}$ using $\tilde{O}(\frac{\sqrt{q}}{\varepsilon})$ products with probability $1 - \delta$.*

The proof of [Theorem 20](#) follows intuition very closely and mostly worries about fine-tuning parameters. For the log-dependencies in the rates, and the formal proof, see [Section 2.6](#).

[104] provides examples of more functions that are well approximated by polynomials. The same basic approach as used in [Algorithm 4](#) should extend easily to functions like these to get a $\tilde{O}(\frac{s}{\varepsilon})$ query complexity, where s is the degree of the approximating polynomial.

2.4.2 A Lower Bound

We now provide a lower bound by reduction. In particular, we reduce the problem of estimating $\text{tr}(\mathbf{A}^q)$ to estimating the trace of a Wishart Matrix, which we know is hard (See: [Theorem 17](#)).

Notably, the rate of the lower bound does not match the rate of the upper bound in its dependence on q . Designing an improved lower bound capable of growing with $\Omega(\sqrt{q})$ is an active and open problem.

Theorem 21. Fix parameter $q \in \mathbb{N}$. Then, for any small enough ε , any Matrix-Vector Oracle Algorithm that approximates $\text{tr}(\mathbf{A}^q)$ to relative error ε must use $\Omega(\frac{1}{\varepsilon})$ queries in the worst-case.

We will rely on the following two concentration claims:

Lemma 22. Fix q . Let $\mathbf{W} \in \mathbb{R}^{d \times d}$ be a Wishart matrix. Then, for all $k \in [q]$,

$$\text{tr}(\mathbf{W}^k) \asymp C_{k,E} d^{k+1}$$

and

$$|\text{tr}(\mathbf{W}^k) - \mathbb{E}[\text{tr}(\mathbf{W}^k)]| \lesssim C_{k,V} d^k$$

hold with high probability. Further, $C_E := \sum_{k=1}^q \left(\frac{e}{k}\right)^k C_{k,E} = O(1)$ and $C_V := \sum_{k=2}^q \left(\frac{e}{k}\right)^k C_{k,V} = O(1)$.

We now prove **Theorem 21**:

Proof. We start by defining our hard instance. Let $\mathbf{W} \in \mathbb{R}^{d \times d}$ be a Wishart Matrix, $\eta > 0$ to be defined later, and let $\mathbf{A} := \mathbf{I} + \eta \mathbf{W}$. Notably, oracle access to $\mathbf{A}$ is trivially equivalent to oracle access to $\mathbf{W}$. Then, we can expand

$$\text{tr}(\mathbf{A}^q) = d + q\eta \text{tr}(\mathbf{W}) + \sum_{k=2}^q \binom{q}{k} \eta^k \text{tr}(\mathbf{W}^k)$$

First, note that if $\eta = \frac{1}{qd}$, then

$$q\eta \text{tr}(\mathbf{W}) \asymp q\eta d^2 = d. \quad (2.4)$$

Second, we bound

$$\begin{aligned} \text{tr}(\mathbf{A}^q) &= \sum_{k=0}^q \binom{q}{k} \eta^k \text{tr}(\mathbf{W}^k) \asymp d + \sum_{k=1}^q \left(\frac{qe}{k}\right)^k \eta^k C_{k,E} d^{k+1} \\ &= d + d \sum_{k=1}^q \left(\frac{qe\eta d}{k}\right)^k C_{k,E} = d + d \sum_{k=1}^q \left(\frac{e}{k}\right)^k C_{k,E} \\ &= d + dC_E \asymp d. \end{aligned} \quad (2.5)$$

Third, we bound

$$\begin{aligned}
|(\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})) - \mathbb{E}[\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})]| &\leq \sum_{k=2}^q \binom{q}{k} \eta^k |\text{tr}(\mathbf{W}^k) - \mathbb{E}[\text{tr}(\mathbf{W}^k)]| \\
&\lesssim \sum_{k=2}^q \left(\frac{qe}{k}\right)^k \eta^k C_{k,V} d^k \\
&= \sum_{k=2}^q \left(\frac{qe\eta d}{k}\right)^k C_{k,V} \\
&= \sum_{k=2}^q \left(\frac{e}{k}\right)^k C_{k,V} = C_V \asymp 1 \quad (2.6)
\end{aligned}$$

We have now laid the tracks to prove the theorem. For any $\tilde{t}_0 \in (1 \pm \varepsilon) \text{tr}(\mathbf{A}^q)$, let $\tilde{t}_1 := \tilde{t}_0 - \mathbb{E}[\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})]$, and let $\tilde{t}_2 := \frac{1}{q\eta} \tilde{t}_1$. Then,

$$\begin{aligned}
q\eta |\tilde{t}_2 - \text{tr}(\mathbf{W})| &= |\tilde{t}_1 - q\eta \text{tr}(\mathbf{W})| \\
&= |\tilde{t}_0 - q\eta \text{tr}(\mathbf{W}) - \mathbb{E}[\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})]| \\
&\leq |\tilde{t}_0 - \text{tr}(\mathbf{A}^q)| + |\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W}) - \mathbb{E}[\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})]| \\
&\leq \varepsilon \text{tr}(\mathbf{A}^q) + |(\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})) - \mathbb{E}[\text{tr}(\mathbf{A}^q) - q\eta \text{tr}(\mathbf{W})]| \\
&\hspace{25em} (\text{Defn of } \tilde{t}_0) \\
&\lesssim \varepsilon \text{tr}(\mathbf{A}^q) + 1 \hspace{10em} (\text{Equation (2.6)}) \\
&\lesssim \varepsilon d + 1 \hspace{10em} (\text{Equation (2.5)}) \\
&\lesssim \varepsilon d \hspace{10em} (\text{If we take } d \asymp \frac{1}{\varepsilon})
\end{aligned}$$

Dividing by $q\eta = \frac{1}{d}$ on both side, we get

$$|\tilde{t}_2 - \text{tr}(\mathbf{W})| \lesssim \varepsilon d^2 \asymp \varepsilon \text{tr}(\mathbf{W})$$

where the last bound follows from [Theorem 22](#). That is, given a relative error estimate $\tilde{t}_0 \in (1 \pm \varepsilon) \text{tr}(\mathbf{A}^q)$, we can construct $\tilde{t}_2$ to estimate the trace of a Wishart $\mathbf{W}$ that has $O(\varepsilon)$ relative error when $d \asymp \frac{1}{\varepsilon}$. So, by [Theorem 17](#), it takes $\Omega(\frac{1}{\varepsilon})$ matvecs to construct $\tilde{t}_0$ with constant probability. $\square$

2.4.3 Proving [Theorem 22](#)

Proof. Proving the mean bound $C_{k,E}$.

Note that $\text{tr}(\mathbf{W}) = \|\mathbf{G}\|_F^2 \sim \chi_{d^2}^2$, so $\text{tr}(\mathbf{W}) \geq C_{tr}d^2$. Further, $\|\mathbf{W}\|_2 = (\sigma_1(\mathbf{G}))^2 \leq C_{op}(\sqrt{d})^2 = C_{op}d$ [101]. So, we know that $\sum_i \lambda_i(\mathbf{W}) \asymp d^2$ while $\max_i \lambda_i \lesssim d$. Therefore, $\mathbf{W}$ has at least $m := \frac{C_{tr}}{2C_{op}-C_{tr}}d = C_md$ eigenvalues that are $\geq C_{eig}d$ where $C_{eig} = \frac{C_{tr}}{2C_{op}}$ ². And so, we can see that

$$\text{tr}(\mathbf{W}^k) = \sum_{i=1}^d (\lambda_i(\mathbf{W}))^k \leq d(\lambda_1(\mathbf{W}))^k \leq C_{op}^k d^{k+1}$$

and

$$\text{tr}(\mathbf{W}^k) = \sum_{i=1}^d (\lambda_i(\mathbf{W}))^k \geq \sum_{i=1}^m (\lambda_i(\mathbf{W}))^k \geq m(C_{eig}d)^k \geq C_m C_{eig}^k d^{k+1}.$$

Then, notice that

$$\sum_{k=1}^q \left(\frac{e}{k}\right)^k C_{op}^k = \sum_{k=1}^q \left(\frac{eC_{op}}{k}\right)^k = O(1)$$

and

$$\sum_{k=1}^q \left(\frac{e}{k}\right)^k C_m C_{eig}^k = C_m \sum_{k=1}^q \left(\frac{eC_{eig}}{k}\right)^k = O(1).$$

This completes the proof for concentrating $\text{tr}(\mathbf{W}^k)$ for all $q \in \mathbb{N}$ simultaneously.

Proving the variance bound $C_{k,V}$

We use Corollary 1.8 from [49]:

Imported Theorem 23. *Let $\mathbf{W} \in \mathbb{R}^{d \times d}$ be a Wishart matrix. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) := f(x^2)$ is Lipschitz. Then, with probability at least $1 - \delta$, we have*

$$|\text{tr}(f(\mathbf{W})) - \mathbb{E}[\text{tr}(f(\mathbf{W}))]| \leq 2|g|_{Lip} \sqrt{C \ln(2/\delta)},$$

where $|g|_{Lip}$ is the Lipschitz constant for g on $\mathbb{R}$.

Recall that with high probability, $\|\mathbf{W}\|_2 \leq C_{op}d$ for some constant C_{op} .

We would like to just appeal to this theorem with $f(x) = x^k$. However, this f

²Abstracting this out, suppose $1 > a_1 > \dots > a_n > 0$, and $\sum_i a_i > cn$. Let m be the number a_i s.t. $a_i > \gamma n$ for $\gamma \in (0, 1)$. Then $cn \leq \sum_i a_i = \sum_{i=1}^m a_i + \sum_{m+1}^n a_i \leq m + (n-m)r$. Solving this gets $m \geq \frac{c-\gamma}{1-\gamma}n$, which passes the sanity check of $m \leq n$ when $c < 1$. Taking $r = c/2$ gets $m \geq \frac{1}{2/c-1}$. Removing the abstraction, we have $a_i = \lambda_i(\mathbf{W})/C_{op}d$ and $c = C_{tr}/C_{op}$ and $d = n$. So, we just need to ensure $c < 1$. However, if this bound on c didn't hold, then we would have $\sum_i \lambda_i > \text{tr}(\mathbf{W})$, which isn't possible. So, we don't even have to certify this condition.

invokes a g with unbounded Lipschitz constant. So, we define

$$f_{clip}(x) := \begin{cases} 0 & x < 0 \\ x^k & x \in [0, C_{op}d] \\ C_{op}^k d^k & x > C_{op}d \end{cases}$$

In particular, we have that $f_{clip}(\mathbf{W}) = \mathbf{W}^k$ with high probability. We then apply the imported theorem to the function f_{clip} , getting the following bound with high probability:

$$|\text{tr}(\mathbf{W}^k) - \mathbb{E}[\text{tr}(f_{clip}(\mathbf{W}))]| \leq C_{var} |g_{clip}|_{Lip}$$

for some constant C_{var} that depends only on failure probability. It remains to bound $|g_{clip}|_{Lip}$. In particular, since $g_{clip}(x) = f_{clip}(x^2)$ is differentiable almost everywhere, we have $|g_{clip}|_{Lip} = \max_{x: g'(x) \text{ exists}} |g'(x)|$. Since

$$g'(x) = f'(x^2) \cdot 2x = \begin{cases} 0 & x < 0 \\ 2kx^k & x \in [0, C_{op}d] \\ 0 & x > C_{op}d \end{cases}$$

we find that $|g_{clip}|_{Lip} = 2kC_{op}^k d^k$. Therefore:

$$|\text{tr}(\mathbf{W}^k) - \mathbb{E}[\text{tr}(f_{clip}(\mathbf{W}))]| \leq 2kC_{var}C_{op}^k d^k.$$

This concludes the concentration we need for the lemma. We lastly remark that $C_{k,V} = 2kC_{var}C_{op}^k$ indeed satisfies

$$\sum_{k=2}^q \left(\frac{e}{k}\right)^k C_{k,V} = \sum_{k=2}^q \left(\frac{e}{k}\right)^k 2kC_{var}C_{op}^k = 2C_{var} \sum_{k=2}^q \left(\frac{eC_{op}}{k^{1-1/k}}\right)^k = O(1).$$

□

2.5 Conclusion

We presented results that work towards provably worst-case optimal algorithms in the Matrix-Vector Oracle Model for trace and trace-monomial estimation. While the proverbial book seems closed for trace estimation, there remains many interesting open problems for trace-monomial and other spectral-sum problems. For examples:

- Can we sharpen the $\text{tr}(\mathbf{A}^q)$ lower bound to be $\Omega(\frac{\sqrt{q}}{\varepsilon})$?
- How well can non-adaptive algorithms perform in this setting? Can we even achieve a $O(\frac{q}{\varepsilon})$ rate?
- If there exists an adaptivity gap for trace-monomial estimation, then how do we prove it?

Remark: Statistical hypothesis testing has been able to prove lower bounds for non-adaptive algorithms in the past, so this seems like a reasonable approach [83].

- Many interesting spectral sums are well approximated by polynomials. Can we extend this lower bound to estimate $\text{tr}(f(\mathbf{A}))$ when $f(\lambda_i) \approx p(\lambda_i)$ for some polynomial $p(t)$?

Recent result: With this approach we can show $\text{tr}(e^{\mathbf{A}})$ requires $\Omega(\frac{1}{\sqrt{\varepsilon}})$ queries in the worst-case. While not matching the upper bound, it is a nice proof-of-concept.

2.6 Proof of **Theorem 20**

Theorem 20 Restated. In **Algorithm 4**, if $k = O(\frac{\sqrt{q} \log(d)}{\sqrt{\log q}}) = \tilde{O}(\sqrt{q})$, $\ell = O(\frac{\log(\frac{q}{\delta} \log(\frac{dq}{\varepsilon}))}{\varepsilon}) = \tilde{O}(\frac{1}{\varepsilon})$, and $p(t)$ is chosen to make **Equation (2.3)** hold for $s = O(\sqrt{q \log(\frac{dq}{\varepsilon})}) = \tilde{O}(\sqrt{q})$, then **Algorithm 4** returns a $(1 \pm \varepsilon)$ estimate to $\text{tr}(\mathbf{A}^q)$ for any $\mathbf{A} \succeq \mathbf{0}$ using $\tilde{O}(\frac{\sqrt{q}}{\varepsilon})$ products.

Proof. We start by reparameterizing **Algorithm 4**, and showing that these rates on k , ℓ , and s result from that. Suppose we run Krylov Iteration until we get an estimate $\tilde{\lambda}_1 \in (1 \pm \Delta_0)\lambda_1(\mathbf{A})$, which requires $O(\frac{\log d}{\sqrt{\Delta_0}})$ iterations [85]. Further, suppose that $p(t)$ satisfies **Equation (2.2)** with error parameter Δ_1 .

We first characterize the eigenvalues of $\mathbf{B}$. Our assumption on Krylov Iteration implies that $\gamma := (1 + \Delta_0)\tilde{\lambda}_1$ has $\gamma \geq \lambda_1(\mathbf{A})$. So $\mathbf{B} = \frac{1}{\gamma}\mathbf{A}$ must have $0 \preceq \mathbf{B} \preceq \mathbf{I}$. Further, since $\mathbf{A} = (1 + \Delta_0)\tilde{\lambda}_1\mathbf{B}$, we can write

$$\tilde{\lambda}_1 \leq (1 + \Delta_0)\lambda_1(\mathbf{A}) = (1 + \Delta_0)^2 \tilde{\lambda}_1 \cdot \lambda_1(\mathbf{B})$$

So, we have $\lambda_1(\mathbf{B}) \geq \frac{1}{(1 + \Delta_0)^2}$. Overall, this means $\lambda_i(\mathbf{B}) \in [0, 1]$ and $\lambda_1(\mathbf{B}) \in [\frac{1}{(1 + \Delta_0)^2}, 1]$.

Next, we show that an accurate trace estimate of $\text{tr}(\mathbf{B}^q)$ implies an accurate trace estimate of $\text{tr}(\mathbf{A}^q)$: Let $\tilde{t} \in (1 \pm \varepsilon) \text{tr}(\mathbf{B}^q)$. Then, $Z := \gamma^q \tilde{t}$ enjoys

$$Z = \gamma^q \tilde{t} \in (1 \pm \varepsilon) \gamma^q \text{tr}(\mathbf{B}^q) = (1 \pm \varepsilon) \text{tr}((\gamma \mathbf{B})^q) = (1 \pm \varepsilon) \text{tr}(\mathbf{A}^q)$$

So, if we can show that $\tilde{t} := \sum_{j=0}^s u_j \tilde{t}_j$ satisfies $\tilde{t} \in (1 \pm \varepsilon) \text{tr}(\mathbf{B}^q)$, then we are done with the analysis.

By union-bounding [Theorem 11](#) s many times, we have that each trace estimate $\tilde{t}_j \in (1 \pm \varepsilon) \text{tr}(\mathbf{B}^j)$ if $\ell = O(\frac{\log(s/\delta)}{\varepsilon})$. Then, by linearity of trace and by design of $p(t)$,

$$\begin{aligned} \tilde{t} &:= \sum_{j=0}^s u_j \tilde{t}_j \\ &\leq (1 + \varepsilon) \sum_{j=0}^s u_j \text{tr}(\mathbf{B}^j) \\ &= (1 + \varepsilon) \sum_{j=0}^s u_j \sum_{i=1}^d \lambda_i^j \\ &= (1 + \varepsilon) \sum_{i=1}^d \sum_{j=0}^s u_j \lambda_i^j \\ &= (1 + \varepsilon) \sum_{i=1}^d p(\lambda_i) \\ &\leq (1 + \varepsilon) \sum_{i=1}^d (\lambda_i^q + \Delta_1) \\ &= (1 + \varepsilon) (\text{tr}(\mathbf{B}^q) + \Delta_1 d) \\ &= \text{tr}(\mathbf{B}^q) + (\varepsilon \text{tr}(\mathbf{B}^q) + (1 + \varepsilon) \Delta_1 d) \end{aligned}$$

Similarly, we get $\tilde{t} \geq \text{tr}(\mathbf{B}^q) - (\varepsilon \text{tr}(\mathbf{B}^q) + (1 + \varepsilon)\Delta_1 d)$. Then, we can write

$$\begin{aligned}
|\tilde{t} - \text{tr}(\mathbf{B}^q)| &\leq \varepsilon \text{tr}(\mathbf{B}^q) + (1 + \varepsilon)\Delta_1 d \\
&= \left(\varepsilon + \frac{(1 + \varepsilon)\Delta_1 d}{\text{tr}(\mathbf{B}^q)} \right) \text{tr}(\mathbf{B}^q) \\
&\leq \left(\varepsilon + \frac{(1 + \varepsilon)\Delta_1 d}{(\lambda_1(\mathbf{B}))^q} \right) \text{tr}(\mathbf{B}^q) \\
&\leq \left(\varepsilon + \frac{(1 + \varepsilon)\Delta_1 d}{((1 + \Delta_0)^{-2})^q} \right) \text{tr}(\mathbf{B}^q) \\
&= \left(\varepsilon + \Delta_1 ((1 + \varepsilon)d(1 + \Delta_0)^{2q}) \right) \text{tr}(\mathbf{B}^q) \\
&\leq 2\varepsilon \text{tr}(\mathbf{B}^q)
\end{aligned}$$

Where the last line takes $\Delta_1 = \frac{\varepsilon}{(1+\varepsilon)d(1+\Delta_0)^{2q}}$. So, this guarantees that $\tilde{t} \in (1 \pm 2\varepsilon) \text{tr}(\mathbf{B}^2)$. All that remains is to tune the values of Δ_0 and Δ_1 to ensure that *Algorithm 4* has low query complexity.

Recall that $s = O(\sqrt{q \log(1/\Delta_1)})$, so plugging in $\Delta_1 = O(\frac{\varepsilon}{d(1+\Delta_0)^{2q}})$ yields $s = O(\sqrt{q \log(\frac{d(1+\Delta_0)^{2q}}{\varepsilon})})$. Notably, for any constant Δ_0 , this yields $s = \tilde{O}(q)$. So we instead enforce $(1 + \Delta_0)^{2q} = q^2$, which is equivalent to $\Delta_0 = q^{1/q} - 1 = \Theta(\frac{\ln q}{q})$.

With Δ_0 and Δ_1 set this way, we find that $s = O(\sqrt{q \log(\frac{dq^2}{\varepsilon})}) = \tilde{O}(\sqrt{q})$. Further, we run $k = O(\frac{\log d}{\sqrt{\Delta_0}}) = O(\frac{\sqrt{q} \log(d)}{\sqrt{\log(q)}}) = \tilde{O}(\sqrt{q})$ Krylov Iterations. So, overall, we use a total of $k + s\ell = \tilde{O}(\sqrt{q} + \frac{\sqrt{q}}{\varepsilon}) = \tilde{O}(\frac{\sqrt{q}}{\varepsilon})$ matrix-vector products.

Lastly, note that $\gamma = (1 + \Delta_0)^2 \tilde{\lambda}_1 = q^{1/q} \tilde{\lambda}_1$, explaining Line 2 of *Algorithm 4*. $\square$

Chapter 3

Trace Estimation in the Kronecker-Matrix-Vector Oracle Model

3.1 Introduction

A ubiquitous problem in scientific computing is that of estimating the trace of a $D \times D$ matrix $\mathbf{A}$ that can be only accessed via matrix-vector products. In other words, we are given access to an oracle that can evaluate $\mathbf{A}\mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^D$, and our goal is to approximate $\text{tr}(\mathbf{A})$ with as few oracle queries as possible. This *matrix-vector oracle model* or *implicit matrix model* is a common computational framework used in much of the numerical linear algebra community [113, 21, 51, 8, 94]. However, in many applications, we cannot efficiently evaluate $\mathbf{A}\mathbf{x}$ for all vectors $\mathbf{x} \in \mathbb{R}^D$. Instead, only some matrix vector products can be efficiently computed.

In this paper, we study trace estimation in the *Kronecker-matrix-vector oracle model*, also known as the *rank-one vector model*, where we can evaluate $\mathbf{A}\mathbf{x}$ for any vector $\mathbf{x}$ that can be written as a Kronecker product of smaller vectors, i.e. $\mathbf{x} = \mathbf{x}_1 \otimes \mathbf{x}_2 \otimes \cdots \otimes \mathbf{x}_k$ for some vectors $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^d$. Here, $\otimes$ denotes the *Kronecker product*, which is a common primitive in many tensor-based algorithms and in much of quantum physics. The number of vectors (k) and the number of entries (d) in each of $\mathbf{x}_1, \dots, \mathbf{x}_k$ are parameters of the model.

Our study is motivated by recent literature on entanglement estimation in quantum states [39]. This computational model has also been studied in other applied domains [21, 66]. An important setting where we can efficiently compute Kronecker-

matrix-vector products is when $\mathbf{A}$ is the sum of a small number of matrices, where each summand is a Kronecker product of k matrices. Such matrices occur as Hamiltonians of various quantum models [125]. Another instance in which we can efficiently compute Kronecker-matrix-vector products is when $\mathbf{A}$ is the flattening of a tensor network. This is the case in [39], where the oracle is efficiently implemented when $\mathbf{A}$ describes a density operator that represents certain systems of quantum particles. Recently, there has been some effort to find algorithms that can efficiently solve linear algebra problems using this oracle, although the papers do not explicitly define such a Kronecker-matrix-vector oracle [19, 6, 3, 114]. In this paper, we focus on the fundamental problem of trace estimation.

The trace estimation problem in the Kronecker-matrix-vector oracle model can be exactly solved using $D = d^k$ oracle queries. Since each standard basis vector $\mathbf{e}_1, \dots, \mathbf{e}_D$ obeys Kronecker structure, we can exactly look up each diagonal entry of $\mathbf{A}$ and evaluate $\text{tr}(\mathbf{A}) = \sum_{i=1}^D \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i$. So, our goal is to estimate the trace of $\mathbf{A}$ to $(1 \pm \varepsilon)$ relative error using $o(d^k)$ matrix-vector products. In the standard non-Kronecker setting, the most fundamental algorithm for trace estimation is Hutchinson's Estimator [43, 60]:

$$H_\ell(\mathbf{A}) := \frac{1}{\ell} \sum_{j=1}^{\ell} \mathbf{x}^{(j)\top} \mathbf{A} \mathbf{x}^{(j)} \quad \text{where} \quad \mathbf{x}^{(j)} \stackrel{iid}{\sim} \mathcal{D}.$$

If the distribution of random vectors has $\mathbb{E}[\mathbf{x}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = \mathbf{I}$, then $\mathbb{E}[H_\ell(\mathbf{A})] = \text{tr}(\mathbf{A})$. Further, if $\mathcal{D}$ is a Gaussian or Rademacher distribution, then $\text{Var}[H_\ell(\mathbf{A})] \leq \frac{2}{\ell} \|\mathbf{A}\|_F^2$. In particular, under the common assumption that $\mathbf{A}$ is PSD, so that $\|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})$, we then find that the standard deviation of Hutchinson's estimator is

$$\sqrt{\text{Var}[H_\ell(\mathbf{A})]} \leq \sqrt{\frac{2}{\ell}} \|\mathbf{A}\|_F \leq \sqrt{\frac{2}{\ell}} \text{tr}(\mathbf{A}) \leq \varepsilon \text{tr}(\mathbf{A})$$

when we take $\ell \geq \Omega(\frac{1}{\varepsilon^2})$ queries. For large dimension D , this very simple algorithm is much more efficient than exactly computing the diagonal entries of $\mathbf{A}$, since the estimator has no dependence on the size of $\mathbf{A}$ at all.

This estimator can be naturally generalized to the Kronecker-matrix-vector oracle model by simply changing the distribution $\mathcal{D}$. Instead of taking $\mathcal{D}$ to be a distribution over Gaussian or Rademacher vectors, we now take $\mathcal{D}$ to be the Kronecker of k vectors drawn from some other isotropic zero-mean distribution $\mathcal{D}'$:

$$\mathbf{x} = \mathbf{x}_1 \otimes \mathbf{x}_2 \otimes \dots \otimes \mathbf{x}_k \quad \text{where} \quad \mathbf{x}_i \stackrel{iid}{\sim} \mathcal{D}'$$

Notably, if $\mathbb{E}[\mathbf{x}_i] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] = \mathbf{I}$, then we also immediately have that $\mathbb{E}[\mathbf{x}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{x} \mathbf{x}^\top] = \mathbf{I}$, so that $\mathbb{E}[H_\ell(\mathbf{A})] = \text{tr}(\mathbf{A})$. This simple estimator, which we refer to as the Kronecker-Hutchinson Estimator, fits the Kronecker-matrix-vector oracle model, and we may hope that it is efficient enough to estimate the trace of $\mathbf{A}$ very quickly. At a high level, we answer this in the negative, proving novel and very tight bounds on the variance of the Kronecker-Hutchinson Estimator, showing that the sample complexity of the estimator has an unavoidable exponential dependence on k .

3.1.1 Prior Work

We first remark there is a great deal of literature on trace estimation without the Kronecker constraint. The design of Hutchinson's Estimator is attributed to [43, 60], and some of the core papers that analyze the estimator are [7, 100, 29].

Although relatively a new topic, trace estimation in the Kronecker-matrix-vector oracle model has recently been explored in the literature. From the physics literature, the works of [39] and [59] analyze trace estimators when $\mathbf{A} = \otimes_{i=1}^k \mathbf{A}_i$, and shows some empirical results for their performance. From a theoretical perspective, [122] provides bounds that can be used to analyze the Kronecker-Hutchinson Estimator. However, the paper provides very coarse bounds for this problem, showing that if $k = O(d)$ then we have $\mathbf{x}^\top \mathbf{A} \mathbf{x} \in \text{tr}(\mathbf{A}) \pm O(kd^{k-1} \|\mathbf{A}\|_2)$, where $\|\mathbf{A}\|_2$ is the operator norm of $\mathbf{A}$. To obtain a relative error $(1 \pm \varepsilon)$, this suggests an overall sample complexity of $\ell = O(\frac{kd^{k-1} \|\mathbf{A}\|_2}{\varepsilon^2 \text{tr}(\mathbf{A})}) = O(\frac{D \|\mathbf{A}\|_2}{\varepsilon^2 \text{tr}(\mathbf{A})})$, which is generally more than D queries.

The research of [19] directly analyzes the Kronecker-Hutchinson Estimator when $k = 2$, showing first that $\ell = O(\frac{1}{\varepsilon^2})$ samples suffice to ensure that $H_\ell(\mathbf{A}) \geq (1 - \varepsilon) \text{tr}(\mathbf{A})$, and second that $\ell = O(\frac{d \|\mathbf{A}\|_2}{\varepsilon^2 \text{tr}(\mathbf{A})})$ samples suffice to ensure that $H_\ell(\mathbf{A}) \leq (1 + \varepsilon) \text{tr}(\mathbf{A})$. Lastly, Remark 2 of the arXiv version of [3] is a result on using a Kronecker-Johnson-Lindenstrauss transform, and implies that $O(\frac{3^k}{\varepsilon^2})$ samples suffice for the Kronecker-Hutchinson Estimator with Rademacher vectors to achieve relative error $(1 \pm \varepsilon)$. While this worst-case bound is tight, the paper gives no way to understand when this bound is loose, and for what matrices fewer samples are needed. Further, all the aforementioned theoretical papers rely on heavy-duty analysis, for instance involving Gaussian and Rademacher Chaos theory. In contrast, our contributions give tighter results using simpler techniques.

We also note that [3] does also include a Johnson-Lindenstrauss style guarantee for working with Kronecker-shaped data using a sketch that has $O(\frac{k}{\varepsilon^2})$ rows. However, this sketch does not actually operate in the Kronecker-matrix-vector oracle model.

3.1.2 Our Contributions

Our core technical contribution is an exact expression of the variance of Hutchinson’s Estimator when our query vectors are formed as a Kronecker product of Gaussian vectors. This also provides an upper bound on the variance when the estimator instead uses a Kronecker product of Rademacher vectors. For PSD matrices, our results imply a worst-case upper bound of $\ell = O(\frac{3^k}{\varepsilon^2})$ on the number of samples needed to estimate the trace of $\mathbf{A}$ to relative error $(1 \pm \varepsilon)$. We match this upper bound with a lower bound showing that the Kronecker-Hutchinson Estimator with iid zero-mean unit-variance vectors has to use $\ell = \Omega(\frac{3^k}{\varepsilon^2})$ samples to achieve standard deviation $\varepsilon \text{tr}(\mathbf{A})$. Lastly, we show that if we are allowed to use complex Kronecker-matrix-vector products instead of real matrix-vector products (i.e. if $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ where $\mathbf{x}_i \in \mathbb{C}^d$), then this rate improves to $\Theta(\frac{2^k}{\varepsilon^2})$.

To more concretely state our core technical contribution, we recall the *partial trace* operator from quantum physics. In quantum physics, a Hermitian *density matrix* $\mathbf{A} \in \mathbb{C}^{d^k \times d^k}$ is roughly speaking used to represent the joint probability distribution for a set of k particles. Suppose these particles are partitioned into two sets, so that $[k] = \mathcal{S} \cup \bar{\mathcal{S}}$. Further suppose that Alice can observe the particles in $\mathcal{S}$, and that Bob can observe the particles in $\bar{\mathcal{S}}$, but they cannot observe each other’s particles. Then, the partial trace of $\mathbf{A}$ with respect to $\mathcal{S}$, denoted $\text{tr}_{\mathcal{S}}(\mathbf{A})$, describes the marginal distribution of the particles that Bob observes. This is often called “tracing out the subsystem $\mathcal{S}$ ”.

We include a general formal definition of the partial trace in [Section 3.2](#), but a rough intuition is easily gained by examining the $k = 2$ case. Here, we can break down $\mathbf{A} = \mathbb{R}^{d^2 \times d^2}$ as a block matrix full of $\mathbf{A}_{ij} \in \mathbb{R}^{d \times d}$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \cdots & \mathbf{A}_{1d} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \cdots & \mathbf{A}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{d1} & \mathbf{A}_{d2} & \cdots & \mathbf{A}_{dd} \end{bmatrix}.$$

Then, the partial trace of $\mathbf{A}$ with respect to $\{1\}$ is the sum of the diagonal blocks, i.e. $\text{tr}_{\{1\}}(\mathbf{A}) = \sum_{i=1}^d \mathbf{A}_{ii}$, and the partial trace of $\mathbf{A}$ with respect to $\{2\}$ is the matrix

that contains the trace of each block matrix, i.e.

$$\text{tr}_{\{2\}}(\mathbf{A}) = \begin{bmatrix} \text{tr}(\mathbf{A}_{11}) & \text{tr}(\mathbf{A}_{12}) & \dots & \text{tr}(\mathbf{A}_{1d}) \\ \text{tr}(\mathbf{A}_{21}) & \text{tr}(\mathbf{A}_{22}) & \dots & \text{tr}(\mathbf{A}_{2d}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{tr}(\mathbf{A}_{d1}) & \text{tr}(\mathbf{A}_{d2}) & \dots & \text{tr}(\mathbf{A}_{dd}) \end{bmatrix}.$$

We will also need to recall the *partial transpose* operator from quantum physics. This operator is often used in quantum physics, but lacks an intuitive physical meaning analogous to what the partial trace enjoys. Like the partial trace, the partial transpose can also be taken over any subset of particles $\mathcal{V} \subseteq [k]$. We denote the partial transpose of $\mathbf{A}$ with respect to $\mathcal{V}$ as $\mathbf{A}^{\tau_{\mathcal{V}}}$, and call this “transposing the subsystem $\mathcal{V}$ ”. We have a formal definition of the partial transpose in [Section 3.2](#). But, when $k = 2$, the two partial transposes of $\mathbf{A}$ are:

$$\mathbf{A}^{\tau_{\{1\}}} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{21} & \dots & \mathbf{A}_{d1} \\ \mathbf{A}_{12} & \mathbf{A}_{22} & \dots & \mathbf{A}_{d2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{1d} & \mathbf{A}_{2d} & \dots & \mathbf{A}_{dd} \end{bmatrix} \quad \text{and} \quad \mathbf{A}^{\tau_{\{2\}}} = \begin{bmatrix} \mathbf{A}_{11}^{\top} & \mathbf{A}_{12}^{\top} & \dots & \mathbf{A}_{1d}^{\top} \\ \mathbf{A}_{21}^{\top} & \mathbf{A}_{22}^{\top} & \dots & \mathbf{A}_{2d}^{\top} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{d1}^{\top} & \mathbf{A}_{d2}^{\top} & \dots & \mathbf{A}_{dd}^{\top} \end{bmatrix}.$$

Having setup both the partial trace and the partial transpose, we can now state our core theorem:

Theorem 24. Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is independent and is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Let $\bar{\mathbf{A}} := \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \mathbf{A}^{\tau_{\mathcal{V}}}$ be the average of all partial transposes of $\mathbf{A}$. Then, $\text{Var}[\mathbf{x}^{\top} \mathbf{A} \mathbf{x}] \leq \sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F^2$. Furthermore, if all $\mathbf{x}_i$ are $\mathcal{N}(\mathbf{0}, \mathbf{I})$, then this expression is an equality.

The matrix $\bar{\mathbf{A}}$ seems odd at a glance, so we first justify why it is natural. For the classical $k = 1$ setting with Gaussian vectors, we can write that $\text{Var}[\mathbf{x}^{\top} \mathbf{A} \mathbf{x}] = 2 \|\mathbf{A}\|_F^2$ for all symmetric matrices $\mathbf{A}$ [7, 80]. For asymmetric matrices, this equality does not hold. Instead, we let $\hat{\mathbf{A}} := \frac{\mathbf{A} + \mathbf{A}^{\top}}{2}$, and note that $\mathbf{x}^{\top} \mathbf{A} \mathbf{x} = \mathbf{x}^{\top} \hat{\mathbf{A}} \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^D$. Since $\hat{\mathbf{A}}$ is symmetric, we then find that $\text{Var}[\mathbf{x}^{\top} \mathbf{A} \mathbf{x}] = 2 \|\hat{\mathbf{A}}\|_F^2$. [Theorem 24](#) generalizes this idea to all possible ways to partially transpose across all possible subsystems $\mathcal{V} \subseteq [k]$.

Next, note that $\|\hat{\mathbf{A}}\|_F \leq \|\mathbf{A}\|_F$, and that symmetric matrices have $\mathbf{A} = \hat{\mathbf{A}}$. So, we know that $\text{Var}[\mathbf{x}^{\top} \mathbf{A} \mathbf{x}] = 2 \|\hat{\mathbf{A}}\|_F^2 \leq 2 \|\mathbf{A}\|_F^2$ holds for all matrices $\mathbf{A}$, with equality for symmetric $\mathbf{A}$. This bound can be easier to work with, since it does not

require worrying about $\hat{\mathbf{A}}$. For large k , we similarly bound $\|\text{tr}_S(\bar{\mathbf{A}})\|_F \leq \|\text{tr}_S(\mathbf{A})\|_F$, providing a more approachable version of [Theorem 24](#):

Theorem 25. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is independent and is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq \sum_{S \subseteq [k]} 2^{k-|S|} \|\text{tr}_S(\mathbf{A})\|_F^2$. Furthermore, if all $\mathbf{x}_i$ are $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and if $\mathbf{A} = \bar{\mathbf{A}}$, then this expression is an equality.*

Note that many reasonable matrices will have $\mathbf{A} = \bar{\mathbf{A}}$. For instance, recall that the Kronecker-matrix-vector oracle model may be used when $\mathbf{A}$ is the sum of a small number of matrices, where each summand is a Kronecker product of k matrices. If each summand is the Kronecker product of *symmetric* matrices, then we have $\mathbf{A} = \bar{\mathbf{A}}$.

We demonstrate the utility of [Theorem 25](#) by using it to show the following guarantee on estimating the trace of a random rank-one matrix:

Theorem 26. *Fix $d, k \in \mathbb{N}$. Let $\mathbf{A} = \mathbf{g} \mathbf{g}^\top$ where $\mathbf{g} \in \mathbb{R}^{d^k}$ is a vector of iid standard normal Gaussians. Then, with probability $\frac{3}{4}$, we know that $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x} \mid \mathbf{g}] \leq O((2 + \frac{1}{d})^k (\text{tr}(\mathbf{A}))^2)$. In particular, with probability $\frac{2}{3}$, we know that the Kronecker-Hutchinson Estimator returns a $(1 \pm \varepsilon)$ relative error estimate to the trace of $\mathbf{A}$ if $\ell = O(\frac{1}{\varepsilon^2}(2 + \frac{1}{d})^k)$.*

This rate of $(2 + \frac{1}{d})^k$ is obviously expensive, but our sample complexity is lower than what is achievable with any other technique that we are aware of. Nevertheless, even without $\bar{\mathbf{A}}$, this sum of partial traces may sometimes still be too unwieldy to work with. For such cases, we provide a worst-case bound for PSD matrices:

Theorem 27. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ be a PSD matrix. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq (3^k \text{tr}(\mathbf{A}))^2$.*

In particular, by Chebyshev's Inequality, [Theorem 27](#) implies that $\ell = O(\frac{3^k}{\varepsilon^2})$ samples suffice to estimate $\text{tr}(\mathbf{A})$ to relative error $(1 \pm \varepsilon)$. For the case of [19], when $k = 2$, we get the following:

Corollary 28. *Let $\mathbf{A} \in \mathbb{R}^{d^2 \times d^2}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \mathbf{x}_2$, where each $\mathbf{x}_i$ is independent and is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Let $\bar{\mathbf{A}} = \frac{1}{4}(\mathbf{A} + \mathbf{A}^{\top_{\{1\}}} + \mathbf{A}^{\top_{\{2\}}} + \mathbf{A}^\top)$. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq 2 \|\text{tr}_{\{1\}}(\bar{\mathbf{A}})\|_F^2 + 2 \|\text{tr}_{\{2\}}(\bar{\mathbf{A}})\|_F^2 + 4 \|\bar{\mathbf{A}}\|_F^2$. If $\mathbf{A}$ is PSD, with probability $\frac{2}{3}$, we know that the Kronecker-Hutchinson Estimator returns a $(1 \pm \varepsilon)$ relative error estimate to the trace of $\mathbf{A}$ if $\ell = O(\varepsilon^{-2})$.*

This success probability can be boosted to $1 - \delta$ for an arbitrary δ by picking up a $\log(\frac{1}{\delta})$ dependence in the sample complexity by returning the median of $O(\log(\frac{1}{\delta}))$ iid trials of $H_\ell(\mathbf{A})$. We further show that the expensive rate of $O(\frac{3^k}{\varepsilon^2})$ is essentially tight in the worst case for the Kronecker-Hutchinson Estimator:

Theorem 29. *Fix $d, k \in \mathbb{N}$. Consider Hutchinson's Estimator run with vectors $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ where $\mathbf{x}_i$ are vectors of iid real zero-mean unit-variance entries. Then, there exists a PSD matrix $\mathbf{A}$ such that Hutchinson's Estimator needs $\ell = \Omega(\frac{(3-\frac{2}{d})^k}{\varepsilon^2})$ samples in order to have standard deviation $\leq \varepsilon \text{tr}(\mathbf{A})$.*

Complex Kronecker-Matrix-Vector Products

Although certainly cheaper than exactly computing the trace with d^k queries to the diagonal of $\mathbf{A}$, the rate of $\frac{3^k}{\varepsilon^2}$ is prohibitively expensive for large k . Our next set of results show that the bound improves if we can issue complex queries instead of real ones, that is when $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ where $\mathbf{x}_i \in \mathbb{C}^d$. It seems realistic to say that any software implementation of the Kronecker-matrix-vector oracle could handle complex queries instead of real ones.

Note that for $k = 1$, there is almost no distinction between the real and complex oracle models, as any complex matrix-vector product can be simulated by running two real matrix-vector products. However, this is no longer the case for large values of k and real simulation becomes expensive. To see this, note that the real part of the Kronecker product of complex vectors might not have a Kronecker structure. Because of this, it may take 2^k real-valued Kronecker-matrix-vector products to simulate a single complex-valued Kronecker-matrix-vector product. In a vague intuitive sense, the complex Kronecker-structured vectors are exponentially less constrained than the real Kronecker-structured vectors. Running with this intuition in mind, we prove the following analogues to [Theorems 24 to 27](#) and [29](#):

Theorem 30. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is either a complex Rademacher vector or a complex Gaussian vector. That is, we have $\mathbf{x}_i = \frac{1}{\sqrt{2}}\mathbf{r}_i + \frac{i}{\sqrt{2}}\mathbf{m}_i$ where $\mathbf{r}_i$ and $\mathbf{m}_i$ are iid real-valued vectors of either Rademacher or standard normal Gaussian entries. Let $\bar{\mathbf{A}} := \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \mathbf{A}^{\text{tr}_{\mathcal{V}}}$ be the average of all partial transposes of $\mathbf{A}$. Then, $\text{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}] \leq \sum_{\mathcal{S} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F^2 \leq \sum_{\mathcal{S} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2$. If $\mathbf{x}_i$ are complex Gaussians, then the first expression is an equality. If we further know that $\mathbf{A} = \bar{\mathbf{A}}$, then both expression are equalities. Further, if $\mathbf{A}$ is PSD, then we have $\text{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}] \leq (2^k \text{tr}(\mathbf{A}))^2$.*

Theorem 31. Fix $d, k \in \mathbb{N}$. Let $\mathbf{A} = \mathbf{g}\mathbf{g}^\top$ where $\mathbf{g} \in \mathbb{R}^{d^k}$ is a vector of iid standard normal Gaussians. Then, with probability $\frac{3}{4}$, we know that $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x} \mid \mathbf{g}] \leq O((1 + \frac{1}{d})^k (\text{tr}(\mathbf{A}))^2)$. In particular, with probability $\frac{2}{3}$, we know that the Kronecker-Hutchinson Estimator returns a $(1 \pm \varepsilon)$ relative error estimate to the trace of $\mathbf{A}$ if $\ell = O(\frac{1}{\varepsilon^2}(1 + \frac{1}{d})^k)$.

Theorem 32. Fix $d, k \in \mathbb{N}$. Consider Hutchinson's Estimator run with vectors $\mathbf{x} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_k$ where $\mathbf{x}_i$ are vectors of iid complex zero-mean unit-variance entries. Then, there exists a PSD matrix $\mathbf{A}$ such that Hutchinson's Estimator needs $\ell = \Omega(\frac{(2 - \frac{1}{d})^k}{\varepsilon^2})$ samples in order to have standard deviation $\leq \varepsilon \text{tr}(\mathbf{A})$.

By applying Chebyshev's Inequality to [Theorems 27](#) and [30](#), we see an *exponential speedup* from $\ell = O(\frac{3^k}{\varepsilon^2})$ queries in the real-valued case to $\ell = O(\frac{2^k}{\varepsilon^2})$ queries in the complex-valued case. For large k , this gap from 3^k to 2^k is a very significant speedup, and we are not aware of any other prior works which discuss this potential.

Another way to view this speedup is from comparing the Kronecker-Hutchinson Estimator to just summing the diagonal of $\mathbf{A}$, which takes exactly d^k queries. When $d = 2$, the real-valued Kronecker-Hutchinson Estimator is exponentially more expensive than just summing the diagonal directly. However, the complex-valued Kronecker-Hutchinson Estimator is only $O(\frac{1}{\varepsilon^2})$ times more expensive in the worst case. This is another way to frame the power of the complex-valued estimator – the complex-valued estimator is at most polynomially slower than summing the diagonal, while the real-valued estimator can be exponentially slower.

We can also directly compare [Theorems 26](#) and [31](#). In the real-valued case, the sample complexity grows as $\ell = O(\frac{1}{\varepsilon^2}(2 + \frac{1}{d})^k)$, while in the complex-valued case, the sample complexity grows as $\ell = O(\frac{1}{\varepsilon^2}(1 + \frac{1}{d})^k)$. Notably, if $k = O(d)$ then this rate grows as $O(\frac{2^k}{\varepsilon^2})$ for real-valued queries and $O(\frac{1}{\varepsilon^2})$ for complex-valued queries. This shows a power exponential gap between our bounds, where we are only leveraging the difference between complex-valued and real-valued vectors.

Additional Results and Hope for a Faster Algorithm

We complement the aforementioned results with a few informative additions. First, we show a lower bound against all Kronecker-matrix-vector algorithms. By adapting the techniques of [17] and [62], we show that any Kronecker-matrix-vector algorithm that estimates the trace of $\mathbf{A}$ with standard deviation $\varepsilon \text{tr}(\mathbf{A})$ must make at least $\Omega(\frac{\sqrt{k}}{\varepsilon})$ oracle queries.

Second, we note that the lower bounds in [Theorems 29](#) and [32](#) both use construction of $\mathbf{A}$ such that $\mathbf{A} = \mathbf{A}_1 \otimes \cdots \otimes \mathbf{A}_k$. That is, the Kronecker-Hutchinson Estimator converges slowest for matrices that have a Kronecker structure. However, if we truly are given $\mathbf{A}$ with the promise that $\mathbf{A} = \mathbf{A}_1 \otimes \cdots \otimes \mathbf{A}_k$ for some unknown matrices $\mathbf{A}_1, \dots, \mathbf{A}_k$, then in [Section 3.7](#) we show that the trace of $\mathbf{A}$ can be recovered exactly using just $kd + 1$ Kronecker-matrix-vector products. Similarly, [Theorems 26](#) and [31](#) suggest that the Kronecker-Hutchinson Estimator converges exponentially slowly for random rank-one matrices. But in this rank-one case, we show that we can exactly recover the trace of $\mathbf{A}$ with a single Kronecker-matrix-vector query!

This trend, where the matrices that make the Kronecker-Hutchinson Estimator converge slowly can have their traces efficiently estimated by another algorithm, is reminiscent of the setup for the analysis in [83]. In that paper, the authors recall the analysis of the standard deviation of Hutchinson’s Estimator:

$$\sqrt{\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}]} \leq \sqrt{\frac{2}{\ell}} \|\mathbf{A}\|_F \leq \sqrt{\frac{2}{\ell}} \text{tr}(\mathbf{A}) \leq \varepsilon \text{tr}(\mathbf{A})$$

where the first inequality uses that $\|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})$ for PSD $\mathbf{A}$, and the second inequality is forced to pick the sample complexity $\ell \geq \frac{2}{\varepsilon^2}$. Since the first inequality is only tight for matrices $\mathbf{A}$ that are nearly low-rank, the authors propose an algorithm which combines Hutchinson’s Estimator with a low-rank approximation method that can efficiently compute the trace of nearly-low-rank matrices efficiently. This way, their algorithm has a worst-case matrix-vector complexity of $\Theta(\frac{1}{\varepsilon})$, while Hutchinson’s Estimator is stuck with $\Theta(\frac{1}{\varepsilon^2})$.

Our analysis similarly characterizes two cases when the Kronecker-Hutchinson Estimator converges slowly – when $\mathbf{A}$ is either low-rank or has Kronecker structure. We also show that we can compute the trace such matrices very efficiently. This suggests that a more involved algorithm can interpolate between these low query complexities and perhaps even avoid any exponential dependence on k at all. We leave this as an important problem for future work.

Lastly, we remark that the analysis from [83] crucially uses a tight bound for the variance of Hutchinson’s Estimator in terms of $\|\mathbf{A}\|_F$. We believe that our core results, [Theorems 24](#) and [25](#), can serve an analogous role in constructing a faster trace-estimating algorithm.

3.2 Preliminaries

We use capital bold letter to denote matrices, lowercase bold letters to denote vectors, and lowercase non-bold letters to denote scalars. $\mathbb{R}$ is the set of reals, and $\mathbb{C}$ is the set of complex numbers. $\Re[\mathbf{x}]$ and $\Im[\mathbf{x}]$ denote the real and imaginary parts of a vector or scalar. $\mathbf{x}^\top$ denotes the transpose of a vector, while $\mathbf{x}^H$ denotes the conjugate transpose of a matrix or vector. $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ denotes our input matrix. We use the bracket notation $[\mathbf{B}]_{ij}$ to denote the i, j entry of matrix $\mathbf{B}$. We let $\mathbf{e}_i$ denote the i^{th} standard basis vector, whose exact dimension may depend on the context. We let $\mathbf{I} \in \mathbb{R}^{d \times d}$ always refer to the $d \times d$ identity matrix. We let $\|\mathbf{A}\|_F$ denote the Frobenius norm, $\text{tr}(\mathbf{A})$ the trace, $\text{tr}_S(\mathbf{A})$ the partial trace, and $\mathbf{A}^{\top\nu}$ the partial transpose. We let $[k] = \{1, \dots, k\}$ be the set of integers from 1 to k .

3.2.1 Kronecker product

The Kronecker product of two matrices $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$ is $\mathbf{A} \otimes \mathbf{B} \in \mathbb{R}^{np \times mq}$, where

$$\mathbf{A} \otimes \mathbf{B} := \begin{bmatrix} [\mathbf{A}]_{11}\mathbf{B} & [\mathbf{A}]_{12}\mathbf{B} & \dots & [\mathbf{A}]_{1m}\mathbf{B} \\ [\mathbf{A}]_{21}\mathbf{B} & [\mathbf{A}]_{22}\mathbf{B} & \dots & [\mathbf{A}]_{2m}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ [\mathbf{A}]_{n1}\mathbf{B} & [\mathbf{A}]_{n2}\mathbf{B} & \dots & [\mathbf{A}]_{nm}\mathbf{B} \end{bmatrix}$$

We also use the shorthands $\otimes_{i=1}^k \mathbf{x}_i := \mathbf{x}_1 \otimes \mathbf{x}_2 \otimes \dots \otimes \mathbf{x}_k$, as well as $\mathbf{A}^{\otimes k} := \otimes_{i=1}^k \mathbf{A}$.

The Kronecker product has three key properties that we repeatedly use:

Lemma 33 (Mixed-Product Property (Folklore)). $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$

Lemma 34 (Transposing (Folklore)). $(\mathbf{A} \otimes \mathbf{B})^\top = (\mathbf{A}^\top \otimes \mathbf{B}^\top)$

Lemma 35 (Fact 7.4.1 in [14]). Let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$. Then, $\mathbf{x} \otimes \mathbf{y} = (\mathbf{I}_n \otimes \mathbf{y})\mathbf{x} = (\mathbf{x} \otimes \mathbf{I}_m)\mathbf{y}$.

At one point, we will also require a short lemma about the squared Frobenius norm and the Kronecker product. This lemma is roughly equivalent to noting that $\|\mathbf{A}\|_F^2 = \sum_{i,j=1}^d \|\mathbf{A}_{ij}\|_F^2$ in the block matrix decomposition of $\mathbf{A}$ when $k = 2$ in [Section 3.1.2](#).

Lemma 36. Let $\mathbf{C}, \mathbf{E} \in \mathbb{R}^{d^{k-i} \times d^{k-i}}$ and $\mathbf{D} \in \mathbb{R}^{d^i \times d^i}$ for any $i \in [k]$. Then, we have

$$\sum_{\hat{i}, \hat{j}=1}^{d^i} \|(\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{D} (\mathbf{E} \otimes \mathbf{e}_{\hat{j}})\|_F^2 = \|(\mathbf{C} \otimes \mathbf{I})^\top \mathbf{D} (\mathbf{E} \otimes \mathbf{I})\|_F^2.$$

Proof. First, recall there exists a permutation matrix $\mathbf{P}$ such that $\mathbf{C} \otimes \mathbf{y} = \mathbf{P}(\mathbf{y} \otimes \mathbf{C})\mathbf{P}^\top$ for all $\mathbf{y} \in \mathbb{R}^{d^i}$. Now let $\hat{\mathbf{D}} := \mathbf{P}\mathbf{D}\mathbf{P}$. Then, by expand $\hat{\mathbf{D}}$ as a block matrix of matrices $\hat{\mathbf{D}}_{\hat{i},\hat{j}} \in \mathbb{R}^{d^{k-i} \times d^{k-i}}$, and noting that $(\mathbf{e}_{\hat{i}} \otimes \mathbf{C})^\top = \begin{bmatrix} \mathbf{0} & \cdots & \mathbf{0} & \mathbf{C}^\top & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}$. We then get that

$$(\mathbf{e}_{\hat{i}} \otimes \mathbf{C})^\top \hat{\mathbf{D}} (\mathbf{e}_{\hat{j}} \otimes \mathbf{E}) = \begin{bmatrix} \mathbf{0} & \cdots & \mathbf{0} & \mathbf{C}^\top & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{D}}_{11} & \cdots & \hat{\mathbf{D}}_{1d} \\ \vdots & \ddots & \vdots \\ \hat{\mathbf{D}}_{d1} & \cdots & \hat{\mathbf{D}}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{E} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} = \mathbf{C}^\top \hat{\mathbf{D}}_{\hat{i}\hat{j}} \mathbf{E}.$$

We can also notice that

$$(\mathbf{I} \otimes \mathbf{C})^\top \hat{\mathbf{D}} (\mathbf{I} \otimes \mathbf{E}) = \begin{bmatrix} \mathbf{C}^\top & & \\ & \ddots & \\ & & \mathbf{C}^\top \end{bmatrix} \begin{bmatrix} \hat{\mathbf{D}}_{11} & \cdots & \hat{\mathbf{D}}_{dd} \\ \vdots & \ddots & \vdots \\ \hat{\mathbf{D}}_{d1} & \cdots & \hat{\mathbf{D}}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{E} & & \\ & \ddots & \\ & & \mathbf{E} \end{bmatrix} = \begin{bmatrix} \mathbf{C}^\top \hat{\mathbf{D}}_{11} \mathbf{E} & \cdots & \mathbf{C}^\top \hat{\mathbf{D}}_{1d} \mathbf{E} \\ \vdots & \ddots & \vdots \\ \mathbf{C}^\top \hat{\mathbf{D}}_{d1} \mathbf{E} & \cdots & \mathbf{C}^\top \hat{\mathbf{D}}_{dd} \mathbf{E} \end{bmatrix}.$$

Then, noting that the Frobenius norm is invariant to permutation, we find that

$$\begin{aligned} \sum_{\hat{i},\hat{j}=1}^{d^i} \|(\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{D} (\mathbf{E} \otimes \mathbf{e}_{\hat{j}})\|_F^2 &= \sum_{\hat{i},\hat{j}=1}^{d^i} \|\mathbf{P}^\top (\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{P} \mathbf{D} \mathbf{P} (\mathbf{e}_{\hat{j}} \otimes \mathbf{E}) \mathbf{P}^\top\|_F^2 \\ &= \sum_{\hat{i},\hat{j}=1}^{d^i} \|(\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \hat{\mathbf{D}} (\mathbf{e}_{\hat{j}} \otimes \mathbf{E})\|_F^2 \\ &= \sum_{\hat{i},\hat{j}=1}^{d^i} \|\mathbf{C}^\top \hat{\mathbf{D}}_{\hat{i}\hat{j}} \mathbf{E}\|_F^2 \\ &= \|(\mathbf{I} \otimes \mathbf{C})^\top \hat{\mathbf{D}} (\mathbf{I} \otimes \mathbf{E})\|_F^2 \\ &= \|(\mathbf{C} \otimes \mathbf{I})^\top \hat{\mathbf{D}} (\mathbf{E} \otimes \mathbf{I})\|_F^2, \end{aligned}$$

where the last equality comes from again permuting the matrices inside the norm. $\square$

3.2.2 The Partial Trace Operator

We take a moment to formalize the partial trace operator and characterize some useful properties of it. We start by defining the partial trace with respect to a single

subsystem:

Definition 37. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Then, the partial trace of $\mathbf{A}$ with respect to its i^{th} subsystem is

$$\text{tr}_{\{i\}}(\mathbf{A}) := \sum_{j=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_j \otimes \mathbf{I}^{\otimes k-1})^\top \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_j \otimes \mathbf{I}^{\otimes k-1}) \in \mathbb{R}^{d^{k-1} \times d^{k-1}}. \quad (3.1)$$

where these identity matrices are $d \times d$ matrices, and where $\mathbf{e}_j \in \mathbb{R}^d$ is the j^{th} standard basis vector.

Equivalently, the partial trace operator can be defined as the unique linear operator which guarantees that for all $\mathbf{A} \in \mathbb{R}^{d^{i-1} \times d^{i-1}}$, $\mathbf{B} \in \mathbb{R}^{d \times d}$, and $\mathbf{C} \in \mathbb{R}^{d^{k-i} \times d^{k-i}}$, we have that $\text{tr}_{\{i\}}(\mathbf{A} \otimes \mathbf{B} \otimes \mathbf{C}) = \text{tr}(\mathbf{B}) \cdot (\mathbf{A} \otimes \mathbf{C})$.

We will often consider the partial trace of $\mathbf{A}$ with respect to a set of subsystems $\mathcal{S} \subseteq [k]$. We can think of this as either composing the operation in Equation (3.1) $|\mathcal{S}|$ many times, or as extending the summation in Equation (3.1) to sum over all $d^{|\mathcal{S}|}$ basis vectors that span the product of subspaces defined in $\mathcal{S}$. To see this equivalence, consider the partial trace of $\mathbf{A}$ with respect to $\mathcal{S} = \{1, 2\}$:

$$\begin{aligned} \text{tr}_{\{1,2\}}(\mathbf{A}) &= \text{tr}_{\{1\}}(\text{tr}_{\{2\}}(\mathbf{A})) \\ &= \sum_{j=1}^d (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-1})^\top \left(\sum_{\hat{j}=1}^d (\mathbf{I} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2})^\top \mathbf{A} (\mathbf{I} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2}) \right) (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-1}) \\ &= \sum_{j=1}^d \sum_{\hat{j}=1}^d (\mathbf{e}_j \otimes \mathbf{I} \otimes \mathbf{I}^{\otimes k-2})^\top (\mathbf{I} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2})^\top \mathbf{A} (\mathbf{I} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2}) (\mathbf{e}_j \otimes \mathbf{I} \otimes \mathbf{I}^{\otimes k-2}) \\ &= \sum_{j=1}^d \sum_{\hat{j}=1}^d (\mathbf{e}_j \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2})^\top \mathbf{A} (\mathbf{e}_j \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-2}). \end{aligned} \quad (\text{Theorem 33})$$

Next, note that $\mathbf{e}_j \otimes \mathbf{e}_{\hat{j}} \in \mathbb{R}^{d^2}$ is a standard basis vector in $\mathbb{R}^{d^2}$ uniquely identified by the pair $(j, \hat{j})$. That is, we can replace this double sum above with a single sum over standard basis vectors in $\mathbb{R}^{d^2}$:

$$\text{tr}_{\{1,2\}}(\mathbf{A}) = \sum_{j=1}^{d^2} (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-2})^\top \mathbf{A} (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-2}). \quad (\mathbf{e}_j \in \mathbb{R}^{d^2} \text{ now})$$

Which now looks remarkably similar to Theorem 37, but with a larger definition of “subsystem” that the standard basis vector is acting on. Formally, we define the

partial trace of $\mathbf{A}$ with respect to a set $\mathcal{S}$ as following:

Definition 38. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$, and let $\mathcal{S} \subseteq [k]$. Let $i_1, \dots, i_{|\mathcal{S}|}$ be the indices in $\mathcal{S}$, sorted so that $i_1 < i_2 < \dots < i_{|\mathcal{S}|}$. Then, the partial trace of $\mathbf{A}$ with respect to the subsystem $\mathcal{S}$ is

$$\text{tr}_{\mathcal{S}}(\mathbf{A}) := \text{tr}_{\{i_1\}}(\text{tr}_{\{i_2\}}(\dots(\text{tr}_{\{i_{|\mathcal{S}|}\}}(\mathbf{A})))) \in \mathbb{R}^{d^{k-|\mathcal{S}|} \times d^{k-|\mathcal{S}|}}.$$

In the special case that $\mathcal{S} = \{i, i+1, \dots, k\}$, we denote this partial trace as $\text{tr}_i(\mathbf{A})$.

In principle, the order of subscripts does not really matter. Tracing out the first subsystem and then tracing out the second subsystem produces the exact same matrix as tracing out the second subsystem and then tracing out the first subsystem. However, notationally, a non-ascending order becomes messy since removing the second subsystem changes the indices. To see this messiness, note that the above statement would be formalized as $\text{tr}_{\{1\}}(\text{tr}_{\{2\}}(\mathbf{A})) = \text{tr}_{\{1\}}(\text{tr}_{\{1\}}(\mathbf{A}))$. To avoid this issue, we always expand the partial traces in $\mathcal{S}$ in sorted order as in [Theorem 38](#).

Moving on, a vital property of the partial trace is that it is trace-preserving:

Lemma 39 (Folklore). For all $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ and $\mathcal{S} \subseteq [k]$, we have $\text{tr}(\text{tr}_{\mathcal{S}}(\mathbf{A})) = \text{tr}(\mathbf{A})$.

3.2.3 Post-Measurement Reduced Density Matrices

Our analysis also heavily involves post-measurement reduced density matrices, which we now define, and whose long name originates from quantum physics. As mentioned in [Section 3.1.2](#), if $\mathbf{A}$ is a matrix that describes the joint probability distribution of many particles (a density matrix), then $\text{tr}_{\mathcal{S}}(\mathbf{A})$ describes the joint probability distribution of the particles that Bob can view. The Post-measurement Reduced Density Matrix (PMRDM) of $\mathbf{A}$ with respect to vector $\mathbf{x}$ describes the joint distribution of the particles that Bob can view *given that* Alice measured her particles in the direction of the vector $\mathbf{x}$.¹ For brevity, we refer to this matrix as the PMRDM.

Definition 40. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$, $i \in [k]$, and $\mathbf{x}_i \in \mathbb{R}^d$. Then the PMRDM of $\mathbf{A}$ after measuring $\mathbf{x}$ along the i^{th} subsystem is

$$\mathcal{M}_{\{i\}}(\mathbf{A}) := (\mathbf{I}^{\otimes i-1} \otimes \mathbf{x}_i \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{x}_i \otimes \mathbf{I}^{\otimes k-i}) \in \mathbb{R}^{d^{k-1} \times d^{k-1}}.$$

¹For the purpose of understanding the paper, there is no particular need to understand this intuition. Those interested in learning more are recommended to read through Chapter 6 of [1].

Further let $\mathbf{x}_1, \dots, \mathbf{x}_i \in \mathbb{R}^d$. Then, the PMRDM of $\mathbf{A}$ after measuring according to $\mathbf{x}_1, \dots, \mathbf{x}_i$ in the first i subsystems of $[k]$ is

$$\begin{aligned}\mathcal{M}_{:i}(\mathbf{A}) &:= \mathcal{M}_{\{1\}}(\mathcal{M}_{\{2\}}(\dots(\mathcal{M}_{\{i\}}(\mathbf{A})))) \\ &= (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{A} (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i}) \in \mathbb{R}^{d^{k-i} \times d^{k-i}}.\end{aligned}$$

where $\mathbf{x}_{:i} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_i \in \mathbb{R}^{d^i}$.

Notice that the notation for the PMRDM $\mathcal{M}_{\{i\}}(\mathbf{A})$ does not explicitly show the vector $\mathbf{x}_i$ that define the measurement. This is done for visual clarity, and is safe because the meaning of the vector $\mathbf{x}_i$ will always be clear from context.² Further, while we could define the PMRDM with respect to an arbitrary set of subsystems $\mathcal{S}$, as done in [Theorem 38](#), our proofs only need to examine the PMRDM when looking at the first i subsystems, so for simplicity we only define the PMRDM for that case.

We need one important property from the PMRDM, which relates it to the partial trace:

Lemma 41. *Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$, $i \in [k]$, and $\mathbf{x}_1, \dots, \mathbf{x}_i \in \mathbb{R}^d$. Define $\mathbf{x}_{:i} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_i$. Then, $\text{tr}(\mathcal{M}_{:i}(\mathbf{A})) = \mathbf{x}_{:i}^\top \text{tr}_{i+1:}(\mathbf{A}) \mathbf{x}_{:i}$.*

Proof. We directly expand and simplify:

$$\begin{aligned}\text{tr}(\mathcal{M}_{:i}(\mathbf{A})) &= \text{tr}((\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{A} (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i})) \\ &= \sum_{j=1}^{d^{k-i}} \mathbf{e}_j^\top (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{A} (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i}) \mathbf{e}_j \\ &= \sum_{j=1}^{d^{k-i}} \mathbf{x}_{:i}^\top (\mathbf{I}^{\otimes i} \otimes \mathbf{e}_j)^\top \mathbf{A} (\mathbf{I}^{\otimes i} \otimes \mathbf{e}_j) \mathbf{x}_{:i} \quad (\text{Theorem 35}) \\ &= \mathbf{x}_{:i}^\top \text{tr}_{i+1:}(\mathbf{A}) \mathbf{x}_{:i}\end{aligned}$$

□

3.2.4 The Partial Transpose Operator

Finally, we formalize the partial transpose operator, and characterize a useful property of it. We start by defining the partial transpose of $\mathbf{A}$ with respect to a single subsystem.

²Namely, we always take $\mathbf{x}_i$ to be the i^{th} term in the expansion of $\mathbf{x} = \otimes_{i=1}^k \mathbf{x}_i$, where $\mathbf{x}_i$ has either iid Gaussian or Rademacher entries.

Definition 42. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ and $i \in [k]$. Then the partial transpose of $\mathbf{A}$ with respect to the i^{th} subsystem is

$$\mathbf{A}^{\top_{\{i\}}} := \sum_{\hat{i}, \hat{j}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{I}^{\otimes k-i}).$$

Equivalently, the partial transpose operator is the unique linear operator which guarantees that for all $\mathbf{B} \in \mathbb{R}^{d^{i-1} \times d^{i-1}}$, $\mathbf{C} \in \mathbb{R}^{d \times d}$, $\mathbf{D} \in \mathbb{R}^{d^{k-i} \times d^{k-i}}$, we have that $(\mathbf{B} \otimes \mathbf{C} \otimes \mathbf{D})^{\top_{\{i\}}} = (\mathbf{B} \otimes \mathbf{C}^{\top} \otimes \mathbf{D})$. To verify this equivalence, notice that we can express the transpose of a matrix as

$$\sum_{\hat{i}, \hat{j}=1}^d \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \mathbf{C} \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} = \sum_{\hat{i}, \hat{j}=1}^d [\mathbf{C}]_{\hat{j}\hat{i}} \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} = \mathbf{C}^{\top}.$$

Then, we see that

$$\begin{aligned} \sum_{\hat{i}, \hat{j}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) (\mathbf{B} \otimes \mathbf{C} \otimes \mathbf{D}) (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) &= \sum_{\hat{i}, \hat{j}=1}^d (\mathbf{B} \otimes [\mathbf{C}^{\top}]_{\hat{j}\hat{i}} \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{D}) \\ &= (\mathbf{B} \otimes \mathbf{C}^{\top} \otimes \mathbf{D}). \end{aligned}$$

Like the partial trace, we will often be interested in taking the partial transpose of $\mathbf{A}$ with respect to a set of subsystems $\mathcal{V} \subseteq [k]$. We extend the definition of the partial transpose directly to this case:

Definition 43. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ and $\mathcal{V} \subseteq [k]$. Let $i_1, \dots, i_{|\mathcal{V}|}$ be the indices in $\mathcal{V}$. Then, the partial transpose of $\mathbf{A}$ with respect to the subsystem $\mathcal{V}$ is

$$\mathbf{A}^{\top_{\mathcal{V}}} := (((\mathbf{A}^{\top_{i_1}})^{\top_{i_2}}) \dots)^{\top_{i_{|\mathcal{V}|}}}.$$

Notably, the order of partial transposes does not matter in this case. That is, we have $(\mathbf{A}^{\top_{\{i\}}})^{\top_{\{j\}}} = (\mathbf{A}^{\top_{\{j\}}})^{\top_{\{i\}}}$. We next establish two useful properties of the partial transpose:

Lemma 44. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ and $i \in [k]$. Then, $\text{tr}_{\{i\}}(\mathbf{A}^{\top_{\{i\}}}) = \text{tr}_{\{i\}}(\mathbf{A})$.

Proof. We directly compute

$$\begin{aligned}
\text{tr}_{\{i\}}(\mathbf{A}^{\top\{i\}}) &= \sum_{\hat{i}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-i})^{\top} \mathbf{A}^{\top\{i\}} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-i}) \\
&= \sum_{\hat{i}, \hat{j}, \hat{\ell}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}}^{\top} \mathbf{e}_{\hat{j}} \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \\
&= \sum_{\hat{i}, \hat{j}, \hat{\ell}=1}^d \mathbb{1}_{[\hat{i}=\hat{j}]} \mathbb{1}_{[\hat{i}=\hat{\ell}]} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{I}^{\otimes k-i}) \\
&= \sum_{\hat{i}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-i})^{\top} \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-i}) \\
&= \text{tr}_{\{i\}}(\mathbf{A}).
\end{aligned}$$

□

Lemma 45. Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$, $i \in [k]$, $\mathbf{B} \in \mathbb{R}^{d^{i-1} \times d^{i-1}}$, and $\mathbf{C} \in \mathbb{R}^{d^{k-i} \times d^{k-i}}$. Then, for all $\hat{i}, \hat{j} \in [d]$ we have

$$(\mathbf{B} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{C})^{\top} \mathbf{A} (\mathbf{B} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{C}) = (\mathbf{B} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{C})^{\top} \mathbf{A}^{\top\{i\}} (\mathbf{B} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{C})$$

Proof. We expand the right-hand-side:

$$\begin{aligned}
&(\mathbf{B} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{C})^{\top} \mathbf{A}^{\top\{i\}} (\mathbf{B} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{C}) \\
&= (\mathbf{B} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{C})^{\top} \left(\sum_{\hat{\ell}, \hat{m}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{\ell}} \mathbf{e}_{\hat{m}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{\ell}} \mathbf{e}_{\hat{m}}^{\top} \otimes \mathbf{I}^{\otimes k-i}) \right) (\mathbf{B} \otimes \mathbf{e}_{\hat{j}} \otimes \mathbf{C}) \\
&= \sum_{\hat{\ell}, \hat{m}=1}^d (\mathbf{B}^{\top} \otimes \mathbf{e}_{\hat{i}}^{\top} \mathbf{e}_{\hat{\ell}} \mathbf{e}_{\hat{m}}^{\top} \otimes \mathbf{C}^{\top}) \mathbf{A} (\mathbf{B} \otimes \mathbf{e}_{\hat{\ell}} \mathbf{e}_{\hat{m}}^{\top} \otimes \mathbf{C}) \\
&= (\mathbf{B}^{\top} \otimes \mathbf{e}_{\hat{j}}^{\top} \otimes \mathbf{C}^{\top}) \mathbf{A} (\mathbf{B} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{C}),
\end{aligned}$$

where we remove sum by noting that $\mathbf{e}_{\hat{i}}^{\top} \mathbf{e}_{\hat{\ell}}$ and $\mathbf{e}_{\hat{m}}^{\top} \mathbf{e}_{\hat{j}}$ are zero unless $\hat{\ell} = \hat{i}$ and $\hat{m} = \hat{j}$. □

We now have enough tools to exactly understand the variance of the Kronecker-Hutchinson Estimator.

3.3 The Exact Variance of Kronecker-Hutchinson Estimator

We begin by recalling the exact variance of the classical ($k = 1$) Hutchinson's Estimator:

Lemma 46 ([43, 60, 7, 80]). *Let $\mathbf{A} \in \mathbb{R}^{D \times D}$ and let $\mathbf{g} \in \mathbb{R}^D$ be a vector of either iid Rademacher or standard normal Gaussian entries. Let $\hat{\mathbf{A}} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$. Then, $\mathbb{E}[\mathbf{g}^\top \mathbf{A} \mathbf{g}] = \text{tr}(\mathbf{A})$ and $\text{Var}[\mathbf{g}^\top \mathbf{A} \mathbf{g}] \leq 2 \|\hat{\mathbf{A}}\|_F^2 \leq 2 \|\mathbf{A}\|_F^2$. The first inequality is tight if $\mathbf{g}$ is Gaussian, and the second is tight if $\mathbf{A}$ is symmetric.*

This factor of 2 is important for our analysis, and is the base of the exponent $2^{k-|S|}$ in [Theorem 24](#). The key to our proof technique is to use [Theorem 35](#) to expand

$$\begin{aligned} \mathbf{x}^\top \mathbf{A} \mathbf{x} &= (\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k)^\top \mathbf{A} (\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k) \\ &= \mathbf{x}_k^\top (\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_{k-1} \otimes \mathbf{I})^\top \mathbf{A} (\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_{k-1} \otimes \mathbf{I}) \mathbf{x}_k \\ &= \mathbf{x}_k^\top \mathcal{M}_{:k-1}(\mathbf{A}) \mathbf{x}_k \end{aligned} \tag{3.2}$$

which then is just a classical ($k = 1$) Hutchinson's Estimate of the trace of $\mathcal{M}_{:k-1}(\mathbf{A})$. [Theorem 46](#) then gives (for Gaussians) an exact understanding of the variance of this estimator. As a warm-up, we first use this proof technique to show that the Kronecker-Hutchinson Estimator is unbiased, though shorter proofs are certainly possible.

Theorem 47. *Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^d$ be iid Rademacher or standard normal Gaussian vectors, and let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$. Then, $\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{tr}(\mathbf{A})$.*

Proof. For any $i \in [k]$,

$$\begin{aligned} \mathbb{E}[\mathbf{x}_{:i}^\top \text{tr}_{i+1:}(\mathbf{A}) \mathbf{x}_{:i}] &= \mathbb{E}[\mathbf{x}_i^\top \mathcal{M}_{:i-1}(\text{tr}_{i+1:}(\mathbf{A})) \mathbf{x}_i] && \text{(Equation (3.2))} \\ &= \mathbb{E}_{\mathbf{x}_{:i-1}} \mathbb{E}_{\mathbf{x}_i}[\mathbf{x}_i^\top \mathcal{M}_{:i-1}(\text{tr}_{i+1:}(\mathbf{A})) \mathbf{x}_i \mid \mathbf{x}_{:i-1}] && \text{(Tower Rule)} \\ &= \mathbb{E}_{\mathbf{x}_{:i-1}} [\text{tr}(\mathcal{M}_{:i-1}(\text{tr}_{i+1:}(\mathbf{A})))] && \text{(Theorem 46)} \\ &= \mathbb{E}[\mathbf{x}_{:i-1}^\top \text{tr}_i(\mathbf{A}) \mathbf{x}_{:i-1}]. && \text{(Theorem 41)} \end{aligned}$$

Repeating this argument k times in a row, and letting $\text{tr}_\emptyset(\mathbf{A}) := \mathbf{A}$, we conclude by [Theorem 39](#) that

$$\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \mathbb{E}[\mathbf{x}_{:k}^\top \text{tr}_\emptyset(\mathbf{A}) \mathbf{x}_{:k}] = \dots = \mathbb{E}[\mathbf{x}_1^\top \text{tr}_2(\mathbf{A}) \mathbf{x}_1] = \text{tr}(\text{tr}_2(\mathbf{A})) = \text{tr}(\mathbf{A})$$

which completes the proof. □

Analyzing the variance of the estimator requires a bit more effort due to the Frobenius norm term and $\bar{\mathbf{A}}$ matrix that appear in [Theorem 46](#). More concretely, we approach the analysis of the variance of $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ via the classical formula

$$\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] - (\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}])^2.$$

Since we already know that $(\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}])^2 = (\text{tr}(\mathbf{A}))^2$, we turn our attention to evaluating $\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2]$. Note that rearranging [Theorem 46](#) gives that

$$\mathbb{E}[(\mathbf{g}^\top \mathbf{A} \mathbf{g})^2] \leq (\text{tr}(\mathbf{A}))^2 + 2 \|\bar{\mathbf{A}}\|_F^2, \quad (3.3)$$

with equality if $\mathbf{g}$ is Gaussian. Then, noting that $\frac{1}{2}(\mathcal{M}_{:k-1}(\mathbf{A}) + \mathcal{M}_{:k-1}(\mathbf{A})^\top) = \mathcal{M}_{:k-1}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))$, and taking the same approach as in the bias analysis, we see that

$$\begin{aligned} \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] &= \mathbb{E}[(\mathbf{x}_k^\top \mathcal{M}_{:k-1}(\mathbf{A}) \mathbf{x}_k)^2] && \text{(Equation (3.2))} \\ &= \mathbb{E}_{\mathbf{x}_{:k-1}} \mathbb{E}_{\mathbf{x}_k}[(\mathbf{x}_k^\top \mathcal{M}_{:k-1}(\mathbf{A}) \mathbf{x}_k)^2 \mid \mathbf{x}_{:k-1}] && \text{(Tower Rule)} \\ &= \mathbb{E}_{\mathbf{x}_{:k-1}} [(\text{tr}(\mathcal{M}_{:k-1}(\mathbf{A})))^2 + 2 \|\mathcal{M}_{:k-1}(\mathbf{B})\|_F^2] && \text{(Equation (3.3))} \\ &= \mathbb{E}_{\mathbf{x}_{:k-1}} [(\mathbf{x}_{:k-1}^\top \text{tr}_{\{k\}}(\mathbf{A}) \mathbf{x}_{:k-1})^2] + 2 \mathbb{E}_{\mathbf{x}_{:k-1}} [\|\mathcal{M}_{:k-1}(\mathbf{B})\|_F^2], && \text{(Theorem 41)} \end{aligned}$$

where we let $\mathbf{B} := \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$. The first term admits a recursive relationship just like in the proof of [Theorem 47](#), so we can easily handle that. The second term however requires more effort. The key observation we use is that the squared Frobenius norm of the PMRDM is in fact a sum of the squares of many Kronecker-Hutchinson

Estimators:

$$\begin{aligned}
\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2 &= \sum_{\hat{i}, \hat{j}=1}^{d^{k-i}} [\mathcal{M}_{:i}(\mathbf{B})]_{\hat{i}\hat{j}}^2 \\
&= \sum_{\hat{i}, \hat{j}=1}^{d^{k-i}} (\mathbf{e}_{\hat{i}}^\top \mathcal{M}_{:i}(\mathbf{B}) \mathbf{e}_{\hat{j}})^2 \\
&= \sum_{\hat{i}, \hat{j}=1}^{d^{k-i}} (\mathbf{e}_{\hat{i}}^\top (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{B} (\mathbf{x}_{:i} \otimes \mathbf{I}^{k-i}) \mathbf{e}_{\hat{j}})^2 \\
&= \sum_{\hat{i}, \hat{j}=1}^{d^{k-i}} (\mathbf{x}_{:i}^\top (\mathbf{I}^{\otimes i} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{I}^{\otimes i} \otimes \mathbf{e}_{\hat{j}}) \mathbf{x}_{:i})^2 \quad (\text{Theorem 35})
\end{aligned}$$

In particular, this last line is the sum of squares of many Kronecker-Hutchinson Estimators for a wide variety of matrices. Despite these being trace estimates for different matrices, we can just carefully analyze the resulting terms using the Mixed-Product Property (Theorem 33), Theorem 35, and Equation (3.3). We get the following recurrence relation:

Lemma 48. Fix $i \in [k]$, and let $\mathbf{B} \in \mathbb{R}^{d^j \times d^j}$ for some $j \geq i$. Then,

$$\mathbb{E}_{\mathbf{x}_{:i}} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] \leq \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\text{tr}_i(\mathbf{B}))\|_F^2 + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\top\{i\}}))\|_F^2,$$

with equality if $\mathbf{x}_i$ is a standard normal Gaussian vector.

Proof. We first extract the vector $\mathbf{x}_i$ via the same technique as in Equation (3.2). Recall that we always take $\mathbf{I} = \mathbf{I}_d \in \mathbb{R}^{d \times d}$ to be the identity in d dimensions.

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x}_{:i}} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] \\
&= \sum_{\hat{i}, \hat{j}} \mathbb{E}_{\mathbf{x}_{:i}} [(\mathbf{e}_{\hat{i}}^\top \mathcal{M}_{:i}(\mathbf{B}) \mathbf{e}_{\hat{j}})^2] \\
&= \sum_{\hat{i}, \hat{j}} \mathbb{E}_{\mathbf{x}_{:i}} [(\mathbf{e}_{\hat{i}}^\top (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes j-i})^\top \mathbf{B} (\mathbf{x}_{:i} \otimes \mathbf{I}^{\otimes j-i}) \mathbf{e}_{\hat{j}})^2] \\
&= \sum_{\hat{i}, \hat{j}} \mathbb{E}_{\mathbf{x}_{:i}} [((\mathbf{x}_{:i-1} \otimes \mathbf{x}_i \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{x}_i \otimes \mathbf{e}_{\hat{j}}))^2] \\
&= \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}} \mathbb{E}_{\mathbf{x}_i} [(\mathbf{x}_i^\top (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{j}}) \mathbf{x}_i)^2 \mid \mathbf{x}_{:i-1}]
\end{aligned}$$

We take a moment to examine the variance of this Hutchinson's sample. Letting

$\mathbf{C} := \mathbf{x}_{:i-1} \otimes \mathbf{I}$, we get that the variance of the sample is at most

$$2 \left\| \frac{1}{2} ((\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{C} \otimes \mathbf{e}_{\hat{j}}) + (\mathbf{C} \otimes \mathbf{e}_{\hat{j}})^\top \mathbf{B} (\mathbf{C} \otimes \mathbf{e}_{\hat{i}})) \right\|_F^2.$$

By [Theorem 45](#), we know the term on the right is equivalent to $(\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B}^{\top\{i\}} (\mathbf{C} \otimes \mathbf{e}_{\hat{j}})$, so that the norm above becomes $2 \left\| (\mathbf{C} \otimes \mathbf{e}_{\hat{i}})^\top \frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) (\mathbf{C} \otimes \mathbf{e}_{\hat{j}}) \right\|_F^2$. [Theorem 36](#) further tells us the the sum of this variance across all $\hat{i}, \hat{j} \in [d^{j-i}]$ is $2 \left\| (\mathbf{C} \otimes \mathbf{I})^\top \frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) (\mathbf{C} \otimes \mathbf{I}) \right\|_F^2$. Returning to our overall analysis, we get

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_{:i}} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] \\ & \leq \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}} \left[(\text{tr}((\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{j}})))^2 + 2 \left\| (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{i}})^\top \frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{j}}) \right\|_F^2 \right] \\ & = \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}=1} \left[(\text{tr}((\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{j}})))^2 \right] + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| (\mathbf{x}_{:i-1} \otimes \mathbf{I}^{\otimes 2})^\top \frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) (\mathbf{x}_{:i-1} \otimes \mathbf{I}) \right\|_F^2 \\ & = \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}} \left[\left(\sum_{\ell=1}^d \mathbf{e}_\ell^\top (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{i}})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{I} \otimes \mathbf{e}_{\hat{j}}) \mathbf{e}_\ell \right)^2 \right] + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| \mathcal{M}_{:i-1} \left(\frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) \right) \right\|_F^2 \\ & = \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}} \left[\left(\mathbf{e}_{\hat{i}}^\top \sum_{\ell=1}^d (\mathbf{x}_{:i-1} \otimes \mathbf{e}_\ell \otimes \mathbf{I})^\top \mathbf{B} (\mathbf{x}_{:i-1} \otimes \mathbf{e}_\ell \otimes \mathbf{I}) \mathbf{e}_{\hat{j}} \right)^2 \right] + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| \mathcal{M}_{:i-1} \left(\frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) \right) \right\|_F^2 \\ & = \mathbb{E}_{\mathbf{x}_{:i-1}} \sum_{\hat{i}, \hat{j}} \left[\left(\mathbf{e}_{\hat{i}}^\top \mathcal{M}_{:i-1}(\text{tr}_{\{i\}}(\mathbf{B})) \mathbf{e}_{\hat{j}} \right)^2 \right] + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| \mathcal{M}_{:i-1} \left(\frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) \right) \right\|_F^2 \\ & = \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| \mathcal{M}_{:i-1}(\text{tr}_i(\mathbf{B})) \right\|_F^2 + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \left\| \mathcal{M}_{:i-1} \left(\frac{1}{2} (\mathbf{B} + \mathbf{B}^{\top\{i\}}) \right) \right\|_F^2 \end{aligned}$$

Note that the only inequality applies [Equation \(3.3\)](#), which is an equality if $\mathbf{x}_i$ is a standard normal Gaussian. □

Notice that this lemma takes in an arbitrary matrix $\mathbf{B}$, instead of the specific choice of $\mathbf{B} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ from earlier. This is useful, as we apply [Theorem 48](#) repeatedly in the following proof:

Lemma 49. Fix $i \in [k]$, and let $\mathbf{B} \in \mathbb{R}^{d^j \times d^j}$ for some $j \geq i$. Let $\bar{\mathbf{B}} = \frac{1}{2^i} \sum_{\nu \subseteq [i]} \mathbf{B}^{\top \nu}$. Then,

$$\mathbb{E}_{\mathbf{x}_{:i}} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] \leq \sum_{S \subseteq [i]} 2^{i-|S|} \|\text{tr}_S(\bar{\mathbf{B}})\|_F^2,$$

with equality if $\mathbf{x}_1, \dots, \mathbf{x}_i$ are iid standard normal Gaussian vectors.

Proof. We prove this by induction over $i = 1, \dots, j$. For notational clarity in this induction, we let $f_i(\mathbf{B}) := \frac{1}{2^i} \sum_{\nu \subseteq [i]} \mathbf{B}^{\tau_\nu}$. Then, our goal is to prove that $\mathbb{E}_{\mathbf{x}_i} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] \leq \sum_{S \subseteq [i]} 2^{i-|S|} \|\text{tr}_S(f_i(\mathbf{B}))\|_F^2$ for all i and $\mathbf{B}$. We start with the base case $i = 1$. First note that $\text{tr}_{\{1\}}(\mathbf{B}) = \text{tr}_{\{1\}}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{1\}}})) = \text{tr}_{\{1\}}(f_1(\mathbf{B}))$ by [Theorem 44](#). Then [Theorem 48](#) gives us that

$$\begin{aligned} \mathbb{E}_{\mathbf{x}_1} [\|\mathcal{M}_{\{1\}}(\mathbf{B})\|_F^2] &\leq \|\text{tr}_{\{1\}}(\mathbf{B})\|_F^2 + 2 \left\| \frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{1\}}}) \right\|_F^2 \\ &= 2^{1-1} \|\text{tr}_{\{1\}}(f_1(\mathbf{B}))\|_F^2 + 2^{1-0} \|f_1(\mathbf{B})\|_F^2 \\ &= \sum_{S \subseteq [1]} 2^{1-|S|} \|\text{tr}_S(f_1(\mathbf{B}))\|_F^2. \end{aligned}$$

Next, for the induction case, we assume the lemma holds for $i-1$. Since $f_i(\mathbf{B})$ is the average of the all partial transposes in the first i subsystems of $\mathbf{B}$, we can expand $f_i(\mathbf{B}) = f_{i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))$. So, for instance, by our induction hypothesis we have that

$$\mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))\|_F^2 \leq \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_S(f_{i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))\|_F^2 = \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_S(f_i(\mathbf{B}))\|_F^2.$$

Therefore, by [Theorem 48](#), by our inductive hypothesis, and by the composition of partial traces, we have that

$$\begin{aligned} \mathbb{E}_{\mathbf{x}_i} [\|\mathcal{M}_{:i}(\mathbf{B})\|_F^2] &\leq \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\text{tr}_{\{i\}}(\mathbf{B}))\|_F^2 + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))\|_F^2 \\ &= \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\text{tr}_{\{i\}}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))\|_F^2 + 2 \mathbb{E}_{\mathbf{x}_{:i-1}} \|\mathcal{M}_{:i-1}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))\|_F^2 \\ &\leq \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_S(\text{tr}_{\{i\}}(f_i(\mathbf{B})))\|_F^2 + 2 \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_S(f_i(\mathbf{B}))\|_F^2 \\ &= \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_{S \cup \{i\}}(f_i(\mathbf{B}))\|_F^2 + 2 \sum_{S \subseteq [i-1]} 2^{i-1-|S|} \|\text{tr}_S(f_i(\mathbf{B}))\|_F^2 \\ &= \sum_{S \subseteq [i-1]} 2^{i-(|S|+1)} \|\text{tr}_{S \cup \{i\}}(f_i(\mathbf{B}))\|_F^2 + \sum_{S \subseteq [i-1]} 2^{i-|S|} \|\text{tr}_S(f_i(\mathbf{B}))\|_F^2 \end{aligned} \tag{3.4}$$

$$= \sum_{S' \subseteq [i]} 2^{i-|S'|} \|\text{tr}_{S'}(f_i(\mathbf{B}))\|_F^2 \tag{3.5}$$

The second inequality applies the induction hypotheses to the matrix $\text{tr}_{\{i\}}(\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}}))$ on the left term, and applies the induction hypothesis to the matrix $\frac{1}{2}(\mathbf{B} + \mathbf{B}^{\tau_{\{i\}}})$ on the right term.

Further, Equation (3.5) holds by comparing the first and second terms in Equation (3.4), where the left sum covers all $\mathcal{S}' \subseteq [i]$ such that $i \in \mathcal{S}'$, and where the right term covers all $\mathcal{S}' \subseteq [i]$ such that $i \notin \mathcal{S}'$. Note that the only inequalities used are Equation (3.3) and Theorem 48, as well as the induction hypothesis, which are all equalities when considering iid standard normal Gaussian vectors. $\square$

We now have all the tools needed to prove the bound on the variance of the Kronecker-Hutchinson Estimator.

Theorem 24 Restated. Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is independent and is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Let $\bar{\mathbf{A}} := \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \mathbf{A}^{\text{tr}_{\mathcal{V}}}$ be the average of all partial transposes of $\mathbf{A}$. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq \sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F^2$. Furthermore, if all $\mathbf{x}_i$ are $\mathcal{N}(\mathbf{0}, \mathbf{I})$, then this expression is an equality.

Proof. We start again with the observation in Equation (3.2), which we use to prove the following claim by induction over $i = 1, \dots, k$:

$$\mathbb{E}_{\mathbf{x}_{:i}}[(\mathbf{x}_{:i}^\top \text{tr}_{i+1}(\mathbf{A}) \mathbf{x}_{:i})^2] \leq (\text{tr}(\mathbf{A}))^2 + 2 \|\text{tr}_2(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))\|_F^2 + 2 \sum_{j=1}^{i-1} \mathbb{E}_{\mathbf{x}_{:j}} \|\mathcal{M}_{:j}(\text{tr}_{j+2}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)))\|_F^2, \quad (3.6)$$

where we interpret $\text{tr}_{k+1}(\mathbf{A}) := \mathbf{A}$. The base case for Equation (3.6) is when $i = 1$, where we immediately have

$$\mathbb{E}_{\mathbf{x}_1}[(\mathbf{x}_1^\top \text{tr}_2(\mathbf{A}) \mathbf{x}_1)^2] \leq (\text{tr}(\text{tr}_2(\mathbf{A})))^2 + 2 \|\text{tr}_2(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))\|_F^2 = (\text{tr}(\mathbf{A}))^2 + 2 \|\text{tr}_2(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))\|_F^2$$

which completes the base case. Now we assume Equation (3.6) holds for $i - 1$, and we show that it holds for i :

$$\begin{aligned} & \mathbb{E}_{\mathbf{x}_{:i}}[(\mathbf{x}_{:i}^\top \text{tr}_{i+1}(\mathbf{A}) \mathbf{x}_{:i})^2] \\ &= \mathbb{E}_{\mathbf{x}_{:i-1}} \mathbb{E}_{\mathbf{x}_i}[(\mathbf{x}_i^\top \mathcal{M}_{:i-1}(\text{tr}_{i+1}(\mathbf{A})) \mathbf{x}_i)^2 \mid \mathbf{x}_{:i-1}] \\ &\leq \mathbb{E}_{\mathbf{x}_{:i-1}} \left[(\text{tr}(\mathcal{M}_{:i-1}(\text{tr}_{i+1}(\mathbf{A}))))^2 + 2 \|\mathcal{M}_{:i-1}(\text{tr}_{i+1}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)))\|_F^2 \right] \quad (\text{Theorem 46}) \\ &= \mathbb{E}_{\mathbf{x}_{:i-1}} [(\mathbf{x}_{:i-1}^\top \text{tr}_i(\mathbf{A}) \mathbf{x}_{:i-1})^2] + \mathbb{E}_{\mathbf{x}_{:i-1}} 2 \|\mathcal{M}_{:i-1}(\text{tr}_{i+1}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)))\|_F^2 \\ &\leq (\text{tr}(\mathbf{A}))^2 + 2 \|\text{tr}_2(\mathbf{A})\|_F^2 + 2 \sum_{j=1}^{i-1} \mathbb{E}_{\mathbf{x}_{:j}} \|\mathcal{M}_{:j}(\text{tr}_{j+2}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)))\|_F^2 \end{aligned}$$

where the last line applies the induction hypothesis. This completes the proof by induction of [Equation \(3.6\)](#). We can then appeal to [Theorem 49](#) and merge terms together. First, let $\mathbf{C}_i := \frac{1}{2^i} \sum_{\mathcal{V} \subseteq [i]} (\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))^{\top \mathcal{V}}$, and note that $\mathbf{C}_0 = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$. Then,

$$\begin{aligned} \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] &\leq (\text{tr}(\mathbf{A}))^2 + 2 \|\text{tr}_2: (\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))\|_F^2 + 2 \sum_{i=1}^{k-1} \mathbb{E}_{\mathbf{x}:i} \|\mathcal{M}_{:i}(\text{tr}_{i+2:}(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)))\|_F^2 \\ &\leq (\text{tr}(\mathbf{A}))^2 + 2 \|\text{tr}_2: (\mathbf{C}_0)\|_F^2 + 2 \sum_{i=1}^{k-1} \sum_{S \subseteq [i]} 2^{i-|S|} \|\text{tr}_{S \cup \{i+2, \dots, k\}}(\mathbf{C}_i)\|_F^2 \\ &= (\text{tr}(\mathbf{A}))^2 + \sum_{i=0}^{k-1} \sum_{S \subseteq [i]} 2^{i+1-|S|} \|\text{tr}_{S \cup \{i+2, \dots, k\}}(\mathbf{C}_i)\|_F^2 \\ &= (\text{tr}(\mathbf{A}))^2 + \sum_{i=1}^k \sum_{S \subseteq [i-1]} 2^{i-|S|} \|\text{tr}_{S \cup \{i+1, \dots, k\}}(\mathbf{C}_{i-1})\|_F^2 \end{aligned}$$

We proceed by simplifying the partial trace $\text{tr}_{S \cup \{i+1, \dots, k\}}(\mathbf{C}_{i-1})$. First, note that $\mathbf{C}_{i-1}$ is the average of all partial transposes of $\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ in the first $i-1$ subsystems. By [Theorem 44](#), we can extend this to be the average of all partial transposes of $\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ in across all subsystems $\mathcal{V} \subset [i-1] \cup \{i+1, \dots, k\}$, so that

$$\begin{aligned} \text{tr}_{i+1:}(\mathbf{C}_{i-1}) &= \text{tr}_{i+1:} \left(\frac{1}{2^{k-1}} \sum_{\mathcal{V} \subseteq [k] \setminus \{i\}} (\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))^{\top \mathcal{V}} \right) \\ &= \text{tr}_{i+1:} \left(\frac{\frac{1}{2^{k-1}} \sum_{\mathcal{V} \subseteq [k] \setminus \{i\}} \mathbf{A}^{\top \mathcal{V}} + \frac{1}{2^{k-1}} \sum_{\mathcal{V} \subseteq [k] \setminus \{i\}} (\mathbf{A}^\top)^{\top \mathcal{V}}}{2} \right). \end{aligned}$$

The left term above is the average of all partial transposes of $\mathbf{A}$ that do not include subsystem i . Since $(\mathbf{A}^\top)^{\top \mathcal{V}} = \mathbf{A}^{\top \bar{\mathcal{V}}}$ where $\bar{\mathcal{V}} := [k] \setminus \mathcal{V}$, the right term above is the average of all partial transposes of $\mathbf{A}$ that do include i . So, we find that $\text{tr}_{i+1:}(\mathbf{C}_{i-1}) = \text{tr}_{i+1:}(\frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \mathbf{A}^{\top \mathcal{V}}) = \text{tr}_{i+1:}(\bar{\mathbf{A}})$. We are therefore left with the expression

$$\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] \leq (\text{tr}(\mathbf{A}))^2 + \sum_{i=1}^k \sum_{S \subseteq [i-1]} 2^{i-|S|} \|\text{tr}_{S \cup \{i+1, \dots, k\}}(\bar{\mathbf{A}})\|_F^2. \quad (3.7)$$

We now focus on the double summation in [Equation \(3.7\)](#). We start by understanding

the inner summation in terms of the index i from the outer iteration. Let

$$F(i) := \sum_{\mathcal{S} \subseteq [i-1]} 2^{i-|\mathcal{S}|} \|\text{tr}_{\mathcal{S} \cup \{i+1, \dots, k\}}(\bar{\mathbf{A}})\|_F^2$$

so that

$$\sum_{i=1}^k \sum_{\mathcal{S} \subseteq [i-1]} 2^{i-|\mathcal{S}|} \|\text{tr}_{\mathcal{S} \cup \{i+1, \dots, k\}}(\bar{\mathbf{A}})\|_F^2 = \sum_{i=1}^k F(i).$$

We see that $F(i)$ is a weighted sum of $\|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2$ across all $\hat{\mathcal{S}} \subseteq [k]$ of the form $\hat{\mathcal{S}} = \mathcal{S} \cup \{i+1, \dots, k\}$ where $\mathcal{S} \subseteq [i-1]$. That is, we consider all $\hat{\mathcal{S}}$ that satisfies two properties:

1. Indices $i+1, \dots, k$ are all included: $\{i+1, \dots, k\} \subseteq \mathcal{S}$
2. Index i is not included: $i \notin \hat{\mathcal{S}}$

We can intuitively think of i as a “Gap” that precedes the “Suffix” $\{i+1, \dots, k\}$. Crucially, if $\|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2$ appears in a term of $F(i)$, then $\|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2$ *does not* appear in a term of any other $F(j)$. This is because each $\hat{\mathcal{S}}$ has a unique gap: for any $\hat{\mathcal{S}}$, let $i \in [k]$ be the smallest value such that $\{i+1, \dots, k\} \subseteq \hat{\mathcal{S}}$. Then i is the gap of $\hat{\mathcal{S}}$. We note an edge-case, where if $\hat{\mathcal{S}} = \{1, \dots, k\}$ then the gap is $i = 0$ and we recover the term $\|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2 = (\text{tr}(\bar{\mathbf{A}}))^2$ in [Equation \(3.7\)](#).

Next, note that every $\hat{\mathcal{S}}$ is scaled by $2^{i-|\mathcal{S}|}$. Since $|\hat{\mathcal{S}}| = |\mathcal{S} \cup \{i+1, \dots, k\}| = |\mathcal{S}| + k - i$, we know $|\mathcal{S}| - i = |\hat{\mathcal{S}}| - k$, so we can write $2^{i-|\mathcal{S}|} = 2^{k-|\hat{\mathcal{S}}|}$. So, we have shown that every $\hat{\mathcal{S}} \subseteq [k]$ appears exactly once in [Equation \(3.7\)](#), and each term $\|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2$ is scaled by $2^{k-|\hat{\mathcal{S}}|}$. That is,

$$\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] \leq \sum_{\hat{\mathcal{S}} \subseteq [k]} 2^{k-|\hat{\mathcal{S}}|} \|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2,$$

which completes the proof since

$$\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] - (\mathbb{E}[\mathbf{x}^\top \mathbf{A} \mathbf{x}])^2 = \sum_{\hat{\mathcal{S}} \subseteq [k]} 2^{k-|\hat{\mathcal{S}}|} \|\text{tr}_{\hat{\mathcal{S}}}(\bar{\mathbf{A}})\|_F^2.$$

Note that all inequalities in the proof become equalities if we know that $\mathbf{x}_1, \dots, \mathbf{x}_k$ are all iid standard normal Gaussians. $\square$

3.3.1 Proof of Theorem 25

The variance expression in Theorem 24 is extremely tight, being an equality in the case that our query vectors are Kroneckers of Gaussian vectors. However, it is not immediately clear how we should interpret this expression. We now prove Theorem 25, which is a more approachable upper bound of Theorem 24. We start with a short lemma:

Lemma 50. *Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ and $\mathcal{V} \subseteq [k]$. Then, $\|\mathbf{A}^{\top \mathcal{V}}\|_F = \|\mathbf{A}\|_F$.*

Proof. Consider any $i \in \mathcal{V}$. Then, recall that each standard basis $\mathbf{e}_i \in \mathbb{R}^{d^k}$ can be decomposed as $\mathbf{e}_i = \mathbf{e}_{\hat{j}} \otimes \mathbf{e}_{\hat{k}} \otimes \mathbf{e}_{\hat{\ell}}$ where $\mathbf{e}_{\hat{j}} \in \mathbb{R}^{d^{i-1}}$, $\mathbf{e}_{\hat{k}} \in \mathbb{R}^d$, and $\mathbf{e}_{\hat{\ell}} \in \mathbb{R}^{d^{k-i}}$. Then, we can expand the norm of $\mathbf{A}$ as

$$\begin{aligned}
\|\mathbf{A}^{\top \{i\}}\|_F^2 &= \sum_{\hat{i}_1, \hat{i}_2=1}^{d^k} (\mathbf{e}_{\hat{i}_1}^\top \mathbf{A}^{\top \{i\}} \mathbf{e}_{\hat{i}_2})^2 \\
&= \sum_{\hat{j}_1, \hat{j}_2=1}^{d^{i-1}} \sum_{\hat{k}_1, \hat{k}_2=1}^d \sum_{\hat{\ell}_1, \hat{\ell}_2=1}^{d^{k-1}} ((\mathbf{e}_{\hat{j}_1} \otimes \mathbf{e}_{\hat{k}_1} \otimes \mathbf{e}_{\hat{\ell}_1})^\top \mathbf{A}^{\top \{i\}} (\mathbf{e}_{\hat{j}_2} \otimes \mathbf{e}_{\hat{k}_2} \otimes \mathbf{e}_{\hat{\ell}_2}))^2 \\
&= \sum_{\hat{j}_1, \hat{j}_2=1}^{d^{i-1}} \sum_{\hat{k}_1, \hat{k}_2=1}^d \sum_{\hat{\ell}_1, \hat{\ell}_2=1}^{d^{k-1}} \left(\sum_{\hat{m}, \hat{n}=1}^d (\mathbf{e}_{\hat{j}_1}^\top \otimes \mathbf{e}_{\hat{k}_1}^\top \mathbf{e}_{\hat{m}} \mathbf{e}_{\hat{n}}^\top \otimes \mathbf{e}_{\hat{\ell}_1}^\top) \mathbf{A} (\mathbf{e}_{\hat{j}_2} \otimes \mathbf{e}_{\hat{m}} \mathbf{e}_{\hat{n}}^\top \mathbf{e}_{\hat{k}_2} \otimes \mathbf{e}_{\hat{\ell}_2}) \right)^2 \\
&= \sum_{\hat{j}_1, \hat{j}_2=1}^{d^{i-1}} \sum_{\hat{k}_1, \hat{k}_2=1}^d \sum_{\hat{\ell}_1, \hat{\ell}_2=1}^{d^{k-1}} ((\mathbf{e}_{\hat{j}_1}^\top \otimes \mathbf{e}_{\hat{k}_2}^\top \otimes \mathbf{e}_{\hat{\ell}_1}^\top) \mathbf{A} (\mathbf{e}_{\hat{j}_2} \otimes \mathbf{e}_{\hat{k}_1} \otimes \mathbf{e}_{\hat{\ell}_2}))^2 \\
&= \sum_{\hat{i}_1, \hat{i}_2=1}^{d^k} \mathbf{e}_{\hat{i}_1}^\top \mathbf{A} \mathbf{e}_{\hat{i}_2} \\
&= \|\mathbf{A}\|_F^2,
\end{aligned}$$

where we remove the sum over $\hat{m}$ and $\hat{n}$ by noting that $\mathbf{e}_{\hat{k}_1}^\top \mathbf{e}_{\hat{m}}$ and $\mathbf{e}_{\hat{n}}^\top \mathbf{e}_{\hat{k}_2}$ are zero unless $\hat{m} = \hat{k}_1$ and $\hat{n} = \hat{k}_2$. We then complete the lemma by decomposing $\mathbf{A}^{\top \mathcal{V}} = (((\mathbf{A}^{\top \{i_1\}})^\top \mathbf{e}_{\hat{i}_2}) \dots)^\top \mathbf{e}_{\hat{i}_1}$, where $\mathcal{V} = \{i_1, \dots, i_{|\mathcal{V}|}\}$. $\square$

We now prove the theorem:

Theorem 25 Restated. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is independent and is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq \sum_{S \subseteq [k]} 2^{k-|S|} \|\text{tr}_S(\mathbf{A})\|_F^2$. Furthermore, if all $\mathbf{x}_i$ are $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and if $\mathbf{A} = \bar{\mathbf{A}}$, then this expression is an equality.*

Proof. We prove that $\|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F \leq \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F$, which implies the theorem. First, recall that the partial trace is linear, so that by the triangle inequality we can write

$$\|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F = \left\| \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \text{tr}_{\mathcal{S}}(\mathbf{A}^{\top \mathcal{V}}) \right\|_F \leq \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\mathbf{A}^{\top \mathcal{V}})\|_F.$$

Next, we examine the norm $\|\text{tr}_{\mathcal{S}}(\mathbf{A}^{\top \mathcal{V}})\|$. First, by [Theorem 44](#), we can assume without loss of generality that $\mathcal{V} \cap \mathcal{S} = \emptyset$. Then, for notational simplicity, suppose that $\mathcal{V} = \{1, \dots, |\mathcal{V}|\}$. We can then write $\text{tr}_{\mathcal{S}} \mathbf{A}^{\top \mathcal{V}} = (\text{tr}_{\mathcal{S}}(\mathbf{A}))^{\top \mathcal{V}}$. To see this, fix any $i \in \mathcal{V}$ and $j \in \mathcal{S}$. By our without-loss-of-generality claim, we know that $i < j$, and can therefore expand

$$\begin{aligned} \text{tr}_{\{j\}}(\mathbf{A}^{\top \{i\}}) &= \sum_{\hat{i}=1}^d (\mathbf{I}^{\otimes j-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-j})^{\top} \mathbf{A}^{\top \{i\}} (\mathbf{I}^{\otimes j-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-j}) \\ &= \sum_{\hat{i}, \hat{j}, \hat{\ell}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{j}} \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{j-i-1} \otimes \mathbf{e}_{\hat{i}}^{\top} \otimes \mathbf{I}^{\otimes k-j}) \mathbf{A} (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{j}} \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{j-i-1} \otimes \mathbf{e}_{\hat{i}} \otimes \mathbf{I}^{\otimes k-j}) \\ &= \sum_{\hat{i}, \hat{j}, \hat{\ell}=1}^d (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{j}} \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{k-i}) \text{tr}_{\{j\}}(\mathbf{A}) (\mathbf{I}^{\otimes i-1} \otimes \mathbf{e}_{\hat{j}} \mathbf{e}_{\hat{\ell}}^{\top} \otimes \mathbf{I}^{k-i}) \\ &= (\text{tr}_{\{j\}}(\mathbf{A}))^{\top \{i\}}. \end{aligned}$$

So, we can pull all of the partial transposes from inside of the partial trace to outside the partial trace. Then, by [Theorem 50](#), we can conclude that

$$\begin{aligned} \|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F &\leq \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\mathbf{A}^{\top \mathcal{V}})\|_F \\ &= \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \|(\text{tr}_{\mathcal{S}}(\mathbf{A}))^{\top \mathcal{V}}\|_F \\ &= \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F \\ &= \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F. \end{aligned}$$

□

While [Theorem 25](#) is more approachable than [Theorem 24](#), it is still not immediately clear how to think of this rate. In order to get a better understanding of this variance, we turn to the worst-case performance of this estimator for the class of PSD matrices.

3.4 Cost of the Kronecker-Hutchinson Estimator for PSD Matrices

Trace estimation is often studied under the assumption that the input matrix $\mathbf{A}$ is PSD. This assumption is very common and reasonable in practice, and without such an assumption, it is impossible for an algorithm such as Hutchinson's Estimator to achieve a $(1 \pm \varepsilon)$ relative error approximation to the trace of $\mathbf{A}$. This is because an indefinite matrix $\mathbf{A}$ can have $\text{tr}(\mathbf{A}) = 0$ while having large Frobenius norm, meaning that Hutchinson's Estimator would have to perfectly recover the trace of $\mathbf{A}$ despite having a nonzero variance. This would incur infinite sample complexity. However, if we assume that $\mathbf{A}$ is PSD, then $\|\mathbf{A}\|_F \leq \text{tr}(\mathbf{A})$, and so we know that the standard deviation of the classical Hutchinson's Estimator is $\sqrt{\frac{2}{\ell}} \|\mathbf{A}\|_F \leq \sqrt{\frac{2}{\ell}} \text{tr}(\mathbf{A})$. That is, $\ell = O(\frac{1}{\varepsilon^2})$ samples suffice to achieve $(1 \pm \varepsilon)$ relative error with at least constant probability.

We now make the same assumption and study the Kronecker-Hutchinson Estimator, and show that $\ell = O(\frac{3^k}{\varepsilon^2})$ samples suffices to estimate the trace of a PSD matrix to $(1 \pm \varepsilon)$ relative error. To start, we need a basic property of the partial trace:

Lemma 51. *Let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ be a PSD matrix, and let $\mathcal{S} \subseteq [k]$. Then, $\text{tr}_{\mathcal{S}}(\mathbf{A})$ is also PSD.*

Proof. Recall that if $\mathbf{A}$ is PSD, then $\mathbf{B}^\top \mathbf{A} \mathbf{B}$ is also PSD for any matrix $\mathbf{B}$ of compatible dimension. Further recall that if $\mathbf{A}$ and $\mathbf{C}$ are both PSD, then $\mathbf{A} + \mathbf{C}$ is also PSD. Since $\text{tr}_{\mathcal{S}}(\mathbf{A})$ is a composition of such operations on $\mathbf{A}$, we get that $\text{tr}_{\mathcal{S}}(\mathbf{A})$ is also PSD. $\square$

Lemma 52 (Folklore). *Let $\mathbf{B}$ be a PSD matrix. Then, $\|\mathbf{B}\|_F \leq \text{tr}(\mathbf{B})$.*

We now prove our next core theorem:

Theorem 27 Restated. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ be a PSD matrix. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is either a Rademacher vector or a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian vector. Then, $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq (3^k \text{tr}(\mathbf{A}))^2$.*

Proof. We first apply [Theorems 25, 39](#) and [52](#) to get

$$\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] \leq \sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2 \leq \sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} (\text{tr}(\text{tr}_{\mathcal{S}}(\mathbf{A})))^2 = (\text{tr}(\mathbf{A}))^2 \sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|}.$$

So we turn our attention to bounding this sum. If we let $\sigma_i := \mathbb{1}_{[i \notin \mathcal{S}]}$, then we can write $k - |\mathcal{S}| = \sum_i \sigma_i$, and so

$$\sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} = \sum_{\sigma \in \{0,1\}^k} 2^{\sum_i \sigma_i} = \sum_{\sigma_1 \in \{0,1\}} 2^{\sigma_1} \left(\dots \left(\sum_{\sigma_{k-1} \in \{0,1\}} 2^{\sigma_{k-1}} \left(\sum_{\sigma_k \in \{0,1\}} 2^{\sigma_k} \right) \right) \right) = 3^k.$$

We then conclude that $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] \leq \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] \leq 3^k (\text{tr}(\mathbf{A}))^2$. $\square$

In particular, if we take ℓ samples in the Kronecker-Hutchinson Estimator, we get a standard deviation of $\sqrt{\text{Var}[H_\ell(\mathbf{A})]} \leq \sqrt{\frac{3^k}{\ell} (\text{tr}(\mathbf{A}))}$, which is less than $\varepsilon \text{tr}(\mathbf{A})$ for $\ell \geq \frac{3^k}{\varepsilon^2}$. More formally, by Chebyshev's inequality, we find that $\ell = O(\frac{3^k}{\varepsilon^2})$ samples suffices to get

$$(1 - \varepsilon) \text{tr}(\mathbf{A}) \leq H_\ell(\mathbf{A}) \leq (1 + \varepsilon) \text{tr}(\mathbf{A})$$

with constant probability.

3.4.1 A Matching Lower Bound

We might hope that the painful rate of $\ell \geq \frac{3^k}{\varepsilon^2}$ results from a loose analysis of the Kronecker-Hutchinson Estimator. Or, perhaps, that some more clever choice of random vector distribution can avoid this painful exponential dependence. We show that this is not the case, at least not among vectors with real iid mean-zero unit-variance components:

Theorem 29 Restated. *Fix $d, k \in \mathbb{N}$. Consider Hutchinson's Estimator run with independent vectors $\mathbf{x} = \mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_k$ where $\mathbf{x}_i$ are vectors of real iid zero-mean unit-variance entries. Then, there exists a PSD matrix $\mathbf{A}$ such that Hutchinson's Estimator needs $\ell = \Omega(\frac{(3-\frac{2}{d})^k}{\varepsilon^2})$ samples in order to have standard deviation $\leq \varepsilon \text{tr}(\mathbf{A})$.*

Proof. We let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$ be the all-ones matrix. If we let $\mathbf{e} \in \mathbb{R}^d$ denote the all-ones vector, then we can write $\mathbf{A} = \otimes_{i=1}^k \mathbf{e} \mathbf{e}^\top$. This is helpful because we can then use the Mixed-Product property ([Theorem 33](#)) to show that

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \otimes_{i=1}^k \mathbf{x}_i^\top \mathbf{e} \mathbf{e}^\top \mathbf{x}_i = \prod_{i=1}^k (\mathbf{e}^\top \mathbf{x}_i)^2,$$

where we can replace the Kronecker product with a scalar product because the Kronecker product of scalars is just the scalar product. Notably, we see that $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is a product of k iid terms. Since we are interested in the variance of the estimator, we

actually want to examine

$$\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] = \prod_{i=1}^k \mathbb{E}[(\mathbf{e}^\top \mathbf{x}_i)^4] = (\mathbb{E}[(\mathbf{e}^\top \mathbf{x}_1)^4])^k.$$

So, letting $\mathbf{y} := \mathbf{x}_1$ in order to simplify notation, we turn our attention to lower bounding the expectation of $(\mathbf{e}^\top \mathbf{y})^4 = (\sum_{i=1}^d y_i)^4$ where $\mathbf{y}$ is a vector of iid zero-mean unit-variance terms. We expand this sum as

$$\mathbb{E}[(\sum_{i=1}^d y_i)^4] = \sum_{i,j,m,n=1}^d \mathbb{E}[y_i y_j y_m y_n].$$

This expectation is zero if any random variable appears exactly once, like how $\mathbb{E}[y_1 y_2^3] = \mathbb{E}[y_1] \mathbb{E}[y_2^3] = 0$ by independence. So, we can reduce the sum above to terms that either look like $\mathbb{E}[y_i^4]$ or like $\mathbb{E}[y_i^2 y_j^2]$. Terms like $\mathbb{E}[y_i^4]$ occur exactly d times. Terms like $\mathbb{E}[y_i^2 y_j^2]$ occur exactly $3d^2 - 3d$ times³. We can then bound $\mathbb{E}[y_i^4] \geq (\mathbb{E}[y_i^2])^2 = 1$ by Jensen's Inequality, and $\mathbb{E}[y_i^2 y_j^2] = \mathbb{E}[y_i^2] \mathbb{E}[y_j^2] = 1$ by independence. So, we get that

$$\mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] = (\mathbb{E}[(\mathbf{e}^\top \mathbf{y})^4])^k \geq (d + 3d^2 - 3d)^k = (3d^2 - 2d)^k.$$

Which, noting that $\text{tr}(\mathbf{A}) = d^k$, in turn gets a variance lower bound of

$$\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \mathbb{E}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2] - (\text{tr}(\mathbf{A}))^2 \geq (3d^2 - 2d)^k - d^{2k}.$$

Then, since $\text{Var}[H_\ell(\mathbf{A})] = \frac{1}{\ell} \text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}]$, in order for the Kronecker-Hutchinson's Estimator to have standard deviation $\varepsilon \text{tr}(\mathbf{A})$, we need to take

$$\ell \geq \frac{\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x}]}{\varepsilon^2 (\text{tr}(\mathbf{A}))^2} \geq \frac{(3d^2 - 2d)^k - d^{2k}}{\varepsilon^2 d^{2k}} = \frac{(3 - \frac{2}{d})^k - 1}{\varepsilon^2} = \Omega\left(\frac{(3 - \frac{2}{d})^k}{\varepsilon^2}\right).$$

□

We note that this lower bound uses a rank-one matrix, which is interesting because we can exactly compute the trace of any rank-one matrix with a single Kronecker-matrix-vector product. In particular, if $\mathbf{A} = \mathbf{u} \mathbf{u}^\top$ for some $\mathbf{u} \in \mathbb{R}^d$, then for any vector $\mathbf{x}$ not orthogonal to $\mathbf{u}$, we have that

$$\frac{\|\mathbf{A} \mathbf{x}\|_2^2}{\mathbf{x}^\top \mathbf{A} \mathbf{x}} = \frac{\mathbf{x}^\top \mathbf{u} \cdot \mathbf{u}^\top \mathbf{u} \cdot \mathbf{u}^\top \mathbf{x}}{\mathbf{x}^\top \mathbf{u} \cdot \mathbf{u}^\top \mathbf{x}} = \mathbf{u}^\top \mathbf{u} = \text{tr}(\mathbf{A}).$$

³This can happen 3 ways: if $i = j \neq m = n$ or if $i = m \neq j = n$ or if $i = n \neq j = m$. Each such pattern occurs $d^2 - d$ times, so the overall pattern occurs $3d^2 - 3d$ times.

Since a Kronecker of Gaussian vectors is not orthogonal to any fixed vector $\mathbf{u}$ with probability 1, we can exactly compute this trace with probability 1. That is to say, the hard instance used in this lower bound is truly only hard for the Kronecker-Hutchinson Estimator, and is not hard in general for the Kronecker-matrix-vector oracle model.

The lower bound is also structured in a second way – the matrix $\mathbf{A} = \otimes_{i=1}^k \mathbf{e}\mathbf{e}^\top$ is not only rank-one but also has Kronecker structure. In [Section 3.7](#) we prove that we can exactly compute the trace of any matrix $\mathbf{A}$ with Kronecker structure by using $kd+1$ Kronecker-matrix-vector products. That is to say, the hard instance used in this lower bound is in a second sense only hard for the Kronecker-Hutchinson’s Estimator, and its class of hard input matrices is not hard in general for the Kronecker-matrix-vector oracle model. That said, we do not have a clear sense of what algorithms may be able to bridge this gap and admit much faster trace estimation.

3.4.2 Estimating the Trace of a Random Rank-One Matrix

To round off our discussion of PSD matrix trace estimation, we show that estimating the trace of a random rank-one matrix still has an exponential dependence on k , but a milder dependence than shown in the worst-case bound of [Theorem 29](#). This proof serves two purposes. First, supposing our upper bound is tight, it suggests that the Kronecker-Hutchinson Estimator can still have an exponentially slow rate of convergence for matrices that are far from having Kronecker structure. Second, it shows how the variance bound from [Theorem 25](#) can be used to get a tightened understanding of the concentration of the Kronecker-Hutchinson Estimator for particular matrices.

We study the random rank-one matrix $\mathbf{A} = \mathbf{g}\mathbf{g}^\top \in \mathbb{R}^{d^k \times d^k}$, where $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. By analyzing the expected squared Frobenius norm of the partial traces of $\mathbf{A}$, we prove the following:

Theorem 26 Restated. *Fix $d, k \in \mathbb{N}$. Let $\mathbf{A} = \mathbf{g}\mathbf{g}^\top$ where $\mathbf{g} \in \mathbb{R}^{d^k}$ is a vector of iid standard normal Gaussians. Then, with probability $\frac{3}{4}$, we know that $\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x} \mid \mathbf{g}] \leq O((2 + \frac{1}{d})^k (\text{tr}(\mathbf{A}))^2)$. In particular, with probability $\frac{2}{3}$, we know that the Kronecker-Hutchinson Estimator returns a $(1 \pm \varepsilon)$ relative error estimate to the trace of $\mathbf{A}$ if $\ell = O(\frac{1}{\varepsilon^2}(2 + \frac{1}{d})^k)$.*

Proof. We start by analyzing the expected squared Frobenius norm of the partial trace of $\mathbf{A}$ with respect to a set of subsystems $\mathcal{S} \subseteq [k]$. Let $i := |\mathcal{S}|$, and note that without loss of generality we can assume that $\mathcal{S} = \{1, \dots, i\}$ when trying to

compute $\mathbb{E}[\|\text{tr}_S(\mathbf{A})\|_F^2]$. We then decompose the vector $\mathbf{g} = \begin{bmatrix} \mathbf{g}_1 \\ \vdots \\ \mathbf{g}_{d^i} \end{bmatrix}$ where $\mathbf{g}_i \in \mathbb{R}^{d^{k-i}}$ are standard normal Gaussian vectors. Then, note that we can decompose $\mathbf{g} = (\mathbf{e}_1 \otimes \mathbf{g}_1) + \dots + (\mathbf{e}_{d^i} \otimes \mathbf{g}_{d^i})$. This lets us write out

$$\mathbf{g}^\top (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-i}) = \sum_{j=1}^{d^i} (\mathbf{e}_j \otimes \mathbf{g}_j)^\top (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-i}) = \sum_{j=1}^{d^i} (\mathbf{e}_j^\top \mathbf{e}_j \otimes \mathbf{g}_j^\top) = \mathbf{g}_j^\top.$$

Then, if we define $\mathbf{G} = \begin{bmatrix} \mathbf{g}_1 & \dots & \mathbf{g}_{d^i} \end{bmatrix} \in \mathbb{R}^{d^{k-i} \times d^i}$, then we see that

$$\text{tr}_S(\mathbf{g}\mathbf{g}^\top) = \sum_{j=1}^{d^i} (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-i})^\top \mathbf{g}\mathbf{g}^\top (\mathbf{e}_j \otimes \mathbf{I}^{\otimes k-i}) = \sum_{j=1}^{d^i} \mathbf{g}_j \mathbf{g}_j^\top = \mathbf{G}\mathbf{G}^\top.$$

Noticing that $\mathbf{G}$ is just a matrix full of iid standard normal Gaussians, we can then bound $\|\text{tr}_S(\mathbf{A})\|_F^2 = \|\mathbf{G}\mathbf{G}^\top\|_F^2 = \|\mathbf{G}\|_4^4$ in expectation by just using known properties of the norms of Gaussians, where $\|\cdot\|_4$ denotes the Schatten 4-norm. For instance, Lemma A.1 from [116] implies that a standard normal Gaussian matrix $\mathbf{G} \in \mathbb{R}^{N \times P}$ has $\mathbb{E}[\|\mathbf{G}\|_4^4] = NP(N + P + 1)$. So,

$$\mathbb{E}[\|\text{tr}_S(\mathbf{A})\|_F^2] = \mathbb{E}[\|\mathbf{G}\|_4^4] = d^k(d^{k-|\mathcal{S}|} + d^{|\mathcal{S}|} + 1).$$

This in turn implies that

$$\begin{aligned} \mathbb{E}_{\mathbf{g}} \left[\sum_{S \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_S(\mathbf{A})\|_F^2 \right] &= \sum_{S \subseteq [k]} 2^{k-|\mathcal{S}|} d^k (d^{k-|\mathcal{S}|} + d^{|\mathcal{S}|} + 1) \\ &= \sum_{i=0}^k \binom{k}{i} 2^{k-i} d^k (d^{k-i} + d^i + 1). \end{aligned}$$

The last line above recognizes that we only depended on $|\mathcal{S}|$, so we could just sum over all values of $i = |\mathcal{S}|$, scaled by the $\binom{k}{i}$ times that $|\mathcal{S}|$ appears. Rearranging terms,

and recalling that $\sum_{i=0}^k \binom{k}{i} c^i = (1+c)^k$, we get

$$\begin{aligned}
\mathbb{E}_{\mathbf{g}} \left[\sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2 \right] &= 2^k d^k \sum_{i=0}^k \binom{k}{i} 2^{-i} (d^{k-i} + d^i + 1) \\
&= 2^k d^k \sum_{i=0}^k \binom{k}{i} \left(d^k \left(\frac{1}{2d}\right)^i + \left(\frac{d}{2}\right)^i + \left(\frac{1}{2}\right)^i \right) \\
&= 2^k d^k \left(d^k \left(1 + \frac{1}{2d}\right)^k + \left(1 + \frac{d}{2}\right)^k + \left(\frac{3}{2}\right)^k \right) \\
&= 2^k d^{2k} \left(\left(1 + \frac{1}{2d}\right)^k + \left(\frac{1}{2} + \frac{1}{d}\right)^k + \left(\frac{3}{2d}\right)^k \right) \\
&\leq 2^k d^{2k} \cdot 3 \left(1 + \frac{1}{2d}\right)^k \\
&= 3d^{2k} \left(2 + \frac{1}{d}\right)^k.
\end{aligned}$$

Then, by Markov's Inequality, we know that

$$\text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x} \mid \mathbf{g}] \leq \mathbb{E}_{\mathbf{g}} \left[\sum_{\mathcal{S} \subseteq [k]} 2^{k-|\mathcal{S}|} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2 \right] \leq 8 \cdot 3d^{2k} \left(2 + \frac{1}{d}\right)^k$$

holds with probability at least $\frac{7}{8}$. Then, since $\text{tr}(\mathbf{A}) = \|\mathbf{g}\|_2^2$ is a chi-square random variable with parameter d^k , we can use Chebyshev's Inequality to bound $\text{tr}(\mathbf{A}) > \frac{1}{2}d^k$ with probability at least $\frac{7}{8}$ if $d^k > 64$. We condition on both of these events, together holding with probability at least $\frac{3}{4}$. Given these events, the variance of the Kronecker-Hutchinson Estimator is therefore

$$\text{Var}[(\mathbf{x}^\top \mathbf{A} \mathbf{x})^2 \mid \mathbf{g}] \leq 24d^{2k} \left(2 + \frac{1}{d}\right)^k \leq 96 \left(2 + \frac{1}{d}\right)^k (\text{tr}(\mathbf{A}))^2.$$

We can now examine how many samples ℓ are needed for the Kronecker-Hutchinson Estimator to achieve a $(1 \pm \varepsilon)$ relative error estimation to $\mathbf{A}$ with constant probability. By Chebyshev's Inequality, we get that $|H_\ell(\mathbf{A}) - \text{tr}(\mathbf{A})| \leq \sqrt{12 \text{Var}[H_\ell(\mathbf{A}) \mid \mathbf{g}]} = \frac{\sqrt{12}}{\sqrt{\ell}} \sqrt{\text{Var}[H_\ell(\mathbf{A}) \mid \mathbf{g}]}$ with probability $\frac{1}{12}$. To make this additive error at most $\varepsilon \text{tr}(\mathbf{A})$, we take

$$\ell = \frac{12 \text{Var}[\mathbf{x}^\top \mathbf{A} \mathbf{x} \mid \mathbf{g}]}{\varepsilon^2 (\text{tr}(\mathbf{A}))^2} \leq \frac{12 \cdot 96 \left(2 + \frac{1}{d}\right)^k (\text{tr}(\mathbf{A}))^2}{\varepsilon^2 (\text{tr}(\mathbf{A}))^2} = 1152 \frac{\left(2 + \frac{1}{d}\right)^k}{\varepsilon^2}.$$

That is, with constant probability, $\ell = O(\frac{1}{\varepsilon^2} (2 + \frac{1}{d})^k)$ samples suffice to get a relative error approximation to a random rank-one matrix with constant probability. By allowing larger d and k values, and by using sharper concentrations, the constant of

1152 can be significantly improved. $\square$

3.5 The Complex-Valued Kronecker-Matrix-Vector Oracle

Until now, we have only considered using a *real-valued* Kronecker-matrix-vector oracle, where we are only allowed to compute $\mathbf{Ax}$ for vectors $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ where $\mathbf{x}_i \in \mathbb{R}^d$. In this section, we consider how our results change when we instead allow $\mathbf{x}_i \in \mathbb{C}^d$. If we did not have any Kronecker constraint on our oracle, then this change of oracles would be banal. If some algorithm computes q complex-valued non-Kronecker matrix-vector products with $\mathbf{A}$, then we can find another algorithm which produces the exact same result, but instead uses $2q$ real-valued non-Kronecker matrix-vector products with $\mathbf{A}$. Each complex-values query gets decomposed as $\mathbf{Ax} = \mathbf{A}\Re[\mathbf{x}] + i\mathbf{A}\Im[\mathbf{x}]$, which can be evaluated with just two real-valued oracle queries.

However, this picture changes with a Kronecker constraint. If $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ for $\mathbf{x}_i \in \mathbb{C}^d$, then the real part of the vector $\Re[\mathbf{x}]$ does not necessarily have Kronecker structure as well, and therefore we cannot reproduce the strategy above. We can see this even with very small vectors:

$$\begin{bmatrix} 1 \\ i \end{bmatrix} \otimes \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} 1 \\ i \\ i \\ -1 \end{bmatrix} \quad \text{which gives} \quad \Re \left(\begin{bmatrix} 1 \\ i \\ i \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad \text{and} \quad \Im \left(\begin{bmatrix} 1 \\ i \\ i \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

The real and imaginary component vectors do not have Kronecker structure. We can still simulate a complex-valued oracle with a real-valued oracle, but this requires far more real-valued oracle queries in the worst case. Specifically, we expand the real and imaginary parts of each vectors $\mathbf{x}_i$. To keep the notation simple, let $\mathbf{x}_i = \mathbf{r}_i + i\mathbf{m}_i$

decompose the real and imaginary parts of $\mathbf{x}_i \in \mathbb{C}^d$, so that $\mathbf{r}_i, \mathbf{m}_i \in \mathbb{R}^d$. Then,

$$\begin{aligned}
\mathbf{x}_1 \otimes \mathbf{x}_2 \otimes \cdots \otimes \mathbf{x}_k &= (\mathbf{r}_1 + i\mathbf{m}_1) \otimes \mathbf{x}_2 \otimes \cdots \otimes \mathbf{x}_k \\
&= (\mathbf{r}_1 \otimes \mathbf{x}_2 \otimes \cdots \otimes \mathbf{x}_k) + i(\mathbf{m}_1 \otimes \mathbf{x}_2 \otimes \cdots \otimes \mathbf{x}_k) \\
&= (\mathbf{r}_1 \otimes \mathbf{r}_2 \otimes \cdots \otimes \mathbf{x}_k) + i(\mathbf{r}_1 \otimes \mathbf{m}_2 \otimes \cdots \otimes \mathbf{x}_k) \\
&\quad + i(\mathbf{m}_1 \otimes \mathbf{r}_2 \otimes \cdots \otimes \mathbf{x}_k) - (\mathbf{m}_1 \otimes \mathbf{m}_2 \otimes \cdots \otimes \mathbf{x}_k) \\
&= \dots
\end{aligned}$$

where we continue to expand this out into a sum of 2^k real-valued Kronecker-structured vectors. So, if some algorithm solves a problem using q complex-valued Kronecker-structure matrix-vector products with $\mathbf{A}$, then we can only immediately guarantee the existence of an algorithm that computes $2^k q$ real-valued Kronecker-structured products with $\mathbf{A}$ that also solves the problem.

Keeping this exponential gap between the real-valued and complex-valued oracles in mind, we now turn our attention back to the Kronecker-Hutchinson Estimator. In the non-Kronecker case, it is well known that the Hutchinson Estimator performs well with complex queries:

Lemma 53. *Let $\mathbf{A} \in \mathbb{R}^{D \times D}$ and let $\mathbf{g} \in \mathbb{C}^D$ be a vector of either iid complex Rademacher or complex standard normal Gaussian entries. That is, we have $\mathbf{g} = \frac{1}{\sqrt{2}}\mathbf{r} + \frac{i}{\sqrt{2}}\mathbf{m}$ where $\mathbf{r}$ and $\mathbf{m}$ are iid real-valued vectors of either Rademacher or standard normal Gaussian entries. Then, $\mathbb{E}[\mathbf{g}^H \mathbf{A} \mathbf{g}] = \text{tr}(\mathbf{A})$ and $\text{Var}[\mathbf{g}^H \mathbf{A} \mathbf{g}] \leq \|\frac{\mathbf{A} + \mathbf{A}^T}{2}\|_F^2 \leq \|\mathbf{A}\|_F^2$, where the first inequality is tight if $\mathbf{g}$ is complex Gaussian, and the second is tight if $\mathbf{A}$ is symmetric.*

Note that the difference between [Theorem 46](#) and [Theorem 53](#) is just a factor of 2 in the variance term. However, this factor of 2 has been extremely influential in all of our proofs so far. Improving this constant to be 1 and following the exact same proofs give the following results:

Theorem 30 Restated. *Fix $d, k \in \mathbb{N}$, and let $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$. Let $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$, where each $\mathbf{x}_i$ is either a complex Rademacher vector or a complex Gaussian vector. Let $\bar{\mathbf{A}} := \frac{1}{2^k} \sum_{\mathcal{V} \subseteq [k]} \mathbf{A}^{\text{tr}_{\mathcal{V}}}$ be the average of all partial transposes of $\mathbf{A}$. Then, $\text{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}] \leq \sum_{\mathcal{S} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\bar{\mathbf{A}})\|_F^2 \leq \sum_{\mathcal{S} \subseteq [k]} \|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2$. If $\mathbf{x}_i$ are complex Gaussians, then the first expression is an equality. If we further know that $\mathbf{A} = \bar{\mathbf{A}}$, then both expressions are equalities. Further, if $\mathbf{A}$ is PSD, then we have $\text{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}] \leq (2^k \text{tr}(\mathbf{A}))^2$.*

Proof. Retrace the proofs of [Theorems 48](#) and [49](#) as well as [Theorems 25](#) and [27](#). Anywhere that a 2 appears, replace it with a 1. $\square$

We can similarly show that this dependence of 2^k is essentially tight:

Theorem 32 Restated. *Fix $d, k \in \mathbb{N}$. Consider Hutchinson's Estimator run with vectors $\mathbf{x} = \mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_k$ where $\mathbf{x}_i$ are vectors of iid complex zero-mean unit-variance entries. Then, there exists a PSD matrix $\mathbf{A}$ such that Hutchinson's Estimator needs $\ell = \Omega(\frac{(2-\frac{1}{d})^k}{\varepsilon^2})$ samples in order to have standard deviation $\leq \varepsilon \text{tr}(\mathbf{A})$.*

Proof. Like in the proof of Theorem 29, we take $\mathbf{A} = \otimes_{i=1}^k \mathbf{e}\mathbf{e}^\top$ where $\mathbf{e} \in \mathbb{R}^d$ is the all-ones vector. In fact, we follow the proof of Theorem 29 rather closely, but we recreate much of it to handle the complex conjugation correctly. First, we expand the product

$$\mathbb{E}[(\mathbf{x}^\mathbf{H} \mathbf{A} \mathbf{x})^2] = \prod_{i=1}^k \mathbb{E}[(\mathbf{x}_i^\mathbf{H} \mathbf{e} \mathbf{e}^\mathbf{H} \mathbf{x}_i)^2] = \left(\mathbb{E}[(\mathbf{e}^\mathbf{H} \mathbf{x}_1)^2 (\overline{\mathbf{e}^\mathbf{H} \mathbf{x}_1})^2] \right)^k.$$

We let $\mathbf{y} := \mathbf{x}_1$ to simplify the notation. Then, we expand this inner expectation:

$$\mathbb{E}[(\mathbf{e}^\mathbf{H} \mathbf{y})^2 (\overline{\mathbf{e}^\mathbf{H} \mathbf{y}})^2] = \mathbb{E}[(\sum_{i,j=1}^d y_i y_j)(\sum_{m,n=1}^d \overline{y_m y_n})] = \sum_{i,j,m,n=1}^d \mathbb{E}[y_i y_j \overline{y_m y_n}].$$

This expectation is zero if any random variable appears exactly once, like how $\mathbb{E}[y_1 y_2 \overline{y_2^2}] = \mathbb{E}[y_1] \mathbb{E}[y_2 \overline{y_2^2}] = 0$ by independence. So, we only need to consider terms in this sum where we do not have any one term appear alone. This can happen in three different ways.

First, if $i = j = m = n$, then by Jensen's inequality, we get $\mathbb{E}[y_i y_j \overline{y_m y_n}] = \mathbb{E}[|y_i|^4] \geq (\mathbb{E}[|y_i|^2])^2 = 1$. This case occurs d times in the sum. Second, if $i = j \neq m = n$, then by the iid distribution of the entry of $\mathbf{y}$, we get $\mathbb{E}[y_i y_j \overline{y_m y_n}] = \mathbb{E}[y_i^2] \mathbb{E}[\overline{y_m^2}] = |\mathbb{E}[y_i^2]|^2 \geq 0$. Lastly, if $i = m \neq j = n$ or $i = n \neq j = m$, then $\mathbb{E}[y_i y_j \overline{y_m y_n}] = \mathbb{E}[|y_i|^2] \mathbb{E}[|y_j|^2] = 1$. This case occurs $2(d^2 - d)$ times in the sum. So, we get that

$$\mathbb{E}[(\mathbf{x}^\mathbf{H} \mathbf{A} \mathbf{x})^2] \geq (2d^2 - 2d + d)^k = (2d^2 - d)^k.$$

Which, noting that $\text{tr}(\mathbf{A}) = d^k$, in turn gets a variance lower bound of

$$\text{Var}[\mathbf{x}^\mathbf{H} \mathbf{A} \mathbf{x}] = \mathbb{E}[(\mathbf{x}^\mathbf{H} \mathbf{A} \mathbf{x})^2] - (\text{tr}(\mathbf{A}))^2 \geq (2d^2 - d)^k - d^{2k}.$$

Then, since $\text{Var}[H_\ell(\mathbf{A})] = \frac{1}{\ell} \text{Var}[\mathbf{x}^\mathbf{H} \mathbf{A} \mathbf{x}]$, in order for the complex Kronecker-Hutchinson's

Estimator to have standard deviation $\varepsilon \operatorname{tr}(\mathbf{A})$, we need to take

$$\ell \geq \frac{\operatorname{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}]}{\varepsilon^2 (\operatorname{tr}(\mathbf{A}))^2} \geq \frac{(2d^2 - d)^k - d^{2k}}{\varepsilon^2 d^{2k}} = \frac{(2 - \frac{1}{d})^k - 1}{\varepsilon^2} = \Omega\left(\frac{(2 - \frac{1}{d})^k}{\varepsilon^2}\right).$$

□

Lastly, we produce a much nicer bound on the sample complexity of the Kronecker-Hutchinson's Estimator when working with random rank-one matrices:

Theorem 31 Restated. *Fix $d, k \in \mathbb{N}$. Let $\mathbf{A} = \mathbf{g}\mathbf{g}^T$ where $\mathbf{g} \in \mathbb{R}^{d^k}$ is a vector of iid standard normal Gaussians. Then, with probability $\frac{3}{4}$, we know that $\operatorname{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x} \mid \mathbf{g}] \leq O((1 + \frac{1}{d})^k (\operatorname{tr}(\mathbf{A}))^2)$. In particular, with probability $\frac{2}{3}$, we know that the Kronecker-Hutchinson Estimator returns a $(1 \pm \varepsilon)$ relative error estimate to the trace of $\mathbf{A}$ if $\ell = O(\frac{1}{\varepsilon^2}(1 + \frac{1}{d})^k)$.*

Proof. Recall from the proof of [Theorem 26](#) that the expected squared Frobenius norm of the partial trace of $\mathbf{A}$ is

$$\mathbb{E}[\|\operatorname{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2] = \mathbb{E}[\|\mathbf{G}\|_4^4] = d^k(d^{k-|\mathcal{S}|} + d^{|\mathcal{S}|} + 1).$$

Then, we can directly bound

$$\begin{aligned} \sum_{\mathcal{S} \subseteq [k]} \mathbb{E}[\|\operatorname{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2] &= \sum_{\mathcal{S} \subseteq [k]} d^k(d^{k-|\mathcal{S}|} + d^{|\mathcal{S}|} + 1) \\ &= \sum_{i=0}^k \binom{k}{i} d^k(d^{k-i} + d^i + 1). \end{aligned}$$

Where the last line recognizes that we only depended on $|\mathcal{S}|$, so we could just sum over all values of $i = |\mathcal{S}|$, scaled by the $\binom{k}{i}$ times that $|\mathcal{S}|$ appears. Rearranging terms,

and recalling that $\sum_{i=0}^k \binom{k}{i} c^i = (1+c)^k$, we get

$$\begin{aligned}
\sum_{\mathcal{S} \subseteq [k]} \mathbb{E}[\|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2] &= d^k \sum_{i=0}^k \binom{k}{i} (d^{k-i} + d^i + 1) \\
&= d^k \sum_{i=0}^k \binom{k}{i} \left(d^k \left(\frac{1}{d}\right)^i + d^i + 1^i \right) \\
&= d^k \left(d^k \left(1 + \frac{1}{d}\right)^k + (1+d)^k + 2^k \right) \\
&= d^{2k} \left(2 \left(1 + \frac{1}{d}\right)^k + \left(\frac{2}{d}\right)^k \right) \\
&\leq 3d^{2k} \left(1 + \frac{1}{d}\right)^k.
\end{aligned}$$

As in the proof of [Theorem 26](#), we know that $\text{tr}(\mathbf{A}) \geq \frac{1}{2}d^k$ with probability $\frac{7}{8}$. Further, by Markov's Inequality, we can bound

$$\begin{aligned}
\text{Var}[(\mathbf{x}^H \mathbf{A} \mathbf{x})^2 \mid \mathbf{g}] &\leq 8 \mathbb{E}_{\mathbf{g}} \left[\sum_{\mathcal{S} \subseteq [k]} \mathbb{E}[\|\text{tr}_{\mathcal{S}}(\mathbf{A})\|_F^2] \right] = 24d^{2k} \left(1 + \frac{1}{d}\right)^k \\
&\leq 96 \left(1 + \frac{1}{d}\right)^k (\text{tr}(\mathbf{A}))^2.
\end{aligned}$$

Following the same probability arguments as in the proof of [Theorem 26](#), we get that taking

$$\ell = \frac{12 \text{Var}[\mathbf{x}^H \mathbf{A} \mathbf{x}]}{\varepsilon^2 (\text{tr}(\mathbf{A}))^2} \leq \frac{12 \cdot 96 \left(1 + \frac{1}{d}\right)^k (\text{tr}(\mathbf{A}))^2}{\varepsilon^2 (\text{tr}(\mathbf{A}))^2} = 1152 \frac{\left(1 + \frac{1}{d}\right)^k}{\varepsilon^2}$$

samples suffices for the complex-valued Kronecker-Hutchinson's Estimator to recover a $(1 \pm \varepsilon)$ relative error approximation to the trace of $\mathbf{A}$ with probability $\frac{2}{3}$. $\square$

Recall that the rate in the real-valued case was $\ell = O(\frac{1}{\varepsilon^2}(2 + \frac{1}{d})^k)$, and the rate here in the complex-valued case is $\ell = O(\frac{1}{\varepsilon^2}(1 + \frac{1}{d})^k)$. In the real-valued case, this rate grows at least as fast as 2^k , preventing any polynomial dependence on k from being possible. However, in the complex valued case, if $k = O(d)$, then the Kronecker-Hutchinson Estimator actually converges in $\ell = O(\frac{1}{\varepsilon^2})$ samples. We only show that the real-valued algorithm converges exponentially slowly, while the complex-valued algorithm converges polynomially quickly!

3.6 Looking Beyond the Kronecker-Hutchinson. Estimator

We have shown that the Kronecker-Hutchinson Estimator has a painful exponential dependence on the k in the worst case. However, we are broadly interested in the class of all algorithms that operate in the Kronecker-matrix-vector oracle model. We show that all trace estimation algorithms in this model must have query complexity that grows with k :

Theorem 54. *Any algorithm that accesses a PSD matrix $\mathbf{A}$ via Kronecker-matrix-vector products $\mathbf{A}\mathbf{x}^{(1)}, \dots, \mathbf{A}\mathbf{x}^{(q)}$, where $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}$ are (possibly adaptively) chosen real-valued vectors, requires $q = \Omega(\frac{\sqrt{k}}{\varepsilon})$ in order to output an estimate $\hat{t}$ such that $\sqrt{\mathbb{E}[(\hat{t} - \text{tr}(\mathbf{A}))^2]} \leq \varepsilon \text{tr}(\mathbf{A})$.*

If we constrain ourselves to estimators $\hat{t}$ that are unbiased (i.e. $\mathbb{E}[\hat{t}] = \text{tr}(\mathbf{A})$), then this lower bound says that $q = \Omega(\frac{\sqrt{k}}{\varepsilon})$ Kronecker-matrix-vector products are required to achieve standard deviation $\leq \varepsilon \text{tr}(\mathbf{A})$. Our analysis builds off of tools in [17, 62], which studies matrix-vector lower bounds without the Kronecker constraint. In particular, we use the following theorem:

Definition 55. *Let $\mathbf{G} \in \mathbb{R}^{d \times d}$ be a matrix of iid standard normal Gaussian entries. Then, the matrix $\mathbf{W} := \mathbf{G}^\top \mathbf{G} \in \mathbb{R}^{d \times d}$ is distributed as a Wishart(d) random matrix.*

Imported Theorem 56 (Lemma 13 from [17]). *Let $\mathbf{W} \sim \text{Wishart}(d)$. Then, for any sequence of (possibly adaptively chosen) matrix-vector queries $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}$ and responses $\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$ (so that $\mathbf{z}^{(j)} = \mathbf{W}\mathbf{x}^{(j)}$), there exists a rotation matrix $\mathbf{V} \in \mathbb{R}^{d \times d}$ constructed as a function of $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}$ such that the matrix $\mathbf{V}\mathbf{W}\mathbf{V}^\top$ can be written as*

$$\mathbf{V}\mathbf{W}\mathbf{V}^\top = \mathbf{\Delta} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \widetilde{\mathbf{W}} \end{bmatrix},$$

where $\widetilde{\mathbf{W}} \in \mathbb{R}^{d-q \times d-q}$ conditioned on the values of $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}, \mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$ is distributed as Wishart matrix. That is, $\widetilde{\mathbf{W}} \mid (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}, \mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}) \sim \text{Wishart}(d - q)$. Further, $\mathbf{\Delta}$ is a PSD matrix that is a deterministic function of $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}, \mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$.

This theorem is powerful because it says that any algorithm that has accessed a Wishart matrix via q matrix-vector products must have a submatrix which the algorithm has absolutely no knowledge of. We note that in [17], they do not state that $\mathbf{\Delta}$ is a deterministic function of $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}, \mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$, but this fact is easily verified by examining by the proof of Lemma 13 in [17].

We start by extending [Theorem 56](#) to the Kronecker-matrix-vector oracle model. We construct our hard instance input matrix as $\mathbf{A} = \otimes_{i=1}^k \mathbf{W}_i \in \mathbb{R}^{d^k \times d^k}$, where $\mathbf{W}_1, \dots, \mathbf{W}_k \in \mathbb{R}^{d \times d}$ are iid Wishart matrices:

Corollary 57. *Let $\mathbf{W}_1, \dots, \mathbf{W}_k$ be iid Wishart(d) matrices, and let $\mathbf{A} := \otimes_{i=1}^k \mathbf{W}_i \in \mathbb{R}^{d^k \times d^k}$. Then, for any sequence of (possibly adaptively chosen) Kronecker-matrix-vector queries $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}$ and responses $\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$ (so that $\mathbf{z}^{(j)} = \mathbf{A}\mathbf{x}^{(j)}$), there exists orthogonal matrices $\mathbf{V}_1, \dots, \mathbf{V}_k \in \mathbb{R}^{d \times d}$ such that $\mathbf{V} = \otimes_{i=1}^k \mathbf{V}_i$ has*

$$\mathbf{V}\mathbf{A}\mathbf{V}^\top = \otimes_{i=1}^k \left(\Delta_i + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \widetilde{\mathbf{W}}_i \end{bmatrix} \right).$$

Let $\mathbf{x}^{(j)} = \otimes_{i=1}^k \mathbf{x}_i^{(j)}$ and $\mathbf{z}^{(j)} = \mathbf{W}_i \mathbf{x}_i^{(j)}$ be decompositions of the query and response vectors. Then, each $\widetilde{\mathbf{W}}_i \in \mathbb{R}^{d-q \times d-q}$ conditioned on the values of $\mathbf{x}_i^{(1)}, \dots, \mathbf{x}_i^{(q)}, \mathbf{z}_i^{(1)}, \dots, \mathbf{z}_i^{(q)}$ is distributed as a Wishart matrix. That is, $\widetilde{\mathbf{W}}_i \mid (\mathbf{x}_i^{(1)}, \dots, \mathbf{x}_i^{(q)}, \mathbf{z}_i^{(1)}, \dots, \mathbf{z}_i^{(q)}) \sim \text{Wishart}(d-q)$. Further, each Δ_i is a PSD matrix that is a deterministic function of $\mathbf{x}_i^{(1)}, \dots, \mathbf{x}_i^{(q)}, \mathbf{z}_i^{(1)}, \dots, \mathbf{z}_i^{(q)}$.

With this setup, we can now form a lower bound against any trace estimating algorithm. We first prove in [Theorem 58](#) that, in terms of minimizing mean squared error, no estimator can outperform the conditional expectation $\tilde{t} = \mathbb{E}[\text{tr}(\mathbf{A}) \mid \mathbf{V}_1, \dots, \mathbf{V}_k, \Delta_1, \dots, \Delta_k]$. Next, in [Theorem 59](#), we compute the exact variance of this estimator. Lastly, in the final proof of [Theorem 54](#), we compare this variance to the trace of $\mathbf{A}$, to conclude that $q \geq \Omega(\frac{\sqrt{k}}{\epsilon})$.

To simplify notation, we let $\mathcal{W} = (\widetilde{\mathbf{W}}_1, \dots, \widetilde{\mathbf{W}}_k)$ denote the list of hidden Wishart matrices, and let $\mathcal{V} := (\mathbf{V}_1, \dots, \mathbf{V}_k, \Delta_1, \dots, \Delta_k)$ be the variables that $\tilde{t}$ is conditioned on. Note that the algorithm only observes the response vectors $\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$, and that the algorithm does not exactly know the vectors $\mathbf{z}_i^{(j)} = \mathbf{W}_i \mathbf{x}_i^{(j)}$. Therefore, the algorithm might not observe all of $\mathcal{V}$, but $\mathcal{V}$ definitely covers everything that the algorithm does see.

Lemma 58. *Consider an algorithm that interacts with $\mathbf{A}$ via q Kronecker-matrix-vector products as in the setting of [Theorem 57](#). Let $\hat{t}$ be the output of that algorithm, and let $\tilde{t} := \mathbb{E}[\text{tr}(\mathbf{A}) \mid \mathcal{V}]$. Then, $\tilde{t}$ has better means squared error than $\hat{t}$:*

$$\mathbb{E}[(\hat{t} - \text{tr}(\mathbf{A}))^2] \geq \mathbb{E}[(\tilde{t} - \text{tr}(\mathbf{A}))^2].$$

Proof. Note that the algorithm only observes $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}, \mathbf{z}^{(1)}, \dots, \mathbf{z}^{(q)}$, so the estimator $\hat{t}$ has to be a (possibly randomized) function of $\mathcal{V}$. We then expand the mean

squared error of $\hat{t}$ by using the tower rule:

$$\begin{aligned}\mathbb{E}[(\hat{t} - \text{tr}(\mathbf{A}))^2] &= \mathbb{E}[\mathbb{E}_{\mathcal{W}}[(\hat{t} - \text{tr}(\mathbf{A}))^2 \mid \mathcal{V}]] \\ &\geq \mathbb{E}[\mathbb{E}_{\mathcal{W}}[(\tilde{t} - \text{tr}(\mathbf{A}))^2 \mid \mathcal{V}]] \\ &= \mathbb{E}[(\tilde{t} - \text{tr}(\mathbf{A}))^2],\end{aligned}$$

where the inequality comes from $\tilde{t}$ being the minimum mean square error estimator for that particular conditional expectation. $\square$

For any specific instantiation of $\mathcal{V}$, we can ask what the expected mean squared error of $\tilde{t}$ is:

$$\mathbb{E}_{\mathcal{W}}[(\text{tr}(\mathbf{A}) - \tilde{t})^2 \mid \mathcal{V}] = \mathbb{E}_{\mathcal{W}}[(\text{tr}(\mathbf{A}) - \mathbb{E}[\text{tr}(\mathbf{A}) \mid \mathcal{V}])^2 \mid \mathcal{V}] = \text{Var}_{\mathcal{W}}[\text{tr}(\mathbf{A}) \mid \mathcal{V}].$$

We do not really care about this value for a specific choice of $\mathcal{V}$, but instead care about it on average across all values of $\mathcal{V}$. That is, we care about the squared error $\mathbb{E}[(\tilde{t} - \text{tr}(\mathbf{A}))^2] = \mathbb{E}_{\mathcal{V}}[\text{Var}[\text{tr}(\mathbf{A}) \mid \mathcal{V}]]$. We then want to ensure that the error is less than $\varepsilon \mathbb{E}[\text{tr}(\mathbf{A})]$. So, we want to find out what value of q ensures that

$$\sqrt{\mathbb{E}[\text{Var}[\text{tr}(\mathbf{A}) \mid \mathcal{V}]]} \leq \varepsilon \mathbb{E}[\text{tr}(\mathbf{A})].$$

We start by computing the variance inside the square root on the left.

Lemma 59. *The mean squared error of $\tilde{t}$ is $\mathbb{E}[\text{Var}[\text{tr}(\mathbf{A}) \mid \mathcal{V}]] = (d^4 + 2d^2)^k - (d^4 + 2d^2 - 2(d - q)^2)^k$.*

Proof. Let $X_i := \text{tr}(\mathbf{W}_i)$ and $X := \text{tr}(\mathbf{A})$, so that $X = \prod_{i=1}^k X_i$ is a product of k independent random variables. Then, since these terms are independence, we know that

$$\text{Var}[\prod_{i=1}^k X_i] = \prod_{i=1}^k \mathbb{E}[X_i^2] - \prod_{i=1}^k (\mathbb{E}[X_i])^2.$$

Further, since $X_i \mid \mathcal{V}$ is independent of $X_j \mid \mathcal{V}$ for $i \neq j$, we know that

$$\begin{aligned}\mathbb{E}[\text{Var}[\prod_{i=1}^k X_i \mid \mathcal{V}]] &= \mathbb{E} \left[\prod_{i=1}^k \mathbb{E}[X_i^2 \mid \mathcal{V}] - \prod_{i=1}^k (\mathbb{E}[X_i \mid \mathcal{V}])^2 \right] \\ &= \prod_{i=1}^k \mathbb{E}[X_i^2] - \prod_{i=1}^k \mathbb{E}[(\mathbb{E}[X_i \mid \mathcal{V}])^2].\end{aligned}$$

The last expectation on the right can be simplified a bit further. Since $\text{Var}[X_i | \mathcal{V}] = \mathbb{E}[X_i^2 | \mathcal{V}] - (\mathbb{E}[X_i | \mathcal{V}])^2$, we can write

$$\mathbb{E}[(\mathbb{E}[X_i | \mathcal{V}])^2] = \mathbb{E}[\mathbb{E}[X_i^2 | \mathcal{V}] - \text{Var}[X_i | \mathcal{V}]] = \mathbb{E}[X_i^2] - \mathbb{E}[\text{Var}[X_i | \mathcal{V}]],$$

which in turn means we have

$$\mathbb{E}[\text{Var}[\prod_{i=1}^k X_i | \mathcal{V}]] = \prod_{i=1}^k \mathbb{E}[X_i^2] - \prod_{i=1}^k (\mathbb{E}[X_i^2] - \mathbb{E}[\text{Var}[X_i | \mathcal{V}]]) . \quad (3.8)$$

Now we can directly analyze these means and expectations. First, note that $X_i = \text{tr}(\mathbf{W}_i) = \text{tr}(\mathbf{G}_i^\top \mathbf{G}_i) = \|\mathbf{G}_i\|_F^2 \sim \chi_{d^2}^2$ is a chi-squared random variable. Therefore, $\mathbb{E}[X_i^2] = \text{Var}[X_i] + (\mathbb{E}[X_i])^2 = 2d^2 + d^4$. Next, we focus on $\mathbb{E}[\text{Var}[X_i | \mathcal{V}]]$. By the linearity and cyclic property of the trace, and since $\mathbf{V}_i \mathbf{V}_i^\top = \mathbf{I}$, we have that

$$X_i = \text{tr}(\mathbf{W}_i) = \text{tr}(\mathbf{W}_i \mathbf{V}_i \mathbf{V}_i^\top) = \text{tr}(\mathbf{V}_i^\top \mathbf{W}_i \mathbf{V}_i) = \text{tr}(\mathbf{\Delta}_i) + \text{tr}(\widetilde{\mathbf{W}}_i).$$

Therefore the only term in $\mathcal{V}$ that X_i depends on is $\mathbf{\Delta}_i$, and so we can expand

$$\mathbb{E}[\text{Var}[X_i | \mathcal{V}]] = \mathbb{E}[\text{Var}[\text{tr}(\mathbf{\Delta}_i) + \text{tr}(\widetilde{\mathbf{W}}_i) | \mathbf{\Delta}_i]] = \mathbb{E}[\text{Var}[\text{tr}(\widetilde{\mathbf{W}}_i)]] = \text{Var}[\text{tr}(\widetilde{\mathbf{W}}_i)] = 2(d - q)^2,$$

where the last equality notes that $\text{tr}(\widetilde{\mathbf{W}}_i) = \text{tr}(\widetilde{\mathbf{G}}_i^\top \widetilde{\mathbf{G}}_i) = \|\widetilde{\mathbf{G}}_i\|_F^2 \sim \chi_{(d-q)^2}$ is also a chi-squared random variable. Plugging our expectations back into [Equation \(3.8\)](#), we find that

$$\mathbb{E}[\text{Var}[\prod_{i=1}^k X_i | \mathcal{V}]] = (d^4 + 2d^2)^k - (d^4 + 2d^2 - 2(d - q)^2)^k,$$

which completes the proof. □

We now have built up the tools we need to prove the central lower bound:

Theorem 54 Restated. *Any algorithm that accesses a PSD matrix $\mathbf{A}$ via Kronecker-matrix-vector products $\mathbf{A}\mathbf{x}^{(1)}, \dots, \mathbf{A}\mathbf{x}^{(q)}$, where $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(q)}$ are (possibly adaptively) chosen real-valued vectors, requires $q = \Omega(\frac{\sqrt{k}}{\varepsilon})$ in order to output an estimate $\hat{t}$ such that $\sqrt{\mathbb{E}[(\hat{t} - \text{tr}(\mathbf{A}))^2]} \leq \varepsilon \text{tr}(\mathbf{A})$.*

Proof. We consider $\mathbf{A}$ as in [Theorem 57](#). By [Theorems 58](#) and [59](#), we know that

$$\mathbb{E}[(\hat{t} - \text{tr}(\mathbf{A}))^2] \geq \mathbb{E}[(\tilde{t} - \text{tr}(\mathbf{A}))^2] = (d^4 + 2d^2)^k - (d^4 + 2d^2 - 2(d - q)^2)^k.$$

We then want to show that this term on the right hand side is at most $\varepsilon^2(\mathbb{E}[\text{tr}(\mathbf{A})])^2$. First, we quickly note that $\mathbb{E}[\text{tr}(\mathbf{A})] = \prod_{i=1}^k \mathbb{E}[\text{tr}(\mathbf{W}_i)] = d^{2k}$, since $\text{tr}(\mathbf{W}_i) = \text{tr}(\mathbf{G}_i^\top \mathbf{G}_i) = \|\mathbf{G}_i\|_F^2 \sim \chi_{d^2}^2$ is a chi-squared random variable. So, we want to find what values of q ensure that

$$(d^4 + 2d^2)^k - (d^4 + 2d^2 - 2(d - q)^2)^k \leq \varepsilon^2 d^{4k}.$$

In fact, it will be mathematically convenient to demand slightly less accuracy, which will still give a meaningful lower bound:

$$(d^4 + 2d^2)^k - (d^4 + 2d^2 - 2(d - q)^2)^k \leq \varepsilon^2 (d^4 + 2d^2)^k.$$

We can rearrange this inequality to produce

$$(d - q)^2 \leq \frac{1}{2}(d^4 + 2d^2)(1 - (1 - \varepsilon^2)^{1/k}).$$

Note that for $\varepsilon \in (0, \frac{1}{2})$ and $k \geq 1$, we have that $(1 - \frac{2\varepsilon^2}{k})^k = (1 - \frac{2\varepsilon^2}{k})^{\frac{k}{2\varepsilon^2} \cdot 2\varepsilon^2} \leq e^{-2\varepsilon^2} \leq 1 - \varepsilon^2$, so that $(1 - \varepsilon^2)^{1/k} \geq 1 - \frac{2\varepsilon^2}{k}$. So, by additionally bounding $d^4 + 2d^2 \leq 2d^4$ for $d \geq 2$, we can simplify the above bound to be

$$(d - q)^2 \leq \frac{1}{2}(d^4 + 2d^2)(1 - (1 - \varepsilon^2)^{1/k}) \leq \frac{2d^4 \varepsilon^2}{k}.$$

Solving this inequality for q gives $q \geq d - \frac{\sqrt{2\varepsilon}}{\sqrt{k}} d^2$. Optimizing over d gives $q \geq \frac{4\sqrt{k}}{\sqrt{2\varepsilon}} = \Omega(\frac{\sqrt{k}}{\varepsilon})$, completing the proof. $\square$

3.7 Exactly Computing the Trace of $\mathbf{A} = \otimes_{i=1}^k \mathbf{A}_i$ in $kd + 1$ Queries

In this section, we are promised that $\mathbf{A} = \mathbf{A}_1 \otimes \cdots \otimes \mathbf{A}_k$ for some unknown matrices $\mathbf{A}_1, \dots, \mathbf{A}_k \in \mathbb{R}^{d \times d}$. We show that by using exactly $kd + 1$ Kronecker-matrix-vector products, we can recover a set of matrices $\mathbf{B}_1, \dots, \mathbf{B}_k \in \mathbb{R}^{d \times d}$ such that $\otimes_{i=1}^k \mathbf{B}_i = \mathbf{A}$. Note that we cannot exactly recover the matrices $\mathbf{A}_1, \dots, \mathbf{A}_k$ because the Kronecker product does not admit a unique decomposition: we have $\alpha \mathbf{A} \otimes \mathbf{B} = \mathbf{A} \otimes \alpha \mathbf{B}$ for all scalars α and matrices $\mathbf{A}, \mathbf{B}$. However, we are able to recover the trace exactly since $\text{tr}(\mathbf{A}_1 \otimes \mathbf{A}_2) = \text{tr}(\mathbf{A}_1) \text{tr}(\mathbf{A}_2)$, we can then compute $\text{tr}(\mathbf{A}) = \prod_{i=1}^k \text{tr}(\mathbf{B}_i)$.

Theorem 60. *Let $\mathbf{A} = \mathbf{A}_1 \otimes \cdots \otimes \mathbf{A}_k \in \mathbb{R}^{d^k \times d^k}$. Then, with probability 1, [Algorithm 5](#) computes exactly $kd+1$ Kronecker-matrix-vector products with $\mathbf{A}$ and returns matrices $\mathbf{B}_1, \dots, \mathbf{B}_k$ such that $\mathbf{A} = \mathbf{B}_1 \otimes \cdots \otimes \mathbf{B}_k$.*

Algorithm 5 Exact Kronecker Recovery**input:** Kronecker-matrix-vector oracle access to $\mathbf{A} \in \mathbb{R}^{d^k \times d^k}$.**output:** Factor matrices $\mathbf{B}_1, \dots, \mathbf{B}_k \in \mathbb{R}^{d \times d}$.

- 1: Sample $\mathcal{N}(\mathbf{0}, \mathbf{I})$ vectors $\mathbf{g}_1, \dots, \mathbf{g}_k \in \mathbb{R}^d$.
- 2: Let $\bar{\mathbf{x}} = \otimes_{i=1}^k \mathbf{g}_i$.
- 3: Compute $\gamma := \bar{\mathbf{x}}^\top \mathbf{A} \bar{\mathbf{x}}$.
- 4: If $\gamma = 0$, then **return** $\mathbf{B}_1, \dots, \mathbf{B}_k = \mathbf{0}$
- 5: For all $i \in [k], m \in [d]$, let $\mathbf{x}_i^{(m)} = (\otimes_{j=1}^{i-1} \mathbf{g}_j) \otimes \mathbf{e}_m \otimes (\otimes_{j=i+1}^k \mathbf{g}_j)$.
- 6: For all $i \in [k], m \in [d]$, compute $\mathbf{A} \mathbf{x}_i^{(m)}$.
- 7: For all $i \in [k], m \in [d], n \in [d]$, store $[\mathbf{B}_i]_{m,n} = \gamma^{-(1-\frac{1}{k})} \mathbf{x}_i^{(m)\top} \mathbf{A} \mathbf{x}_i^{(n)}$.
- 8: **return** $\mathbf{B}_1, \dots, \mathbf{B}_k$.

Proof. At a high level, the quadratic form $\mathbf{x}_i^{(m)\top} \mathbf{A} \mathbf{x}_i^{(n)}$ is trying to recover the (m, n) entry of $\mathbf{A}_i$. To see why, notice that

$$\mathbf{x}_i^{(m)\top} \mathbf{A} \mathbf{x}_i^{(n)} = (\otimes_{j=1}^{i-1} \mathbf{g}_j^\top \mathbf{A}_j \mathbf{g}_j) \otimes \mathbf{e}_m^\top \mathbf{A}_i \mathbf{e}_n \otimes (\otimes_{j=i+1}^k \mathbf{g}_j^\top \mathbf{A}_j \mathbf{g}_j).$$

If we let $\gamma_i := \mathbf{g}_i^\top \mathbf{A}_i \mathbf{g}_i$ denote the various components of $\gamma = \otimes_{i=1}^k \mathbf{g}_i^\top \mathbf{A}_i \mathbf{g}_i = \prod_{i=1}^k \gamma_i$, then we can write the above expression as

$$\mathbf{x}_i^{(m)\top} \mathbf{A} \mathbf{x}_i^{(n)} = (\prod_{j \neq i} \gamma_j) \mathbf{e}_m^\top \mathbf{A}_i \mathbf{e}_n = \frac{\gamma}{\gamma_i} [\mathbf{A}_i]_{m,n}$$

So, we get that the matrix $\mathbf{B}_i$ has $[\mathbf{B}_i]_{m,n} = \frac{1}{\gamma_i} [\mathbf{A}_i]_{m,n}$. We therefore conclude that

$$\otimes_{i=1}^k \mathbf{B}_i = \otimes_{i=1}^k \left(\frac{1}{\gamma_i} \mathbf{A}_i \right) = \left(\prod_{i=1}^k \frac{1}{\gamma_i} \right) \mathbf{A} = \frac{\gamma}{\gamma} \mathbf{A} = \mathbf{A}.$$

This analysis implicitly assumes that $\gamma_i = \mathbf{g}_i^\top \mathbf{A}_i \mathbf{g}_i \neq 0$ for all $i \in [k]$. Since $\mathbf{g}_i$ are random Gaussian vectors, we know that $\mathbf{g}_i$ does not lie in the nullspace of $\mathbf{A}_i$ with probability 1 so long as $\mathbf{A}_i$ is not the all-zeros matrix. So, [Algorithm 5](#) is correct so long as no factor $\mathbf{A}_i$ is the all-zeros matrix. If some $\mathbf{A}_i$ is the all-zeros matrix, then we get that $\mathbf{A}$ is also the all-zeros matrix, so that $\gamma = 0$, and therefore our output will have $\mathbf{B}_1, \dots, \mathbf{B}_k$ all being all-zeros matrices, which is correct.

To count the number of Kronecker-matrix-vector products computed, note that [Algorithm 5](#) only computes such products on lines 3 and 6. Line 3 computes exactly 1 product, and line 6 computes exactly kd products. $\square$

3.8 Conclusion

In this paper, we produced new and tight bounds on the rate of convergence for the Kronecker-Hutchinson Estimator. We show that the estimator generally has an exponential dependence on k , and we showed an expression for the exact variance of the estimator when using the Kronecker of Gaussian vectors.

There are several natural future directions for this research. It is worth exploring the variance of more classes of matrices under [Theorem 27](#). In particular, is there a natural property of matrices which exactly characterizes the input matrices for which the Kronecker-Hutchinson Estimator has a polynomial rate of convergence? If so, this would be a significant step towards a sub-exponential runtime for trace estimation in the Kronecker-matrix-vector oracle.

It is also worth looking at algorithms in the Kronecker-matrix-vector oracle model beyond the Kronecker-Hutchinson Estimator. For instance, when $\mathbf{A}$ has a quickly decaying spectrum, could a low-rank approximation method like ones used in the non-Kronecker case provide good trace estimation guarantees [70, 105]?

There is also potential for newer and stronger lower bounds. We provide a lower bound against possibly adaptive algorithms, which is very general, but is also perhaps too general to be descriptive of most algorithms people are currently using. Is there perhaps a lower bound against all sketching-based methods which is exponential in k ? Or, even more narrowly, is there a lower bound against all quadratic-form based algorithms, in the vain of [131]?

Lastly, there is also plenty of potential to study more problems in the Kronecker-matrix-vector oracle model. Is it possible to accurately and efficiently compute the top eigenvalue or eigenvector of a matrix from Kronecker-matrix-vector products? How about solving a linear system? These problems all need to be explored in great detail.

Chapter 4

Single Vector Krylov

4.1 Introduction

Krylov subspace methods have been studied since the 1950s and remain our most reliable algorithms for approximating eigenvectors and singular vectors of large matrices. Krylov methods access a matrix $\mathbf{A}$ via repeated matrix multiplications (each considered an iteration of the method) either with a single vector or a block of vectors. There has been significant interest in analyzing how many iterations are required to obtain accurate eigenvector or singular vector approximations. Classic work studies both single vector [64, 90] and block methods [31, 63, 102, 103].

More recently, there has been interest in analyzing Krylov subspace methods specifically for the downstream task of *low-rank approximation*. Since the top k singular vectors can be used to obtain an optimal rank- k approximation, the goal is to understand how many iterations are required to compute approximate singular vectors that yield a near-optimal rank- k approximation [99, 53, 132]. This problem differs from classical work because convergence to the actual top singular vectors is sufficient but *not necessary* for obtaining an accurate low-rank approximation [34].

Prototypical single vector and block Krylov methods for low-rank approximation are shown in [Algorithm 6](#) and [Algorithm 7](#).¹ For an $d \times k$ input $\mathbf{A}$, both methods returns an $d \times k$ orthogonal $\mathbf{Q}$ so that the rank- k matrix $\mathbf{Q}\mathbf{Q}^\top\mathbf{A}$ is a good approximation to $\mathbf{A}$. Ideally, it is nearly as good as $\mathbf{A}$'s optimal rank k approximation, $\mathbf{A}_k$,

¹[Algorithms 6](#) and [7](#) are examples of the simplest possible implementations of Krylov methods for low-rank approximation. In practice, various optimizations like the Lanczos recurrence are often applied, and additional care is necessary to ensure that the orthogonal basis for the Krylov subspace $\mathbf{K}$ is computed in a numerically stable way [103]. While an important topic, this paper is not focused on the numerical stability of Lanczos methods. All derivations assume computation in the Real RAM model of arithmetic.

Algorithm 6 Single Vector Krylov Method for Low-Rank Approximation

input: Matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$. Target rank k . Starting vector $\mathbf{x} \in \mathbb{R}^d$. Number of iterations t .

output: Orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$.

- 1: Compute an orthonormal basis $\mathbf{Z}$ for $\mathbf{K} = [\mathbf{x}, (\mathbf{A}\mathbf{A}^\top)\mathbf{x}, (\mathbf{A}\mathbf{A}^\top)^2\mathbf{x}, \dots, (\mathbf{A}\mathbf{A}^\top)^t\mathbf{x}]$.
 - 2: Compute $\mathbf{U}_k$, the k top eigenvectors of $\mathbf{M} = \mathbf{Z}^\top \mathbf{A}\mathbf{A}^\top \mathbf{Z}$
 - 3: **return** $\mathbf{Q} = \mathbf{Z}\mathbf{U}_k$.
-

Algorithm 7 Block Krylov Method for Low-Rank Approximation

input: Matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$. Target rank k . Starting block $\mathbf{B} \in \mathbb{R}^{d \times \ell}$. Number of iterations t .

output: Orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$.

- 1: Compute an orthonormal basis $\mathbf{Z}$ for $\mathbf{K} = [\mathbf{B}, (\mathbf{A}\mathbf{A}^\top)\mathbf{B}, (\mathbf{A}\mathbf{A}^\top)^2\mathbf{B}, \dots, (\mathbf{A}\mathbf{A}^\top)^t\mathbf{B}]$.
 - 2: Compute $\mathbf{U}_k$, the k top eigenvectors of $\mathbf{M} = \mathbf{Z}^\top \mathbf{A}\mathbf{A}^\top \mathbf{Z}$
 - 3: **return** $\mathbf{Q} = \mathbf{Z}\mathbf{U}_k$.
-

which is given via projection onto $\mathbf{A}$'s top k singular vectors.²

4.1.1 Large block methods and gap-free bounds

Most recent work on Krylov methods for low-rank approximation focuses on “large block” methods, where ℓ in [Algorithm 7](#) is chosen to be $\geq k$ [99, 52, 48, 85, 115, 133, 33, 115]. In this regime, block methods are known to quickly converge to a near-optimal low-rank approximation. For example, in just $O\left(\frac{\log(d/\varepsilon)}{\sqrt{(\sigma_k - \sigma_{\ell+1})/\sigma_k}}\right)$ iterations, [Algorithm 7](#) initialized with an i.i.d. random Gaussian matrix $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ with block size $\ell \geq k$, achieves with high probability the bound

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad (4.1)$$

for any $\varepsilon > 0$ and $\|\cdot\|_\xi$ being either the Frobenius or spectral norm [85]. That is, convergence is linear with a rate depending on the square root of the relative gap from the k^{th} singular value, σ_k , to the $(\ell + 1)^{\text{st}}$ singular value, $\sigma_{\ell+1}$. Even for ℓ mildly larger than k , this gap is often quite large. For example, [53] recommends setting $\ell = k + 5$ or $k + 10$.

²For simplicity, we focus on computing an approximate left singular vector subspace spanned by $\mathbf{Q} \in \mathbb{R}^{d \times k}$. If we instead care about computing right singular vectors, [Algorithms 6](#) and [7](#) can be applied to $\mathbf{A}^\top$ instead.

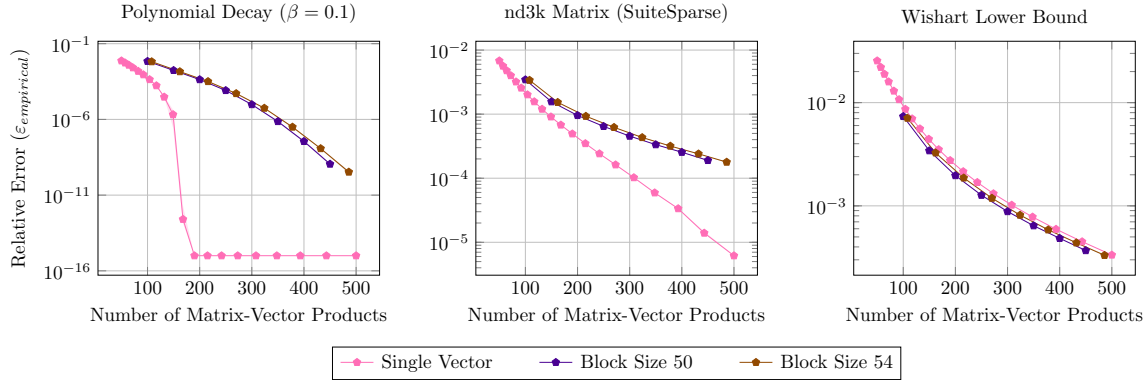


Figure 4.1: Comparison of the number of matrix-vector products needed for Krylov iteration to converge to an accurate rank 50 approximation under different block sizes. The left figure uses a matrix with singular values decaying polynomially; the middle figure uses a matrix from the SuiteSparse library; the right figure uses a worst-case matrix from the literature. See [Section 4.6.5](#) for more details. In the first two plots, single vector Krylov outperforms large block methods. Even in the adversarially chosen hard instant on the right, it does not perform much worse.

Beyond such spectrum dependent guarantees, another advantage of large block Krylov methods is that they enjoy *gap-independent* bounds, which do not involve any terms depending on $\mathbf{A}$'s spectrum. For example, a now standard result is that [Algorithm 7](#) achieves [Equation \(4.1\)](#) in just $O(\frac{1}{\sqrt{\epsilon}} \log(\frac{d}{\epsilon}))$ iterations [85].³ Further, this bound is essentially optimal among all methods that access $\mathbf{A}$ only through matrix-vector products [109, 9]. Bounds where the iteration complexity does not depend on properties of $\mathbf{A}$, are called “universal” guarantees [120]. Universal bounds are useful in applications where properties like large spectral gaps cannot be ensured, but where worst-case accuracy guarantees are still desired [54, 69, 110].

In contrast to large block sizes, it is *impossible* to prove gap-independent guarantees for single vector or small block Krylov iteration. To see why, consider $\mathbf{A} \in \mathbb{R}^{k \times n}$ that is all zeros, except that $\mathbf{A}_{ii} = 1$ for $i = 1, \dots, k$. I.e., $\mathbf{A} = \begin{bmatrix} \mathbf{I}_k & \mathbf{0} \end{bmatrix}$, where $\mathbf{I}_k$ denotes the $k \times k$ identity matrix. If we run [Algorithm 7](#) on this matrix with block size $\ell < k$, then it can be checked that the Krylov subspace $\mathbf{K}$ will have rank $\ell < k$, and thus any low-rank approximation obtained from the subspace cannot be near-optimal. In general, bounds for small block methods must depend *inversely* on the gaps between sequential singular values. In the above example, these gaps are equal to 0.

³Randomly initialized block power method with block size k gives a similar bound, but with a suboptimal $1/\epsilon$ rather than $1/\sqrt{\epsilon}$ dependence on the error [99, 53].

4.1.2 Main contribution: the virtue of small block Krylov methods

The inability of single vector and small block Krylov methods to offer gap-independent bounds has been a point of concern for the use of these methods in computing low-rank approximations [69, 85, 71]. At the same time, in practice, low-rank approximation is frequently solved using iterative eigensolvers based on single vector or small block methods. Such methods are the standard in MATLAB, Julia, Python, and essentially all languages used for matrix computations [68, 76, 108]. These methods often perform very well, converging quickly to good low-rank approximations. In fact, in our experience, they typically outperform large block methods in terms of the number of matrix-vector products required to achieve a desired level of accuracy – see [Figure 4.1](#).⁴

The main goal of this paper is to explain this phenomenon. We ask:

For low-rank approximation, when and why do small block Krylov methods require the same or fewer matrix-vector multiplications than large block Krylov methods?

We answer this question in a strong way by proving that small block methods nearly match or even improve on all known theoretical guarantees on the convergence of large-block methods for low-rank approximation. In particular, up to a *logarithmic dependence* on the smallest gap between singular values, the trade-off between accuracy and number of matrix-vector products achieved by small block methods matches the trade-off achieved by large block methods. Since there are a variety of guarantees known for large block methods, this claim is broken down as a number of results throughout our paper. We state one such result as a concrete example:

Theorem 61. *For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_i}$ be the smallest relative gap among the top k singular values. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 6](#) initialized with an i.i.d. mean zero Gaussian vector $\mathbf{x}$ and run for $t = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{\varepsilon}} \log(\frac{d}{\varepsilon\delta}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$, letting $\|\cdot\|_{\xi}$ be the spectral or Frobenius norm,*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_{\xi} \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_{\xi}.$$

⁴The number of matrix-vector products used by an algorithm does not necessarily translate directly into the computational cost of the algorithm. For example, in many computing systems, it is faster to multiply a matrix $\mathbf{A}$ by a block of k vectors all at once, than to multiply by k vectors chosen in sequence. Nevertheless, matrix-vector products are still a valuable measure of complexity for many problems where they dominate other runtime costs.

As discussed, [85] prove that [Algorithm 7](#) with block size k achieves an identical error bound in $O\left(\frac{\log(d/\varepsilon\delta)}{\sqrt{\varepsilon}}\right)$ iterations. This translates to $O\left(\frac{k \log(d/\varepsilon\delta)}{\sqrt{\varepsilon}}\right)$ matrix-vector products, which [Theorem 61](#) matches, except for the dependence on $\log(1/g_{\min})$. At the same time, [Theorem 61](#) improves on the large block bound by separating the $\log(d/\varepsilon\delta)$ and k terms.

Remark. Since it is a logarithmic instead of polynomial dependence, we consider the $\log(1/g_{\min})$ term to be mild for typical problems. In experiments, it appears to have little impact on the observed convergence of the single vector Krylov method (see [Section 4.6](#)). Indeed, except in adversarial cases, such as the identity matrix, where $g_{\min}$ truly equals 0, in finite precision, we cannot expect to resolve singular value gaps to accuracy better than machine precision. So, it is reasonable to think that in practice, this term should be at most a moderate constant. We make this intuition formal in [Section 4.5](#), showing that the dependence on $g_{\min}$ can be eliminated in a smoothed analysis setting (i.e., when the input is perturbed by a small random matrix).

Our proof for [Theorem 61](#) (with some additional results) is given in [Section 4.3](#). Our approach is via a black-box reduction to the existing analysis for large block methods. In particular, we view the single-vector method of [Algorithm 6](#) as a block Krylov method in disguise. We observe that the span of the single-vector Krylov subspace $\mathbf{K}$ (Line 1 of [Algorithm 6](#)) is exactly equivalent to the span of a block Krylov subspace generated from a specific starting matrix. Concretely, suppose [Algorithm 6](#) is run for $t \geq k$ iterations and let $\mathbf{S}_k \in \mathbb{R}^{d \times k}$ equal the first k columns of $\mathbf{K}$. I.e.,

$$\mathbf{S}_k := \begin{bmatrix} \mathbf{x} & \mathbf{A}\mathbf{A}^\top \mathbf{x} & (\mathbf{A}\mathbf{A}^\top)^2 \mathbf{x} & \dots & (\mathbf{A}\mathbf{A}^\top)^{k-1} \mathbf{x} \end{bmatrix}. \quad (4.2)$$

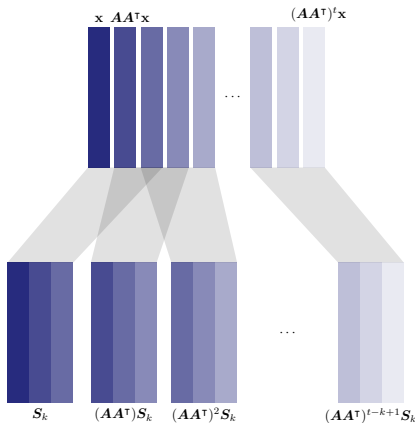


Figure 4.2: Our main analysis is based on the simple observation that the span of the Krylov subspace generated by a single vector Krylov method after t iterations is exactly equivalent to that generated by a block Krylov method run for $t - k + 1$ iterations with starting block $\mathbf{S}_k = [\mathbf{x}, (\mathbf{A}\mathbf{A}^\top)\mathbf{x}, \dots, (\mathbf{A}\mathbf{A}^\top)^{k-1}\mathbf{x}]$. This observation allows us to take advantage of existing results on block methods in a black-box way.

Then we can check that for $q = t - k + 1$:

$$\text{span}(\mathbf{K}) = \text{span} \left(\begin{bmatrix} \mathbf{S}_k & \mathbf{A}\mathbf{A}^\top \mathbf{S}_k & (\mathbf{A}\mathbf{A}^\top)^2 \mathbf{S}_k & \dots & (\mathbf{A}\mathbf{A}^\top)^q \mathbf{S}_k \end{bmatrix} \right). \quad (4.3)$$

This equivalence is visualized in [Figure 4.2](#). Since both [Algorithm 6](#) and [Algorithm 7](#) only depend on the *span* of the Krylov subspace they generate (through $\mathbf{Z}$), the single vector method thus matches the block Krylov method run for $k - 1$ fewer iterations, with the specific starting block $\mathbf{S}_k$.

With this perspective, a naive hope might be to directly appeal to prior results on block Krylov iteration to analyze the single vector method. Unfortunately, these results rely on the fact that the starting matrix $\mathbf{B}$ is chosen at random, typically with i.i.d. Gaussian or sub-Gaussian entries [53, 85, 8]. In contrast, $\mathbf{S}_k$ is far from a random Gaussian matrix. Its columns are highly dependent on each other. To understand just how far $\mathbf{S}_k$ is from an ideal starting matrix, note that, generally, a block Krylov subspace with q blocks will have rank qk when $\mathbf{B}$ is a random Gaussian matrix. In contrast, the block Krylov subspace $\begin{bmatrix} \mathbf{S}_k & \mathbf{A}\mathbf{A}^\top \mathbf{S}_k & \dots & (\mathbf{A}\mathbf{A}^\top)^q \mathbf{S}_k \end{bmatrix}$ only has rank $t = q + k - 1$.

Surprisingly, however, we are still able to show that $\mathbf{S}_k$ provides a (barely) good enough starting matrix for the block Krylov method in [Algorithm 7](#) to succeed. To do so, we consider a natural definition of what it means to be a “good” starting matrix. At a high-level, we need $\mathbf{S}_k$ to have non-negligible inner product with all top k singular vectors of $\mathbf{A}$. While $\mathbf{S}_k$ is exponentially worse in terms of starting inner product than a random $\mathbf{B}$, this is made up for by the fact that it is far cheaper (in terms of matrix-vector products) to build a block Krylov subspace with $\mathbf{S}_k$; we can compute a degree q subspace using just $q + k$ matrix-vector products with $\mathbf{A}$. In contrast, computing a degree q block Krylov subspace with a random starting block $\mathbf{B} \in \mathbb{R}^{d \times k}$ requires qk matrix-vector products. The detailed proof is presented in [Section 4.3](#).

4.1.3 Results and Paper Organization

In addition to [Theorem 61](#), we prove several other convergence bounds for single vector and small block Krylov methods, which we briefly summarize below. [Section 4.6](#) contains experiments which demonstrate that our bounds are predictive of the performance of these methods in practice.

Spectrum adaptive bounds, [Section 4.4.1](#). The black-box nature of [Theo-](#)

rem 61’s proof allows us to similarly adapt other results on large block Krylov methods to the single vector setting. For example, we show that Algorithm 6 matches known “spectrum dependent” bounds for large block methods. As discussed in Section 4.1.1, the convergence rate of these bounds depends on $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$, the gap between the k^{th} and $(\ell + 1)^{\text{st}}$ singular values. Since this gap increases with ℓ , there is a natural tradeoff: a larger block means more matrix-vector products per iteration, but fewer iterations. Our proof shows that single vector methods match any block size $\ell \geq k$ (up to a dependence on $\log(1/g_{\min})$). I.e., they automatically match the complexity of the method with *best choice of large block size*, without need for parameter tuning.

Schatten p-norm low-rank approximation, Sections 4.4.2 and 4.4.3. Recently, a combination of spectrum dependent and spectrum independent bounds have been used to give faster convergence rates for Schatten p-norm low rank approximation. In particular, for constant p , [8] show how to find a low-rank approximation achieving $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_p \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_p$ using just $\tilde{O}(k/\varepsilon^{1/3})$ matrix-vector products with $\mathbf{A}$. For the Frobenius norm (i.e., $p = 2$), this is an improvement on the $\tilde{O}(k/\sqrt{\varepsilon})$ required by [85]. Their method requires running Algorithm 7 with multiple choices of block size, and optimizing over the best Krylov subspace. We show that, again up to a logarithmic dependence on $1/g_{\min}$, the exact same guarantees can be obtained by simply running a single vector Krylov method. In concurrent work, [9] show a similar result without a dependence on $\log(1/g_{\min})$ or $\log(d)$, although only for the special case of rank-1 approximation.

Beyond block size 1, Section 4.4.4. While the above results focus on the single vector Krylov method, our bounds naturally generalize to other small block sizes between 1 and k (e.g. 4, 10, or $k - 1$). For general small block size b , we show that dependence on $g_{\min}$ can be replaced with a dependence on the smallest “ b^{th} order” gap $g_{\min,b} := \min_{i \in \{1, \dots, k-b\}} \frac{\sigma_i - \sigma_{i+b}}{\sigma_i}$.

Removing the gap dependence, Section 4.5. While a dependence on singular value gaps is unavoidable for small block methods in the worst-case, the parameter seems to rarely have an impact in practice. We take a step towards explaining this observation via a *smoothed-analysis* result [111, 106]. Specifically, we leverage work in random matrix theory on *eigenvalue repulsion*, which shows that small spectral gaps in a matrix are brittle: adding a tiny amount of random noise to any matrix ensures that its singular value gaps are at worst inverse polynomial in the problem parameters. Using this fact, we present bounds that replace the dependence on $\log(1/g_{\min})$ in our prior results with a dependence on $\log(\frac{d\kappa_k}{\delta\varepsilon})$ for randomly perturbed matrices, where

$\kappa_k = \sigma_1/\sigma_k$ measures the conditioning of the top k singular values. From an algorithm design perspective, the $\log(1/g_{\min})$ can be removed even in the worst case by explicitly adding a random diagonal perturbation to $\mathbf{A}$.

4.1.4 Related Work

We briefly discuss additional prior work on low-rank approximation and Krylov methods, though since the literature is rich, so we cannot cover all relevant prior work. As discussed, early analyses of Krylov methods for approximating eigenvectors consider both single vector and block methods [102, 45, 65]. However, this work does not directly provide strong bounds for low-rank approximation, since convergence to the top singular vectors is not required to accurately solve the problem [34].

For low-rank approximation, large block methods have been more popular. In addition to prior work already discussed, this includes work on randomized sketching methods, which can be viewed as large block Krylov methods run for one or two iterations [74, 26, 25, 36, 75]. Sketching methods have become a mainstay technique in randomized numerical linear algebra.

Work on single vector or small block methods for low-rank approximation has been more sparse. [127] experimentally study small block methods, and suggest that large blocks are only worthwhile when singular value gaps are very small, when low precision suffices, or when making many passes over a matrix is expensive. [133] theoretically studies the related problem of singular value approximation for all block sizes, and as in our work, obtains linear convergence rates depending on $g_{k \rightarrow \ell}$. They also show superlinear rates when $\mathbf{A}$ has a sufficiently quickly decaying spectrum. While it is difficult to directly compare their results to ours on low-rank approximation, it would be interesting to consider such spectra in our setting. Finally, we note that [5] proves a result similar to [Theorem 61](#) using an algorithm that in some ways is a single vector Krylov method. However, because the method iteratively restarts k times with k randomly chosen starting vectors, it ultimately returns a solution from a block Krylov subspace.

Related to our results in [Section 4.5](#), we note that adding small random perturbations to avoid small singular value gaps or other conditioning issues is a technique that has been employed in several recent works focused on worst-case runtime bounds for linear algebraic problems [16, 93, 10].

4.2 Notation

We use capital bold letters to denote matrices, lowercase bold letters to denote vectors, and lowercase non-bold letters to denote scalars. For a matrix $\mathbf{Q}$, we let $\mathbf{q}_i$ denote the i^{th} column, $\text{span}(\mathbf{Q})$ denote its column span, and $\mathbf{Q}^\top$ denote its transpose. Typically, $\mathbf{A} \in \mathbb{R}^{d \times n}$ denotes our input matrix. We let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ denote the SVD of $\mathbf{A}$, with $\mathbf{U} \in \mathbb{R}^{d \times d}$, $\mathbf{\Sigma} \in \mathbb{R}^{d \times d}$, $\mathbf{V} \in \mathbb{R}^{d \times n}$. We let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d \geq 0$ denote the singular values of $\mathbf{A}$ (the diagonal entries of $\mathbf{\Sigma}$). We let $\mathbf{U}_k \in \mathbb{R}^{d \times k}$ and $\mathbf{V}_k \in \mathbb{R}^{n \times k}$ denote the first k columns of $\mathbf{U}$ and $\mathbf{V}$, and let $\mathbf{\Sigma}_k \in \mathbb{R}^{k \times k}$ denote the top $k \times k$ principal submatrix of $\mathbf{\Sigma}$. Then $\mathbf{A}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$ is the best rank- k approximation to $\mathbf{A}$ in any unitarily invariant norm. When $\mathbf{A}$ is square, we let $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ be the eigendecomposition of $\mathbf{A}$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$ are the eigenvalues of $\mathbf{A}$ (the diagonal entries of $\mathbf{\Lambda}$). We often work with symmetric positive semi-definite (PSD) matrices, which have all non-negative eigenvalues. In this case, the singular values equal the eigenvalues. We also work with matrix polynomials. If $p(x) = \sum_{i=1}^q c_i x^i$ is a polynomial and if $\mathbf{A}$ is square, $p(\mathbf{A}) := \sum_{i=1}^q c_i \mathbf{A}^i$.

We let $\|\mathbf{x}\|_2$ denote the vector ℓ_2 norm, $\|\mathbf{A}\|_2$ the spectral norm, $\|\mathbf{A}\|_F$ the Frobenius norm, and $\|\mathbf{A}\|_p = (\sum_{i=1}^p \sigma_i^p)^{1/p}$ the Schatten p -norm. Wherever $\|\mathbf{A}\|_\xi$ is used, the equation holds for both the spectral and Frobenius norms. We let $[n] = \{1, \dots, n\}$ be the set of integers between 1 and n . Finally, we let $\mathcal{N}(\mathbf{0}, \mathbf{I})$ denote the distribution over vectors whose entries are i.i.d. mean zero unit variance Gaussians. The dimension will be clear from context.

4.3 Proof of **Theorem 61**

In this section, we prove **Theorem 61** by showing that $\mathbf{S}_k$ as described in **Section 4.1.2** is a good enough starting matrix for block Krylov iteration. This proof serves as a foundation for all additional results in the paper. Throughout, we assume that $\mathbf{A} \in \mathbb{R}^{d \times d}$ is square and positive semidefinite. We shown in **Section 4.7** that this is without loss of generality: running **Algorithm 6** or **Algorithm 7** on a matrix $\mathbf{C} \in \mathbb{R}^{d \times n}$ with SVD $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ yields an identical output to running the method on the PSD matrix $\mathbf{A} = (\mathbf{C}\mathbf{C}^\top)^{1/2} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top$. Further, all low-rank approximation and singular value approximation results guaranteed for the returned matrix $\mathbf{Q}$ directly carry over from $\mathbf{A}$ to $\mathbf{C}$.

4.3.1 A naive approach that actually works.

As discussed in [Section 4.1.2](#), our main approach is to view the single-vector method of [Algorithm 6](#) as a block Krylov method in disguise. Specifically, recall the matrix $\mathbf{S}_k$ and the Krylov subspace from [Equations \(4.2\) and \(4.3\)](#). Since we now assume that $\mathbf{A}$ is PSD, we have $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^2$, so we can write

$$\mathbf{S}_k := \begin{bmatrix} \mathbf{x} & \mathbf{A}^2\mathbf{x} & \mathbf{A}^4\mathbf{x} & \dots & \mathbf{A}^{2(k-1)}\mathbf{x} \end{bmatrix} \quad (4.4)$$

and

$$\text{span}(\mathbf{K}) = \text{span} \left(\begin{bmatrix} \mathbf{S}_k & \mathbf{A}^2\mathbf{S}_k & \dots & \mathbf{A}^{2q}\mathbf{S}_k \end{bmatrix} \right), \quad (4.5)$$

where $q := t - k + 1$. This matches the subspace spanned by $\mathbf{K}$ in [Algorithm 7](#) with starting block $\mathbf{B} = \mathbf{S}_k$. Since the output of [Algorithm 6](#) and [Algorithm 7](#) depend only on the *span* of the Krylov subspace they construct, we will use this equivalence to appeal to prior results on block Krylov methods to analyze [Algorithm 6](#). To do so, we need to show that, even though $\mathbf{S}_k$ is very much unlike the i.i.d. random starting matrices used in prior work, it still provides a good enough starting matrix for convergence to a near-optimal low-rank approximation.

Towards that end, we use a natural definition of what it means to be a “good” starting matrix that specifically will allow us to leverage results on block Krylov methods from [85] in a black-box way. That same definition suffices for other results as well [132, 33]. Intuitively, we require that a starting matrix $\mathbf{B}$ has nontrivial inner product with all of the top k singular vectors of $\mathbf{A}$:

Definition 62 ((k, L) -good Starting Matrix). *Let $\mathbf{A} \in \mathbb{R}^{d \times n}$ be a matrix with top k left singular vectors $\mathbf{U}_k \in \mathbb{R}^{d \times k}$. A matrix $\mathbf{B} \in \mathbb{R}^{d \times k}$ is a (k, L) -good starting matrix for $\mathbf{A}$ if, letting $\mathbf{Q} \in \mathbb{R}^{d \times k}$ be an orthonormal basis for $\text{span}(\mathbf{B})$, $\mathbf{U}_k^\top \mathbf{Q}$ is invertible and $\|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 \leq L$.*

The condition above is equivalent to requiring that all singular values of $\mathbf{U}_k^\top \mathbf{Q}$ are at least $\frac{1}{\sqrt{L}}$, or that all principle angles between $\text{span}(\mathbf{U}_k)$ and $\text{span}(\mathbf{B})$ have $\cos(\theta_i) \geq \frac{1}{\sqrt{L}}$ [33].

Using [Theorem 62](#), we can immediately obtain bounds on the low-rank approximation error of a subspace $\mathbf{Q}$ returned by [Algorithm 7](#) when run with any (k, L) -good starting matrix. We will use such bounds to analyze the single vector Krylov method of [Algorithm 6](#), after proving that $\mathbf{S}_k$ is (k, L) -good. Consider the following bound, which depends logarithmically on L :

Imported Theorem 63 (Theorem 1 of [85]). *Let $\mathbf{B} \in \mathbb{R}^{d \times k}$ be any (k, L) -good starting matrix ([Theorem 62](#)) matrix for $\mathbf{A}$. If we run Block Krylov iteration ([Algorithm 7](#)) for $q = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, then the output $\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad (4.6)$$

and, letting $\mathbf{q}_i$ be the i^{th} column of $\mathbf{Q}$,

$$|\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2. \quad (4.7)$$

[Theorem 63](#) is implicit in [85], although, as stated in that work, it is specialized to when $\mathbf{B}$ is a matrix with i.i.d. Gaussian entries. In [Section 4.12](#) we discuss how the more general result stated above follows from [85]. [Theorem 63](#) gives two different guarantees. The first bounds low-rank approximation error, in both the Frobenius and spectral norms. The second shows that the columns of $\mathbf{Q}$ can be used to estimate the top singular values of $\mathbf{A}$. This can be called a “Ritz value guarantee” or a “singular value guarantee”.

A random Gaussian matrix $\mathbf{B}$ can be shown to be an $(k, O(dk))$ -good starting matrix with high probability (see Lemma 4 in [85] or [101]). This is intuitive, since $\mathbf{B}$ will span a uniformly random subspace, which has non-negligible inner product with any other fixed subspace, including the one spanned by $\mathbf{U}_k$. For a Gaussian starting block, [Theorem 63](#) therefore gives a bound of $O(\frac{1}{\sqrt{\varepsilon}} \log(d/\varepsilon))$ iterations to achieve [Equations \(4.6\) and \(4.7\)](#).

The fact that $\mathbf{S}_k$ satisfies [Theorem 62](#) is less clear. Our main technical contribution is to prove that it does, albeit with a much larger value of L than in the Gaussian case: in [Section 4.3.2](#) we show that, with probability at least $1 - \delta$, $\mathbf{S}_k$ is a (k, L) -good starting matrix for $L = O(\text{poly}(d/\delta g_{\min}^k))$. Here $g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_i}$ is the minimum gap between $\mathbf{A}$ ’s top k singular values. Since [Theorem 63](#) depends logarithmically on L , it follows that block Krylov with starting block $\mathbf{S}_k$ needs $O\left(\frac{k}{\sqrt{\varepsilon}} \log\left(\frac{1}{g_{\min}}\right) + \frac{1}{\sqrt{\varepsilon}} \log\left(\frac{d}{\varepsilon \delta}\right)\right)$ iterations to achieve [Equations \(4.6\) and \(4.7\)](#) – yielding [Theorem 61](#).

In other words, we require k times as many iterations when starting with $\mathbf{S}_k$ instead of a fully random $\mathbf{B}$. However, notice that running [Algorithm 7](#) for q iterations with starting block $\mathbf{S}_k$ only requires $q + k$ iterations of the single vector Krylov method from [Algorithm 6](#), and thus $q + k$ matrix-vector products. In contrast, running the method with a random $\mathbf{B}$ requires qk matrix-vector products. Ultimately, this allows

us to achieve in *Theorem 61* a total complexity (in terms of matrix-vector products) that matches, and in some cases improves, the block Krylov method initialized with a Gaussian starting matrix, up to the dependence on $\log(1/g_{\min})$.

4.3.2 Main Technical Analysis

Formally, we prove the following (k, L) -good guarantee for $\mathbf{S}_k$. In combination with *Theorem 63* and *Equation (4.5)*, this bound yields *Theorem 61*. It will also serve as the basis for all of the other results discussed in *Section 4.1.3*.

Theorem 64. *Fix any PSD matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ with singular values $\sigma_1 \geq \dots \geq \sigma_d$, let $\mathbf{g}$ be a vector with i.i.d. mean zero Gaussian entries, and let $\mathbf{S}_k = \begin{bmatrix} \mathbf{g} & \mathbf{A}^2 \mathbf{g} & \mathbf{A}^4 \mathbf{g} & \dots & \mathbf{A}^{2(k-1)} \mathbf{g} \end{bmatrix}$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, $\mathbf{S}_k$ is a (k, L) -good starting matrix for $\mathbf{A}$ for $L = \frac{cdk^3 \log(d/\delta)}{\delta^2 g_{\min}^4}$. Here $c = 2.5\pi$ is a fixed constant and $g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$.*

To prove *Theorem 64*, we will need a simple bound on the minimum of k independent Gaussian random variables:

Lemma 65. *Let $g_1, \dots, g_k \sim \mathcal{N}(0, 1)$ and independent. Then, with probability at least $\geq 1 - \delta$, $\min_i g_i^2 \geq \frac{2\delta^2}{\pi k^2}$.*

Proof. We have that:

$$\Pr[\min_i g_i^2 \geq t] = (1 - \Pr[g_1^2 \leq t])^k = (1 - 2\Pr[0 \leq g_1 \leq \sqrt{t}])^k \geq (1 - \sqrt{1 - e^{-2t/\pi}})^k.$$

The last line uses the bound $\Pr[0 \leq g_1 \leq \sqrt{t}] \leq \frac{1}{2}\sqrt{1 - e^{-2t/\pi}}$ from [23]. Setting the right hand side equal to $1 - \delta$ and solving for t , we get:

$$t = \frac{\pi}{2} \ln \frac{1}{1 - (1 - \delta)^{1/k}} \geq \frac{\pi}{2} (1 - (1 - \delta)^{1/k})^2 \geq \frac{\pi}{2} \left(\frac{\delta}{k}\right)^2 = \frac{\pi \delta^2}{2k^2}.$$

In the first inequality we used that $\ln \left(\frac{1}{1-x}\right) \geq x$. In the second we used that $1 - (1 - x)^{1/k} \geq \frac{x}{k}$. $\square$

Proof of Theorem 64. We first argue that $\mathbf{U}_k^\top \mathbf{S}_k$ is invertible. Observe that for any $\mathbf{x} \in \mathbb{R}^k$, $\mathbf{S}_k \mathbf{x} = \hat{p}(\mathbf{A}^2) \mathbf{g}$ for some degree $k-1$ polynomial $\hat{p}$ with coefficients determined by the entries in $\mathbf{x}$. We let $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{U}^\top$ be the SVD of $\mathbf{A}$. Then

$$\mathbf{U}_k^\top \mathbf{S}_k \mathbf{x} = \mathbf{U}_k^\top \hat{p}(\mathbf{A}^2) \mathbf{g} = \mathbf{U}_k^\top \mathbf{U} \hat{p}(\mathbf{\Sigma}^2) \mathbf{U}^\top \mathbf{g} = \hat{p}(\mathbf{\Sigma}_k^2) \tilde{\mathbf{g}},$$

where Σ_k contains the top left k elements of Σ and $\tilde{\mathbf{g}} := \mathbf{U}_k^\top \mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ by rotational invariance of the Gaussian distribution. Note by [Theorem 65](#) that $\min_{i \in \{1, \dots, k\}} \tilde{g}_i^2 \geq \frac{2\delta^2}{\pi k^2}$ with probability at least $1 - \delta$. So altogether, we can write

$$\|\mathbf{U}_k^\top \mathbf{S}_k \mathbf{x}\|_2^2 = \|\mathbf{U}_k^\top \hat{p}(\mathbf{A}^2) \tilde{\mathbf{g}}\|_2^2 = \sum_{i=1}^k (\hat{p}(\sigma_i^2))^2 \tilde{g}_i^2 \geq \frac{2\delta^2}{\pi k^2} \sum_{i=1}^k (\hat{p}(\sigma_i^2))^2. \quad (4.8)$$

Since $\hat{p}$ has degree $k - 1$, if none of the top k singular values are repeated (i.e., $g_{\min} > 0$), we have that the right hand side is nonzero for any nonzero $\mathbf{x}$. Thus, $\mathbf{U}_k^\top \mathbf{S}_k$ is invertible.

Now, let $\mathbf{Q} \in \mathbb{R}^{d \times k}$ be any orthonormal basis for $\text{span}(\mathbf{S}_k)$, so that $\mathbf{S}_k = \mathbf{Q}\mathbf{C}$ for some invertible matrix $\mathbf{C} \in \mathbb{R}^{k \times k}$ (observe that $\mathbf{S}_k$ must be full rank since $\mathbf{U}_k^\top \mathbf{S}_k$ is invertible). Since $\mathbf{U}_k^\top \mathbf{S}_k$ is invertible, we then also know that $\mathbf{U}_k^\top \mathbf{Q}$ is invertible, as required by [Theorem 62](#). We also have $\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1} = \mathbf{Q}\mathbf{C}(\mathbf{U}_k^\top \mathbf{Q}\mathbf{C})^{-1} = \mathbf{Q}(\mathbf{U}_k^\top \mathbf{Q})^{-1}$ and therefore $\|\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1}\|_2^2 = \|\mathbf{Q}(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 = \|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2$. So, to prove the theorem, it suffices to bound

$$\|\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1}\|_2^2 = \max_{\mathbf{x}} \frac{\|\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1} \mathbf{x}\|_2^2}{\|\mathbf{x}\|_2^2} = \max_{\mathbf{x}} \frac{\|\mathbf{S}_k \mathbf{x}\|_2^2}{\|\mathbf{U}_k^\top \mathbf{S}_k \mathbf{x}\|_2^2} = \max_{\deg(\hat{p}) \leq k-1} \frac{\|\hat{p}(\mathbf{A}^2) \mathbf{g}\|_2^2}{\|\mathbf{U}_k^\top \hat{p}(\mathbf{A}^2) \mathbf{g}\|_2^2}. \quad (4.9)$$

We already bounded the denominator in (4.8). Thus, we turn to the numerator. Since $\mathbf{g}$ has i.i.d. mean zero, unit variance Gaussian entries we have for each i , $g_i^2 \leq 1 + 4 \log(1/\delta)$ with probability at least $1 - \delta$ by standard concentration bounds for chi-squared random variables [67]. So, by a union bound, $\max_i g_i^2 \leq 5 \log(d/\delta)$ for $d > 2$. We thus have:

$$\|\hat{p}(\mathbf{A}^2) \mathbf{g}\|_2^2 \leq 5 \log(d/\delta) \sum_{i=1}^d (\hat{p}(\sigma_i^2))^2 \leq 5d \log(d/\delta) \cdot \max_{i \in \{1, \dots, d\}} (\hat{p}(\sigma_i^2))^2. \quad (4.10)$$

Combining (4.10) and (4.8), we conclude that

$$\|\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1}\|_2^2 \leq \frac{5\pi dk^2 \log(d/\delta)}{2\delta^2} \cdot \max_{\deg(\hat{p}) \leq k-1} \frac{\max_{i \in \{1, \dots, d\}} (\hat{p}(\sigma_i^2))^2}{\sum_{i=1}^k (\hat{p}(\sigma_i^2))^2}. \quad (4.11)$$

We now focus on bounding the maximum in (4.11). Observe that if there were no gap between two of the top k singular values, then some nonzero polynomial $\hat{p}$ could make the denominator zero by equaling zero on the at most $k - 1$ unique values in $\sigma_1, \dots, \sigma_k$. So, any bound on (4.11) must depend on the minimum gap between

singular values. Second, note that if the maximum in the numerator is achieved for $i \leq k$, then the overall ratio is trivially at most 1. So, without loss of generality, we only consider $\max_{i \in \{k+1, \dots, d\}} (\hat{p}(\sigma_i^2))^2$ in the numerator.

To bound this ratio, we follow a similar broad approach to [102], who bounds a related term by expanding $\hat{p}$ as an interpolating polynomial. Formally, we write $\hat{p}$ as a Lagrange interpolating polynomial over $\sigma_1^2, \dots, \sigma_k^2$:

$$\hat{p}(x) = \sum_{i=1}^k \phi_i \ell_i(x) \quad \text{where} \quad \phi_i := \hat{p}(\sigma_i^2), \quad \ell_i(x) := \prod_{\substack{j \in \{1, \dots, k\} \\ j \neq i}} \frac{x - \sigma_j^2}{\sigma_i^2 - \sigma_j^2}, \quad i \in \{1, \dots, k\}.$$

For any $0 \leq x \leq \sigma_k^2$ we have

$$|\ell_i(x)| \leq \prod_{\substack{j \in \{1, \dots, k\} \\ j \neq i}} \frac{\sigma_j^2}{|\sigma_i^2 - \sigma_j^2|} \leq \prod_{\substack{j \in \{1, \dots, k\} \\ j \neq i}} \frac{\sigma_j^2}{|\sigma_i - \sigma_j|^2} \leq \frac{1}{g_{\min}^{2k}},$$

where the second inequality uses that $|\sigma_i^2 - \sigma_j^2| \geq |\sigma_i - \sigma_j|^2$ for all $\sigma_i, \sigma_j \geq 0$. Next, we write $\phi = [\phi_1 \dots \phi_k] = [\hat{p}(\sigma_1^2) \dots \hat{p}(\sigma_k^2)]$ and obtain:

$$\begin{aligned} \frac{\max_{i \in \{k+1, \dots, d\}} |\hat{p}(\sigma_i^2)|^2}{\sum_{i=1}^k (\hat{p}(\sigma_i^2))^2} &\leq \frac{\max_{x \in [0, \sigma_k^2]} |\hat{p}(x)|^2}{\sum_{i=1}^k (\hat{p}(\sigma_i^2))^2} = \frac{\max_{x \in [0, \sigma_k^2]} |\sum_{i=1}^k \phi_i \ell_i(x)|^2}{\sum_{i=1}^k (\hat{p}(\sigma_i^2))^2} \\ &\leq \frac{\max_{x \in [0, \sigma_k^2]} (\|\phi\|_1 \max_i |\ell_i(x)|)^2}{\|\phi\|_2^2} \\ &= \frac{\|\phi\|_1^2}{\|\phi\|_2^2} \max_{x \in [0, \sigma_k^2], i \in \{1, \dots, k\}} |\ell_i(x)|^2 \\ &\leq \frac{k}{g_{\min}^{4k}}. \end{aligned}$$

Finally, we plug back into (4.11) to conclude that $\|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 = \|\mathbf{S}_k(\mathbf{U}_k^\top \mathbf{S}_k)^{-1}\|_2^2 \leq \frac{5\pi dk^3}{2g_{\min}^{4k} \delta^2} \log(\frac{d}{\delta})$, which completes the proof. $\square$

4.3.3 Proof of **Theorem 61**

As mentioned, **Theorem 61** follows directly by combining **Theorem 63** and **Theorem 64**. In particular, since $\mathbf{S}_k$ is an (k, L) -good starting matrix for $L \leq \left(\frac{d}{\delta g_{\min}^k}\right)^c$ for a constant c , if we run the block Krylov method initialized with $\mathbf{S}_k$, then with

probability at least $1 - \delta$, we obtain $\mathbf{Q}$ achieving the guarantees of [Theorem 61](#) after

$$q = O\left(\frac{1}{\sqrt{\varepsilon}} \log\left(\frac{dL}{\varepsilon}\right)\right) = O\left(\frac{k}{\sqrt{\varepsilon}} \log\left(\frac{1}{g_{\min}}\right) + \frac{1}{\sqrt{\varepsilon}} \log\left(\frac{d}{\varepsilon\delta}\right)\right) \text{ iterations.}$$

Moreover, by [Equation \(4.5\)](#), the $\mathbf{Q}$ returned by [Algorithm 7](#) with $\mathbf{S}_k$ as the starting block, is exactly the same as the $\mathbf{Q}$ returned by the single vector Krylov method ([Algorithm 6](#)) after $q + k$ iterations. Overall, we get the following full version of [Theorem 61](#), which includes both the low-rank approximation and singular value guarantees from [Theorem 63](#):

Theorem 61 Restated. *For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_i}$. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 6](#) initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{\varepsilon}} \log(\frac{d}{\varepsilon\delta}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

4.4 Additional Applications of Main Result

In this section, we leverage our analysis of the starting block $\mathbf{S}_k$ to give several results beyond [Theorem 61](#). In [Section 4.4.1](#), we start by generalizing from using $\mathbf{S}_k$ to using $\mathbf{S}_\ell$, which has $\ell \geq k$ columns, and lets us obtain faster rates of convergence when the spectrum of $\mathbf{A}$ decays between singular values k and $\ell + 1$. Next, in [Section 4.4.2](#), we use the results of [Theorem 61](#) and [Section 4.4.1](#) to get a faster rate of convergence in the Frobenius norm, simplifying the algorithm of [8]. In [Section 4.4.3](#), we generalize from the Frobenius norm to Schatten p -norms, also simplifying [8]. Lastly, in [Section 4.4.4](#), we generalize from single vector Krylov with block size $b = 1$ to small-block Krylov with any block size $b < k$.

4.4.1 Faster Convergence with Spectral Decay

As discussed in [Section 4.1](#), in addition to spectrum-independent bounds, we know that large block Krylov methods achieve very fast convergence when using block size $\ell \geq k$ if there is a sufficiently large gap between σ_k and $\sigma_{\ell+1}$. Formally, [85] show the following:

Imported Theorem 66 (Theorem 13 of [85]). *Let $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ be any (ℓ, L) -good starting matrix ([Theorem 62](#)) matrix for $\mathbf{A}$, for some $\ell \geq k$. If we run Block Krylov*

iteration ([Algorithm 7](#)) for $q = O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, where $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$, then the output $\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Our second main result shows that the convergence rate of [Theorem 66](#) applies to single vector Krylov as well. In particular, we can simply apply the same idea as was used to prove [Theorem 61](#), but with “simulated block size” $\ell \geq k$. Letting

$$\mathbf{S}_\ell := \begin{bmatrix} \mathbf{x} & \mathbf{A}^2 \mathbf{x} & \mathbf{A}^4 \mathbf{x} & \dots & \mathbf{A}^{2(\ell-1)} \mathbf{x} \end{bmatrix} \quad \text{and} \quad \text{span}(\mathbf{K}) = \text{span} \left(\begin{bmatrix} \mathbf{S}_\ell & \mathbf{A}^2 \mathbf{S}_\ell & \dots & \mathbf{A}^{2q} \mathbf{S}_\ell \end{bmatrix} \right),$$

we observe that single vector Krylov run for t iterations exactly computes $\text{span}(\mathbf{K})$ for $q = t - \ell + 1$. This lets us show that single vector Krylov run for t iterations essentially matches the convergence rate of block Krylov run for $\approx t/\ell$ iterations:

Theorem 67 (Spectral Decay Convergence). *For $\mathbf{A} \in \mathbb{R}^{d \times n}$ and $\ell \geq k$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ and $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 6](#) initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{\ell}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{d}{\delta \varepsilon}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Proof. By [Theorem 64](#), we know that $\mathbf{S}_\ell$ is an (ℓ, L) -good starting matrix for block Krylov iteration on $\mathbf{A}$, where $L \leq (\frac{d}{\delta g_{\min}^\ell})^c$ for a constant c . So, by [Theorem 66](#), we find that block Krylov iteration with starting block $\mathbf{S}_\ell$ converges in

$$q = O \left(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log \left(\frac{dL}{\varepsilon} \right) \right) = O \left(\frac{\ell}{\sqrt{g_{k \rightarrow \ell}}} \log \left(\frac{1}{g_{\min}} \right) + \frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log \left(\frac{d}{\delta \varepsilon} \right) \right)$$

iterations. Moreover, by [Equation \(4.5\)](#), the $\mathbf{Q}$ returned by [Algorithm 7](#) with $\mathbf{S}_\ell$ as the starting block is exactly the same as the $\mathbf{Q}$ returned by the single vector Krylov method ([Algorithm 6](#)) after $q + \ell$ iterations, which completes the proof. $\square$

Comparison to Prior Bounds. In terms of the number of matrix-vector products computed, [Theorem 67](#) can significantly improve upon the prior work for large k . [85] require using a block size $\ell \geq k$ for $t = O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{d}{\delta \varepsilon}))$ iterations, or equivalently for $O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \cdot \ell \log(\frac{d}{\delta \varepsilon}))$ matrix-vector products. Assuming that $g_{\min}$ is a constant, our guarantee for single vector Krylov methods only requires $O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} (\ell + \log(\frac{d}{\varepsilon \delta})))$

matrix-vector products. That is, we reduce the product between ℓ and $\log(\frac{d}{\delta\varepsilon})$ into a sum, suggesting a nearly ℓ -fold speedup when we want high precision results. Further, we obtain the above guarantee for any $\ell \geq k$, meaning that single vector Krylov automatically competes with the bound for the *best possible choice* of block size, without knowing it in advance. We observe in [Section 4.6.5](#) that the above theoretical advantages translate into practice – for matrices with decaying singular value spectra, we find that single vector Krylov methods substantially outperform large block methods.

Comparison to Lower Bounds. It is worth comparing [Theorem 67](#) to the lower bound of [109], who show that finding an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that $\sum_{i=1}^k \mathbf{q}_i^\top \mathbf{A} \mathbf{q}_i \geq (1 - \varepsilon) \sum_{i=1}^k \sigma_i(\mathbf{A})$ requires at least $\Omega(\frac{k \log d}{\sqrt{g_{k \rightarrow k}}})$ matrix-vector products when $\varepsilon = g_{k \rightarrow k} = \frac{\sigma_k - \sigma_{k+1}}{\sigma_k}$. In comparison, applying [Theorem 67](#) with $\ell = k$ yields an upper bound of roughly $O(\frac{k \log(1/g_{\min})}{\sqrt{g_{k \rightarrow k}}} + \frac{\log d}{\sqrt{g_{k \rightarrow k}}})$, which only does not violate the lower bound of [109] since $\frac{1}{g_{\min}} = \text{poly}(d)$ for their input. I.e., their input suffers from very small singular value gaps.

When $k = 1$, the intuition for the $\log d$ dependence is that since a random start vector has $\frac{1}{\text{poly}(d)}$ inner product with the top singular vector, it requires $\log d$ iterations to converge. The matching upper and lower bounds of $\Theta(\frac{k \log d}{\sqrt{g_{k \rightarrow k}}})$ from [85] and [109] suggest that the cost of rank- k approximation is simply k times the cost of rank-1 approximation, i.e., roughly $k \log d$. In fact, the situation is more nuanced. Our work suggests that, unless gaps between singular values are very small, the cost of rank- k approximation is much actually cheaper.

4.4.2 Improved Results for Frobenius Norm Low-Rank Approximation

[Theorem 67](#) shows that single vector Krylov methods achieve strong spectrum-adaptive guarantees, converging at essentially the same rate as the best choice of block size, if not faster. As a concrete application of this observation, we show that single vector Krylov methods automatically match a recent result of [8] on Frobenius norm low-rank approximation. [8] propose an algorithm that combines the results of running block Krylov iteration twice, once with block size k for $\tilde{O}(\frac{1}{\varepsilon^{1/3}})$ iterations and once with block size $O(\frac{k}{\varepsilon^{1/3}})$ for $\tilde{O}(1)$ iterations. Their algorithm obtains a $(1 + \varepsilon)$ optimal low-rank approximation in the Frobenius norm using $\tilde{O}(\frac{k}{\varepsilon^{1/3}})$ matrix-vector products instead of $\tilde{O}(\frac{k}{\sqrt{\varepsilon}})$, as required by [Theorem 61](#). Since [Theorem 61](#) and [Theorem 67](#) show that single vector Krylov methods match the convergence rates of both block size k and block size $O(\frac{k}{\varepsilon^{1/3}})$, they therefore match the result of [8]. Formally we have:

Theorem 68. For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ where $\ell = \Theta(\frac{k}{\varepsilon^{1/3}})$. For any $\varepsilon, \delta \in (0, 1)$, **Algorithm 6** initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\varepsilon^{1/3}} \log(\frac{1}{g_{\min}}) + \frac{1}{\varepsilon^{1/3}} \log(\frac{d}{\delta\varepsilon}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$, we have

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F.$$

We formally prove **Theorem 68** in **Section 4.8**, by generalizing and formalizing an analysis stated in the introduction of [8]. We note that, in concurrent work, [9] show a similar result which has no $g_{\min}$ dependence and further removes the $\log d$ dependence. However, their bound only applies to the special case of rank-1 approximation.

4.4.3 Improved Rates for Schatten Norm Low-Rank Approximation

The work of [8] also gives bounds for low-rank approximation in general Schatten p -Norms for any $p \geq 1$. They show that by running Krylov iteration 4 times, on both $\mathbf{A}$ and $\mathbf{A}^\top$ and with both a relatively small block size $\ell = k$ and a relatively large block size $\ell = \tilde{O}(\frac{k}{\varepsilon^{1/3}})$, they can recover a low-rank approximation to $\mathbf{A}$ in the Schatten p -norm using $\tilde{O}(\frac{kp^{1/6}}{\varepsilon^{1/3}})$ matrix-vector multiplications. As in the case of Frobenius norm low-rank approximation, we show that we can match this entire process with a single instantiation of single vector Krylov, yielding:

Theorem 69 (Schatten- p Norm Low-Rank Approximation). For $\mathbf{A} \in \mathbb{R}^{d \times n}$ and $p \geq 1$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ where $\ell = \Theta(\frac{k}{\varepsilon^{1/3}p^{1/3}})$. For any $\varepsilon, \delta \in (0, 1)$, let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ be the result of running **Algorithm 6** on $\mathbf{A}^\top$ initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for

$$t = O\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} \log\left(\frac{1}{g_{\min}}\right) + (\sqrt{p} + \frac{p^{1/6}}{\varepsilon^{1/3}}) \log\left(\frac{np}{\delta\varepsilon}\right)\right) = \tilde{O}\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} + \sqrt{p}\right)$$

iterations. Let $\mathbf{Z} \in \mathbb{R}^{d \times k}$ be an orthonormal basis for $\mathbf{A}\mathbf{Q}$. Then, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Z}\mathbf{Z}^\top \mathbf{A}\|_p \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_p,$$

where $\|\mathbf{A}\|_p := (\sum_{i=1}^d \sigma_i(\mathbf{A})^p)^{1/p}$ is the Schatten p -norm.

We prove **Theorem 69** in **Section 4.9**. Unlike our previous results, **Theorem 69** first uses single vector Krylov to compute an orthonormal basis $\mathbf{Q}$, but then outputs $\mathbf{Z}$, which is an orthonormal basis for $\mathbf{A}\mathbf{Q}$. We suspect that this two-step process is an

artifact of the analysis in [8], and that simply returning the output of a single vector Krylov method suffices.

Note that since the same single vector Krylov method obtains the result of [Theorem 69](#) for any choice of p , and setting $p = O(\frac{\log d}{\varepsilon})$ closely approximates the Schatten- ∞ norm (i.e., the spectral norm), we achieve the best known result for outputting a low-rank approximation *simultaneously* in all Schatten p -norms:

Corollary 70. *For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ where $\ell = \Theta(\frac{k}{\log^{1/3}(d)})$. For any $\varepsilon, \delta \in (0, 1)$, let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ be the result of running [Algorithm 6](#) on $\mathbf{A}^\top$ initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for*

$$t = O\left(\frac{k \log^{\frac{1}{6}}(d)}{\sqrt{\varepsilon}} \log\left(\frac{1}{g_{\min}}\right) + \frac{\sqrt{\log(d)}}{\sqrt{\varepsilon}} \log\left(\frac{d}{\delta \varepsilon}\right)\right) = \tilde{O}\left(\frac{d}{\sqrt{\varepsilon}}\right)$$

iterations. Let $\mathbf{Z} \in \mathbb{R}^{d \times k}$ be an orthonormal basis for $\mathbf{A}\mathbf{Q}$. Then, for $\varepsilon_p := \varepsilon \cdot \min\{1, \frac{\sqrt{p\varepsilon}}{\log(d)}\} \leq \varepsilon$, with probability at least $1 - \delta$, simultaneously for all $p \geq 1$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_p \leq (1 + \varepsilon_p) \|\mathbf{A} - \mathbf{A}_k\|_p.$$

A similar but weaker guarantee is available from [8]. They show that running four Block Krylov methods with block size choices depending on p suffices to obtain a low-rank approximation in the Schatten q -norm for all $q \leq p$. If we let $p = O(\frac{\log d}{\varepsilon})$, so that $\|\mathbf{A}\|_p$ approximates the spectral norm, then, their result shows that $\tilde{O}(\frac{k}{\sqrt{\varepsilon}})$ matrix-vector products suffice to output a $(1 + \varepsilon)$ relative error low-rank approximation in all Schatten norms, including the Frobenius norm. In contrast, [Theorem 68](#) shows that running single vector Krylov for $\tilde{O}(\frac{k}{\sqrt{\varepsilon}})$ iterations actually gives a better relative error of $(1 + \varepsilon^{3/2})$ in the Frobenius norm.

4.4.4 General Bounds for Small-Block Methods

Our approach to analyzing single vector Krylov also extends to ‘small-block’ Krylov methods, which use block size $1 < b < k$. Such methods are common in practice as they help avoid slow convergence due to nearby singular values: intuitively, we expect a block size of b to be effective even when the input $\mathbf{A}$ has clusters of at most b very close singular values. Additionally, parallelism often lets us compute multiple matrix-vector products with $\mathbf{A}$ just as quickly as computing a single matrix-vector product, with incentives the use of a block size > 1 .

We outline results for small block methods in this section, but defer proofs to

Section 4.11. To start, we generalize the notion of a gap between singular values:

Definition 71. Fix block size $b \in [k]$. For each $i \in [k]$, we let $\mathcal{N}_i \subset [k] \setminus \{i\}$ be the indices of the $b-1$ singular values other than i that minimize $|\frac{\sigma_i - \sigma_j}{\sigma_j}|$. Then let $g_{\min,b}$ be the b^{th} -order gap of $\mathbf{A}$:

$$g_{\min,b} = \min_{i \in [k]} \min_{j \in [k] \setminus \mathcal{N}_i, j \neq i} \left| \frac{\sigma_i - \sigma_j}{\sigma_j} \right|.$$

If $b = k$, then the sets $\mathcal{N}_i$ have no terms, and we define $g_{\min,b} := 1$.

If $b = 1$, then we see that the sets $\mathcal{N}_i$ are empty, and we recover $g_{\min,b} = g_{\min}$ as defined in [Theorem 61](#). If $b = k$, then $g_{\min,b}$ is just 1, which matches the fact that block size k does not require a gap dependence [85]. If e.g., $b = 2$ and $\mathbf{A}$ has two identical singular values, then we may still have $g_{\min,b} > 0$, and so an algorithm can depend on $\frac{1}{g_{\min,b}}$ without risking total failure. With this characterization of small-block gaps, we prove the following generalization of [Theorem 64](#) in [Section 4.11](#):

Theorem 72. Fix any PSD $\mathbf{A} \in \mathbb{R}^{d \times d}$ and block size $b \in \{1, \dots, k\}$. Let $\mathbf{G} \in \mathbb{R}^{d \times b}$ be a matrix with i.i.d. $\mathcal{N}(0, 1)$ entries, and let $\mathbf{S}_r = \begin{bmatrix} \mathbf{G} & \mathbf{A}^2 \mathbf{G} & \mathbf{A}^4 \mathbf{G} & \dots & \mathbf{A}^{2(r-1)} \mathbf{G} \end{bmatrix}$, where $r = k - b + 1$. For any $\delta \in (0, 1)$, with probability at least $(1 - \delta)$, there exists a matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$ that lies in the span of $\mathbf{S}_r$ and is a (k, L) -good starting matrix for $\mathbf{A}$ for $L = \frac{cb^2 k^2 d \log(1/\delta)}{\delta^2 g_{\min,b}^{4(k-b)}}$. Here, c is a fixed constant.

Above, the starting block $\mathbf{S}_r$ has $rb \approx b(k-b)$ columns, which can be larger than k . [Theorem 72](#) shows that there exists a matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$ in the span of $\mathbf{S}_r$ which satisfies the (k, L) -good property of [Theorem 62](#). Since the Krylov subspace generated by $\mathbf{Q}$ lies within the Krylov subspace generated by $\mathbf{S}_r$, we can thus use [Theorem 63](#) and [Theorem 66](#) to obtain guarantees for the subspace generated by $\mathbf{S}_r$. For a formal argument, see [Section 4.11](#). Overall, by plugging in the value of $L = \text{poly}(\frac{d}{\delta g_{\min,b}^{k-b}})$ into these theorems, we achieve similar guarantees as for single vector Krylov, but with a dependence on $g_{\min,b}$. For instance, we generalize [Theorem 61](#), obtaining:

Theorem 73. For $\mathbf{A} \in \mathbb{R}^{d \times n}$ and $b \leq k$, let $g_{\min,b}$ as in [Theorem 71](#). For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 7](#) initialized with i.i.d. $\mathcal{N}(0, 1)$ matrix $\mathbf{G} \in \mathbb{R}^{d \times b}$ and run for $t = O(\frac{k-b}{\sqrt{\varepsilon}} \log(\frac{1}{g_{\min,b}}) + \frac{1}{\sqrt{\varepsilon}} \log(\frac{d}{\delta \varepsilon}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

In particular, this requires $O(\frac{b(k-b)}{\sqrt{\varepsilon}} \log(\frac{1}{g_{\min,b}}) + \frac{b}{\sqrt{\varepsilon}} \log(\frac{d}{\delta\varepsilon}))$ matrix-vector products with $\mathbf{A}$.

For constant $b = O(1)$, the above theorem recovers the same asymptotic matrix-vector complexity as [Theorem 61](#) but with an improved dependence on singular value gaps. For $k - b = O(1)$, it nearly matches the matrix-vector complexity of block Krylov with block size k , but with a very mild gap dependence. When $b = \frac{k}{2}$, we get the worst of both worlds, needing $m = \tilde{O}(\frac{k^2}{\sqrt{\varepsilon}})$ matrix-vector products. This may be a limitation of our proof techniques: it could be possible to show that the matrix-vector complexity scales linearly in k for any block size.

Further, observe that by using [Theorem 72](#), the spectrum-adaptive results of [Section 4.4.1](#), the fast Frobenius results of [Section 4.4.2](#), and the fast Schatten norm results of [Section 4.4.3](#) also apply in the general block size $b \leq k$ case. Wherever those theorems have $g_{\min}$, we replace this with $g_{\min,b}$, where $g_{\min,b}$ is defined now across the top $\ell - b$ singular values instead of the top $k - b$.

4.5 Random Perturbation Analysis

[Theorems 61, 67, 69](#) and [73](#) all show that single vector or small-block Krylov methods match or improve the existing bounds for large block methods, up to a factor of $\log(\frac{1}{g_{\min}})$. Single vector methods cannot avoid this dependence in general: if one of $\mathbf{A}$'s top singular values is repeated, so that $g_{\min} = 0$, then the Krylov subspace can be rank deficient and fail to converge to the top subspace of $\mathbf{A}$. To address this possible point of failure, we take a smoothed-analysis approach, showing that if our input instance is subjected to a small random perturbation, the dependence on $g_{\min}$ can be removed. The key tool we leverage is a line of work from random matrix theory called *eigenvalue repulsion*, which shows that small eigenvalue gaps are brittle: by adding a tiny amount of random noise to any matrix, we can ensure that its eigenvalues are well separated [84, 89, 13].

4.5.1 Gap-Independent Bounds for PSD Matrices

Our first result shows that we can remove the dependence on $g_{\min}$ when our input matrix $\mathbf{A}$ is PSD. In [Section 4.5.2](#), we show that this result gives some bounds for non-PSD matrices as well. Our result leverages the following eigenvalue repulsion bound, which we derive in [Section 4.10.1](#) using a result of [84].

Lemma 74. Fix symmetric matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, $\delta \in (0, 1)$, and $\Delta \leq \|\mathbf{A}\|_2$. Let $\mathbf{D} \in \mathbb{R}^{d \times d}$ be a diagonal matrix whose entries are uniformly distributed in $[-\Delta, +\Delta]$. Then, letting $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{D}$ and letting C denote some universal constant, with probability at least $1 - \delta$,

$$\min_{i \in [d-1]} \frac{|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})|}{|\lambda_{i+1}(\tilde{\mathbf{A}})|} \geq \frac{\delta}{Cd^2} \cdot \frac{\Delta^2}{\|\mathbf{A}\|_2^2}.$$

That is, by adding a small amount of noise to the diagonal of $\mathbf{A}$, we can ensure its eigenvalue gaps are polynomially large in the problem parameters. While [Theorem 74](#) holds for general symmetric matrices, we will apply it specifically to PSD $\mathbf{A}$ in our analysis, since for indefinite matrices, having large eigenvalue gaps does not necessarily imply having *singular value gaps*, which are required for our convergence results for single vector Krylov methods. It would be interesting to prove an analogous result to [Theorem 74](#) that applies directly to singular values and use it to generalize our results beyond PSD matrices.

Naturally, the larger the perturbation parameter Δ in [Theorem 74](#), the larger the resulting gaps. Thus, Δ gives a tradeoff between runtime and accuracy: higher Δ leads to larger gaps and thus faster convergence. However, it will also make the result of our algorithms less accurate. In [Section 4.10.2](#) we show that picking Δ such that $\|\mathbf{D}\|_2 \leq \frac{\varepsilon \sigma_{k+1}(\mathbf{A})}{d}$ suffices to guarantee that a near-optimal low-rank approximation of $\tilde{\mathbf{A}}$ is also near optimal for $\mathbf{A}$:

Lemma 75. Let $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{D}$ where $\|\mathbf{D}\|_2 \leq \frac{\varepsilon}{3d} \sigma_{k+1}(\mathbf{A})$ and $\varepsilon \in (0, 1)$. Fix any $\mathbf{Q} \in \mathbb{R}^{d \times k}$ with orthonormal columns $\mathbf{q}_1, \dots, \mathbf{q}_k$. Then, with probability at least $1 - \delta$,

1. If $|\mathbf{q}_i^\top \tilde{\mathbf{A}} \tilde{\mathbf{A}}^\top \mathbf{q}_i - \sigma_i(\tilde{\mathbf{A}})^2| \leq \varepsilon \sigma_{k+1}(\tilde{\mathbf{A}})^2$, then $|\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq 8\varepsilon \sigma_i(\mathbf{A})^2$.
2. If $\|\tilde{\mathbf{A}} - \mathbf{Q} \mathbf{Q}^\top \tilde{\mathbf{A}}\|_2 \leq (1+\varepsilon) \|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_2$, then $\|\mathbf{A} - \mathbf{Q} \mathbf{Q}^\top \mathbf{A}\|_2 \leq (1+2\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_2$.
3. If $\|\tilde{\mathbf{A}} - \mathbf{Q} \mathbf{Q}^\top \tilde{\mathbf{A}}\|_F \leq (1+\varepsilon) \|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_F$, then $\|\mathbf{A} - \mathbf{Q} \mathbf{Q}^\top \mathbf{A}\|_F \leq (1+4\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_F$.

In particular, since we pick $\mathbf{D}$ to be diagonal, we have $\|\mathbf{D}\|_2 \leq \Delta \leq \frac{\varepsilon \sigma_{k+1}(\mathbf{A})}{3d}$. [Theorems 74](#) and [75](#) then together imply the following gap-independent variant of [Theorem 61](#) for PSD matrices.

Corollary 76 (Gap-Independent Convergence). For PSD $\mathbf{A} \in \mathbb{R}^{d \times d}$, let $\kappa_k = \frac{\sigma_1}{\sigma_k}$ and $\Delta = \frac{\varepsilon \sigma_{k+1}}{3d}$. For any $\varepsilon, \delta \in (0, \frac{1}{2})$, let $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{D}$ where $\mathbf{D} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with entries drawn uniformly and i.i.d. from $[-\Delta, \Delta]$. Then, [Algorithm 6](#) run on $\tilde{\mathbf{A}}$ initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{d\kappa_k}{\delta\varepsilon}))$ iterations returns

orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Proof. Proving this result simply requires showing that [Theorem 74](#) implies gaps between $\mathbf{A}$'s singular values. This is not immediate since, even if we assume $\mathbf{A}$ is PSD, $\tilde{\mathbf{A}}$ might not be, so could have negative eigenvalues. An eigenvalue at η and another eigenvalue at $-\eta$, which would give a singular value gap of 0, which we need to rule out. To do so, note that $\|\mathbf{D}\|_2 = \max_i |\mathbf{D}_{i,i}| \leq \Delta < \varepsilon \sigma_{k+1}(\mathbf{A})$. So by assuming that $\mathbf{A}$ is PSD, we know that for any negative eigenvalue $\lambda_i(\tilde{\mathbf{A}})$ of $\tilde{\mathbf{A}}$,

$$|\lambda_i(\tilde{\mathbf{A}})| \leq |\lambda_i(\tilde{\mathbf{A}}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{D}\|_2 < \varepsilon \sigma_{k+1}(\mathbf{A}) \leq \sigma_{k+1}(\tilde{\mathbf{A}}).$$

The last inequality assumes $\varepsilon \leq \frac{1}{2}$ to say that $\sigma_{k+1}(\tilde{\mathbf{A}}) \geq \sigma_{k+1}(\mathbf{A}) - \|\mathbf{D}\|_2 \geq (1 - \varepsilon) \sigma_{k+1}(\mathbf{A}) \geq \varepsilon \sigma_{k+1}(\mathbf{A})$. So, we know that if $\lambda_i(\tilde{\mathbf{A}}) < 0$ then any singular value associated with $\lambda_i(\tilde{\mathbf{A}})$ is not one of the top k singular values of $\tilde{\mathbf{A}}$. So, the top k singular values of $\tilde{\mathbf{A}}$ must all be associated with nonnegative eigenvalues. That is $\sigma_i(\tilde{\mathbf{A}}) = \lambda_i(\tilde{\mathbf{A}})$ for $i \in [k]$. And so, we find that

$$g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{|\sigma_i(\tilde{\mathbf{A}}) - \sigma_{i+1}(\tilde{\mathbf{A}})|}{\sigma_{i+1}(\tilde{\mathbf{A}})} = \min_{i \in \{1, \dots, k-1\}} \frac{|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})|}{|\lambda_{i+1}(\tilde{\mathbf{A}})|} \geq \frac{\delta}{Cn^2} \cdot \frac{\Delta^2}{\|\mathbf{A}\|_2^2}.$$

To complete the theorem, we then plug in $\Delta = \frac{\varepsilon \sigma_{k+1}}{3d}$, getting

$$g_{\min} \geq \frac{\delta}{Cn^2} \cdot \frac{\frac{\varepsilon^2 \sigma_{k+1}^3}{9d^2}}{\sigma_1^2} = \frac{\delta \varepsilon^2}{9Cd^2 \kappa_k^2}.$$

We then appeal to [Theorem 61](#) to get the final iteration complexity of $t = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{\varepsilon}} \log(\frac{d}{\varepsilon \delta})) = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{d \kappa_k}{\delta \varepsilon}))$. $\square$

Up to a logarithmic dependence on κ_k , [Theorem 76](#) exactly matches the gap-independent low-rank approximation for the block Krylov method [85]. The same approach can be generalized to give analogs to [Theorem 67](#), [Theorem 68](#), [Theorem 69](#), and [Theorem 70](#) with a dependence on κ_k instead of $g_{\min}$.

[Theorem 76](#) can be interpreted in several ways. In practice, it is unlikely that adding random noise is in fact needed to break small singular value gaps. Noise inherent in the input matrix or due to roundoff error will generally suffice to rule out the existence of tiny singular value gaps. Thus, the corollary can be thought of as a smoothed-analysis result [111, 106], showing that single vector Krylov methods

display gap-independent convergence even on input instances which are tiny random perturbations of potentially worst-case instances. Alternatively, when rigorous worst-case guarantees are required, we could actually run [Algorithm 6](#) on the true input matrix $\mathbf{A}$ with a random diagonal perturbation added. Since $\mathbf{D}$ is diagonal, the runtime of matrix-vector products with $\tilde{\mathbf{A}}$ will generally be dominated by the runtime of matrix-vector products with $\mathbf{A}$, so this is very efficient. We experimentally explore the convergence of this perturbed iteration on a PSD matrix in [Section 4.6.4](#).

4.5.2 Gap-Independent Bounds for Rectangular and Indefinite Inputs

We next observe that [Theorem 76](#) gives results for non-PSD matrices as well, at least for low-rank approximation with respect to the spectral norm. In this case, we can run a single vector Krylov method on a perturbation of PSD matrix $\mathbf{A}\mathbf{A}^\top$. This suffices because the spectral norm guarantee has the property that a near-optimal basis for approximating $\mathbf{A}\mathbf{A}^\top$ is also a near-optimal for $\mathbf{A}$:

Lemma 77. *Let $\mathbf{A} \in \mathbb{R}^{d \times n}$. Let $\mathbf{Q} \in \mathbb{R}^{d \times k}$ be a matrix with orthonormal columns such that $\|\mathbf{A}\mathbf{A}^\top - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\mathbf{A}^\top\|_2 \leq (1+\varepsilon) \|\mathbf{A}\mathbf{A}^\top - (\mathbf{A}\mathbf{A}^\top)_k\|_2$. Then, $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_2 \leq (1+\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_2$.*

Proof. Let $\mathbf{P} := \mathbf{I} - \mathbf{Q}\mathbf{Q}^\top$. Observe that $\|\mathbf{A}\mathbf{A}^\top - (\mathbf{A}\mathbf{A}^\top)_k\|_2 = \sigma_{k+1}(\mathbf{A})^2$, so we are given that $\|\mathbf{P}\mathbf{A}\mathbf{A}^\top\|_2 \leq (1+\varepsilon) \sigma_{k+1}(\mathbf{A})^2$. Next note that $\mathbf{P}$ is a projection matrix, which can only decrease spectral norms, so we have

$$\|\mathbf{P}\mathbf{A}\|_2^2 = \|(\mathbf{P}\mathbf{A})(\mathbf{P}\mathbf{A})^\top\|_2 = \|\mathbf{P}\mathbf{A}\mathbf{A}^\top\mathbf{P}\|_2 \leq \|\mathbf{P}\mathbf{A}\mathbf{A}^\top\|_2 \leq (1+\varepsilon) \sigma_{k+1}(\mathbf{A})^2.$$

Taking the square root of both sides, and noting that $\sqrt{1+\varepsilon} \leq 1+\varepsilon$, we conclude that $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_2 \leq (1+\varepsilon) \sigma_{k+1}(\mathbf{A}) = (1+\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_2$. $\square$

Combining [Theorem 77](#) with [Theorem 77](#) we obtain the following result:

Corollary 78. *For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $\kappa_k = \frac{\sigma_1}{\sigma_k}$ and $\Delta = \frac{\varepsilon \sigma_{k+1}^2}{3d}$. For any $\varepsilon, \delta \in (0, \frac{1}{2})$, let $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{A}^\top + \mathbf{D}$ where $\mathbf{D} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with entries drawn uniformly and i.i.d. from $[-\Delta, \Delta]$. Then, [Algorithm 6](#) run on $\tilde{\mathbf{A}}$ initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\sqrt{\varepsilon}} \log(\frac{d\kappa_k}{\delta\varepsilon}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_2 \leq (1+\varepsilon) \|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_2.$$

Note that it is not clear that [Theorem 77](#) extends beyond the spectral norm, e.g., to the Frobenius norm. Nevertheless, we suspect that a result comparable to [Theorem 78](#) should hold for all other error metrics considered in this paper.

4.6 Numerical Experiments

In this section we validate the core findings of our theoretical results with numerical experiments. We focus on four key findings. In [Section 4.6.2](#), we verify that the dependence of single vector Krylov on the sequential gap size $g_{\min}$ is in fact logarithmic, matching the theoretical bounds of [Sections 4.3](#) and [4.4](#). In [Section 4.6.3](#), we show that using a small block size $b > 1$ can ameliorate this gap dependence, by replacing $\log(1/g_{\min})$ with $\log(1/g_{\min,b})$ as shown in [Theorem 73](#). Relatedly, in [Section 4.6.4](#) we show that a small random perturbation of the input matrix can break up overlapping singular values and lead to much faster convergence of the single vector Krylov method, matching our theoretical findings from [Section 4.5](#). In [Section 4.6.5](#) we compare single vector and large block Krylov methods. We find that for a wide range of matrices, single vector Krylov methods significantly outperform larger block methods. However, for some very specific worst case instances, large block methods can perform better.

Finally, while our theoretical bounds ignore issues of numerical stability, in [Section 4.6.6](#), we observe empirically that small block methods tend to have significantly more issues with stability than large block methods. Exploring this issue further in future work would be very interesting.

4.6.1 Experimental Set Up

To control against stability issues in our primary experiments, we implement algorithms [Algorithm 6](#) and [Algorithm 7](#) using a full reorthogonalization strategy to keep the Krylov subspace close to orthogonal. At every iteration of the (block) Krylov iteration, we orthogonalize the most recently generated column (resp. block) of the Krylov subspace against all previous columns (resp. blocks) using the modified Gram-Schmidt process. At the next iteration, this column (resp. block) is multiplied by $\mathbf{A}$ to produce a new column (resp. block) of the Krylov subspace. See [Section 4.6.6](#) for more details or our code, which is available on GitHub⁵.

We also note that, like most standard implementations, including those based on

⁵<https://github.com/RaphaelArkadyMeyerNYU/SingleVectorKrylov>

the Lanczos recurrence, our code implements [Algorithm 6](#) run for t iterations using $t + 1$ matrix-vector products with $\mathbf{A}\mathbf{A}^\top$. To see why this is possible, note that the matrix $\mathbf{A}\mathbf{A}^\top\mathbf{Z}$ computed on Line 2 of the algorithm can be formed “on-the-fly” as we generate the Krylov subspace. Let $\mathbf{z}_i$ be the i^{th} column in $\mathbf{Z}$. At each iteration we already compute $\mathbf{A}\mathbf{A}^\top\mathbf{z}_i$ for column $\mathbf{z}_i$ to form column $\mathbf{z}_{i+1}$, so can just store this result to form $\mathbf{A}\mathbf{A}^\top\mathbf{Z}$. Similarly, implementing [Algorithm 7](#) requires $(t + 1)\ell$ matrix-vector products with $\mathbf{A}\mathbf{A}^\top$.

Throughout our experiments, we only report low-rank approximation error in terms of the Frobenius norm, as we found convergence in the spectral and Frobenius norms typically matched quite closely. In particular, letting $\mathbf{Q}$ be the output of [Algorithm 6](#) or [Algorithm 7](#), we report

$$\varepsilon_{\text{empirical}} := \frac{\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_F - \|\mathbf{A} - \mathbf{A}_k\|_F}{\|\mathbf{A} - \mathbf{A}_k\|_F}.$$

Our theory describes how $\varepsilon_{\text{empirical}}$ should change as a function of the number of iterations, the block size, and the singular value gaps of $\mathbf{A}$.

Lastly, we note that, since Krylov methods initialized with random Gaussian vectors are invariant to rotation, without loss of generality we test on diagonal input matrices for all synthetic data experiments. Each matrix’s diagonal entries correspond to its singular values.

4.6.2 Verifying Gap Dependence

We first empirically show that single vector Krylov has a logarithmic dependence on the minimum sequential gap size $g_{\min} = \min_{i \in [k-1]} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$, as predicted by the bounds of [Sections 4.3](#) and [4.4](#). We consider an exponentially decaying spectrum with parameter $\alpha = 1.1$ whose singular values are all nearly repeated, with gap sizes varying between $g_{\min} \in [10^{-10}, 1]$. That is, letting our vector of singular values be denoted $\boldsymbol{\sigma} = [\sigma_1, \dots, \sigma_d]$ so that $\mathbf{A} = \text{diag}(\boldsymbol{\sigma})$, and fixing $d = 1000$, we let

$$\boldsymbol{\sigma} = \left[1, \quad \frac{1}{1+g_{\min}}, \quad \alpha^{-1}, \quad \frac{\alpha^{-1}}{1+g_{\min}}, \quad \alpha^{-2}, \quad \frac{\alpha^{-2}}{1+g_{\min}}, \quad \dots \quad \alpha^{-499}, \quad \frac{\alpha^{-499}}{1+g_{\min}} \right]. \quad (4.12)$$

Since this matrix has a fast decay of singular values, we expect the performance of single vector Krylov to follow the spectral decay rate of [Theorem 67](#). That is, fixing the dimension d and failure probability δ , we should expect the number of iterations to scale as:

$$t \propto \frac{\ell}{\sqrt{g_{k \rightarrow \ell}}} \log \left(\frac{1}{g_{\min}} \right) + \frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log \left(\frac{1}{\varepsilon} \right).$$

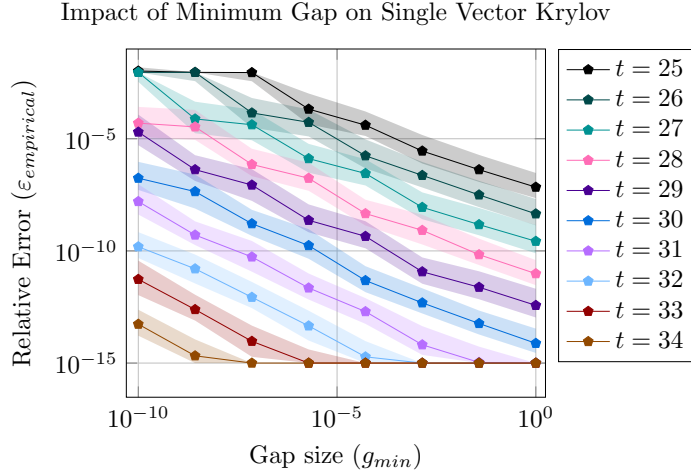


Figure 4.3: Low-rank approximation error vs. minimum gap size for diagonal $\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ with singular values σ as described in Equation (4.12). For 8 different gap sizes logarithmically spaced in $[10^{-10}, 1]$, we run single vector Krylov for $t = 25, 26, \dots, 34$ iterations with target rank $k = 10$. The median of $\varepsilon_{\text{empirical}}$ over 500 independent trials is plotted, with the 25th and 75th quartiles shaded in. When $\varepsilon_{\text{empirical}} < 10^{-15}$, we plot it at 10^{-15} so that the log/log plot does not degenerate. As expected given the theoretical bound of Theorem 67, we see a linear relationship between the log relative error and log gap size.

Rearranging this expression, we can equivalently expect

$$\begin{aligned} \log(\varepsilon) &= -t\sqrt{g_{k \rightarrow \ell}} - \ell \log(g_{\min}) \\ &= -C_t - \ell \log(g_{\min}) \end{aligned}$$

where $C_t := t\sqrt{g_{k \rightarrow \ell}}$ is independent of $g_{\min}$. So, if we plot $\varepsilon_{\text{empirical}}$ versus $g_{\min}$ on a log/log plot for a fixed value of t , we should see a line with negative slope. Further, since the vertical offset of these lines are $C_t \propto t$, we should expect that increasing t should shift these lines downwards and proportionally to t . We see this behavior exactly in Figure 4.3.

4.6.3 Verifying the Effect of Block Size on Gap Dependence

We next show that when $\mathbf{A}$ has very small singular value gaps (or even exactly overlapping singular values), the dependence on $\log(1/g_{\min})$ can be avoided by using a small constant block size b . This lets us instead depend on $\log(1/g_{\min,b})$, as in the analysis of Theorem 73.

We focus on when $\mathbf{A}$ has pairs, but not triplets, of exactly overlapping singular

values. In this case, block Krylov with block size $b = 2$ should perform well, since it should not suffer due to the overlapping singular values. Further, it should match or outperform larger block methods. To show this, we construct an exponentially decaying spectrum with parameter $\alpha = 1.005$ and whose top $k = 50$ singular values are each repeated, with sequential gap size $g_{\min} = 0$. Formally, we choose 1000 singular values as follows:

$$\begin{aligned}\sigma_A &= [1 \quad 1 \quad \alpha^{-1} \quad \alpha^{-1} \quad \alpha^{-2} \quad \alpha^{-2} \quad \dots \quad \alpha^{-25} \quad \alpha^{-25}] \\ \sigma_B &= [\alpha^{-26} \quad \alpha^{-27} \quad \alpha^{-28} \quad \dots \quad \alpha^{-975}] \\ \sigma &= [\sigma_A \quad \sigma_B].\end{aligned}\tag{4.13}$$

In theory, single vector Krylov should completely fail in this case, only capturing a $k/2$ -dimensional subspace of the span of the top k singular vectors. Due to finite precision roundoff, the method nevertheless converges. However, it is still significantly handicapped by the repeated singular values.

In [Figure 4.4](#) we plot the low-rank approximation error vs. number of matrix-vector products of single vector Krylov and block Krylov with block sizes 2, 3, and 50, for target rank $k = 50$. We show both y-linear and y-logarithmic plots to highlight the performance at early and later iterations. We see that block size 2 performs the best across the board, and that block size 3 is only mildly worse. Due to the repeated singular values, single vector Krylov performs worse, especially for the early iterations. It becomes competitive with block size 3 eventually. In contrast, the full block size k method converges much more slowly.

4.6.4 Verifying the Effect of Random Perturbations on Gap Dependence

Next, we show that adding a small amount of random noise to break up small singular value gaps can also make single vector Krylov converge more quickly, verifying the results of [Section 4.5](#). We use the same matrix as in [Section 4.6.3](#), with spectrum given in [Equation \(4.13\)](#). In [Figure 4.5](#), we show that adding noise to $\mathbf{A}$ the order of 10^{-6} leads to single vector Krylov converging nearly as quickly as the optimal $b = 2$ block Krylov method as seen in [Section 4.6.3](#). This noise does limit our eventual accuracy at convergence, which can be seen clearly in our logarithmic error plot. Changing the magnitude of the noise lets us interpolate between fast convergence and high accuracy.

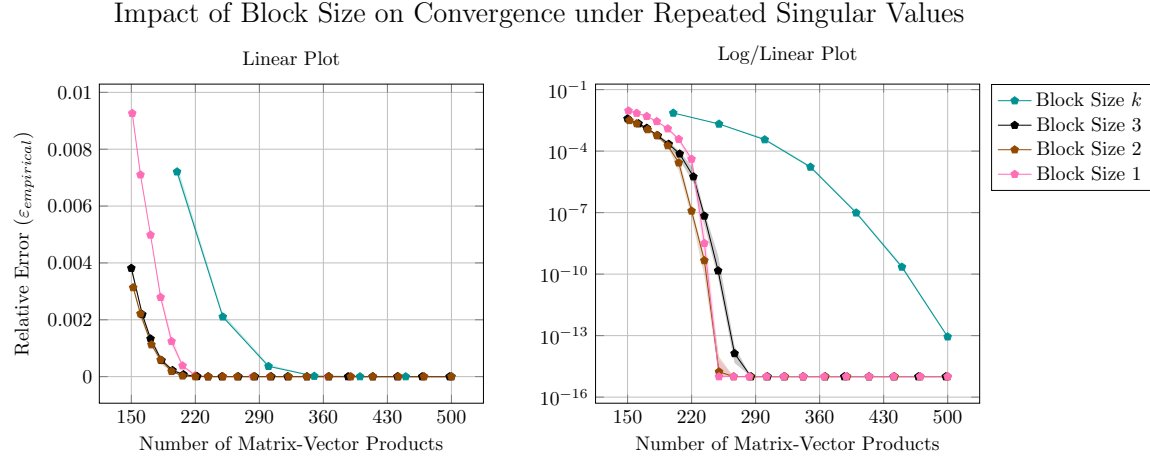


Figure 4.4: Low-rank approximation error vs. number of matrix-vector products for diagonal $\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ with repeated top singular values as in Equation (4.13). We run Krylov iteration with target rank $k = 50$ for block sizes $b = 1, 2, 3, 50$. The median of $\varepsilon_{\text{empirical}}$ over 10 independent trials is plotted, with the 25th and 75th quartiles shaded in. Note that these quartiles are very close together in this example. When $\varepsilon_{\text{empirical}} < 10^{-15}$, we plot it at 10^{-15} so that the plot does not degenerate. The plot on the left has a linear y-axis and highlights the performance for early iterations, while the plot on the right uses a logarithmic y-axis and highlights the performance for later iterations. Overall, we see that block size $b = 2$, which is just large enough to avoid the repeated pairs of singular values in $\mathbf{A}$'s spectrum, performs best. Single vector Krylov is handicapped by the repeated singular values, especially at early iterations. Large block Krylov converges at a significantly slower rate than the small block variants.

4.6.5 Effect of Block Size in Convergence

We next present a wider comparison of how the choice of block size for Krylov iteration effects convergence to a near-optimal low-rank approximation. We fix target rank $k = 50$ and compare block sizes 1, 2, 3, 50, and 54. Block size 1 corresponds to the single vector Krylov method. Block sizes 2 and 3 should be more resilient to a pairs or triplets of very close singular values, respectively. Block sizes k and $k + 4$ are recommended by prior theoretical work on Krylov Iteration for low-rank approximation [85, 115]. We consider eight input matrices. All synthetic inputs are 1000×1000 and diagonal.

1. **Exponential Decay:** $\sigma_i = \alpha^{-i}$ for $\alpha \in \{1.001, 1.01, 1.1\}$
2. **Polynomial Decay:** $\sigma_i = i^{-\beta}$ for $\beta \in \{0.1, 0.5, 1.5\}$
3. **Repeated Singular Values:** A matrix with each of its top k singular values repeated, as defined in Equation (4.13).

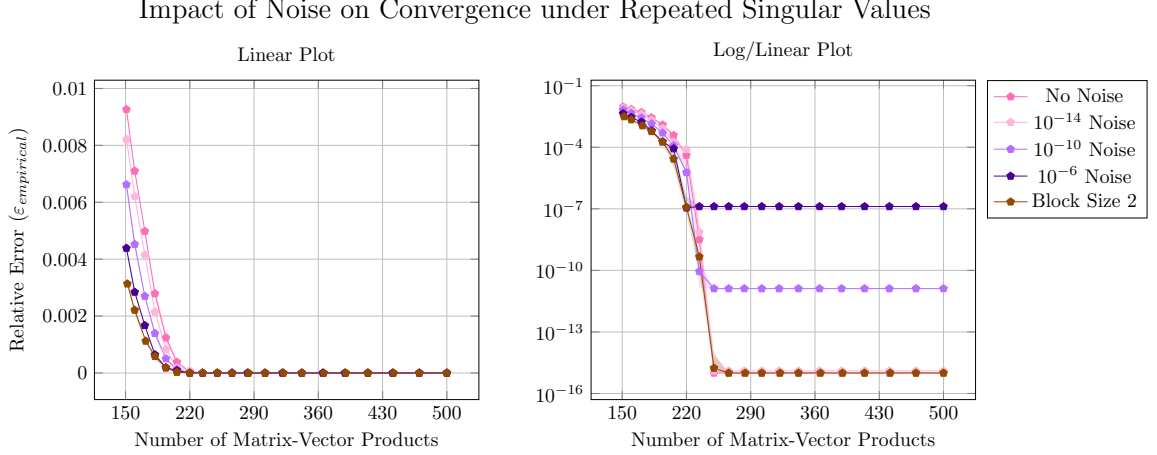


Figure 4.5: Low-rank approximation error vs. number of matrix-vector products for diagonal $\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ with repeated top singular values as described in Equation (4.13). We run the single vector Krylov method with target rank $k = 50$, with varying levels of noise added to $\mathbf{A}$ to separate its repeated singular values. In particular, following Theorem 76, we run the method on $\mathbf{A} + \mathbf{D}$ where $\mathbf{D} \in \mathbb{R}^{1000 \times 1000}$ is a random diagonal matrix with entries drawn uniformly on $[-\Delta, \Delta]$, for $\Delta \in \{10^{-6}, 10^{-10}, 10^{-14}, 0\}$. Low-rank approximation error is measured with respect to the original input $\mathbf{A}$. For comparison, we also show performance of block Krylov with block size $b = 2$ with no noise added. This was the optimal block size for this spectrum as seen in Figure 4.4. The median of $\varepsilon_{\text{empirical}}$ over 10 independent trials is plotted, with the 25th and 75th quartiles shaded in. Note that these quartiles are very close together in this example. When $\varepsilon_{\text{empirical}} < 10^{-15}$, we plot it at 10^{-15} so that the plot does not degenerate. The plot on the left has a linear y-axis and highlights the performance for early iterations, while the plot on the right uses a logarithmic y-axis and highlights the performance for later iterations. We can see that by adding a small random perturbation, we can improve the convergence of single vector Krylov to nearly match that of block Krylov with $b = 2$. Larger noise leads to faster convergence but also larger final error.

4. **Wishart Lower Bound:** $\sigma_i = \sqrt{1 - (\frac{i}{1000})^2}$. This is an approximation of the spectrum of $\mathbf{I} - \frac{1}{5d} \mathbf{G}^\top \mathbf{G}$ where $\mathbf{G}^{1000 \times 1000}$ has i.i.d $\mathcal{N}(0, 1)$ entries. This matrix is used as a lower bound instance for rank-1 low-rank approximation in [8].

5. **nd3k, appu, human_gene_2, exdata_1:** Various real-world matrices arbitrarily chosen from SuiteSparse [32].

We can see the results of these experiments in Figure 4.6. We see that for all except the repeated singular value and Wishart lower bound matrices, single vector Krylov dominates. For the repeated singular value matrix, as in Section 4.6.3, we see that block size 2 dominates again. For the Wishart lower bound matrix from [8], we see

that large block methods marginally (though consistently) outperform small block methods. This lower bound instance is designed to force Krylov methods to converge at a rate of $\frac{1}{\varepsilon^{1/3}}$, instead of at the spectral decay rate $\frac{1}{\sqrt{g_{k \rightarrow \ell}}}$. This seems to make the rate of convergence of single vector Krylov slower than block Krylov, since we pay a $\log(1/g_{\min})$ dependence, while only benefiting a small amount from separating the k and $\log(d/\varepsilon)$ dependence (see [Theorem 61](#)). In contrast, the other figures show matrices where the spectral decay rate controls convergence, where single vector Krylov still pays $\log(1/g_{\min})$ but seems to see performance gains from being able to simulate general block sizes and from separating the $\log(d/\varepsilon)$ dependence from the (simulated) block size dependence (see [Theorem 67](#)).

4.6.6 Block Size and Numerical Stability

It is well known that Krylov methods can suffer from numerical stability issues [46, 78]. In particular, the iterates $(\mathbf{A}\mathbf{A}^\top)^t \mathbf{g}$ approach the same vector (the top singular vector of $\mathbf{A}$) as t grows large. So, $\mathbf{K}$ becomes ill-conditioned. So far, we have focused on convergence guarantees and ignored numerical stability. As discussed, our implementations use orthogonalization to keep $\mathbf{K}$ well-conditioned at all iterations. That is, for single vector Krylov, at every iteration, we compute $(\mathbf{A}\mathbf{A}^\top)\mathbf{k}_{t-1}$ where $\mathbf{k}_{t-1}$ is the last column in the Krylov matrix $\mathbf{K}$. Then we project $(\mathbf{A}\mathbf{A}^\top)\mathbf{k}_{t-1}$ away from all of the previous columns $\mathbf{k}_1, \dots, \mathbf{k}_{t-1}$ via modified Gram-Schmidt, and store the resulting vector as $\mathbf{k}_t$. We do the same for block Krylov, where we compute $(\mathbf{A}\mathbf{A}^\top) \begin{bmatrix} \mathbf{k}_{t-b-1} & \dots & \mathbf{k}_{t-1} \end{bmatrix}$ and add the resulting b columns to $\mathbf{K}$ iteratively via modified Gram-Schmidt.

In practice, Krylov implementations typically spend less effort orthogonalizing at each step. For example, they are commonly implemented via the Lanczos method, where $\mathbf{k}_t$ is only projected away from $\mathbf{k}_{t-1}$ and $\mathbf{k}_{t-2}$. In infinite precision, this is equivalent to projecting away from all previous columns [46]. Similar ideas can be applied to block Krylov methods [99, 102]. While such methods are highly efficient, when using them, $\mathbf{K}$ can lose orthogonality. This can lead to slower convergence or necessitate modifications such as restarts or reorthogonalization [20, 91, 92].

Intuitively, comparing single vector or small block Krylov to large block Krylov with a fixed size Krylov subspace $\mathbf{K}$, we expect that single vector and small block Krylov will be more susceptible to conditioning issues, since they require more iterations to reach the same sized subspace. Thus, we should expect partial orthogonalization methods like Lanczos to lead to slower convergence for these methods as compared to large block methods. With full orthogonalization, we should instead

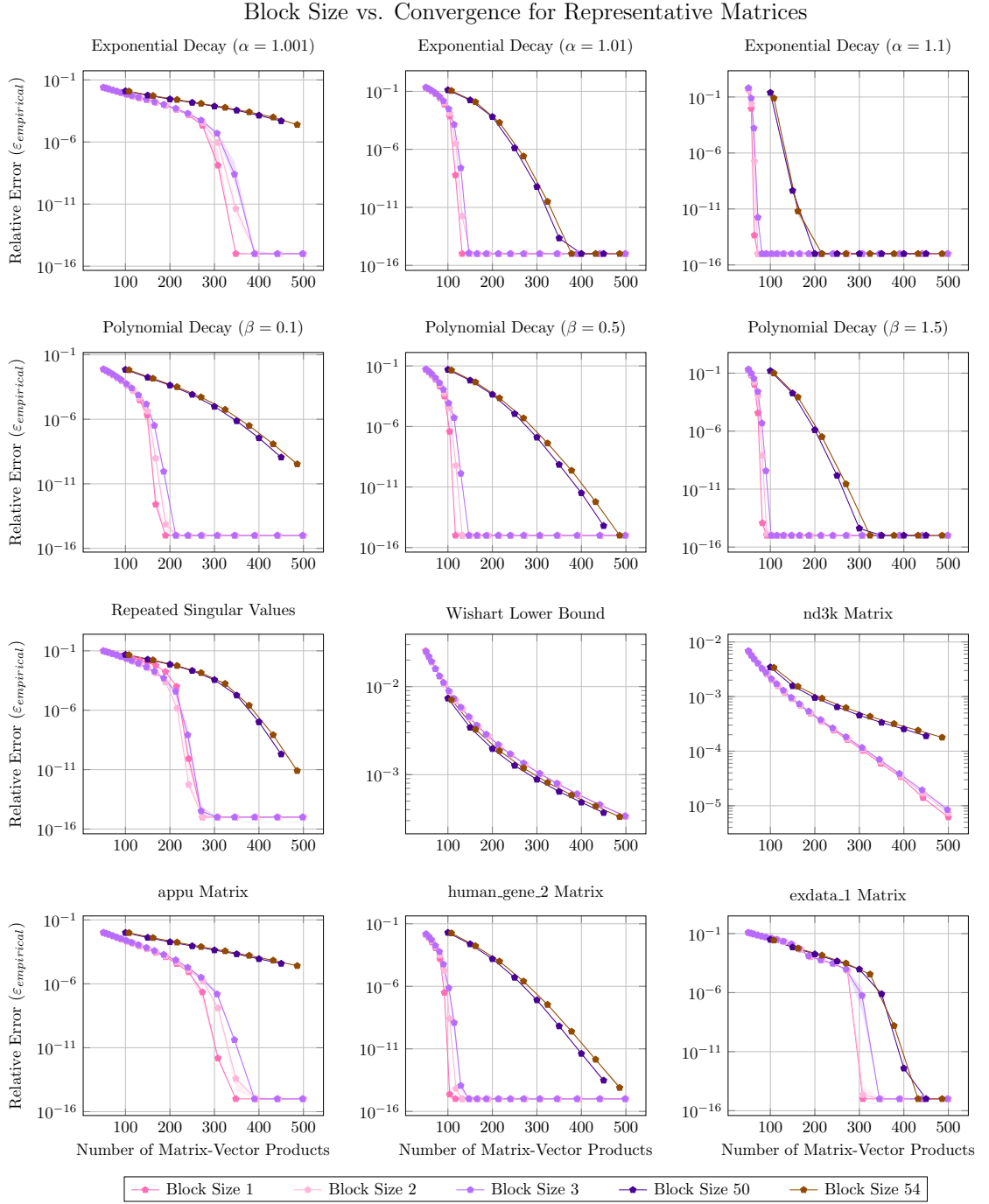


Figure 4.6: Low-rank approximation error vs. number of matrix-vector products for Krylov iteration with various block sizes on synthetic and real-world input matrices as described in Section 4.6.5. In all cases, the target rank is set to $k = 50$. The median of $\varepsilon_{\text{empirical}}$ is plotted over 10 independent trials, with the 25th and 75th quartiles shaded in. Note that these quartiles are very close together in most plots. When $\varepsilon_{\text{empirical}} < 10^{-15}$, we plot it at 10^{-15} so that the plot does not degenerate.

Impact of Block Size and Orthogonalization on Convergence

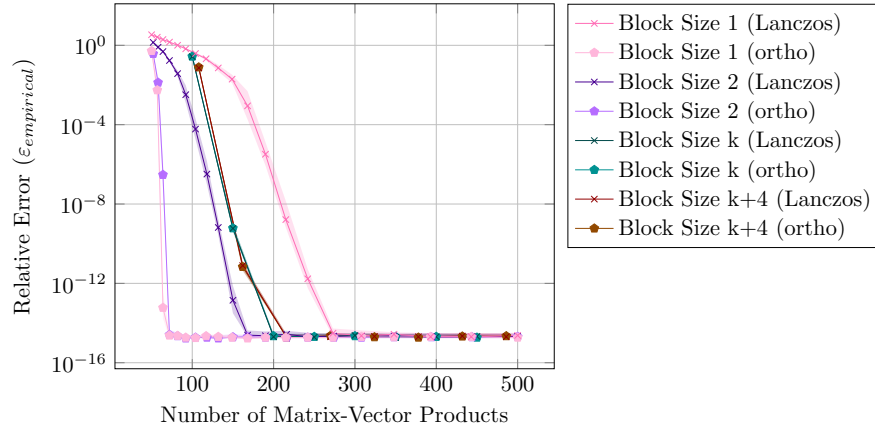


Figure 4.7: Low-rank approximation error vs. number of matrix-vector products for diagonal $\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ with singular values $\sigma_i = 1.1^{-i}$ and target rank $k = 50$. We run for block sizes $b \in \{1, 2, k, k + 4\}$ both with Lanczos and with full orthogonalization. The median of $\varepsilon_{\text{empirical}}$ over 100 independent trials is plotted, with the 25th and 75th quartiles shaded in. Notice that for this experiment, single vector Krylov converges the fastest with full orthogonalization and slowest with Lanczos, and that large block methods have no real gap between full orthogonalization and Lanczos. I.e., large block methods are much more stable even with partial orthogonalization.

expect to see small block methods dominate. We see this trend exactly in Figure 4.7. An interesting extension to our work would be to more closely study the stability of Krylov methods for low-rank approximation, and to develop a more clear theoretical understanding of the advantages of large block methods in this regard.

4.7 Reduction to the Positive Semidefinite Case

In our analysis, we can assume without loss of generality that the input $\mathbf{A}$ is a square PSD matrix. To see why, for any $\mathbf{C} \in \mathbb{R}^{d \times n}$, let $\mathbf{A} = (\mathbf{C}\mathbf{C}^\top)^{1/2}$. Observe that $\mathbf{A} \in \mathbb{R}^{d \times d}$ is PSD. Further, observe that since $\mathbf{A}^2 = \mathbf{A}\mathbf{A}^\top = \mathbf{C}\mathbf{C}^\top$, Algorithm 6 and Algorithm 7 yield identical outputs for $\mathbf{A}$ and $\mathbf{C}$.

Expanding the SVD $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, we can have $\mathbf{A} = (\mathbf{C}\mathbf{C}^\top)^{1/2} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top$. Thus, $\mathbf{A}$ and $\mathbf{C}$ have identical singular values and $\|\mathbf{A} - \mathbf{A}_k\|_\xi = \|\mathbf{C} - \mathbf{C}_k\|_\xi$ for any unitarily invariant norm $\|\cdot\|_\xi$ (including the spectral and Frobenius norms) and any k . Additionally, for any $\mathbf{Q} \in \mathbb{R}^{d \times k}$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top\mathbf{A}\|_\xi = \|\mathbf{U}\mathbf{\Sigma} - \mathbf{Q}\mathbf{Q}^\top\mathbf{U}\mathbf{\Sigma}\|_\xi = \|\mathbf{C} - \mathbf{Q}\mathbf{Q}^\top\mathbf{C}\|_\xi.$$

Thus, any bound on $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi$ in terms of $\|\mathbf{A} - \mathbf{A}_k\|_\xi$ holds identically for $\mathbf{C}$. Finally, for any $\mathbf{q} \in \mathbb{R}^n$, $\mathbf{q}^\top \mathbf{C}\mathbf{C}^\top \mathbf{q} = \mathbf{q}^\top \mathbf{A}^2 \mathbf{q}$. Thus, any bound on $\mathbf{q}^\top \mathbf{A}^2 \mathbf{q}$ in terms of $\sigma_i(\mathbf{A})$ holds identically for $\mathbf{C}$.

4.8 Frobenius Low-Rank Approximation with $\varepsilon^{-1/3}$ Dependence

In this section, we prove [Theorem 68](#). This analysis closely follows the intuition given in the introduction of [8]. We have the following:

Theorem 68 Restated. For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ where $\ell = \Theta(\frac{k}{\varepsilon^{1/3}})$. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 6](#) initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\varepsilon^{1/3}} \log(\frac{1}{g_{\min}}) + \frac{1}{\varepsilon^{1/3}} \log(\frac{d}{\delta\varepsilon}))$ iterations returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F.$$

Proof. We first recall that we have two different guarantees for [Algorithm 6](#). Fix some $\gamma > 0$. By [Theorem 61](#), if we run [Algorithm 6](#) for $t = O(\frac{k}{\sqrt{\gamma}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{\gamma}} \log(\frac{d}{\delta\gamma}))$ iterations, then with probability at least $1 - \delta$,

$$|\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i^2| \leq \gamma \sigma_{k+1}^2.$$

Fix some $\ell \geq k$, and let $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$. Then, by [Theorem 67](#), if we run [Algorithm 6](#) for $t = O(\frac{\ell}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{d}{\delta\eta}))$ iterations, then with probability at least $1 - \delta$ we again have

$$|\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i^2| \leq \eta \sigma_{k+1}^2.$$

We will show that we can get error ε in Frobenius norm by taking $\gamma = \varepsilon^{2/3}$ and $\eta = \frac{\varepsilon}{k}$. In particular, we run a case-analysis between either large-tailed or small-tailed spectra of $\mathbf{A}$.

Small Tailed Case: Suppose $\|\mathbf{A} - \mathbf{A}_k\|_F^2 \leq \frac{k}{\varepsilon^{1/3}} \sigma_{k+1}^2$. Then $\mathbf{A}$ must have a fast spectral decay. In particular, let $\ell = (1 + \frac{4}{\varepsilon^{1/3}})k = O(\frac{k}{\varepsilon^{1/3}})$. Then σ_ℓ is substantially smaller than σ_k :

$$\frac{k}{\varepsilon^{1/3}} \sigma_{k+1}^2 \geq \|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i=k+1}^d \sigma_i^2 \geq (\ell - k) \sigma_\ell^2 = \frac{4k}{\varepsilon^{1/3}} \sigma_\ell^2.$$

That is, $\sigma_\ell \leq \frac{\sigma_{k+1}}{2}$, so that $\sqrt{g_{k \rightarrow \ell}} = \sqrt{\frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}} \geq \frac{1}{\sqrt{2}}$, so the spectral-decay analysis of [Theorem 67](#) says that $t = O(\frac{k}{\varepsilon^{1/3}} \log(\frac{1}{g_{min}}) + \log(\frac{d}{\delta\eta}))$ iterations suffice to get the singular value guarantee $\|\mathbf{q}_i^\top \mathbf{A}\|_2^2 \in \sigma_i^2 \pm \eta \sigma_{k+1}^2$. Since $\mathbf{Q}\mathbf{Q}^\top = \sum_{i=1}^k \mathbf{q}_i \mathbf{q}_i^\top$ is a sum of orthogonal projection matrices,

$$\begin{aligned}
\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F^2 &= \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \|\mathbf{q}_i \mathbf{q}_i^\top \mathbf{A}\|_F^2 && \text{(Matrix Pythagoras)} \\
&= \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \|\mathbf{q}_i^\top \mathbf{A}\|_2^2 \\
&\leq \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \sigma_i^2 + \eta k \sigma_{k+1}^2 && \text{(Singular Value Guarantee)} \\
&= \|\mathbf{A} - \mathbf{A}_k\|_F^2 + \eta k \sigma_{k+1}^2 \\
&\leq \|\mathbf{A} - \mathbf{A}_k\|_F^2 + \eta k \|\mathbf{A} - \mathbf{A}_k\|_F^2 \\
&= (1 + \eta k) \|\mathbf{A} - \mathbf{A}_k\|_F^2.
\end{aligned}$$

So, taking $\eta = \frac{\varepsilon}{k}$ for a total iteration count of $t = O(\frac{k}{\varepsilon^{1/3}} \log(\frac{1}{g_{min}}) + \log(\frac{d}{\delta\varepsilon}))$ suffices in this case.

Large Tailed Case: Suppose $\|\mathbf{A} - \mathbf{A}_k\|_F^2 > \frac{k}{\varepsilon^{1/3}} \sigma_{k+1}^2$. Since the tail is large, even a low-accuracy singular value guarantee still ensures a good Frobenius norm guarantee. In particular, we take the gap-independent analysis of [Theorem 61](#) with $\gamma = \varepsilon^{2/3}$, so that $\|\mathbf{q}_i^\top \mathbf{A}\|_2^2 \in \sigma_i^2 \pm \varepsilon^{2/3} \sigma_{k+1}^2$, and we get

$$\begin{aligned}
\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F^2 &= \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \|\mathbf{q}_i \mathbf{q}_i^\top \mathbf{A}\|_F^2 && \text{(Matrix Pythagoras)} \\
&= \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \|\mathbf{q}_i^\top \mathbf{A}\|_2^2 \\
&\leq \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \sigma_i^2 + \varepsilon^{2/3} k \sigma_{k+1}^2 && \text{(Singular Value Guarantee)} \\
&= \|\mathbf{A} - \mathbf{A}_k\|_F^2 + \varepsilon^{2/3} k \sigma_{k+1}^2 \\
&\leq \|\mathbf{A} - \mathbf{A}_k\|_F^2 + \varepsilon \|\mathbf{A} - \mathbf{A}_k\|_F^2 && (\sigma_{k+1}^2 < \frac{\varepsilon^{1/3}}{k} \|\mathbf{A} - \mathbf{A}_k\|_F^2) \\
&= (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_F^2.
\end{aligned}$$

So, since $\gamma = \varepsilon^{2/3}$ here, we achieve error ε under Frobenius norm with $t = O(\frac{k}{\varepsilon^{1/3}} \log(\frac{1}{g_{min}}) + \frac{1}{\varepsilon^{1/3}} \log(\frac{d}{\delta\varepsilon}))$.

Putting it Together. So, in either case, running $t = \tilde{O}(\frac{k}{\varepsilon^{1/3}})$ iterations suffices to obtain a $(1 + \varepsilon)$ optimal low-rank approximation in the Frobenius norm. Further, the algorithm used in the two cases is identical, so (unlike [8]) we do not have to detect which case we are in and alter the algorithm accordingly. We simply run single vector Krylov. $\square$

4.9 Schatten Norm Low-Rank Approx. with $\varepsilon^{-1/3}$ Dependence

In this section, we prove [Theorem 69](#) by arguing that running [Algorithm 6](#) once effectively simulates running Algorithm 5.4 from [8]. We have the following:

Theorem 69 Restated. For $\mathbf{A} \in \mathbb{R}^{d \times n}$ and $p \geq 1$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ where $\ell = \Theta(\frac{k}{\varepsilon^{1/3} p^{1/3}})$. For any $\varepsilon, \delta \in (0, 1)$, let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ be the result of running [Algorithm 6](#) on $\mathbf{A}^\top$ initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for

$$t = O\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} \log\left(\frac{1}{g_{\min}}\right) + (\sqrt{p} + \frac{p^{1/6}}{\varepsilon^{1/3}}) \log\left(\frac{np}{\delta\varepsilon}\right)\right) = \tilde{O}\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} + \sqrt{p}\right)$$

iterations. Let $\mathbf{Z} \in \mathbb{R}^{d \times k}$ be an orthonormal basis for $\mathbf{A}\mathbf{Q}$. Then, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Z}\mathbf{Z}^\top \mathbf{A}\|_p \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_p,$$

where $\|\mathbf{A}\|_p := (\sum_{i=1}^d \sigma_i(\mathbf{A})^p)^{1/p}$ is the Schatten p -norm.

Proof. Note that [8] outputs orthonormal $\mathbf{Q} \in \mathbb{R}^{n \times k}$ with $\|\mathbf{A} - \mathbf{A}\mathbf{Q}\mathbf{Q}^\top\|$ bounded, rather than $\mathbf{Q} \in \mathbb{R}^{d \times k}$ with $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|$ bounded. However, we can translate their analysis to the later case simply by running their algorithms on $\mathbf{A}^\top$. Thus we consider this case going forward. The first two lines of Algorithm 5.4 in [8] run Block Krylov Iteration twice on $\mathbf{A}^\top$. First, they let $\mathbf{W}_1 \in \mathbb{R}^{n \times k}$ be the result of using block size $\ell_1 = k$ and running until the gap-independent rate gives a singular value guarantee (i.e. [Equation \(4.7\)](#)) with relative error at most $\gamma_1 = \frac{\varepsilon^{2/3}}{p^{1/3}}$. For single vector Krylov, by [Theorem 61](#), this takes

$$O\left(\frac{k}{\sqrt{\gamma_1}} \log\left(\frac{1}{g_{\min}}\right) + \frac{1}{\sqrt{\gamma_1}} \log\left(\frac{n}{\gamma_1 \delta}\right)\right) = O\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} \log\left(\frac{1}{g_{\min}}\right) + \frac{p^{1/6}}{\varepsilon^{1/3}} \log\left(\frac{np}{\varepsilon \delta}\right)\right)$$

iterations. Second, they let $\mathbf{W}_2 \in \mathbb{R}^{n \times k}$ be the result of running with block size $\ell_2 = O(\frac{k}{\varepsilon^{1/3} p^{1/3}})$ for enough iterations so that if $g_{k \rightarrow \ell_2} \geq \frac{1}{p}$, then block Krylov would

achieve error $\gamma_2 = \text{poly}(\frac{\varepsilon}{d})^6$. For single vector Krylov, by [Theorem 67](#), this takes

$$O\left(\frac{\ell_2}{\sqrt{g_{k \rightarrow \ell_2}}} \log\left(\frac{1}{g_{\min}}\right) + \frac{1}{\sqrt{g_{k \rightarrow \ell_2}}} \log\left(\frac{n}{\gamma_2 \delta}\right)\right) = O\left(\frac{kp^{1/6}}{\varepsilon^{1/3}} \log\left(\frac{1}{g_{\min}}\right) + \sqrt{p} \log\left(\frac{n}{\varepsilon \delta}\right)\right)$$

iterations. Note that [Algorithm 6](#) outputs a single matrix $\mathbf{Q}$ that achieves the guarantees needed by both $\mathbf{W}_1$ and $\mathbf{W}_2$.

Next, we consider the third and fourth lines of Algorithm 5.4 in [8]. The third line runs block Krylov on $\mathbf{A}$ directly to estimate several of its singular values. The fourth line uses those estimated singular values to determine if we should return an orthogonal basis for $\mathbf{W}_1^\top \mathbf{A}^\top$ or $\mathbf{W}_2^\top \mathbf{A}^\top$.⁷ Since we have $\mathbf{W}_1 = \mathbf{W}_2 = \mathbf{Q}$, we can ignore the tests in the third and fourth lines, and just always return a basis for $\mathbf{Q}^\top \mathbf{A}^\top$. So, overall, we compute a matrix $\mathbf{Z}$ with the exact same guarantees as the [8] by only using a one instance of single vector Krylov. $\square$

4.10 Eigenvalue Repulsion Corollaries

This appendix covers the proofs needed for [Theorem 76](#). First, we take a result of [84] and use it to prove a gap on the eigenvalues of symmetric matrices. Second, we show that a optimal projection matrix that achieves near-optimal low-rank approximation on the perturbation of $\mathbf{A}$ must also achieve near-optimal low-rank approximation on $\mathbf{A}$ itself.

4.10.1 Proof of [Theorem 74](#)

We first import a result of [84], originally studied in relation to the Wegner Estimate [128]. This result is also given as Equation (1.11) in [4]:

Imported Theorem 79. *Let $\mathbf{A} \in \mathbb{R}^{d \times d}$ be symmetric, and let $\mathbf{D} \in \mathbb{R}^{d \times d}$ be diagonal, with entries drawn i.i.d. from a distribution with pdf $p(\cdot)$. Then, for any interval $\mathcal{I} \subset \mathbb{R}$,*

$$\Pr[\mathbf{A} + \mathbf{D} \text{ has at least 2 eigenvalues in } \mathcal{I}] \leq C(\|p\|_\infty |\mathcal{I}| d)^2,$$

⁶They write $\gamma_2 = \varepsilon$ in the algorithm but in Equation (5.21) we can see they actually want this smaller error. Since gap-dependent rate depends on $\log(\frac{d}{\gamma_2})$, shrinking γ_2 from ε to $\text{poly}(\frac{\varepsilon}{d})$ does not change the asymptotic complexity.

⁷In a personal communication with the authors of [8], we confirmed there is a typo in the current arXiv version of the paper, where the algorithm says to return a matrix $\mathbf{Z}_1$. It should return an orthonormal basis for $\mathbf{W}_1^\top \mathbf{A}^\top$ instead.

for some universal constant C , where $|\mathcal{I}|$ is the length of $\mathcal{I}$, and where $\|p\|_\infty = \max_{t \in \mathbb{R}} p(t)$.

Theorem 74 Restated. Fix symmetric matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, $\delta \in (0, 1)$, and $\Delta \leq \|\mathbf{A}\|_2$. Let $\mathbf{D} \in \mathbb{R}^{d \times d}$ be a diagonal matrix whose entries are uniformly distributed in $[-\Delta, +\Delta]$. Then, letting $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{D}$ and letting C denote some universal constant, with probability at least $1 - \delta$,

$$\min_{i \in [d-1]} \frac{|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})|}{|\lambda_{i+1}(\tilde{\mathbf{A}})|} \geq \frac{\delta}{Cd^2} \cdot \frac{\Delta^2}{\|\mathbf{A}\|_2^2}.$$

Proof. Let $R := 2\|\mathbf{A}\|_2$. Since $\Delta < \|\mathbf{A}\|_2$, we know that $\|\tilde{\mathbf{A}}\|_2 \leq \|\mathbf{A}\|_2 + \|\mathbf{D}\|_2 \leq 2\|\mathbf{A}\|_2 = R$. Let $\gamma > 0$ be a number to be fixed later. Then define

$$\mathcal{I}_i := (-R + i\gamma) \pm \gamma = [(-R + \gamma i) - \gamma, (-R + \gamma i) + \gamma]$$

for $i = 1, \dots, m$, where $m = \frac{4\|\mathbf{A}\|_2}{\gamma} - 1$. These are intervals of width 2γ that overlap and cover the range $[-R, R]$. For instance, we have $\mathcal{I}_1 = [-R, -R + 2\gamma]$, $\mathcal{I}_2 = [-R + \gamma, -R + 2\gamma]$, and $\mathcal{I}_3 = [-R + 2\gamma, R + 4\gamma]$, so that $\mathcal{I}_2$ overlaps with $\mathcal{I}_1$ and $\mathcal{I}_3$. In particular, if $\tilde{\mathbf{A}}$ has two eigenvalues that are additively γ close, so that $|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})| \leq \gamma$, then we know that $\lambda_i(\tilde{\mathbf{A}})$ and $\lambda_{i+1}(\tilde{\mathbf{A}})$ both lie in some $\mathcal{I}_j$. Therefore, we can write

$$\begin{aligned} \Pr[\exists i : |\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})| < \gamma] &\leq \Pr[\text{at least two eigenvalues of } \tilde{\mathbf{A}} \text{ lie in some } \mathcal{I}_j] \\ &\leq \sum_{j=1}^m \Pr[\text{at least two eigenvalues of } \tilde{\mathbf{A}} \text{ lie in } \mathcal{I}_j] \\ &\leq Cm \left(\frac{1}{2\Delta} \cdot 2\gamma \cdot d \right)^2 \quad (\text{Theorem 79}) \\ &\leq \frac{4C\gamma d^2 \|\mathbf{A}\|_2}{\Delta^2} \quad (\text{using that } m \leq \frac{4\|\mathbf{A}\|_2}{\gamma}) \\ &= \delta, \end{aligned}$$

where the last line holds if we fix $\gamma = \frac{\Delta^2 \delta}{4Cd^2 \|\mathbf{A}\|_2}$. That is, with probability at least $1 - \delta$, we know that $|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})| \geq \frac{\Delta^2 \delta}{4Cd^2 \|\mathbf{A}\|_2}$ for all $i = 1, \dots, d-1$. Lastly, we take

$$\frac{|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})|}{|\lambda_{i+1}(\tilde{\mathbf{A}})|} \geq \frac{|\lambda_i(\tilde{\mathbf{A}}) - \lambda_{i+1}(\tilde{\mathbf{A}})|}{\|\tilde{\mathbf{A}}\|_2} \geq \frac{\frac{\Delta^2 \delta}{4Cd^2 \|\mathbf{A}\|_2}}{2\|\mathbf{A}\|_2} = \frac{\Delta^2 \delta}{8Cd^2 \|\mathbf{A}\|_2^2},$$

which completes the proof. $\square$

4.10.2 Proof of Theorem 75

We next show that a small enough perturbation of $\mathbf{A}$ suffices to give approximate SVD results for $\mathbf{A}$ itself.

Theorem 75 Restated. *Let $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{D}$ where $\|\mathbf{D}\|_2 \leq \frac{\varepsilon}{3d}\sigma_{k+1}(\mathbf{A})$ and $\varepsilon \in (0, 1)$. Fix any $\mathbf{Q} \in \mathbb{R}^{d \times k}$ with orthonormal columns $\mathbf{q}_1, \dots, \mathbf{q}_k$. Then, with probability at least $1 - \delta$,*

1. *If $|\mathbf{q}_i^\top \tilde{\mathbf{A}} \tilde{\mathbf{A}}^\top \mathbf{q}_i - \sigma_i(\tilde{\mathbf{A}})^2| \leq \varepsilon \sigma_{k+1}(\tilde{\mathbf{A}})^2$, then $|\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq 8\varepsilon \sigma_i(\mathbf{A})^2$.*
2. *If $\|\tilde{\mathbf{A}} - \mathbf{Q} \mathbf{Q}^\top \tilde{\mathbf{A}}\|_2 \leq (1+\varepsilon) \|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_2$, then $\|\mathbf{A} - \mathbf{Q} \mathbf{Q}^\top \mathbf{A}\|_2 \leq (1+2\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_2$.*
3. *If $\|\tilde{\mathbf{A}} - \mathbf{Q} \mathbf{Q}^\top \tilde{\mathbf{A}}\|_F \leq (1+\varepsilon) \|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_F$, then $\|\mathbf{A} - \mathbf{Q} \mathbf{Q}^\top \mathbf{A}\|_F \leq (1+4\varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_F$.*

Proof.

Singular Value Guarantee. First note that for any real a, b, c such that $|a - b| \leq c$ and $c < b$, we have $|a^2 - b^2| \leq 3bc$. This follows from expanding $(b-c)^2 < a^2 < (b+c)^2$ and applying the AMGM inequality. Then note that $|\sigma_i(\tilde{\mathbf{A}}) - \sigma_i(\mathbf{A})| \leq \|\tilde{\mathbf{A}} - \mathbf{A}\|_2 = \|\mathbf{D}\|_2$. We then find that for $i \leq k+1$,

$$|\sigma_i^2(\tilde{\mathbf{A}}) - \sigma_i^2(\mathbf{A})| \leq 3\sigma_i(\mathbf{A}) \|\mathbf{D}\|_2 \leq \varepsilon \sigma_i(\mathbf{A}) \sigma_{k+1}(\mathbf{A}) \leq \varepsilon \sigma_i(\mathbf{A})^2.$$

Similarly note that $|\|\tilde{\mathbf{A}} \mathbf{q}_i\|_2 - \|\mathbf{A} \mathbf{q}_i\|_2| \leq \|\mathbf{D}\|_2$ and $\|\tilde{\mathbf{A}} \mathbf{q}_i\|_2 \leq 2\sigma_i(\tilde{\mathbf{A}}) \leq 4\sigma_i(\mathbf{A})$, so we have

$$|\|\tilde{\mathbf{A}} \mathbf{q}_i\|_2^2 - \|\mathbf{A} \mathbf{q}_i\|_2^2| \leq 3 \|\tilde{\mathbf{A}} \mathbf{q}_i\|_2 \|\mathbf{D}\|_2 \leq 4\varepsilon \sigma_i(\mathbf{A})^2.$$

Which completes this part by triangle inequality:

$$\begin{aligned} |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| &\leq |\mathbf{q}_i^\top \tilde{\mathbf{A}} \tilde{\mathbf{A}}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| + 4\varepsilon \sigma_i(\mathbf{A})^2 \\ &\leq |\mathbf{q}_i^\top \tilde{\mathbf{A}} \tilde{\mathbf{A}}^\top \mathbf{q}_i - \sigma_i(\tilde{\mathbf{A}})^2| + 7\varepsilon \sigma_i(\mathbf{A})^2 \\ &\leq 8\varepsilon \sigma_i(\mathbf{A})^2. \end{aligned}$$

Spectral Norm Guarantee. Here, first note that $\|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_2 = \sigma_{k+1}(\tilde{\mathbf{A}}) \leq (1 + \varepsilon)\sigma_{k+1}(\mathbf{A})$ by the prior analysis on the singular value guarantee. Next, we use the fact that $\mathbf{I} - \mathbf{Q} \mathbf{Q}^\top$ is a projection to simplify

$$\|\tilde{\mathbf{A}} - \mathbf{Q} \mathbf{Q}^\top \tilde{\mathbf{A}}\|_2 = \|(\mathbf{I} - \mathbf{Q} \mathbf{Q}^\top)(\mathbf{A} + \mathbf{D})\|_2 \geq \|(\mathbf{I} - \mathbf{Q} \mathbf{Q}^\top)\mathbf{A}\|_2 - \|\mathbf{D}\|_2,$$

and since $\|\mathbf{D}\|_2 \leq \varepsilon \sigma_{k+1}(\mathbf{A})$, we have $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_2 \leq \|\tilde{\mathbf{A}} - \mathbf{Q}\mathbf{Q}^\top \tilde{\mathbf{A}}\|_2 + \|\mathbf{D}\|_2 \leq (1 + 2\varepsilon)\sigma_{k+1}(\mathbf{A})$, which completes this part of the lemma.

Frobenius Norm Guarantee. Here, first note that $\|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_F^2 \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_F^2$, since

$$\|\tilde{\mathbf{A}} - \tilde{\mathbf{A}}_k\|_F^2 = \sum_{i=k+1}^d \sigma_i^2(\tilde{\mathbf{A}}) \leq \sum_{i=k+1}^d (\sigma_i(\mathbf{A}) + \|\mathbf{D}\|_2)^2 \leq \|\mathbf{A} - \mathbf{A}_k\|_F^2 + \sum_{i=k+1}^d (\|\mathbf{D}\|_2^2 + 2\sigma_i(\mathbf{A}) \|\mathbf{D}\|_2),$$

where we can further upper bound

$$\sum_{k+1}^d (\|\mathbf{D}\|_2^2 + 2\sigma_i(\mathbf{A}) \|\mathbf{D}\|_2) \leq d\left(\frac{\varepsilon \sigma_{k+1}(\mathbf{A})}{3d}\right)^2 + 2d\sigma_{k+1}(\mathbf{A}) \frac{\varepsilon \sigma_{k+1}(\mathbf{A})}{3d} \leq \varepsilon \sigma_{k+1}^2(\mathbf{A}) \leq \varepsilon \|\mathbf{A} - \mathbf{A}_k\|_F^2.$$

Next, using the fact that $\mathbf{P} = \mathbf{I} - \mathbf{Q}\mathbf{Q}^\top$ is a projection matrix, we simplify

$$\|\tilde{\mathbf{A}} - \mathbf{Q}\mathbf{Q}^\top \tilde{\mathbf{A}}\|_F = \|\mathbf{P}(\mathbf{A} + \mathbf{D})\|_F \geq \|\mathbf{P}\mathbf{A}\|_F - \|\mathbf{P}\mathbf{D}\|_F,$$

and since $\|\mathbf{P}\mathbf{D}\|_F \leq \|\mathbf{P}\|_F \|\mathbf{D}\|_2 \leq \sqrt{d} \frac{\varepsilon}{3d} \sigma_{k+1}(\mathbf{A}) \leq \varepsilon \|\mathbf{A} - \mathbf{A}_k\|_F$, we have

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_F \leq \|\tilde{\mathbf{A}} - \mathbf{Q}\mathbf{Q}^\top \tilde{\mathbf{A}}\|_F + \|\mathbf{P}\mathbf{D}\|_F \leq ((1 + \varepsilon)^2 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_F,$$

which completes the proof since $((1 + \varepsilon)^2 + \varepsilon) \leq 1 + 4\varepsilon$ for $\varepsilon \in (0, 1)$. $\square$

4.11 Krylov Analysis with Small Blocks

This section proves [Theorem 73](#). We first define the starting matrix that we will simulate block Krylov iteration on, analogously to what we use in [Section 4.3](#):

$$\mathbf{S}_r := \begin{bmatrix} \mathbf{G} & \mathbf{A}^2\mathbf{G} & \mathbf{A}^4\mathbf{G} & \dots & \mathbf{A}^{2(r-1)}\mathbf{G} \end{bmatrix} \quad \mathbf{K} \stackrel{\text{span}}{=} \begin{bmatrix} \mathbf{S}_r & \mathbf{A}^2\mathbf{S}_r & \mathbf{A}^4\mathbf{S}_r & \dots & \mathbf{A}^{2q}\mathbf{S}_r \end{bmatrix}$$

where $\ell = rb$ is the simulated block size (so we assume the integer r has $rb > k$) and where $q = t - r + 1$ denotes the number of simulated block-Krylov iterations run. Our proof will set $r = k - b + 1$. Notably, this means that $\mathbf{S}_r$ can have more than k columns, which is not allowed by the definition of (k, L) -good in [Theorem 62](#). So, we first present a generalization of [Theorem 62](#) that also suffices for convergence under [Theorem 63](#) and [Theorem 66](#). In particular, it suffices for $\mathbf{S}_r$ to contain a $d \times k$ size (k, L) -good matrix within its span. The Krylov subspace generated by this matrix will be contained in the subspace generated by $\mathbf{S}_r$, and thus any guarantees that hold

for it apply to $\mathbf{S}_r$ as well. See [Section 4.12](#) for a formal argument.

Definition 80 ((k, L) -good Starting Matrix (Generalized)). *Let $\mathbf{A} \in \mathbb{R}^{d \times n}$ be a matrix with top k left singular vectors $\mathbf{U}_k \in \mathbb{R}^{d \times k}$. A matrix $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ is a (k, L) -good starting matrix for $\mathbf{A}$ if $\|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 \leq L$ for some orthonormal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ whose columns lie in $\text{span}(\mathbf{B})$.*

Observe that when the starting block has exactly $\ell = k$ columns, [Theorem 80](#) matches [Theorem 62](#) exactly. To prove the (k, L) -good guarantee for small block methods, we first note an equivalent formulation of the generalized (k, L) -good guarantee:

Lemma 81. $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ is (k, L) -good for $\mathbf{A} \in \mathbb{R}^{d \times n}$ if and only if there exists a matrix $\mathbf{M} \in \mathbb{R}^{\ell \times k}$ with $\mathbf{U}_k^\top \mathbf{B} \mathbf{M} = \mathbf{I}$ and $\|\mathbf{B} \mathbf{M}\|_2^2 \leq L$.

Proof. Suppose we are given such an $\mathbf{M}$. Since $\text{span}(\mathbf{B} \mathbf{M})$ is a k -dimensional subspace of $\text{span}(\mathbf{B})$, we can let $\mathbf{Q}$ be an orthonormal basis for $\mathbf{B} \mathbf{M}$. Then we have $\mathbf{B} \mathbf{M} = \mathbf{Q} \mathbf{X}$ for some invertible $\mathbf{X} \in \mathbb{R}^{k \times k}$. Since $\mathbf{U}_k^\top \mathbf{Q} \mathbf{X} = \mathbf{U}_k^\top \mathbf{B} \mathbf{M} = \mathbf{I}$, we have $\mathbf{X}^{-1} = \mathbf{U}_k^\top \mathbf{Q}$. And so, we have $\|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 = \|\mathbf{X}\|_2^2 = \|\mathbf{Q} \mathbf{X}\|_2^2 = \|\mathbf{B} \mathbf{M}\|_2^2 \leq L$.

In the other direction, we are given $\mathbf{Q}$ which spans a subspace of $\text{span}(\mathbf{B})$. So, for any invertible matrix $\mathbf{X} \in \mathbb{R}^{k \times k}$, we can write $\mathbf{B} \mathbf{M} = \mathbf{Q} \mathbf{X}$ for some $\mathbf{M} \in \mathbb{R}^{\ell \times k}$. If we take $\mathbf{X} = (\mathbf{U}_k^\top \mathbf{Q})^{-1}$, then we have $\mathbf{U}_k^\top \mathbf{B} \mathbf{M} = \mathbf{U}_k^\top \mathbf{Q} \mathbf{X} = \mathbf{I}$ and $\|\mathbf{B} \mathbf{M}\|_2^2 = \|\mathbf{Q} \mathbf{X}\|_2^2 = \|\mathbf{X}\|_2^2 = \|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 \leq L$. $\square$

With this foundation in place, we prove the following guarantee on $\mathbf{S}_r$:

Theorem 72 Restated. *Fix any PSD matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ with singular values $\sigma_1 \geq \dots \geq \sigma_d$, $k \in [d]$, and $b \in [k]$. Let $r = k - b + 1$. Let $\mathbf{G} \in \mathbb{R}^{d \times b}$ be a matrix i.i.d. $\mathcal{N}(0, 1)$ entries, and let $\mathbf{S}_r = \begin{bmatrix} \mathbf{G} & \mathbf{A}^2 \mathbf{G} & \mathbf{A}^4 \mathbf{G} & \dots & \mathbf{A}^{2(r-1)} \mathbf{G} \end{bmatrix}$. For any $\delta \in (0, 1)$, with probability at least $(1 - \delta)$, $\mathbf{S}_r$ is a (k, L) -good starting matrix for $\mathbf{A}$ for $L = \frac{cb^2 k^2 d \log(1/\delta)}{\delta^2 g_{\min, b}^{4(k-b)}}$. Here c is a fixed constant.*

Proof. We prove the generalized (k, L) -good bound by constructing a matrix $\mathbf{M} \in \mathbb{R}^{\ell \times k}$ with $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{M} = \mathbf{I}$ and then applying [Theorem 81](#). Consider first the matrix $\mathbf{U}^\top \mathbf{S}_r \mathbf{M}$. Writing the SVD $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{U}^\top$, we can define $\hat{\mathbf{G}} := \mathbf{U}^\top \mathbf{G} \in \mathbb{R}^{d \times b}$, which is distributed as a iid $\mathcal{N}(0, 1)$ matrix because of the rotational invariance of the Gaussian. We can then write $\mathbf{U}^\top \mathbf{A}^{2i} \mathbf{G} = \mathbf{U}^\top \mathbf{U} \mathbf{\Sigma}^{2i} \mathbf{U}^\top \mathbf{G} = \mathbf{\Sigma}^{2i} \hat{\mathbf{G}}$. Thus,

$$\mathbf{U}^\top \mathbf{S}_r = \begin{bmatrix} \hat{\mathbf{G}} & \mathbf{\Sigma}^2 \hat{\mathbf{G}} & \mathbf{\Sigma}^4 \hat{\mathbf{G}} & \dots & \mathbf{\Sigma}^{2(r-1)} \hat{\mathbf{G}} \end{bmatrix}.$$

If we let $\hat{\mathbf{g}}_i$ be the i^{th} column of $\hat{\mathbf{G}}$ and permute the columns of $\mathbf{U}^\top \mathbf{G}$, we can write $\mathbf{U}^\top \mathbf{S}_r$ as

$$\mathbf{U}^\top \mathbf{S}_r \mathbf{P} = [\hat{\mathbf{g}}_1 \ \Sigma^2 \hat{\mathbf{g}}_1 \ \dots \ \Sigma^{2(r-1)} \hat{\mathbf{g}}_1 \mid \hat{\mathbf{g}}_2 \ \Sigma^2 \hat{\mathbf{g}}_2 \ \dots \ \Sigma^{2(r-1)} \hat{\mathbf{g}}_2 \mid \dots \mid \hat{\mathbf{g}}_b \ \Sigma^2 \hat{\mathbf{g}}_b \ \dots \ \Sigma^{2(r-1)} \hat{\mathbf{g}}_b].$$

where $\mathbf{P}$ is the permutation matrix that reorders the columns as such. So, for some vector $\mathbf{m}_i$, we can decompose $\mathbf{P}^\top \mathbf{m}_i = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_b \end{bmatrix}$ for $\mathbf{c}_j \in \mathbb{R}^r$, and let $p_{i,j}(t)$ be the degree $r-1$ polynomial with coefficients $\mathbf{c}_j$. Then, we get

$$\mathbf{U}^\top \mathbf{S}_r \mathbf{m}_i = \mathbf{U}^\top \mathbf{S}_r \mathbf{P} \mathbf{P}^\top \mathbf{m}_i = p_{i,1}(\Sigma^2) \hat{\mathbf{g}}_1 + p_{i,2}(\Sigma^2) \hat{\mathbf{g}}_2 + \dots + p_{i,b}(\Sigma^2) \hat{\mathbf{g}}_b. \quad (4.14)$$

And so, the matrix $\mathbf{U}^\top \mathbf{S}_r \mathbf{M}$ is the concatenation of k such vectors $\mathbf{U}^\top \mathbf{S}_r \mathbf{m}_1, \dots, \mathbf{U}^\top \mathbf{S}_r \mathbf{m}_k$.

Observe that $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{M}$ is just the top k rows of $\mathbf{U}^\top \mathbf{S}_r \mathbf{M}$. So to apply [Theorem 81](#), we need to find polynomials $p_{1,1}, \dots, p_{k,b}$ which make the top k rows of $\mathbf{U}^\top \mathbf{S}_r \mathbf{M}$ into an identity matrix. I.e., we need to show $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i = \mathbf{e}_i$ for $i = 1, \dots, k$. We do this by first designing the degree $k-b$ filter polynomials $f_1, \dots, f_k$ by

$$f_i(\sigma_j^2) = \begin{cases} 1 & j = i \\ 0 & j \notin \mathcal{N}_i \cup \{i\} \end{cases}$$

That is, f_i is 1 at σ_i^2 and is 0 on the squares of the $k-b+1$ singular values furthest from σ_i . The squares are there because the Krylov subspace uses polynomial of Σ^2 . We then take

$$p_{i,j}(t) := f_i(t) z_{i,j}$$

for some values of $z_{i,j}$ which will be specified later. Note these polynomials are all degree $r-1 = k-b$, which is why we constrain $r = k-b+1$. If we plug this into [Equation \(4.14\)](#), we get

$$\begin{aligned} \mathbf{U}^\top \mathbf{S}_r \mathbf{m}_i &= p_{i,1}(\Sigma^2) \hat{\mathbf{g}}_1 + p_{i,2}(\Sigma^2) \hat{\mathbf{g}}_2 + \dots + p_{i,b}(\Sigma^2) \hat{\mathbf{g}}_b \\ &= f_i(\Sigma^2) (\hat{\mathbf{g}}_1 z_{i,1} + \hat{\mathbf{g}}_2 z_{i,2} + \dots + \hat{\mathbf{g}}_b z_{i,b}) \\ &= f_i(\Sigma^2) \hat{\mathbf{G}} \mathbf{z}_i, \end{aligned}$$

where $\mathbf{z}_i = [z_{i,1} \ \dots \ z_{i,b}]^\top$. So, we need to find a choice of $\mathbf{z}_i \in \mathbb{R}^b$ such that $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i = \mathbf{e}_i$. Letting $\hat{\mathbf{G}}_k \in \mathbb{R}^{k \times b}$ be the top k rows of $\hat{\mathbf{G}}$, we write

$$\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i = f_i(\Sigma_k^2) \hat{\mathbf{G}}_k \mathbf{z}_i.$$

Recalling that f_i is a filter polynomial, we know that $[\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i]_j = 0$ for $j \notin \mathcal{N}_i \cup \{i\}$. So, we just need to pick $\mathbf{z}_i \in \mathbb{R}^b$ such that $[\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i]_j = 0$ for $j \in \mathcal{N}_i$ and $[\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i]_i = 1$. That is, we need to pick $\mathbf{z}_i$ such that the product $\hat{\mathbf{G}}_k \mathbf{z}_i$ is zero for the $b - 1$ rows in $\mathcal{N}_i$ and is 1 for row i . This is just a linear system, so we pick the unique $\mathbf{z}_i = [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i$, where $[\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}$ just selects the b rows indexed by $\mathcal{N}_i \cup \{i\}$.

Once we have these $\mathbf{z}_i$ vectors, we get $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{m}_i = f_i(\Sigma_k^2) \hat{\mathbf{G}}_k \mathbf{z}_i = \mathbf{e}_i$, and so $\mathbf{U}_k^\top \mathbf{S}_r \mathbf{M} = \mathbf{I}$, as required by [Theorem 81](#). Now we just have to bound $\|\mathbf{S}_r \mathbf{M}\|_2^2$. We do this by bounding $\|\mathbf{S}_r \mathbf{M}\|_2^2 = \|\mathbf{U}^\top \mathbf{S}_r \mathbf{M}\|_2^2 \leq \|\mathbf{U}^\top \mathbf{S}_r \mathbf{M}\|_F^2 \leq \sum_{i=1}^k \|\mathbf{U}^\top \mathbf{S}_r \mathbf{m}_i\|_2^2$.

Let $\Sigma_{-k} \in \mathbb{R}^{(d-k) \times (d-k)}$ be the bottom $d - k$ singular values of $\mathbf{A}$, and let $\hat{\mathbf{G}}_{-k} \in \mathbb{R}^{(d-k) \times b}$ bottom $d - k$ rows of $\hat{\mathbf{G}}$. Then, we can decompose

$$\mathbf{U}^\top \mathbf{S}_r \mathbf{m}_i = f_i(\Sigma^2) \hat{\mathbf{G}} \mathbf{z}_i = \begin{bmatrix} f_i(\Sigma_k^2) \hat{\mathbf{G}}_k \mathbf{z}_i \\ f_i(\Sigma_{-k}^2) \hat{\mathbf{G}}_{-k} \mathbf{z}_i \end{bmatrix} = \begin{bmatrix} \mathbf{e}_i \\ f_i(\Sigma_{-k}^2) \hat{\mathbf{G}}_{-k} [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i \end{bmatrix}.$$

We now bound the lower elements. First, we look at $\hat{\mathbf{G}}_{-k} [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i$, which is the product of a $(d - k) \times b$ Gaussian matrix and inverse of an independent $b \times b$ Gaussian matrix. Since these matrices are independent, we can directly bound

$$\|\hat{\mathbf{G}}_{-k} [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i\|_2 \leq \|\hat{\mathbf{G}}_{-k}\|_2 \cdot \frac{1}{\sigma_{\min}([\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}})} \cdot \|\mathbf{e}\|_2 \leq O\left(\frac{b\sqrt{kd}}{\delta} \sqrt{\log\left(\frac{1}{\delta}\right)}\right), \quad (4.15)$$

where we use the bound $\|\mathbf{G}_{-k}\|_2 \leq O(\sqrt{(d - k)b \ln(\frac{1}{\delta})})$ from Equation (2.3) of [101] and the bound $\sigma_{\min}([\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}) \geq \Omega(\frac{\delta}{k\sqrt{b}})$. This latter bound comes from a union-bound argument using the fact that any square Gaussian $\tilde{\mathbf{G}} \in \mathbb{R}^{b \times b}$ has $\Pr[\sigma_{\min}(\tilde{\mathbf{G}}) \geq \Omega(\frac{\delta}{\sqrt{b}})] \geq 1 - \delta$ [58], and that $[\hat{\mathbf{G}}_k]_{\mathcal{N}_1 \cup \{1\}}, \dots, [\hat{\mathbf{G}}_k]_{\mathcal{N}_k \cup \{k\}}$ are all square Gaussian matrices. Next, we bound $f_i(\Sigma_{-k}^2)$ by writing f_i as a Lagrange interpolating polynomial:

$$f_i(t) = \prod_{j \in [k] \setminus \mathcal{N}_i, j \neq i} \frac{t - \sigma_j^2}{\sigma_i^2 - \sigma_j^2}.$$

We can then use our gap assumption to bound f_i on all singular values below σ_k . That is, for $0 \leq t \leq \sigma_k$ we have

$$|f_i(t)| \leq \prod_{j \in [k] \setminus \mathcal{N}_i, j \neq i} \left| \frac{\sigma_j^2}{\sigma_i^2 - \sigma_j^2} \right| \leq \prod_{j \in [k] \setminus \mathcal{N}_i, j \neq i} \left| \frac{\sigma_j}{\sigma_i - \sigma_j} \right|^2 \leq \frac{1}{g_{\min, b}^{2(k-b)}}.$$

Putting this altogether, we then find that

$$\begin{aligned} \|f_i(\Sigma_{-k}^2) \hat{\mathbf{G}}_{-k} [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i\|_2^2 &\leq \|f_i(\Sigma_{-k}^2)\|_2^2 \|\hat{\mathbf{G}}_{-k} [\hat{\mathbf{G}}_k]_{\mathcal{N}_i \cup \{i\}}^{-1} \mathbf{e}_i\|_2^2 \\ &\leq O\left(\frac{b^2 k d}{g_{\min, b}^{4(k-b)} \delta^2} \log\left(\frac{1}{\delta}\right)\right). \end{aligned}$$

And therefore

$$\|\mathbf{S}_r \mathbf{M}\|_2^2 \leq \|\mathbf{U}^\top \mathbf{S}_r \mathbf{M}\|_F^2 \leq O\left(1 + \frac{b^2 k^2 d}{g_{\min, b}^{4(k-b)} \delta^2} \log\left(\frac{1}{\delta}\right)\right),$$

which completes the proof by [Theorem 81](#). □

4.12 Convergence Results from [Musco Musco '15]

In this section, we show how [Theorem 63](#) and [Theorem 66](#) follow from [85]. We prove the result using a generalization of a (k, L) -good starting block ([Theorem 62](#)), which is useful in the small block analysis of [Section 4.11](#):

Theorem 80 Restated ((k, L) -good Starting Matrix (Generalized)). *Let $\mathbf{A} \in \mathbb{R}^{d \times n}$ be a matrix with top k left singular vectors $\mathbf{U}_k \in \mathbb{R}^{d \times k}$. A matrix $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ is a (k, L) -good starting matrix for $\mathbf{A}$ if $\|(\mathbf{U}_k^\top \mathbf{Q})^{-1}\|_2^2 \leq L$ for some orthonormal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ that lies in $\text{span}(\mathbf{B})$.*

Note that if $\mathbf{B}$ has exactly k columns then this generalized definition exactly matches [Theorem 62](#). This is the case for the analysis of single vector Krylov in [Section 4.3](#), where we take $\mathbf{B} = \mathbf{S}_k$ with exactly k columns and show that $\mathbf{S}_k$ is (k, L) -good. In [Section 4.11](#), we show an equivalent formulation which is easier to use in our analysis:

Theorem 81 Restated. *$\mathbf{B} \in \mathbb{R}^{d \times \ell}$ is (k, L) -good for $\mathbf{A}$ if and only if there exists a matrix $\mathbf{M} \in \mathbb{R}^{\ell \times k}$ with $\mathbf{U}_k^\top \mathbf{B} \mathbf{M} = \mathbf{I}$ and $\|\mathbf{B} \mathbf{M}\|_2^2 \leq L$.*

Given this definition, we will show that [Theorem 63](#) and [Theorem 66](#) follow from [85]. Before diving in, we first state a guarantee on the polynomials used in [85]. They use polynomials of the form $p(x) := \frac{(1+\gamma)\alpha}{T_q(1+\gamma)} T_q\left(\frac{x}{\alpha}\right)$ for some $\alpha > 0$, $\gamma \in (0, 1)$, and $q \in \mathbb{N}$, and where $T_q(x)$ is the degree q Chebyshev polynomial of the first kind. We first show that such polynomials are monotonic for large enough x :

Lemma 82. Let $p(x) = \frac{(1+\gamma)\alpha}{T_q(1+\gamma)} T_q(\frac{x}{\alpha})$ where $\alpha > 0$ and $\gamma \in (0, 1)$. Then, $\max_{x \in [0, \alpha]} p(x) = p(\alpha)$ and $p(x)$ is monotonically increasing on (α, ∞) .

Proof. The result follows from two well-known properties of T_q : that T_q is monotonically increasing on $(1, \infty)$ and that $\max_{t \in [0, 1]} T_q(t) = T_q(1) = 1$. Since $\gamma > 0$, we know that $T_q(1 + \gamma) > T_q(1) = 1$, so we have $\frac{(1+\gamma)\alpha}{T_q(1+\gamma)} > 0$. Therefore, we get that $p(x)$ is monotonically increasing on (α, ∞) , and that $\max_{x \in [0, \alpha]} p(x) = p(\alpha)$. $\square$

In order to relate the generalized (k, L) -good definition to the convergence analysis of [85], we rely on a slight generalization of a theorem used in the appendix of [85]:

Imported Lemma 83 (Lemma 48 of [132]). Let $\mathbf{P} = \mathbf{P}\mathbf{U}\mathbf{U}^\top + \mathbf{E} \in \mathbb{R}^{d \times d}$ be a low-rank factorization of $\mathbf{A} \in \mathbb{R}^{d \times d}$, with $\mathbf{U} \in \mathbb{R}^{d \times k}$ and $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_k$. Let $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ ($\ell \geq k$) be any matrix with $\text{rank}(\mathbf{U}^\top \mathbf{B}) = \text{rank}(\mathbf{U}) = k$. Let $\mathbf{M} \in \mathbb{R}^{\ell \times k}$ with $\mathbf{U}^\top \mathbf{B} \mathbf{M} = \mathbf{I}_k$. Let $\mathbf{C} = \mathbf{A} \mathbf{S} \in \mathbb{R}^{d \times \ell}$. Then,

$$\|\mathbf{P} - \Pi_{\mathbf{C}, k}(\mathbf{P})\|_F^2 \leq \|\mathbf{E}\|_F^2 + \|\mathbf{E} \mathbf{B} \mathbf{M}\|_F^2.$$

Here, $\Pi_{\mathbf{C}, k}(\mathbf{P}) = \mathbf{Y}_{\text{opt}} \mathbf{Y}_{\text{opt}}^\top \mathbf{A} \in \mathbb{R}^{d \times d}$ is the best rank k approximation to $\mathbf{A}$ in the column space of $\mathbf{C}$. So, $\mathbf{Y}_{\text{opt}} \in \mathbb{R}^{d \times k}$ is an orthogonal matrix that lies in the column span of $\mathbf{C}$.

Lemma 48 from [132] is stated with $\mathbf{M} = (\mathbf{U}^\top \mathbf{B})^+$, but the proof only requires that $\mathbf{U}^\top \mathbf{B} \mathbf{M} = \mathbf{I}$. We now prove the imported theorems:

Theorem 63 Restated (Theorem 1 of [85]). Let $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ be any (k, L) -good starting matrix (*Theorem 80*) matrix for $\mathbf{A}$. If we run Block Krylov iteration (*Algorithm 7*) for $q = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, then the output $\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies

$$\|\mathbf{A} - \mathbf{Q} \mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Proof. To recover this guarantee from [85], we recover Properties 1-3 of their Lemma 9. We start by recovering Property 1. We let $p_1(x) = \frac{(1+\gamma)\alpha}{T_q(1+\gamma)} T_q(\frac{x}{\alpha})$ where $\alpha = \sigma_{k+1}(\mathbf{A})$, $\gamma = \frac{\varepsilon}{2}$, and $q = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{dL}{\varepsilon}))$. Let $\mathbf{M} \in \mathbb{R}^{\ell \times k}$ be the matrix guaranteed to exist by *Theorem 81*. We then instantiate *Theorem 83* with $\mathbf{P} = p_1(\mathbf{A})$, $\mathbf{B} = \mathbf{B}$, $\mathbf{M} = \mathbf{M}$, and $\mathbf{U} = \mathbf{U}_k$ being the top k singular vectors of $\mathbf{A}$. By *Theorem 82*, since $\alpha = \sigma_{k+1}(\mathbf{A})$, we know that the top k singular vectors of $\mathbf{A}$ are also the top k singular vectors of $p_1(\mathbf{A})$, and therefore that $\mathbf{P} \mathbf{U} \mathbf{U}^\top = p_1(\mathbf{A})_k$. We then get that

some orthonormal $\mathbf{Y}_1 \in \mathbb{R}^{d \times k}$ in the span of $p(\mathbf{A})\mathbf{B}$ has

$$\begin{aligned}
\|p_1(\mathbf{A}) - \mathbf{Y}_1 \mathbf{Y}_1^\top p_1(\mathbf{A})\|_F^2 &\leq \|p_1(\mathbf{A}) - p_1(\mathbf{A})_k\|_F^2 + \|(p_1(\mathbf{A}) - p_1(\mathbf{A})_k)\mathbf{B}\mathbf{M}\|_F^2 \\
&\leq \|p_1(\mathbf{A}) - p_1(\mathbf{A})_k\|_F^2 + \|p_1(\mathbf{A}) - p_1(\mathbf{A})_k\|_F^2 \|\mathbf{B}\mathbf{M}\|_2^2 \\
&\leq (L+1) \|p_1(\mathbf{A}) - p_1(\mathbf{A})_k\|_F^2 \\
&= (L+1) \sum_{i=k+1}^d p(\sigma_i(\mathbf{A}))^2 \\
&\leq (L+1)d \cdot \frac{4\sigma_{k+1}(\mathbf{A})^2}{2^{2q\sqrt{\gamma}}} \quad (\text{Lemma 5 of [85]}) \\
&\leq \frac{\varepsilon}{2} \sigma_{k+1}(\mathbf{A})^2, \tag{4.16}
\end{aligned}$$

where the last line uses $q \geq \frac{1}{2\sqrt{\gamma}} \log_2(\frac{8(L+1)d}{\varepsilon}) = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{Ld}{\varepsilon}))$. The inequality $\|p_1(\mathbf{A}) - \mathbf{Y}_1 \mathbf{Y}_1^\top p_1(\mathbf{A})\| \leq \frac{\varepsilon}{2} \sigma_{k+1}(\mathbf{A})^2$ is exactly Equation (5) on page 11 of [85]. Once Equation (5) is achieved, the rest of the proof of Property 1 then follows without any alteration.

We next move onto proving properties 2 and 3. The proofs of properties 2 and 3 in [85] both involve defining the matrix $\mathbf{A}_{outer}$. If $\mathbf{A} = \mathbf{U}\Sigma\mathbf{U}^\top$ is the SVD of $\mathbf{A}$, then $\mathbf{A}_{outer} := \mathbf{U}\Sigma_{outer}\mathbf{U}^\top$ where Σ_{outer} contains all the singular values of $\mathbf{A}$ with either $\sigma_i(\mathbf{A}) \geq \sigma_k(\mathbf{A})$ or $\sigma_i(\mathbf{A}) < \frac{1}{1+\varepsilon/2} \sigma_k(\mathbf{A})$. All the other singular values are set to equal zero. Crucially, the top k singular vectors of $\mathbf{A}$ are still the top k singular vectors of $\mathbf{A}_{outer}$.

We let $p_2(x) = \frac{(1+\gamma)\alpha}{T_q(1+\gamma)} T_q(\frac{x}{\alpha})$ where $\alpha = \frac{1}{1+\varepsilon/2} \sigma_{k+1}(\mathbf{A})$, $\gamma = \frac{\varepsilon}{2}$, and $q = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{dL}{\varepsilon}))$. We next instantiate [Theorem 83](#) with $\mathbf{P} = p_2(\mathbf{A}_{outer})$, $\mathbf{B} = \mathbf{B}$, $\mathbf{M} = \mathbf{M}$, and $\mathbf{U} = \mathbf{U}_k$. By [Theorem 82](#), since $\alpha \leq \sigma_k(\mathbf{A}_{outer})$, we know that the top k singular vectors of $\mathbf{A}_{outer}$ are also the top k singular vectors of $p_2(\mathbf{A}_{outer})$, and therefore that $\mathbf{P}\mathbf{U}\mathbf{U}^\top = p_2(\mathbf{A}_{outer})_k$. We then let $\mathbf{Y}_{outer} \in \mathbb{R}^{d \times k}$ be the orthogonal basis that constructs $\Pi_{\mathbf{C},k}(\mathbf{A})$, so that

$$\begin{aligned}
\|p_2(\mathbf{A}_{outer}) - \mathbf{Y}_{outer} \mathbf{Y}_{outer}^\top p_2(\mathbf{A}_{outer})\|_F^2 &\leq \|p_2(\mathbf{A}_{outer}) - p_2(\mathbf{A}_{outer})_k\|_F^2 \\
&\quad + \|(p_2(\mathbf{A}_{outer}) - p_2(\mathbf{A}_{outer})_k)\mathbf{B}\mathbf{M}\|_F^2 \\
&\leq (L+1) \|p_2(\mathbf{A}_{outer}) - p_2(\mathbf{A}_{outer})_k\|_F^2 \\
&\leq \frac{\varepsilon}{2} \sigma_{k+1}(\mathbf{A}_{outer})^2.
\end{aligned}$$

Where the inequalities follow from the same logic as earlier, for [Equation \(4.16\)](#). We again find ourselves at Equation (5) of [85], and follow the rest of the proof on page

11 to get the following guarantee:

$$\|(\mathbf{A}_{outer})_k\|_F^2 - \|\mathbf{Y}_{outer} \mathbf{Y}_{outer}^\top (\mathbf{A}_{outer})_k\|_F^2 \leq \frac{\varepsilon}{2} \sigma_{k+1}(\mathbf{A}_{outer})^2.$$

Since $(\mathbf{A}_{outer})_k = \mathbf{A}_k$, the above inequality then recovers Equation (10) of [85]. Given this proof of Equation (10), the rest of the proof of Property 2 holds as written. Equation (10) is also used on page 14 of [85] to prove Property 3. The rest of the proof of Property 3 also holds without alteration given Equation (10).

Overall, we have recovered the proofs of Properties 1-3 of Lemma 9 of [85]. That is, we have shown that Lemma 9 holds for block Krylov iteration starting from (k, L) -good starting block $\mathbf{B}$ with $q = O(\frac{1}{\sqrt{\varepsilon}} \log(\frac{dL}{\varepsilon}))$ iterations. The proofs in Section 6.2 of [85] then show that Properties 1-3 suffice to achieve the spectral, Frobenius, and singular value guarantees. $\square$

We now move onto **Theorem 66** from [85], which depends on spectral decay. Note there are two different variables ℓ and ℓ_0 in this context. In order to perform rank- k approximation, we want to recover a convergence bound in terms of $g_{k \rightarrow \ell}$ for some $\ell \geq k$. Our starting block $\mathbf{B} \in \mathbb{R}^{d \times \ell_0}$ has $\ell_0 \geq \ell$ columns and is an (ℓ, L) -good starting block. We need to consider this case because, when running block size $b \leq k$ Krylov iteration, the analysis in **Section 4.11** considers a simulated starting block $\mathbf{S}_r$ which uses $\ell_0 \approx b(\ell - b) \geq \ell$ columns to simulate block size ℓ Krylov iteration.

Theorem 66 Restated (Theorem 13 of [85]). *Let $\mathbf{B} \in \mathbb{R}^{d \times \ell_0}$ be any (ℓ, L) -good starting matrix (**Theorem 80**) matrix for $\mathbf{A}$, for some $\ell \geq k$. If we run Block Krylov iteration (**Algorithm 7**) for $q = O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, where $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$, then the output $\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Proof. We again show that Properties 1-3 of Lemma 9 of [85] hold, but now with $q = \frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{dL}{\varepsilon})$. Following Section 7 of [85], we will prove Property 1 for all $l \in [k]$, which in turn implies that Properties 2 and 3 hold. To begin, let $p_3(x) = \frac{(1+\gamma)\alpha}{T_q(1+\gamma)} T_q(\frac{x}{\alpha})$ where $\alpha = \sigma_{\ell+1}(\mathbf{A})$, $\gamma = g_{k \rightarrow \ell}$, and $q = O(\frac{1}{\sqrt{g_{k \rightarrow \ell}}} \log(\frac{dL}{\varepsilon}))$. We let $\mathbf{M} \in \mathbb{R}^{\ell_0 \times \ell}$ be the matrix guaranteed to exist by **Theorem 81**. Then, noticing that the top ℓ singular vectors of $\mathbf{A}$ are also the top ℓ eigenvectors of $p_3(\mathbf{A})$, we again appeal to **Theorem 83** with $\mathbf{P} = p_3(\mathbf{A})$, $\mathbf{B} = \mathbf{B}$, $\mathbf{M} = \mathbf{M}$, and $\mathbf{U} = \mathbf{U}_k$. We get that some orthogonal

$\mathbf{Y}_3 \in \mathbb{R}^{d \times \ell}$ in the span of $p_3(\mathbf{A})\mathbf{B}$ has:

$$\begin{aligned}
\|p_3(\mathbf{A}) - \mathbf{Y}_3 \mathbf{Y}_3^\top p_3(\mathbf{A})\|_F^2 &\leq \|p_3(\mathbf{A}) - p_3(\mathbf{A})_\ell\|_F^2 + \|(p_3(\mathbf{A}) - p_3(\mathbf{A})_\ell)\mathbf{B}\mathbf{M}\|_F^2 \\
&\leq \|p_3(\mathbf{A}) - p_3(\mathbf{A})_\ell\|_F^2 + \|p_3(\mathbf{A}) - p_3(\mathbf{A})_\ell\|_F^2 \|\mathbf{B}\mathbf{M}\|_2^2 \\
&\leq (L+1) \|p_3(\mathbf{A}) - p_3(\mathbf{A})_\ell\|_F^2 \\
&= (L+1) \sum_{i=\ell+1}^d p(\sigma_i(\mathbf{A}))^2 \\
&\leq (L+1)d \cdot \frac{4\sigma_{\ell+1}(\mathbf{A})^2}{2^{2q\sqrt{\gamma}}} \quad (\text{Lemma 5 of [85]}) \\
&\leq \frac{\varepsilon}{2} \sigma_{\ell+1}(\mathbf{A})^2,
\end{aligned}$$

This recovers Equation (5) on page 10 of [85], and from there the rest of the proof of Property 1 holds. As discussed on page 16 of [85], Property 1 holds here for all $l \in [k]$, and therefore Properties 2 and 3 also hold. Then, the analysis in Section 6.2 shows how Properties 1-3 imply the spectral, Frobenius, and singular value guarantees. $\square$

4.13 Single Vector Simultaneous Iteration

We briefly present a single vector algorithm for low-rank approximation based on the standard simultaneous iteration algorithm. Simultaneous iteration, or block power method, extracts a low-rank approximation from the span of $\mathbf{K} = \mathbf{A}^{2q}\mathbf{B}$ where $\mathbf{B}$ is a starting block. We present a prototypical pseudocode for simultaneous iteration in Algorithm 8. [85] show that Algorithm 8 converges from any (k, L) -good starting block $\mathbf{B}$, giving the two following theorems, which follow from the same arguments as in Section 4.12 and [85]:

Algorithm 8 Simultaneous Iteration for Low-Rank Approximation

input: Matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$. Target rank k . Starting block $\mathbf{B} \in \mathbb{R}^{d \times \ell}$. Number of iterations t .

output: Orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$.

- 1: Compute an orthonormal basis $\mathbf{Z}$ for $\mathbf{K} = (\mathbf{A}\mathbf{A}^\top)^t\mathbf{B}$.
 - 2: Compute $\mathbf{U}_k$, the k top eigenvectors of $\mathbf{M} = \mathbf{Z}^\top \mathbf{A}\mathbf{A}^\top \mathbf{Z}$
 - 3: **return** $\mathbf{Q} = \mathbf{Z}\mathbf{U}_k$.
-

Imported Theorem 84 (Theorem 1 of [85]). *Let $\mathbf{B} \in \mathbb{R}^{d \times \ell}$ be any (k, L) -good starting matrix (Theorem 80) matrix for $\mathbf{A}$. If we run Simultaneous Iteration (Algorithm 8) for $q = O(\frac{1}{\varepsilon} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, then the output*

Algorithm 9 Single Vector Simultaneous Iteration for Low-Rank Approximation

input: Matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$. Target rank k . Starting vector $\mathbf{x} \in \mathbb{R}^d$. Number of iterations t . Memory budget $\ell \geq k$.

output: Orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{d \times k}$.

- 1: Compute an orthonormal basis $\mathbf{Z}$ for $\mathbf{K} = [(\mathbf{A}\mathbf{A}^\top)^{t-\ell+1}\mathbf{x}, (\mathbf{A}\mathbf{A}^\top)^{t-\ell+2}\mathbf{x}, \dots, (\mathbf{A}\mathbf{A}^\top)^t\mathbf{x}]$.
- 2: Compute $\mathbf{U}_k$, the k top eigenvectors of $\mathbf{M} = \mathbf{Z}^\top \mathbf{A}\mathbf{A}^\top \mathbf{Z}$
- 3: **return** $\mathbf{Q} = \mathbf{Z}\mathbf{U}_k$.

$\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A}\mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Imported Theorem 85 (Theorem 13 of [85]). Let $\mathbf{B} \in \mathbb{R}^{d \times \ell_0}$ be any (ℓ, L) -good starting matrix ([Theorem 80](#)) matrix for $\mathbf{A}$, for some $\ell \geq k$. If we run Simultaneous Iteration ([Algorithm 8](#)) for $q = O(\frac{1}{g_{k \rightarrow \ell}} \log(\frac{dL}{\varepsilon}))$ iterations with starting block $\mathbf{B}$, where $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$, then the output $\mathbf{Q} \in \mathbb{R}^{d \times k}$ satisfies

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A}\mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

When compared to block Krylov Iteration, subspace iteration uses less memory, but converges slower. Specifically, in the theorems above we obtain a dependence on $1/\varepsilon$ and $1/g_{k \rightarrow \ell}$ in comparison to $1/\sqrt{\varepsilon}$ and $1/\sqrt{g_{k \rightarrow \ell}}$ in the comparable [Theorem 63](#) and [Theorem 66](#) for block Krylov iteration.

In [Algorithm 9](#), we present a “single vector” variant of simultaneous iteration that similarly saves memory over the single vector Krylov method from [Algorithm 6](#). When run for t iterations, instead of storing the entire length t Krylov subspace as in [Algorithm 6](#), the method only stores the last $\ell \geq k$ columns for a specified memory budget ℓ .

Taking $\ell = k$, we can analyze [Algorithm 9](#) by letting $\mathbf{S}_k = [\mathbf{x} \quad \mathbf{A}^2\mathbf{x} \quad \mathbf{A}^4\mathbf{x} \quad \dots \quad \mathbf{A}^{2(k-1)}\mathbf{x}]$ and noticing that [Algorithm 8](#) run with starting matrix $\mathbf{S}_k$ for q iterations produces the matrix $\mathbf{K} = \mathbf{A}^{2q}\mathbf{S}_k = [\mathbf{A}^{2q}\mathbf{x} \quad \mathbf{A}^{2(q+1)}\mathbf{x} \quad \dots \quad \mathbf{A}^{2(q+k)}\mathbf{x}]$. Since [Theorem 64](#) already tells us that $\mathbf{S}_k$ is (k, L) -good for $\mathbf{A}$, applying [Theorem 84](#) immediately gives the following convergence guarantees for [Algorithm 9](#):

Theorem 86. For $\mathbf{A} \in \mathbb{R}^{d \times n}$, let $g_{\min} = \min_{i \in \{1, \dots, k-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 9](#) initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{k}{\varepsilon} \log(\frac{1}{g_{\min}}) + \frac{1}{\sqrt{\varepsilon}} \log(\frac{d}{\varepsilon\delta}))$ iterations with memory budget k returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with

probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Additionally, if [Algorithm 9](#) is run with memory budget $\ell \geq k$, we can obtain a spectrum dependent convergence bound comparable to the result proven in [Section 4.4.1](#) for single vector Krylov iteration. Specifically, since [Theorem 64](#) already tells us that $\mathbf{S}_\ell$ is (ℓ, L) -good for $\mathbf{A}$, applying [Theorem 85](#) immediately gives the following convergence guarantees for [Algorithm 9](#):

Theorem 87. For $\mathbf{A} \in \mathbb{R}^{d \times n}$ and $\ell \geq k$, let $g_{\min} = \min_{i \in \{1, \dots, \ell-1\}} \frac{\sigma_i - \sigma_{i+1}}{\sigma_{i+1}}$ and $g_{k \rightarrow \ell} = \frac{\sigma_k - \sigma_{\ell+1}}{\sigma_k}$. For any $\varepsilon, \delta \in (0, 1)$, [Algorithm 9](#) initialized with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and run for $t = O(\frac{\ell}{g_{k \rightarrow \ell}} \log(\frac{1}{g_{\min}}) + \frac{1}{g_{k \rightarrow \ell}} \log(\frac{d}{\delta \varepsilon}))$ iterations with memory budget ℓ returns an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times k}$ such that, with probability at least $1 - \delta$,

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^\top \mathbf{A}\|_\xi \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_k\|_\xi \quad \text{and} \quad |\mathbf{q}_i^\top \mathbf{A} \mathbf{A}^\top \mathbf{q}_i - \sigma_i(\mathbf{A})^2| \leq \varepsilon \sigma_{k+1}(\mathbf{A})^2.$$

Chapter 5

funNyström 2

5.1 Introduction

A common task in numerical linear algebra is to approximate a matrix function $f(\mathbf{A})$, or some property, like the trace or diagonal entries, of $f(\mathbf{A})$. Formally, given a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with eigendecomposition $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ and a function f defined on the eigenvalues of $\mathbf{A}$, $f(\mathbf{A})$ is defined as

$$f(\mathbf{A}) = \mathbf{U}f(\mathbf{\Lambda})\mathbf{U}^T, \quad f(\mathbf{\Lambda}) = \begin{bmatrix} f(\lambda_1) & & \\ & \ddots & \\ & & f(\lambda_n) \end{bmatrix}.$$

Exactly computing $f(\mathbf{A})$ requires an eigendecomposition of $\mathbf{A}$, which is prohibitively expensive for large scale problems. Instead, we may seek a cheaper approximation to $f(\mathbf{A})$ that suffices for a desired application. In particular, recent research shows that low-rank approximations can help estimate quantities associated with $f(\mathbf{A})$. For example, [55, 70, 105] use a low-rank approximation to estimate $\text{tr}(f(\mathbf{A}))$ for $f(x) = x$, $\log(1+x)$, and $\frac{1}{x+1}$, and [21, 62, 83, 94] use low-rank approximations of matrix functions as a variance reduction technique for the Girard-Hutchinson trace estimator [29, 118, 119]. Similar techniques can be used to estimate the diagonal of $f(\mathbf{A})$ [12, 98].

While low-rank approximations of $f(\mathbf{A})$ are clearly useful, a challenge in computing them is that most algorithms for low-rank approximation require at least some access to the matrix $f(\mathbf{A})$, for example in the form of matrix-vector products. This is true of all Krylov subspace methods, as well as randomized SVD methods based on sketching [53, 107, 116, 117, 132]. Since we cannot directly compute matrix-vector

products with $f(\mathbf{A})$ (we do not have the matrix in hand) we need to resort to approximating them. This can be done, for example, with the Lanczos method [56, Chapter 13] or rational Krylov subspace methods [50]. However, such methods can require many matrix-vector products with $\mathbf{A}$ to accurately compute a single matrix-vector product with $f(\mathbf{A})$. Consequently, the cost of obtaining a low-rank approximation to $f(\mathbf{A})$ can be significantly higher than obtaining a low-rank approximation to $\mathbf{A}$.

5.1.1 The Nyström and funNyström approximations

Recent work in [95] address the challenge discussed above by showing that, when f is non-decreasing, it is often possible to entirely avoid the need to approximate matrix-vector products with $f(\mathbf{A})$ when computing a low-rank approximation. Indeed, constructing a low-rank Nyström approximation to $\mathbf{A}$ often suffices to get a good low-rank approximation to $f(\mathbf{A})$. Before describing in detail the method of [95], which is called funNyström, we briefly recall the general form of Nyström approximation [75, Section 14]. For a SPSP matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the Nyström approximation with respect to a matrix $\mathbf{Q} \in \mathbb{R}^{n \times \ell}$ is

$$\hat{\mathbf{A}} := \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}, \quad (5.1)$$

where $\dagger$ denotes the Moore-Penrose pseudoinverse.¹ The Nyström approximation depends only on $\text{range}(\mathbf{Q})$ [117, Proposition A.2]. Therefore, without loss of generality we assume that $\mathbf{Q}$ has orthonormal columns. When $\mathbf{Q}$ spans the top ℓ eigenvectors of $\mathbf{A}$, $\hat{\mathbf{A}}$ is an optimal rank ℓ approximation to $\mathbf{A}$. So, the goal is to efficiently find $\mathbf{Q}$ that approximately spans these vectors. E.g., $\mathbf{Q}$ might be chosen to be an orthonormal basis for $\text{range}(\mathbf{A}^{q-1}\mathbf{\Omega})$ or $\text{range}\left(\begin{bmatrix} \mathbf{\Omega} & \mathbf{A}\mathbf{\Omega} & \dots & \mathbf{A}^{q-1}\mathbf{\Omega} \end{bmatrix}\right)$ for some $q \geq 1$ and random sketching matrix $\mathbf{\Omega}$. Analyses for various choices of $\mathbf{Q}$ can be found in [22, 30, 44, 95, 116, 117, 124]. Notably, when seeking a rank k approximation, we often construct $\mathbf{Q}$ with $\ell > k$ columns, meaning that $\hat{\mathbf{A}}$ has rank ℓ instead of rank k . In order to recover an exactly rank k approximation, we would return the best rank k approximation to $\hat{\mathbf{A}}$ instead of $\hat{\mathbf{A}}$ itself. Denoted as $\hat{\mathbf{A}}_{(k)}$, this matrix can be computed efficiently in $O(n\ell^2)$ time, plus the time to multiply $\mathbf{A}$ by ℓ vectors (see

¹In line with prior work [53, 75, 116], we use the term *Nyström approximation* to refer to the general form $\mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$. We do not make any assumptions on how $\mathbf{Q}$ is generated. We note that early work on the Nyström method for low-rank approximation, including the seminal work in [130], as well as many follow up papers [35, 86, 135], use the term Nyström approximation more narrowly to refer to methods where $\mathbf{Q}$ is obtained by subsampling columns of the identity matrix (i.e. $\mathbf{A}\mathbf{Q}$ is a subset of columns from $\mathbf{A}$).

Algorithm 10).

Algorithm 10 Nyström approximation

input: PSD $\mathbf{A} \in \mathbb{R}^{n \times n}$. Orthonormal basis $\mathbf{Q} \in \mathbb{R}^{n \times \ell}$. Rank k .

output: Rank k approx. to $\mathbf{A}$ in factored form $\hat{\mathbf{A}}_{(k)} = \hat{\mathbf{U}}_{:,1:k} \hat{\mathbf{\Lambda}}_{1:k,1:k} \hat{\mathbf{U}}_{:,1:k}^T$.

- 1: Compute an eigendecomposition, $\mathbf{V}\mathbf{D}\mathbf{V}^T$, of $\mathbf{Q}^T \mathbf{A} \mathbf{Q}$.
 - 2: Set $\mathbf{B} = \mathbf{A} \mathbf{Q} \mathbf{V} (\mathbf{D}^{1/2})^\dagger$
 - 3: Compute the SVD of $\mathbf{B} = \hat{\mathbf{U}} \mathbf{\Sigma} \mathbf{W}^T$. Set $\hat{\mathbf{\Lambda}} = \mathbf{\Sigma}^2$
 - 4: **return** $\hat{\mathbf{U}}_{:,1:k}, \hat{\mathbf{\Lambda}}_{1:k,1:k}$.
-

The funNyström method of [95] first forms a Nyström approximation $\hat{\mathbf{A}}$, where $\mathbf{Q} = \text{range}(\mathbf{A}^{q-1} \mathbf{\Omega})$ is an orthonormal basis constructed via subspace iteration. The method² then constructs a low-rank approximation to $f(\mathbf{A})$ by returning $f(\hat{\mathbf{A}})_{(k)}$, which denotes the best k -rank approximation to $f(\hat{\mathbf{A}})$. When f is non-decreasing with $f(0) \geq 0$, and $\hat{\mathbf{A}}_{(k)} = \hat{\mathbf{U}}_{:,1:k} \hat{\mathbf{\Lambda}}_{1:k,1:k} \hat{\mathbf{U}}_{:,1:k}^T$ is available in factored form, as returned by Algorithm 10, the funNyström approximation can be computed as:

$$f(\hat{\mathbf{A}})_{(k)} = \hat{\mathbf{U}}_{:,1:k} f(\hat{\mathbf{\Lambda}}_{1:k,1:k}) \hat{\mathbf{U}}_{:,1:k}^T. \quad (5.2)$$

The intuition behind funNyström is that, if f is a positive non-decreasing function, applying f to an SPSD matrix $\mathbf{A}$ does not change the order of eigenvalues. As a result, if $\hat{\mathbf{A}}_{(k)}$ was an *optimal* rank k approximation to $\mathbf{A}$ in any unitarily invariant norm, then $f(\hat{\mathbf{A}})_{(k)}$ would be an optimal rank k approximation to $f(\mathbf{A})$. Hence, we might hope that if $\hat{\mathbf{A}}_{(k)}$ is a near-optimal rank k approximation to $\mathbf{A}$, then $f(\hat{\mathbf{A}})_{(k)}$ remains a near-optimal rank k approximation to $f(\mathbf{A})$.

Formally, [95] derives error bounds for $\|f(\mathbf{A}) - f(\hat{\mathbf{A}})_{(k)}\|$ for a variety of unitarily invariant norms for a special class of non-decreasing functions: *operator monotone* functions satisfying $f(0) = 0$. A function f is operator monotone if for any symmetric matrices $\mathbf{B} \succeq \mathbf{C}$ we have that $f(\mathbf{B}) \succeq f(\mathbf{C})$, where $\succeq$ denotes the Loewner order. Important examples used in applications include $\log(1+x)$, x^α for $\alpha \in [0, 1]$, and the “ridge function” $\frac{x}{x+\lambda}$ for $\lambda \geq 0$ [40]. Notably, all operator monotone functions are concave, so $\mathbf{A}$ has larger normalized eigenvalue gaps than $f(\mathbf{A})$. An important intuition behind the work of [95] is that, given these larger gaps, subspace iteration run on $\mathbf{A}$ should converge no slower to the top eigenvectors of $f(\mathbf{A})$ than if we had run it on $f(\mathbf{A})$ directly.

²The work in [95] considers a more restricted setting where $f(0) = 0$, in which case $f(\hat{\mathbf{A}})_{(k)}$ simply equals $f(\hat{\mathbf{A}})$. Throughout, we only make the weaker assumption that $f(0) \geq 0$.

5.1.2 Our Contributions

The main limitation of [95] is that its analysis of the funNyström method is specialized to $\hat{\mathbf{A}}$ produced by subspace iteration specifically, i.e. $\mathbf{Q}$ in (5.1) is an orthonormal basis for $\text{range}(\mathbf{A}^{q-1}\mathbf{\Omega})$ where $\mathbf{\Omega}$ is a random matrix. The literature contains a wide variety of other low-rank approximation algorithms, many of which often outperform subspace iteration, including e.g. randomized Krylov methods [79, 85], methods based on sparse or structured random projections [25, 73], frequent directions methods [41], or methods based on random sampling [22, 35, 37]. The analysis in [95] tells us nothing about how good a low-rank approximation $f(\hat{\mathbf{A}})_{(k)}$ would be if $\hat{\mathbf{A}}$ is produced by such methods.

We address this gap by significantly generalizing the results in [95]. In particular, we show that if $\hat{\mathbf{A}}$ is any near-optimal low-rank approximation to $\mathbf{A}$ so that $\mathbf{A} \succeq \hat{\mathbf{A}} \succeq \mathbf{0}$, then for any positive, continuous operator monotone function, $f(\hat{\mathbf{A}})_{(k)}$ is a near-optimal low-rank approximation to $f(\mathbf{A})$, *independently of how $\hat{\mathbf{A}}$ was computed*. Importantly, any Nyström approximation satisfies $\mathbf{A} \succeq \hat{\mathbf{A}} \succeq \mathbf{0}$, so our guarantees extend to all possible Nyström approximations to $\mathbf{A}$, no matter how $\mathbf{Q}$ is obtained. For example, we prove the following.

Theorem 88. *Suppose that $\mathbf{A} \succeq \hat{\mathbf{A}} \succeq \mathbf{0}$. Let $\mathbf{A}_{(k)}$ and $\hat{\mathbf{A}}_{(k)}$ be optimal rank k approximations to $\mathbf{A}$ and $\hat{\mathbf{A}}$, respectively. Further suppose that for $\varepsilon \geq 0$,*

$$\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*,$$

where $\|\cdot\|_$ denotes the nuclear norm. Then for any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$ we have*

$$\|f(\mathbf{A}) - f(\hat{\mathbf{A}})_{(k)}\|_* \leq (1 + \varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_*.$$

The above theorem applies to the nuclear norm $\|\cdot\|_*$, however we also prove similar results when low-rank approximation error is measured in the operator norm (Theorem 100) and the Frobenius norm (Theorem 98).

Further, to facilitate the application of these theorems, we present sufficient conditions for an orthonormal basis $\mathbf{Q}$ to produce a near-optimal Nyström approximation of the form (5.1). For example, we prove the following.

Theorem 89. *Let $\mathbf{A} \succeq \mathbf{0}$ and let $\mathbf{Q}$ be an orthonormal basis so that, for $\varepsilon \geq 0$,*

$$\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*. \quad (5.3)$$

Then if $\widehat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ we have

$$\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*.$$

Similar guarantees are proven in [116], but they are constrained to the case when $\mathbf{Q}$ has exactly k columns. In contrast, our results allow $\mathbf{Q}$ to have more than k columns. Guarantees of the form (5.3) are common in the literature; see e.g. [8, 9, 61, 79, 85]. **Theorem 89** allows us to translate these existing results into guarantees for the Nyström approximation. Then, by using **Theorem 88**, these existing results for $\mathbf{Q}$ translate all the way into results for the funNyström approximation. For example, consider the following known result.

Theorem 90 ([8, Theorem 5.1]). *Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\varepsilon \in (0, 1)$. There is an algorithm which performs $O(\frac{k}{\varepsilon^{1/3}})$ matrix vector products with $\mathbf{A}$ and returns an orthonormal matrix $\mathbf{Q} \in \mathbb{R}^{n \times k}$ such that with probability at least 0.9*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*.$$

By using **Theorems 88** and **89**, we immediately get the following corollary.

Corollary 91. *Let $\mathbf{A} \succeq \mathbf{0}$ and $\varepsilon \in (0, 1)$. Let $f : [0, \infty) \rightarrow [0, \infty)$ be a continuous operator monotone function. There is an algorithm which performs $O(\frac{k}{\varepsilon^{1/3}})$ matrix-vector products with $\mathbf{A}$ and returns a rank k Nyström approximation $\widehat{\mathbf{A}}$ such that with probability at least 0.9*

$$\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\|_* \leq (1 + \varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_*.$$

Alternatively, we can recover the main result from the original funNyström paper, [95], by using the following error bound for subspace iteration, which follows from a simple modification of the proof of [117, Theorem 4.1].

Theorem 92 ([117, Theorem 4.1]). *Let $\mathbf{A} \succeq \mathbf{0}$ and let $\gamma = \lambda_{k+1}/\lambda_k$ denote the k^{th} spectral gap of $\mathbf{A}$. Let $\mathbf{Q}$ be an orthonormal basis for $\text{range}(\mathbf{A}^{q-1}\mathbf{\Omega})$, where $q \geq 1$ and $\mathbf{\Omega}$ is a $n \times (k+p)$ random matrix whose entries are independent identically distributed $N(0, 1)$ random variables. If $\widehat{\mathbf{A}}$ is defined as in (5.1) and $p \geq 2$, then we have*

$$\mathbb{E}\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\|_* \leq \left(1 + \gamma^{2(q-1)} \frac{k}{p-1}\right) \|\mathbf{A} - \mathbf{A}_{(k)}\|_*$$

By using **Theorems 88** and **92** we immediately recover [95, Theorem 3.9].

Corollary 93. *Consider the setting of [Theorem 92](#). Then for any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$, we have*

$$\mathbb{E}\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\|_* \leq \left(1 + \gamma^{2(q-1)} \frac{k}{p-1}\right) \|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_*.$$

As for [Theorem 88](#), we give similar guarantees to [Theorem 89](#) in the Frobenius and operator norm. We also provide guarantees for eigenvalue estimation. As in [95], a strength of our results is that one does not need to know f beforehand. The user can compute and store a Nyström approximation for $\mathbf{A}$ and use it to construct a funNyström approximation to *any* continuous operator monotone function, and our results show that this approximation is always near-optimal as long as the Nyström approximation is near-optimal.

In addition to our main theorems relating Nyström approximation of $\mathbf{A}$ to funNyström approximation of $f(\mathbf{A})$, we investigate if we can relax any of the assumptions in the results. Specifically, we ask: (i) is the assumption that $\mathbf{A} \succeq \widehat{\mathbf{A}}$ necessary? (ii) is the operator monotonicity assumption necessary, or is it sufficient to assume that f is concave and increasing? For some norms, we are able to find counterexamples that show that these assumptions are in fact necessary. Our findings are summarized in [Table 5.1](#). We note, however, that our counterexamples only show that there are cases when $\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\| \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|$, but $\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\| > (1 + \varepsilon + \delta)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|$ for some non-zero δ . They do not, for example, rule out the possibility that $\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\| \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|$ could imply that $\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\| \leq (1 + C\varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|$ for some fixed constant $C \geq 1$ (e.g., $C = 2$ or $C = 10$). Finding stronger counter examples is an interesting direction for future work.

Guarantee	f is operator monotone		f is concave, non-decreasing	
	$\mathbf{A} \succeq \hat{\mathbf{A}}$	$\mathbf{A} \not\succeq \hat{\mathbf{A}}$	$\mathbf{A} \succeq \hat{\mathbf{A}}$	$\mathbf{A} \not\succeq \hat{\mathbf{A}}$
Nuclear Norm	✓ ₉₇	✗ ₁₁₄	✗ ₁₁₂	✗ ₁₁₄
Frobenius Norm	✓ [*] ₉₈	✗ ₁₁₄	?	✗ ₁₁₄
Spectral Norm	✓ ₁₀₀	✓ ₁₀₀	?	✗ ₁₁₃
Eigenvalue	✓ ₁₀₁	✓ ₁₀₁	✓ ₁₀₁	✓ ₁₀₁

Table 5.1: Our current knowledge of what assumptions are required for $\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\| \leq (1+\varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|$ to imply $\|f(\mathbf{A}) - f(\hat{\mathbf{A}})_{(k)}\| \leq (1+\varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|$. A checkmark (✓) indicates that the implication holds under the listed assumptions, and we provide a link to the theorem proving it. A cross (✗) indicates that a counterexample with the listed assumptions violates the guarantee. The question marks (?) indicate open problems. The asterisk for the Frobenius norm denotes that we make a slightly stronger assumption for that guarantee; see [Section 5.2.1](#) for more details.

5.1.3 Notation

We consider a symmetric positive semidefinite (SPSD) matrix $\mathbf{A}$. $\hat{\mathbf{A}}$ usually denotes an approximation to $\mathbf{A}$ so that $\mathbf{A} \succeq \hat{\mathbf{A}} \succeq \mathbf{0}$. For a matrix $\mathbf{B}$, $\sigma_i(\mathbf{B})$ denotes the i^{th} largest singular value and, if $\mathbf{B} \succeq \mathbf{0}$, $\lambda_i(\mathbf{B})$ is the i^{th} largest eigenvalue. For conciseness, we drop the argument and let λ_i denote $\lambda_i(\mathbf{A})$ and $\hat{\lambda}_i$ denote $\lambda_i(\hat{\mathbf{A}})$. For a matrix $\mathbf{B}$ we let $\mathbf{P}_{\mathbf{B}}$ denote orthogonal projector onto $\text{range}(\mathbf{B})$. Hence, if $\mathbf{V}$ is an orthonormal basis for $\text{range}(\mathbf{B})$ we have $\mathbf{P}_{\mathbf{B}} = \mathbf{V}\mathbf{V}^T$. We let $\mathbf{B}_{(k)}$ denote the best rank k approximation to $\mathbf{B}$ obtained by the truncated singular value decomposition. For $p \in [1, \infty]$ we denote by $\|\cdot\|_{(p)}$ the Schatten p -norm. We write $\|\cdot\|_2 = \|\cdot\|_{(\infty)}$, $\|\cdot\|_F = \|\cdot\|_{(2)}$, and $\|\cdot\|_* = \|\cdot\|_{(1)}$ to denote the operator, Frobenius, and nuclear norm respectively.

5.2 Good Nyström approximations imply good fun-Nyström approximations

In this section prove our main results: that near optimal Nyström approximations imply near optimal funNyström approximations for the nuclear, Frobenius, and operator norms. We also prove analogous guarantees for eigenvalue estimation. We begin with establishing a few useful lemmas. We start by recalling some basic properties of operator monotone and concave functions.

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Lemma 94. *Let $f : [0, \infty) \rightarrow [0, \infty)$ be a continuous operator monotone function. Then,*

(i) f is concave;

(ii) $f \in C^\infty(0, \infty)$.

Proof. ((i)) This is due to [15, Theorem V.2.5]. ((ii)) This is due to [15, p.134-135]. $\square$

By the above, the following fact about concave functions also extends to continuous operator monotone functions.

Lemma 95. *Let $f : [0, \infty) \rightarrow [0, \infty)$ be a concave function. Then,*

(i) $\frac{f(x)}{x}$ is decreasing for $x > 0$;

(ii) $f(tx) \leq tf(x)$ for $t \geq 1$;

(iii) $f(tx) \geq tf(x)$ for $0 \leq t \leq 1$.

(iv) For $t \geq 0$, the function $f(x) - f(x - t)$ is decreasing. .

Proof. ((i)): f is concave if and only if the function

$$R(x_1, x_2) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \quad (5.4)$$

is decreasing in x_2 for any fixed x_1 (or vice versa). Hence, since $f(0) \geq 0$ we have that $\frac{f(x)}{x} = R(0, x) + \frac{f(0)}{x}$ is decreasing, as required.

((ii)): For any fixed x we know that the function $h_x(t) = f(tx)$ is concave. Hence, by ((i)) we know that $\frac{h_x(t)}{t}$ is decreasing. Therefore, if $t \geq 1$ we have $\frac{f(tx)}{t} = \frac{h_x(t)}{t} \leq \frac{h_x(1)}{1} = f(x)$, which yields the desired result.

((iii)): Proven in an analogous way to ((ii)) but using $\frac{h_x(t)}{t} \geq \frac{h_x(1)}{1}$ if $t \leq 1$.

((iv)): Since $R(x_1, x_2)$ in (5.4) is decreasing in x_1 for any fixed x_2 and vice versa we have for any $x \geq y \geq t$

$$\frac{f(y) - f(y - t)}{t} = R(y - t, y) \geq R(x - t, y) \geq R(x - t, x) = \frac{f(x) - f(x - t)}{t},$$

which yields the desired result. $\square$

The following lemma provides an upper bound for the Schatten norm difference of two ordered SPSD matrices. In particular, it bounds $\|\mathbf{B} - \mathbf{C}\|_{(p)}^p$ by the difference of the Schatten norms of $\mathbf{B}$ and $\mathbf{C}$. The latter is easier to analyze because it allows us to separate the eigenvalues of $\mathbf{B}$ and $\mathbf{C}$. This will be especially useful for proving results for the nuclear and Frobenius norm.

Lemma 96. *Let $\mathbf{B} \succeq \mathbf{C} \succeq \mathbf{0}$, then for $p \geq 1$*

$$\|\mathbf{B} - \mathbf{C}\|_{(p)} \leq \left(\|\mathbf{B}\|_{(p)}^p - \|\mathbf{C}\|_{(p)}^p \right)^{1/p}.$$

When $p = 1$ the inequality becomes an equality.

Proof. By a result by McCarthy [77, Lemma 2.6] we know that if $\mathbf{X}, \mathbf{Y} \succeq \mathbf{0}$,

$$\text{tr}((\mathbf{X} + \mathbf{Y})^p) \geq \text{tr}(\mathbf{X}^p) + \text{tr}(\mathbf{Y}^p).$$

Setting $\mathbf{X} = \mathbf{B} - \mathbf{C}$ and $\mathbf{Y} = \mathbf{C}$ yields the desired inequality. When $p = 1$, using the fact that $\mathbf{B} \succeq \mathbf{C} \succeq \mathbf{0}$, we find that, as required,

$$\|\mathbf{B} - \mathbf{C}\|_* = \text{tr}(\mathbf{B} - \mathbf{C}) = \text{tr}(\mathbf{B}) - \text{tr}(\mathbf{C}) = \|\mathbf{B}\|_* - \|\mathbf{C}\|_*. \quad \square$$

5.2.1 Frobenius and nuclear norm guarantees

In this section we prove two results. First, we prove [Theorem 88](#).

Theorem 97 ([Theorem 88](#) restated). *Suppose that, for $\varepsilon \geq 0$, $\mathbf{A} \succeq \widehat{\mathbf{A}} \succeq \mathbf{0}$ satisfy*

$$\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\|_* \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_{(k)}\|_*.$$

Then for any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$,

$$\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\|_* \leq (1 + \varepsilon) \|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_*.$$

Additionally, we prove an analogous guarantee for the Frobenius norm.

Theorem 98. *Suppose that, for $\varepsilon \geq 0$, $\mathbf{A} \succeq \widehat{\mathbf{A}} \succeq \mathbf{0}$ satisfy*

$$\|\mathbf{A}\|_F^2 - \|\widehat{\mathbf{A}}_{(k)}\|_F^2 \leq (1 + \varepsilon) \|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2.$$

Then for any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$,

$$\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\|_F^2 \leq (1 + \varepsilon) \|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_F^2.$$

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We note that, by [Theorem 96](#), $\|\mathbf{A}\|_F^2 - \|\widehat{\mathbf{A}}_{(k)}\|_F^2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2$ is a *stronger* assumption than $\|\mathbf{A} - \widehat{\mathbf{A}}\|_F^2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2$. I.e., [Theorem 98](#) requires assuming more than that $\widehat{\mathbf{A}}_{(k)}$ is a near optimal low-rank approximation to $\mathbf{A}$ in the Frobenius norm. However, the assumption is still reasonable because, as we show in [Section 5.3](#), many standard low-rank algorithms return results that satisfy this stronger guarantee.

We prove [Theorems 97](#) and [98](#) as special cases of a single general theorem about Schatten norms. In particular, by [Theorem 96](#) we know that $\|\mathbf{A}\|_* - \|\widehat{\mathbf{A}}\|_* = \|\mathbf{A} - \widehat{\mathbf{A}}\|_*$, so the following theorem about Schatten norms immediately implies both [Theorems 97](#) and [98](#) by taking $p = 1$ and $p = 2$, respectively.

Theorem 99. *Fix $p \in [1, \infty)$. Suppose that, for $\varepsilon \geq 0$, $\mathbf{A} \succeq \widehat{\mathbf{A}} \succeq \mathbf{0}$ satisfy*

$$\|\mathbf{A}\|_{(p)}^p - \|\widehat{\mathbf{A}}_{(k)}\|_{(p)}^p \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_{(p)}^p$$

Then for any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$,

$$\|f(\mathbf{A}) - f(\widehat{\mathbf{A}})_{(k)}\|_{(p)}^p \leq (1 + \varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_{(p)}^p.$$

Proof of [Theorem 99](#). Let λ_i and $\widehat{\lambda}_i$ denote the i^{th} largest eigenvalues of $\mathbf{A}$ and $\widehat{\mathbf{A}}$, respectively. Our assumption on $\widehat{\mathbf{A}}$ implies that

$$\sum_{i=1}^k (\lambda_i^p - \widehat{\lambda}_i^p) \leq \varepsilon \sum_{i=k+1}^n \lambda_i^p. \quad (5.5)$$

By Weyl's monotonicity principle, $\mathbf{A} \succeq \widehat{\mathbf{A}}$ implies that $\lambda_i \geq \widehat{\lambda}_i$ [[57](#), Corollary 7.7.4 (c)]. Let us define $(1 - \delta_i) = \left(\frac{\widehat{\lambda}_i}{\lambda_i}\right)^p$ for $\delta_i \in [0, 1]$ for $i = 1, \dots, k$. Hence, [\(5.5\)](#) implies that

$$\sum_{i=1}^k \delta_i \lambda_i^p \leq \varepsilon \sum_{i=k+1}^n \lambda_i^p. \quad (5.6)$$

By [Theorem 95](#) ((iii)), we know that $f(\widehat{\lambda}_i)^p \geq (1 - \delta_i)f(\lambda_i)^p$, and so

$$\sum_{i=1}^k (f(\lambda_i)^p - f(\widehat{\lambda}_i)^p) \leq \sum_{i=1}^k \delta_i f(\lambda_i)^p. \quad (5.7)$$

[Theorem 95](#) ((i)) also shows that, for all $i = 1, \dots, k$ and $j = k+1, \dots, n$, $\frac{\lambda_j}{\lambda_i} \leq \frac{f(\lambda_j)}{f(\lambda_i)}$.

So, we have,

$$\begin{aligned}
\frac{\lambda_j^p}{\delta_i \lambda_i^p} &\leq \frac{f(\lambda_j)^p}{\delta_i f(\lambda_i)^p} \\
\Rightarrow \frac{\sum_{j=k+1}^n \lambda_j^p}{\delta_i \lambda_i^p} &\leq \frac{\sum_{j=k+1}^n f(\lambda_j)^p}{\delta_i f(\lambda_i)^p} \\
\Rightarrow \frac{\delta_i \lambda_i^p}{\sum_{j=k+1}^n \lambda_j^p} &\geq \frac{\delta_i f(\lambda_i)^p}{\sum_{j=k+1}^n f(\lambda_j)^p} \tag{5.8}
\end{aligned}$$

Combining (5.8) with (5.7) and (5.6) gives

$$\sum_{i=1}^k (f(\lambda_i)^p - f(\hat{\lambda}_i)^p) \leq \sum_{i=1}^k \delta_i f(\lambda_i)^p \leq \frac{\sum_{i=1}^k \delta_i \lambda_i^p}{\sum_{j=k+1}^n \lambda_j^p} \sum_{j=k+1}^n f(\lambda_j)^p \leq \varepsilon \sum_{i=k+1}^n f(\lambda_i)^p. \tag{5.9}$$

Hence, (5.9) implies that

$$\|f(\mathbf{A})\|_{(p)}^p - \|f(\hat{\mathbf{A}})_{(k)}\|_{(p)}^p \leq (1 + \varepsilon) \|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_{(p)}^p.$$

We have $f(\mathbf{A}) \succeq f(\hat{\mathbf{A}})_{(k)} \succeq \mathbf{0}$ since f is operator monotone. The desired inequality follows from applying [Theorem 96](#). $\square$

5.2.2 Operator norm guarantees

We next an analogue to [Theorems 97](#) and [98](#) for the operator norm. Our operator norm result is actually more general since we do not require access to an approximation $\hat{\mathbf{A}}$ satisfying $\hat{\mathbf{A}} \preceq \mathbf{A}$. The result applies to *any* $\mathbf{B}$ such that $\|\mathbf{A} - \mathbf{B}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2$. We do not even require that $\mathbf{B}$ is rank k .

Theorem 100. *Suppose that, for $\varepsilon \geq 0$, $\mathbf{A}, \mathbf{B} \succeq \mathbf{0}$ satisfy*

$$\|\mathbf{A} - \mathbf{B}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2.$$

Let $r = \text{rank}(\mathbf{B})$. For any continuous operator monotone function $f : [0, \infty) \rightarrow [0, \infty)$,

$$\|f(\mathbf{A}) - f(\mathbf{B})_{(r)}\|_2 \leq (1 + \varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_2.$$

Proof. First assume that $f(0) = 0$. As a consequence of this assumption, we have

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$f(\mathbf{B})_{(r)} = f(\mathbf{B})$. We leverage [15, Theorem X.1.1] about operator monotone functions, which says that $\|f(\mathbf{A}) - f(\mathbf{B})\|_2 \leq f(\|\mathbf{A} - \mathbf{B}\|_2)$. Then, recalling from [Theorem 94](#) that f is increasing and concave, we have that

$$\begin{aligned} \|f(\mathbf{A}) - f(\mathbf{B})\|_2 &\leq f(\|\mathbf{A} - \mathbf{B}\|_2) && ([15, \text{Theorem X.1.1}]) \\ &\leq f((1 + \varepsilon)\lambda_{k+1}) \\ &\leq (1 + \varepsilon)f(\lambda_{k+1}) && (\text{Theorem 95 } ((ii))) \\ &= (1 + \varepsilon)\|f(\mathbf{A}) - f(\mathbf{A})_{(k)}\|_2, \end{aligned}$$

which yields the desired inequality for the case when $f(0) = 0$. We now consider the general case when $f(0) \geq 0$. Write $f(x) = g(x) + f(0)$, where $g(x)$ is a continuous operator monotone function satisfying $g(0) = 0$. Let $\mathbf{P}_{\mathbf{B}}$ be the orthogonal projector onto $\text{range}(\mathbf{B})$. We have that $f(\mathbf{B})_{(r)} = g(\mathbf{B}) + f(0)\mathbf{P}_{\mathbf{B}}$, so

$$f(\mathbf{A}) - f(\mathbf{B})_{(r)} = g(\mathbf{A}) - g(\mathbf{B}) + f(0)(\mathbf{I} - \mathbf{P}_{\mathbf{B}}), \quad (5.10)$$

Let $\mathbf{X}$ be any unit vector. Since $g(0) = 0$ we have $|\mathbf{X}^T(g(\mathbf{A}) - g(\mathbf{B}))\mathbf{X}| \leq (1 + \varepsilon)g(\lambda_{k+1})$. Hence, using [\(5.10\)](#) we get

$$\begin{aligned} \mathbf{X}^T(f(\mathbf{A}) - f(\mathbf{B})_{(r)})\mathbf{X} &\leq (1 + \varepsilon)g(\lambda_{k+1}) + f(0)\|(\mathbf{I} - \mathbf{P}_{\mathbf{B}})\mathbf{X}\|_2^2 \leq \\ &(1 + \varepsilon)g(\lambda_{k+1}) + f(0) \leq (1 + \varepsilon)f(\lambda_{k+1}), \end{aligned}$$

and

$$\begin{aligned} \mathbf{X}^T(f(\mathbf{A}) - f(\mathbf{B})_{(r)})\mathbf{X} &\geq -(1 + \varepsilon)g(\lambda_{k+1}) + f(0)\|(\mathbf{I} - \mathbf{P}_{\mathbf{B}})\mathbf{X}\|_2^2 \geq \\ &-(1 + \varepsilon)g(\lambda_{k+1}) - (1 + \varepsilon)f(0) \geq -(1 + \varepsilon)f(\lambda_{k+1}). \end{aligned}$$

We conclude that $\|f(\mathbf{A}) - f(\mathbf{B})_{(r)}\|_2 \leq (1 + \varepsilon)f(\lambda_{k+1})$, as required. $\square$

5.2.3 Eigenvalue guarantees

In this section we establish guarantees for eigenvalue estimation. In particular, we show that if the eigenvalues of a SPSD matrix $\hat{\mathbf{A}}$ are good approximations to the eigenvalues of $\mathbf{A}$, then the eigenvalues of $f(\hat{\mathbf{A}})$ are even better approximations to the eigenvalues of $f(\mathbf{A})$. This result could be combined with results that prove eigenvalue approximation guarantees for algorithms including subspace iteration and block Krylov subspace methods [85].

Theorem 101. *Suppose that, for $\varepsilon \in [0, 1]$, we have estimates $\widehat{\lambda}_1 \geq \widehat{\lambda}_2 \geq \dots \geq \widehat{\lambda}_k \geq 0$ of the k largest eigenvalues of $\mathbf{A}$ satisfying*

$$0 \leq \lambda_i - \widehat{\lambda}_i \leq \varepsilon \lambda_{k+1} \quad \text{for } i = 1, \dots, k.$$

Then for any non-decreasing concave function $f : [0, \infty) \rightarrow [0, \infty)$,

$$0 \leq f(\lambda_i) - f(\widehat{\lambda}_i) \leq \varepsilon f(\lambda_{k+1}) \quad \text{for } i = 1, \dots, k.$$

Proof. Note that by **Theorem 95**(*iv*) the function

$$g(t) = f(t) - f(t - \varepsilon \lambda_{k+1})$$

is decreasing. Hence, for $i = 1, \dots, k$ we have

$$\begin{aligned} 0 &\leq f(\lambda_i) - f(\widehat{\lambda}_i) && (f \text{ is non-decreasing}) \\ &\leq f(\lambda_i) - f(\lambda_i - \varepsilon \lambda_{k+1}) && (f \text{ is non-decreasing}) \\ &\leq f(\lambda_{k+1}) - f((1 - \varepsilon)\lambda_{k+1}) && (g(\lambda_i) \leq g(\lambda_{k+1}) \text{ since } \lambda_i \geq \lambda_{k+1}) \\ &\leq \varepsilon f(\lambda_{k+1}), && (\text{Theorem 95 } ((iii))) \end{aligned}$$

as required. $\square$

The assumption in **Theorem 101** can be weakened to the case when we have small relative errors

$$0 \leq \lambda_i - \widehat{\lambda}_i \leq \varepsilon \lambda_i. \tag{5.11}$$

By the monotonicity of f and **Theorem 95** (*iii*), we have that (5.11) implies

$$f(\lambda_i) - f(\widehat{\lambda}_i) \leq f(\lambda_i) - f((1 - \varepsilon)\lambda_i) \leq \varepsilon f(\lambda_i).$$

Hence, we also have the following result.

Theorem 102. *Suppose that, for $\varepsilon \in [0, 1]$, we have estimates $\widehat{\lambda}_1 \geq \widehat{\lambda}_2 \geq \dots \geq \widehat{\lambda}_k \geq 0$ of the k largest eigenvalues of $\mathbf{A}$ satisfying*

$$0 \leq \lambda_i - \widehat{\lambda}_i \leq \varepsilon \lambda_i \quad \text{for } i = 1, \dots, k.$$

Then for any non-decreasing concave function $f : [0, \infty) \mapsto [0, \infty)$,

$$0 \leq f(\lambda_i) - f(\widehat{\lambda}_i) \leq \varepsilon f(\lambda_i) \quad \text{for } i = 1, \dots, k.$$

5.3 Good projections imply good Nyström approximations

In this section we show that if $\mathbf{Q}$ is an orthonormal basis so that $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$ is a good rank k approximation to $\mathbf{A}$, then $\hat{\mathbf{A}}_{(k)}$ is a *better* rank k approximation to $\mathbf{A}$, where $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ is the Nyström approximation to $\mathbf{A}$. Existing low-rank approximation literature commonly provides guarantees for the error $\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|$, where $\mathbf{Q}$ is the output of some algorithm, see e.g. [8, 9, 48, 61, 79, 85].³ Hence, this result allows us to transform many known low-rank approximation guarantees into low-rank approximation guarantees for the rank k truncated Nyström approximation $\hat{\mathbf{A}}_{(k)}$. Further, by the results in Section 5.2 we therefore extend these guarantees to the funNyström approximation.

We point out that whenever $\mathbf{Q}$ has exactly k columns, many of the results in this section would follow from [116, Lemma 5.2], which shows that $\|\mathbf{A} - \hat{\mathbf{A}}\| \leq \|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|$ for any unitarily invariant norm $\|\cdot\|$. However, often $\mathbf{Q}$ has more than k columns, e.g. when $\mathbf{Q}$ is an orthonormal basis for a Krylov subspace, and we want to establish guarantees when we truncate $\hat{\mathbf{A}}$ back to rank k . Truncation is desirable when the low-rank approximation is needed for downstream applications like data visualization or k -means clustering [97].

We show that $\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\| \leq \|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|$ for the nuclear and Frobenius norms. Perhaps surprisingly, the inequality is false in the operator norm and we provide a counterexample. Lastly, we also provide an analogous guarantee for estimating the eigenvalues of $\mathbf{A}$. Before doing so, we recall a standard fact about Nyström approximation used throughout this section.

Lemma 103 ([44, Equation (4)]). *For any $\mathbf{Q} \in \mathbb{R}^{n \times \ell}$, the Nyström approximation satisfies $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A} = \mathbf{A}^{1/2}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2}$.*

5.3.1 Frobenius norm guarantees

In first prove the following result on Frobenius norm low-rank approximation.

Theorem 104. *Let $\mathbf{A} \succeq \mathbf{0}$ and let $\mathbf{Q}$ be an orthonormal basis so that, for $\varepsilon \geq 0$,*

$$\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_F^2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2. \quad (5.12)$$

³We recall that $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)} = \mathbf{Q}(\mathbf{Q}^T\mathbf{A})_{(k)}$ and $(\mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{Q}\mathbf{Q}^T)_{(k)} = \mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})_{(k)}\mathbf{Q}^T$. Both $\mathbf{Q}(\mathbf{Q}^T\mathbf{A})_{(k)}$ and $\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})_{(k)}\mathbf{Q}^T$ are preferable for computational purposes since we only have to compute the best rank k approximation of a smaller matrix. However, in the following sections we use $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$ and $(\mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{Q}\mathbf{Q}^T)_{(k)}$, since it simplifies our notation.

Then if $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ we have

$$\|\mathbf{A}\|_F^2 - \|\hat{\mathbf{A}}_{(k)}\|_F^2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2.$$

Theorem 104 establishes that the condition needed for **Theorem 98** can be achieved with many low-rank approximation algorithms, including e.g. block Krylov subspace methods, sketching methods, and sampling methods [8, 9, 27, 79, 85, 107, 132]. Further, by **Theorem 96** we know that $\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|_F^2 \leq \|\mathbf{A}\|_F^2 - \|\hat{\mathbf{A}}_{(k)}\|_F^2$, which shows that if (5.12) is satisfied then $\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|_F^2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_F^2$. That is, $\hat{\mathbf{A}}_{(k)}$ is a near-optimal rank k approximation.

We begin by recalling a fact about the best rank k approximation to $\mathbf{A}$ constrained to $\text{range}(\mathbf{Q})$.

Lemma 105. *Let $\mathbf{Q}$ be an orthonormal basis. Then,*

$$\|\mathbf{B} - \mathbf{Q}(\mathbf{Q}^T\mathbf{B})_{(k)}\|_F = \min_{\mathbf{C}: \text{rank}(\mathbf{C}) \leq k} \|\mathbf{B} - \mathbf{Q}\mathbf{C}\|_F^2, \quad (5.13)$$

and

$$\|\mathbf{B} - \mathbf{Q}(\mathbf{Q}^T\mathbf{B}\mathbf{Q})_{(k)}\mathbf{Q}^T\|_F = \min_{\mathbf{C}: \text{rank}(\mathbf{C}) \leq k} \|\mathbf{B} - \mathbf{Q}\mathbf{C}\mathbf{Q}^T\|_F^2. \quad (5.14)$$

Proof. (5.13) was proven in [48, Theorem 3.5] and (5.14) is proven in a similar fashion: Let $\mathbf{C}$ be any matrix so that $\text{rank}(\mathbf{C}) \leq k$. Notice that

$$\langle \mathbf{B} - \mathbf{Q}\mathbf{Q}^T\mathbf{B}\mathbf{Q}\mathbf{Q}^T, \mathbf{Q}\mathbf{Q}^T\mathbf{B}\mathbf{Q}\mathbf{Q}^T - \mathbf{Q}\mathbf{C}\mathbf{Q}^T \rangle = 0$$

Therefore, using the Pythagorean theorem we obtain

$$\begin{aligned} \|\mathbf{B} - \mathbf{Q}\mathbf{C}\mathbf{Q}^T\|_F^2 &= \|\mathbf{B} - \mathbf{Q}\mathbf{Q}^T\mathbf{B}\mathbf{Q}\mathbf{Q}^T + \mathbf{Q}\mathbf{Q}^T\mathbf{B}\mathbf{Q}\mathbf{Q}^T - \mathbf{Q}\mathbf{C}\mathbf{Q}^T\|_F^2 = \\ &= \|\mathbf{B} - \mathbf{Q}\mathbf{Q}^T\mathbf{B}\mathbf{Q}\mathbf{Q}^T\|_F^2 + \|\mathbf{Q}^T\mathbf{B}\mathbf{Q} - \mathbf{C}\|_F^2. \end{aligned}$$

Thus, to minimize $\|\mathbf{B} - \mathbf{Q}\mathbf{C}\mathbf{Q}^T\|_F^2$ we should choose $\mathbf{C} = (\mathbf{Q}^T\mathbf{B}\mathbf{Q})_{(k)}$. $\square$

With **Theorem 105** at hand, we can show that error of the rank k truncated Nyström approximation is sandwiched between the error of two projection based rank k approximations.

Lemma 106. *Let $\mathbf{Q}$ be an orthonormal basis and let $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ and*

suppose $\mathbf{A} \succeq \mathbf{0}$. Then the following holds

$$\begin{aligned} \|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2 &\leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}\mathbf{Q}})_{(k)}\|_F^2 \leq \|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\|_F^2 \leq \\ \|\mathbf{A}\|_F^2 - \|\widehat{\mathbf{A}}_{(k)}\|_F^2 &= \|\mathbf{A} - (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}})_{(k)}\|_F^2 \leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2. \end{aligned}$$

Proof. The first inequality is immediate from (5.13) since $(\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}\mathbf{Q}})_{(k)}$ is a rank k approximation whose range is contained in $\text{range}(\mathbf{A}\mathbf{Q})$. By a similar argument, the second inequality is immediate from (5.14) since $\widehat{\mathbf{A}}_{(k)}$ is a rank k approximation whose range and co-range is contained in $\text{range}(\mathbf{A}\mathbf{Q})$. The third inequality is a consequence of Lemma 96 for $p = 2$ since $\mathbf{A} \succeq \widehat{\mathbf{A}}_{(k)} \succeq \mathbf{0}$.

To prove the equality, note that by Theorem 103 $\widehat{\mathbf{A}} = \mathbf{A}^{1/2}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2} = (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})^T(\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})$. Hence, $\widehat{\mathbf{A}}_{(k)} = (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}^T(\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}$. Then observe that there always exists an orthogonal projection $\mathbf{P}$ so that $\text{range}(\mathbf{P}) \subseteq \text{range}(\mathbf{A}^{1/2}\mathbf{Q})$ and

$$\widehat{\mathbf{A}}_{(k)} = (\mathbf{P}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})^T(\mathbf{P}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2}) = (\mathbf{P}\mathbf{A}^{1/2})^T(\mathbf{P}\mathbf{A}^{1/2}).$$

Hence,

$$\begin{aligned} \|\mathbf{A}\|_F^2 - \|\widehat{\mathbf{A}}_{(k)}\|_F^2 &= \|\mathbf{A}\|_F^2 - \|(\mathbf{P}\mathbf{A}^{1/2})^T(\mathbf{P}\mathbf{A}^{1/2})\|_F^2 = \|\mathbf{A}\|_F^2 - \|\mathbf{P}\mathbf{A}\mathbf{P}\|_F^2 \\ &= \|\mathbf{A} - \mathbf{P}\mathbf{A}\mathbf{P}\|_F^2. \end{aligned}$$

Finally, noting that $\mathbf{P}\mathbf{A}\mathbf{P} = (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}})_{(k)}$ yields the desired equality.

For the last inequality in Theorem 106, we let $\bar{\mathbf{P}}$ be an orthogonal projection so that $\text{range}(\bar{\mathbf{P}}) \subseteq \text{range}(\mathbf{Q})$ and $\bar{\mathbf{P}}\mathbf{A} = (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}$. Note that $\mathbf{A}^{1/2}\bar{\mathbf{P}}\mathbf{A}^{1/2}$ is a rank k approximation to $\mathbf{A}$ whose range and co-range are both contained in $\text{range}(\mathbf{A}^{1/2}\mathbf{Q})$. By (5.14) we have that, as required,

$$\begin{aligned} \|\mathbf{A} - (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}})_{(k)}\|_F^2 &\leq \|\mathbf{A} - \mathbf{A}^{1/2}\bar{\mathbf{P}}\mathbf{A}^{1/2}\|_F^2 = \\ \|\mathbf{A} - \bar{\mathbf{P}}\mathbf{A}\|_F^2 &\leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2. \end{aligned} \quad \square$$

Proof of Theorem 104. The proof of our main result in this section follows immediately from Lemma 106. In particular, we have since

$$\|\mathbf{A}\|_F^2 - \|\widehat{\mathbf{A}}_{(k)}\|_F^2 \leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2 = \|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_F^2. \quad \square$$

Remark 107. We remark on a few additional consequence of Lemma 106 that may be of independent interest.

1. The lemma implies that $\|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2 \leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2$ and that $\|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}\mathbf{Q}})_{(k)}\|_F^2 \leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{Q}})_{(k)}\|_F^2$. Hence, if we approximate $\mathbf{A}$ via either a one-sided or two-sided projection onto $\mathbf{A}\mathbf{Q}$, the error is always better than if we simply project onto $\mathbf{Q}$. In other words, we establish an intuitive fact: that subspace iteration monotonically improves Frobenius norm low-rank approximation error. Via a change of basis, a similar result is true for rectangular matrices. One can show that $\|\mathbf{A} - (\mathbf{P}_{(\mathbf{A}\mathbf{A}^T)^{p/2}\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2 \leq \|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F^2$ for any positive integer p .
2. If one has obtained an orthonormal basis $\mathbf{Q}$ with ℓ columns so that $\|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(k)}\|_F \leq \varepsilon$ then $\|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A}\mathbf{P}_{\mathbf{A}\mathbf{Q}})_{(k)}\|_F \leq \varepsilon$. This implies that a relative error low-rank approximation guarantee for one-sided projection translates to a guarantee for two-sided projection, at the cost of at most ℓ extra matrix-vector products with $\mathbf{A}$ to form $\mathbf{A}\mathbf{Q}$.
3. Given a basis $\mathbf{Q}$, we require ℓ matrix-vector multiplications with $\mathbf{A}$ to either form the one-sided projection $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$ or to form the rank k truncated Nystrom approximation $\hat{\mathbf{A}}_{(k)}$ where $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$. However, truncated Nystrom approximation always provides better error in the Frobenius norm, so should be preferred.

5.3.2 Nuclear norm guarantees

In this section we establish similar guarantees as in the previous section, but for the nuclear norm. Specifically, we prove the following theorem.

Theorem 108 (Theorem 89 restated). *Let $\mathbf{A} \succeq \mathbf{0}$ and let $\mathbf{Q}$ be an orthonormal basis so that, for $\varepsilon \geq 0$,*

$$\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*.$$

Then if $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ we have

$$\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|_* \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*.$$

Proof. Since $\mathbf{A} \succeq \hat{\mathbf{A}}_{(k)} \succeq \mathbf{0}$ we know that

$$\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|_* = \text{tr}(\mathbf{A} - \hat{\mathbf{A}}_{(k)}).$$

Then, since $\widehat{\mathbf{A}}_{(k)} = (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}^T(\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}$ by [Theorem 103](#), we have

$$\text{tr}(\mathbf{A} - \widehat{\mathbf{A}}_{(k)}) = \|\mathbf{A}^{1/2}\|_F^2 - \|(\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}\|_F^2 = \|\mathbf{A}^{1/2} - (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}\|_F^2.$$

Choose an orthogonal projector $\mathbf{P}$ so that $\text{range}(\mathbf{P}) \subseteq \text{range}(\mathbf{Q})$ and $\mathbf{P}\mathbf{A} = (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$. Finally, by [Lemma 106](#) and [Lemma 105](#) we have

$$\begin{aligned} \|\mathbf{A}^{1/2} - (\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}\|_F^2 &\leq \|\mathbf{A}^{1/2} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A}^{1/2})_{(k)}\|_F^2 \leq \\ \|\mathbf{A}^{1/2} - \mathbf{P}\mathbf{A}^{1/2}\|_F^2 &= \text{tr}(\mathbf{A}) - \text{tr}(\mathbf{P}\mathbf{A}\mathbf{P}) = \text{tr}(\mathbf{A}) - \text{tr}(\mathbf{P}\mathbf{A}) \leq \\ \|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_* &\leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_*, \end{aligned}$$

which yields the desired inequality. $\square$

5.3.3 Operator norm guarantees

In this section we consider the operator norm. When $\mathbf{Q}$ has exactly k columns, [\[116\]](#) establishes the following guarantee.

Theorem 109 ([\[116, Lemma 5.2\]](#)). *Let $\mathbf{A} \succeq \mathbf{0}$ and let $\mathbf{Q} \in \mathbb{R}^{n \times k}$ be an orthonormal basis so that, for some $\varepsilon \geq 0$,*

$$\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2.$$

Then if $\widehat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ we have

$$\|\mathbf{A} - \widehat{\mathbf{A}}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2.$$

Ideally, we would extend this guarantee to the case when $\mathbf{Q}$ has $\ell > k$ columns, as in [Theorems 104](#) and [108](#) for the Frobenius and nuclear norms. I.e., we might hope to prove that $\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2$ implies $\|\mathbf{A} - \widehat{\mathbf{A}}_{(k)}\|_2 \leq (1 + \varepsilon)\|\mathbf{A} - \mathbf{A}_{(k)}\|_2$. Interestingly, however, we show that doing so is impossible. In particular, consider the following counterexample.

$$\mathbf{A} = \begin{bmatrix} 9.627 & 1.538 & -0.717 & 1.418 & -0.309 \\ 1.538 & 8.084 & 1.904 & -1.868 & 0.573 \\ -0.717 & 1.904 & 1.353 & -1.538 & -1.300 \\ 1.418 & -1.868 & -1.538 & 2.534 & 0.169 \\ -0.309 & 0.573 & -1.300 & 0.169 & 6.055 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.15)$$

For these matrices, $\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(2)}\|_2 \approx (1 + 2.59 \times 10^{-8})\|\mathbf{A} - \mathbf{A}_{(2)}\|_2$ whereas $\|\mathbf{A} - \hat{\mathbf{A}}_{(2)}\|_2 \approx (1 + 5.75 \times 10^{-3})\|\mathbf{A} - \mathbf{A}_{(2)}\|_2$, so $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$ is a better rank k approximation to $\mathbf{A}$ compared to $\hat{\mathbf{A}}_{(k)}$. As guaranteed by [Theorem 109](#), we do at least have that $3.75 \approx \|\mathbf{A} - \hat{\mathbf{A}}\|_2 < \|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_2 \approx 6.24$. Via the same counterexample, we also have the following.

Remark 110. In [Section 5.3.1](#) we showed that Frobenius norm low-rank approximation error decreases monotonically in the number of subspace iterations (see [Theorem 107](#)). The same is not true in the operator norm. To see this, let $\mathbf{A}$ and $\mathbf{Q}$ be as in [\(5.15\)](#). We can check that

$$\|\mathbf{A} - (\mathbf{P}_{\mathbf{Q}}\mathbf{A})_{(2)}\|_2 \approx 6.449 < 6.455 \approx \|\mathbf{A} - (\mathbf{P}_{\mathbf{A}\mathbf{Q}}\mathbf{A})_{(2)}\|_2.$$

5.3.4 Eigenvalue guarantee

We conclude the theoretical part of the paper by noting a guarantee for eigenvalue estimation. Specifically, if we have a basis $\mathbf{Q}$ so that top k singular values of $\mathbf{Q}\mathbf{Q}^T\mathbf{A}$ are estimates of the eigenvalues of $\mathbf{A}$, then the eigenvalues of the Nyström approximation $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$ can only be better estimates.

Theorem 111. Let $\mathbf{A} \succeq \mathbf{0}$ and let $\mathbf{Q}$ be an orthonormal basis so that $\lambda_i - \varepsilon_i \leq \sigma_i(\mathbf{Q}^T\mathbf{A}) \leq \lambda_i$ for $i = 1, \dots, k$ and for $\varepsilon_1, \dots, \varepsilon_k \geq 0$. Then if $\hat{\lambda}_i$ is the i^{th} eigenvalue of $\hat{\mathbf{A}} = \mathbf{A}\mathbf{Q}(\mathbf{Q}^T\mathbf{A}\mathbf{Q})^\dagger\mathbf{Q}^T\mathbf{A}$, we have $\lambda_i - \varepsilon_i \leq \hat{\lambda}_i \leq \lambda_i$ for $i = 1, \dots, k$.

Proof. Notice that, by [Theorem 103](#), we have

$$\mathbf{Q}^T\hat{\mathbf{A}} = (\mathbf{Q}^T\mathbf{A}^{1/2}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}})\mathbf{A}^{1/2} = \mathbf{Q}^T\mathbf{A}.$$

Therefore, by applying a standard singular value inequality [57, p.452] we have

$$\hat{\lambda}_i = \sigma_i(\hat{\mathbf{A}}) \geq \sigma_i(\mathbf{Q}^T\hat{\mathbf{A}}) = \sigma_i(\mathbf{Q}^T\mathbf{A}) \geq \lambda_i - \varepsilon_i.$$

We complete the proof by noting that $\hat{\lambda}_i \leq \lambda_i$ because $\hat{\mathbf{A}} \preceq \mathbf{A}$ [57, Corollary 7.7.4 (c)]. $\square$

5.4 Numerical experiments

In this section, we numerically verify our theoretical results. Experiments were performed on a MacBook Pro in MATLAB (v. 2020a) and scripts to reproduce our figures are available at <https://github.com/davpersson/funNystrom-v2>.

In all our experiments, we begin with computing an orthonormal basis $\mathbf{Q}$ with $\ell \geq k$ columns. In Sections 5.4.1-5.4.3 we outline three different algorithms for doing so. Next, using the orthonormal basis $\mathbf{Q}$, we construct a Nyström approximation $\hat{\mathbf{A}}$ as defined in (5.1) and the projection based approximation $\mathbf{Q}\mathbf{Q}^T\mathbf{A}$. Note that once $\mathbf{Q}$ is computed, constructing the two approximations comes at the same computational cost. Then, we truncate $\hat{\mathbf{A}}$ and $\mathbf{Q}\mathbf{Q}^T\mathbf{A}$ to rank k to obtain $\hat{\mathbf{A}}_{(k)}$ and $(\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}$. Finally, we compare the following quantities

$$\begin{aligned}\varepsilon_{\text{projection}} &= \frac{\|\mathbf{A} - (\mathbf{Q}\mathbf{Q}^T\mathbf{A})_{(k)}\|}{\|\mathbf{A} - \mathbf{A}_{(k)}\|} - 1; \\ \varepsilon_{\text{Nyström}} &= \frac{\|\mathbf{A} - \hat{\mathbf{A}}_{(k)}\|}{\|\mathbf{A} - \mathbf{A}_{(k)}\|} - 1; \\ \varepsilon_{\text{funNyström}} &= \frac{\|f(\mathbf{A}) - f(\hat{\mathbf{A}}_{(k)})\|}{\|f(\mathbf{A}) - f(\mathbf{A}_{(k)})\|} - 1,\end{aligned}$$

where $\|\cdot\| = \|\cdot\|_*$, $\|\cdot\|_F$ or $\|\cdot\|_2$. For comparing accuracy in estimating eigenvalues we use the alternative metrics:

$$\begin{aligned}\varepsilon_{\text{projection}} &= \max_{i=1,\dots,k} \left\{ \frac{\lambda_i - \sigma_i(\mathbf{Q}^T\mathbf{A})}{\lambda_i} \right\}; \\ \varepsilon_{\text{Nyström}} &= \max_{i=1,\dots,k} \left\{ \frac{\lambda_i - \hat{\lambda}_i}{\lambda_i} \right\}; \\ \varepsilon_{\text{funNyström}} &= \max_{i=1,\dots,k} \left\{ \frac{f(\lambda_i) - f(\hat{\lambda}_i)}{f(\lambda_i)} \right\}.\end{aligned}$$

Our theory suggests that, for the Frobenius norm, nuclear norm, or for eigenvalue estimation, $\varepsilon_{\text{projection}} \geq \varepsilon_{\text{Nyström}} \geq \varepsilon_{\text{funNyström}}$. For the operator norm, we expect the second inequality to hold, but we have shown a counterexample to the first in Section 5.3.3. However, in our experiments we generally observe that $\varepsilon_{\text{projection}} \geq \varepsilon_{\text{Nyström}}$ even when $\|\cdot\| = \|\cdot\|_2$.

5.4.1 Column subset selection

In this experiment we compute the orthonormal basis $\mathbf{Q}$ using the randomly pivoted Cholesky algorithm [22, Algorithm 2.1]. In this setting, $\mathbf{Q} = [\mathbf{e}_{i_1} \ \dots \ \mathbf{e}_{i_\ell}]$ where $\{i_1, \dots, i_\ell\} \subseteq \{1, \dots, n\}$ is an index set returned by the algorithm, and $\mathbf{e}_i$ is the i^{th} standard basis vector. We set $k = 10$ and $\ell = k + q$ for $q = 0, \dots, 6$. We let

$\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ be defined by

$$\mathbf{A}_{ij} = \left(\left(\frac{i}{1000} \right)^{10} + \left(\frac{j}{1000} \right)^{10} \right)^{\frac{1}{10}},$$

where $\mathbf{A}_{ij}$ denotes the (i, j) -entry of $\mathbf{A}$. This example is inspired by the numerical experiments on column subset selection in [28]. We set the matrix function to be $f(x) = \frac{x}{x+1}$, which is operator monotone. The results are presented in Figure 5.1, which shows that $\varepsilon_{\text{projection}} \geq \varepsilon_{\text{Nyström}} \geq \varepsilon_{\text{funNyström}}$ for all norms and q we consider, which confirms our theoretical results.

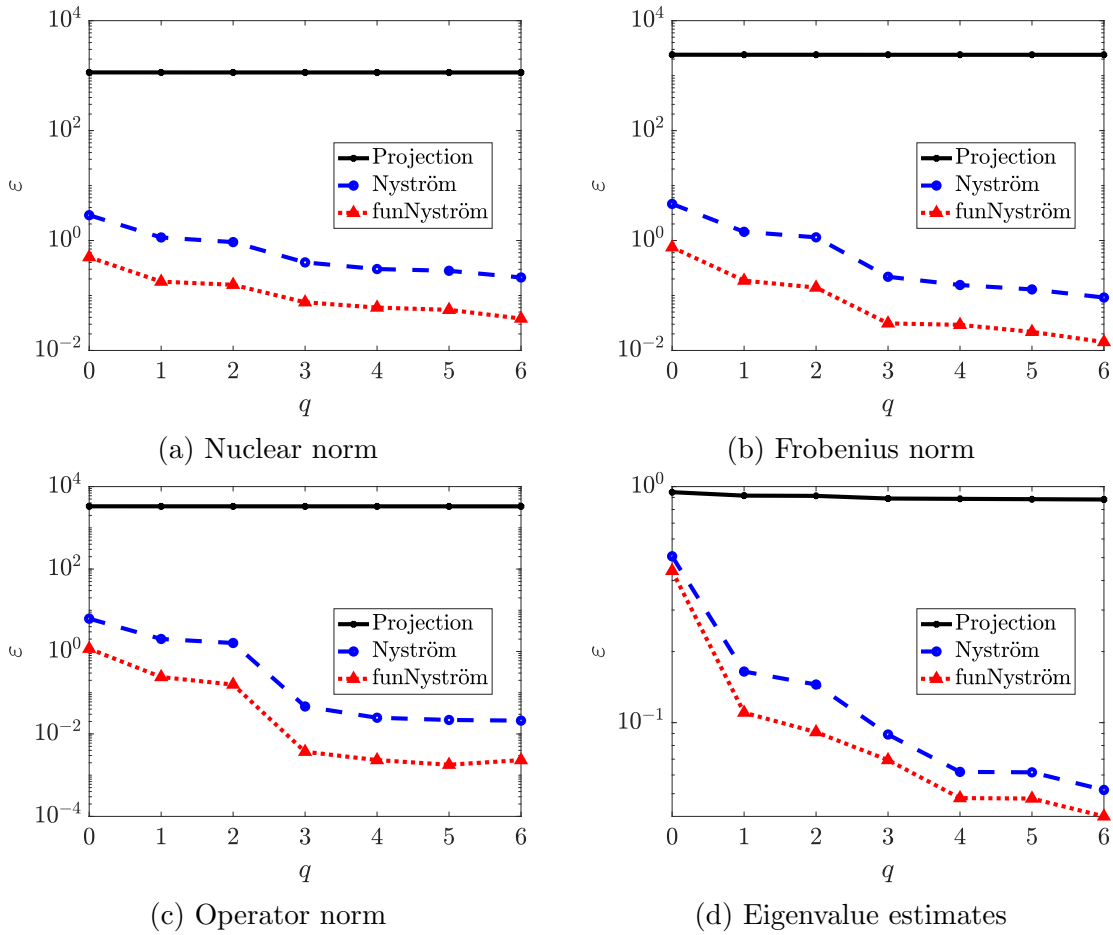


Figure 5.1: Comparing $\varepsilon_{\text{projection}}$, $\varepsilon_{\text{Nyström}}$, and $\varepsilon_{\text{funNyström}}$ for column subset selection. Note that $\varepsilon_{\text{projection}}$ is significantly worse than the $\varepsilon_{\text{Nyström}}$ and $\varepsilon_{\text{funNyström}}$ since the orthogonal projection $\mathbf{Q}\mathbf{Q}^T$ zeros out all except ℓ rows of $\mathbf{A}$. In contrast, the Nyström approximation $\hat{\mathbf{A}} = \mathbf{A}^{1/2}\mathbf{P}_{\mathbf{A}^{1/2}\mathbf{Q}}\mathbf{A}^{1/2}$ effectively performs half a step of subspace iteration on $\mathbf{Q}$, giving a better approximation.

5.4.2 Krylov iteration

In this experiment we set $\mathbf{Q}$ to be an orthonormal basis for the Krylov subspace $\text{range}\left(\begin{bmatrix} \mathbf{\Omega} & \mathbf{A}\mathbf{\Omega} & \dots & \mathbf{A}^q\mathbf{\Omega} \end{bmatrix}\right)$ where $\mathbf{\Omega}$ is a random $3000 \times k$ matrix whose entries are independent identically distributed Gaussian random variables with mean 0 and variance 1. We set $k = 10$ and vary $q = 0, 1, \dots, 6$. We let $\mathbf{A}$ be a 3000×3000 SPSD matrix whose eigenvalues are $\lambda_i = i^{-1}$ for $i = 1, \dots, n$. We set the matrix function to be $f(x) = \log(1 + x)$, which is operator monotone. The results are presented in [Figure 5.2](#). Again, for all norms and all choices of q , we see that $\varepsilon_{\text{projection}} \geq \varepsilon_{\text{Nyström}} \geq \varepsilon_{\text{funNyström}}$.

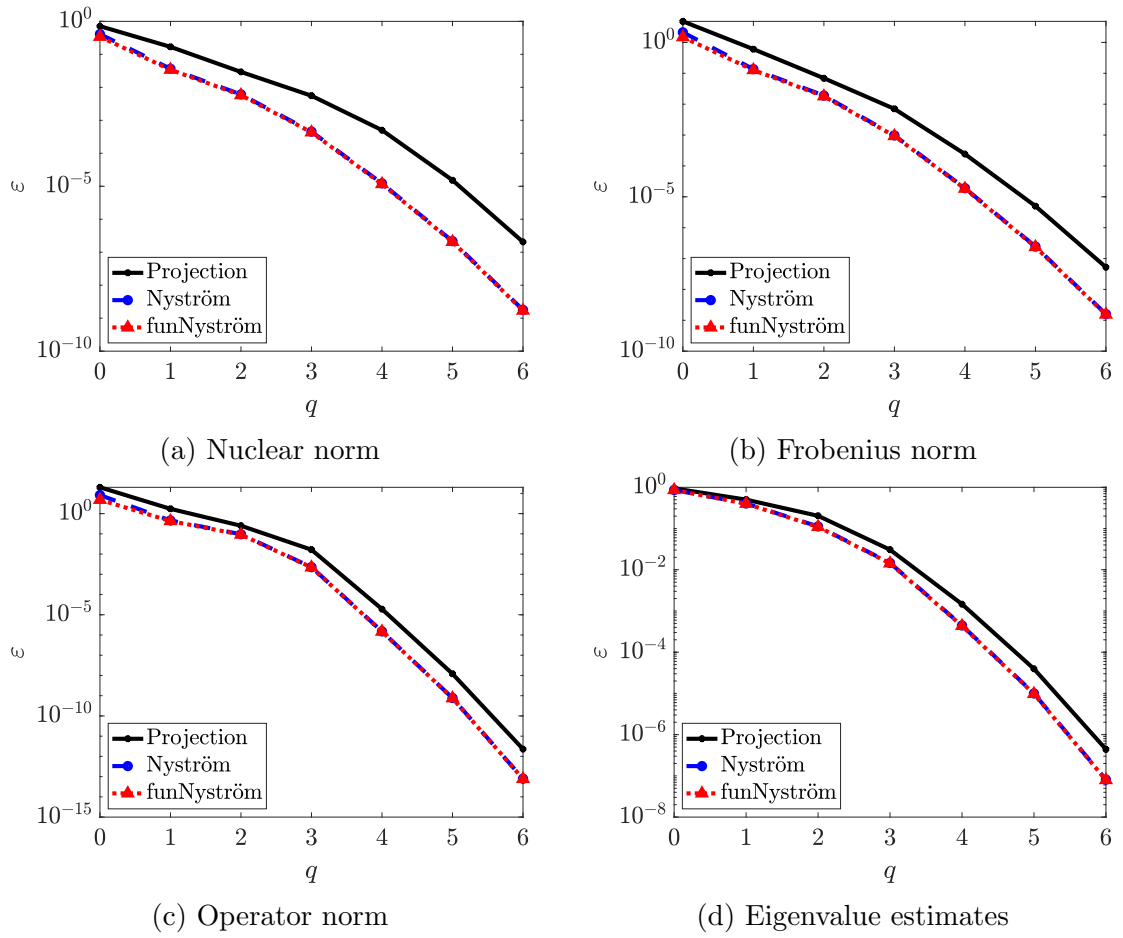


Figure 5.2: Comparing $\varepsilon_{\text{projection}}$, $\varepsilon_{\text{Nyström}}$, and $\varepsilon_{\text{funNyström}}$ for Krylov iteration.

5.4.3 Subspace iteration

Finally, we set $\mathbf{Q}$ to be an orthonormal basis for $\text{range}(\mathbf{A}^q\mathbf{\Omega})$ where $\mathbf{\Omega}$ is a random $3000 \times k$ matrix whose entries are independent identically distributed Gaussian ran-

dom variables with mean 0 and variance 1. We set $k = 10$ and vary $q = 0, 1, \dots, 6$. We let $\mathbf{A}$ be a 3000×3000 SPSP matrix whose eigenvalues are $\lambda_i = e^{-i}$ for $i = 1, \dots, n$. We set the matrix function to be $f(x) = x^{1/2}$, which is operator monotone. The results are presented in Figure 5.3. Once again, for all norms and all choices of q , we see that $\varepsilon_{\text{projection}} \geq \varepsilon_{\text{Nyström}} \geq \varepsilon_{\text{funNyström}}$.

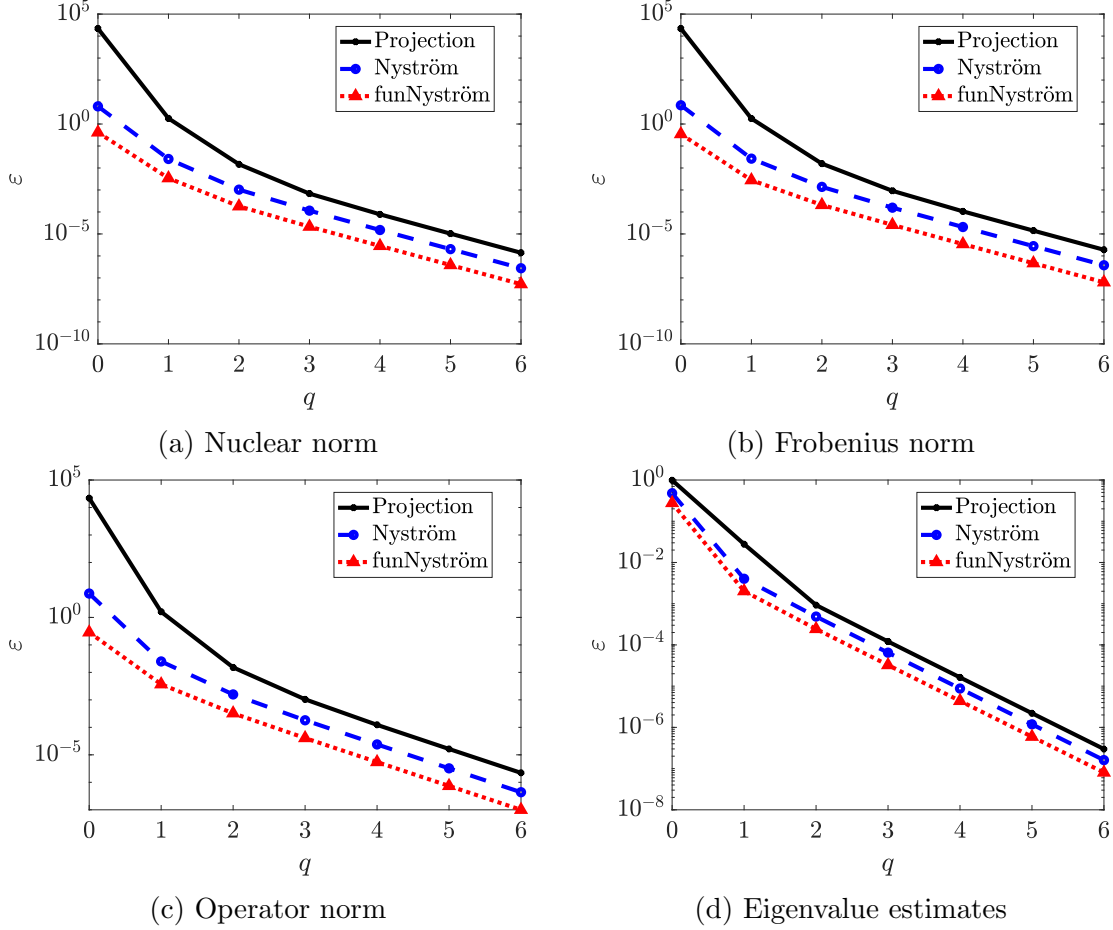


Figure 5.3: Comparing $\varepsilon_{\text{projection}}$, $\varepsilon_{\text{Nyström}}$, and $\varepsilon_{\text{funNyström}}$ for subspace iteration.

5.5 Counterexamples

In this section, we give examples that demonstrate that the assumptions in our results in Section 5.2 are necessary. Recall that in Section 5.2, $\hat{\mathbf{A}}$ is a SPSP rank k approximation of $\mathbf{A}$, not necessarily the Nyström approximation. We define

$$\varepsilon_{\text{original}} = \frac{\|\mathbf{A} - \hat{\mathbf{A}}\|}{\|\mathbf{A} - \mathbf{A}_{(k)}\|} - 1; \quad \varepsilon_{\text{function}} = \frac{\|f(\mathbf{A}) - f(\hat{\mathbf{A}})\|}{\|f(\mathbf{A}) - f(\mathbf{A}_{(k)})\|} - 1,$$

where $\|\cdot\| = \|\cdot\|_*, \|\cdot\|_F$ or $\|\cdot\|_2$. The results in [Section 5.2](#) show that if (i) $\mathbf{A} \succeq \hat{\mathbf{A}}$, and (ii) f is a continuous operator monotone function with $f(0) \geq 0$, then $\varepsilon_{\text{function}} \leq \varepsilon_{\text{original}}$. We might ask if we really have to assume that $\mathbf{A} \succeq \hat{\mathbf{A}}$ or if it is sufficient to only assume that $\hat{\mathbf{A}}$ is SPSPD. Similarly, we might ask if f has to be operator monotone, or if it is sufficient to only assume that f is non-decreasing, continuous, and concave.

We provide counter examples to show that, in most cases, the answer to these questions is no: our current assumptions are necessary. Specifically, we show instances where $\varepsilon_{\text{function}} > \varepsilon_{\text{original}}$ for specific norms when the assumptions are relaxed. Our results are summarized in [Table 5.1](#) in [Section 5.1](#).

Example 112 (Operator monotone is necessary for nuclear norm). *Let $\mathbf{A} = \begin{bmatrix} 1.1 & \\ & 0.1 \end{bmatrix}$ and $\hat{\mathbf{A}} = \mathbf{V}\mathbf{V}^T$ where $\mathbf{V} = \begin{bmatrix} 1 \\ 0.095 \end{bmatrix}$. With $k = 1$ and $f(x) = \min\{1, x\}$, we have that $\varepsilon_{\text{function}} > 1.158\varepsilon_{\text{original}}$ in the nuclear norm. For this example, $\mathbf{A} \succeq \hat{\mathbf{A}}$ and f is non-decreasing and concave, but not operator monotone.*

Example 113 ($\hat{\mathbf{A}} \preceq \mathbf{A}$ or operator monotone is necessary for operator norm). *Let $\mathbf{A} = \begin{bmatrix} 1.01 & \\ & 0.01 \end{bmatrix}$ and $\hat{\mathbf{A}} = \mathbf{V}\mathbf{V}^T$ where $\mathbf{V} = \begin{bmatrix} 1.01 \\ 0.01 \end{bmatrix}$. With $k = 1$ and $f(x) = \min\{1, x\}$, we have $\varepsilon_{\text{function}} > 1.402\varepsilon_{\text{original}}$ in the operator norm. For this example, $\mathbf{A} \not\succeq \hat{\mathbf{A}}$ and f is non-decreasing and concave, but not operator monotone.*

For the two examples above, we can confirm that $f(x) = \min\{1, x\}$ is not operator monotone because it is not differentiable, which violates [Theorem 94](#) ((ii)).

Example 114 ($\hat{\mathbf{A}} \preceq \mathbf{A}$ is necessary for Frobenius and nuclear norms). *Let $\mathbf{A} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$ and $\hat{\mathbf{A}} = \mathbf{V}\mathbf{V}^T\mathbf{A}\mathbf{V}\mathbf{V}^T$ where $\mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Then, with $k = 1$ and $f(x) = \sqrt{x}$ we have $\varepsilon_{\text{function}} > 1.095\varepsilon_{\text{original}}$ in the nuclear norm and $\varepsilon_{\text{function}} > 1.049\varepsilon_{\text{original}}$ in the Frobenius norm. For this example, f is operator monotone, but it can be checked that $\mathbf{A} \not\succeq \hat{\mathbf{A}}$.*

Despite the above counterexamples, gaps still remain in our understanding of what assumptions are necessary for a good low-rank approximation to $\mathbf{A}$ to imply a good low-rank approximation for $f(\mathbf{A})$. For example, we have not found a counter example for the operator and Frobenius norms when $\mathbf{A} \succeq \hat{\mathbf{A}}$ and f is non-decreasing and concave, but not operator monotone. Additionally, we emphasize that our counterexamples do not rule our “approximate” versions of our results on funNyström ([Theorems 97, 98 and 100](#)). For example, while $\varepsilon_{\text{function}}$ can be larger than $\varepsilon_{\text{original}}$, it might be possible to prove that $\varepsilon_{\text{function}} \leq 2 \cdot \varepsilon_{\text{original}}$ under relaxed assumptions.

5.6 Conclusion

In this work, we significantly generalize a result of Persson and Kressner [95] by showing that near-optimal Nyström approximations imply near-optimal funNyström approximations, independently of how the Nyström approximation was obtained. This implies that any method that computes good Nyström approximations can be used to obtain good low-rank approximations of operator monotone matrix functions. We have also shown that good low-rank projections imply good Nyström approximations. This allows us to translate low-rank approximation guarantees for many common algorithms into guarantees for Nyström approximation, and thus for funNyström approximation.

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